Learning to Lens

An Introduction to General Relativity
and Strong Gravitational Lensing

with Mathematica

Based on the published works of

P. Schneider, J. Ehlers & E. E. Falco, *Gravitational Lenses* (1992)

R. Narayan & M. Bartelmann, *Lectures on Gravitational Lensing* (1997)

A. O. Petters, H. Levine & J. Wambsganss, *Singularity Theory and Gravitational Lensing* (2001)

S. Carroll, *Spacetime and Geometry* (2004)

A. B. Congdon & C. R. Keeton, *Principles of Gravitational Lensing* (2018)

M. Meneghetti, *Introduction to Gravitational Lensing* (2021)

P. Saha, D. Sluse, J. Wagner & L. L. R. Williams, *Essentials of Strong Gravitational Lensing* (2024)

— Instructor Edition (with Solutions) —

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with

Claude (Anthropic)
via Claude Code

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About This Document

This tutorial suite was produced by **Rodrigo Córdova Rosado** (Harvard–Smithsonian Center for Astrophysics) with the assistance of **Claude** (Anthropic), an AI language model, via **Claude Code** — Anthropic's command-line tool for collaborative software engineering and technical writing.

The physics content in these notes draws from the following published textbooks and review articles:

- S. Carroll, *Spacetime and Geometry: An Introduction to General Relativity* (Addison Wesley, 2004)
- A. B. Congdon & C. R. Keeton, *Principles of Gravitational Lensing* (Springer, 2018)
- R. Narayan & M. Bartelmann, *Lectures on Gravitational Lensing* (1997; arXiv:astro-ph/9606001)
- P. Saha, D. Sluse, J. Wagner & L. L. R. Williams, *Essentials of Strong Gravitational Lensing*, Space Science Reviews **220**, 12 (2024)
- M. Meneghetti, *Introduction to Gravitational Lensing: With Python Examples* (Springer, 2021)
- P. Schneider, J. Ehlers & E. E. Falco, *Gravitational Lenses* (Springer, 1992)
- A. O. Petters, H. Levine & J. Wambsganss, *Singularity Theory and Gravitational Lensing* (Birkhäuser, 2001)

The workflow was as follows: the author directed Claude to co-write the LaTeX lecture notes, Wolfram Mathematica scripts, and interactive notebooks, drawing on the reference texts above. Claude drafted the text and code; the author reviewed, revised, and approved all content.

Importantly, all mathematical derivations and symbolic results in these notes have been verified deterministically using Wolfram Mathematica's computer algebra system — not generated probabilistically by an LLM. Every equation, identity, and worked solution was checked by running the companion `.wl` scripts (via `wolframscript`), which perform exact symbolic computation. The Mathematica verification outputs are reproducible and accompany each module in the repository.

The AI assistant was used for:

- Drafting expository prose and pedagogical explanations
- Structuring the curriculum and organizing modules
- Writing Mathematica code (subsequently executed and verified)
- Formatting LaTeX and managing cross-references

The AI assistant was **not** used as a source of mathematical truth. All physics content derives from the published works listed above, and all symbolic computations are verified by Mathematica.

This is a **living document**. Additional details, examples, and lessons will be added over time as the author continues to learn and teach this material. The most current version is always available at: https://github.com/rodelcr/Learning_to_Lens

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Notation Guide

Symbol	Meaning	Units / Notes
Coordinates and Angles		
$\boldsymbol{\theta}$	Image position (angular)	radians or arcsec
$\boldsymbol{\beta}$	Source position (angular)	radians or arcsec
$\boldsymbol{\alpha}$	Reduced deflection angle	radians
$\hat{\alpha}$	Physical deflection angle	radians
$\boldsymbol{\xi}$	Position in lens plane	physical (kpc, Mpc)
Distances		
D_{d}	Observer–lens angular diameter distance	Mpc
D_{s}	Observer–source angular diameter distance	Mpc
D_{ds}	Lens–source angular diameter distance	Mpc
Lensing Quantities		
θ_{E}	Einstein radius	arcsec
Σ_{cr}	Critical surface mass density	$\mathrm{g\,cm^{-2}}$ or $M_\odot\,\mathrm{pc}^{-2}$
κ	Convergence $= \Sigma/\Sigma_{\mathrm{cr}}$	dimensionless
γ	Shear magnitude	dimensionless
ψ	Effective lensing potential	dimensionless
μ	Magnification	dimensionless
$\mathcal{A}$	Jacobian (inverse magnification) matrix	2×2
Cosmology		
H_0	Hubble constant	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$
Ω_{m}	Matter density parameter	dimensionless
Ω_Λ	Dark energy density parameter	dimensionless
GR / Metric		
$g_{\mu\nu}$	Metric tensor	
$\Gamma^{\alpha}{}_{\mu\nu}$	Christoffel symbols	
$R^{\alpha}{}_{\beta\mu\nu}$	Riemann curvature tensor	
$R_{\mu\nu}$	Ricci tensor	
R	Ricci scalar	

Mathematica files are cross-linked throughout these notes. Links formatted as `filename.nb` or `filename.wl` point to companion Mathematica notebooks and scripts in the `Mathematica/`

directory of this repository.

Part I

General Relativity Foundations

Chapter 1

Special Relativity and Tensor Basics

> *General relativity (GR) is Einstein's theory of space, time, and gravitation. At heart it is a very simple subject ... The essential idea is perfectly straightforward: while most forces of nature are represented by fields defined on spacetime, gravity is inherent in spacetime itself.* — Sean Carroll, *Spacetime and Geometry*

Before we can understand how gravity bends light, we need the language of special relativity and tensors. This chapter reviews the foundations: Minkowski spacetime, Lorentz transformations, four-vectors, and the index notation that will carry us through the rest of these notes. The emphasis is on building fluency with the notation and the key physical ideas — time dilation, length contraction, the invariant interval — so that the transition to curved spacetime in later modules feels natural.

Companion Mathematica files:

- `lorentz_transforms.wl` — symbolic construction and verification of Lorentz boost matrices
- `four_vectors.nb` — interactive visualizations of worldlines, light cones, and time dilation

1.1 From Newton to Einstein

In Newtonian mechanics, space and time are independent. The gravitational force between two masses M and m separated by a distance r is (cf. Carroll eq. 1.1)

$$\mathbf{F} = -\frac{GMm}{r^2}\,\hat{\mathbf{r}}\,, \tag{1.1}$$

and the gravitational potential Φ satisfies Poisson's equation,

$$\nabla^2\Phi = 4\pi G\rho\,. \tag{1.2}$$

In GR, both of these are replaced by statements about the curvature of spacetime. The key equation we will eventually arrive at is Einstein's field equation (Carroll eq. 1.5):

$$R_{\mu\nu} - \tfrac{1}{2}R\,g_{\mu\nu} = 8\pi G\,T_{\mu\nu}\,. \tag{1.3}$$

But we are getting ahead of ourselves. For now, we focus on *flat* spacetime — the arena of special relativity — and learn the tensor language needed to write down equations like (1.3).

1.2 Spacetime and the Invariant Interval

Special relativity (SR) unifies space and time into a single four-dimensional continuum called **Minkowski space**. An individual point in spacetime is an **event**, and the path of a particle through spacetime is its **worldline**.

Unlike Newtonian mechanics, SR has no absolute notion of simultaneity. Instead, the fundamental invariant is the **spacetime interval** between two events (Carroll eq. 1.10):

$$\boxed{(\Delta s)^2 = -(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} \tag{1.4}$$

where we work in **natural units** with $c = 1$ throughout this chapter (we will restore factors of c when needed for astrophysical applications).

Remark 1.2.1. We adopt the **mostly-plus** sign convention $(-, +, +, +)$, following Carroll. Some references (especially in particle physics) use the opposite convention $(+, -, -, -)$. Be careful when comparing equations across textbooks.

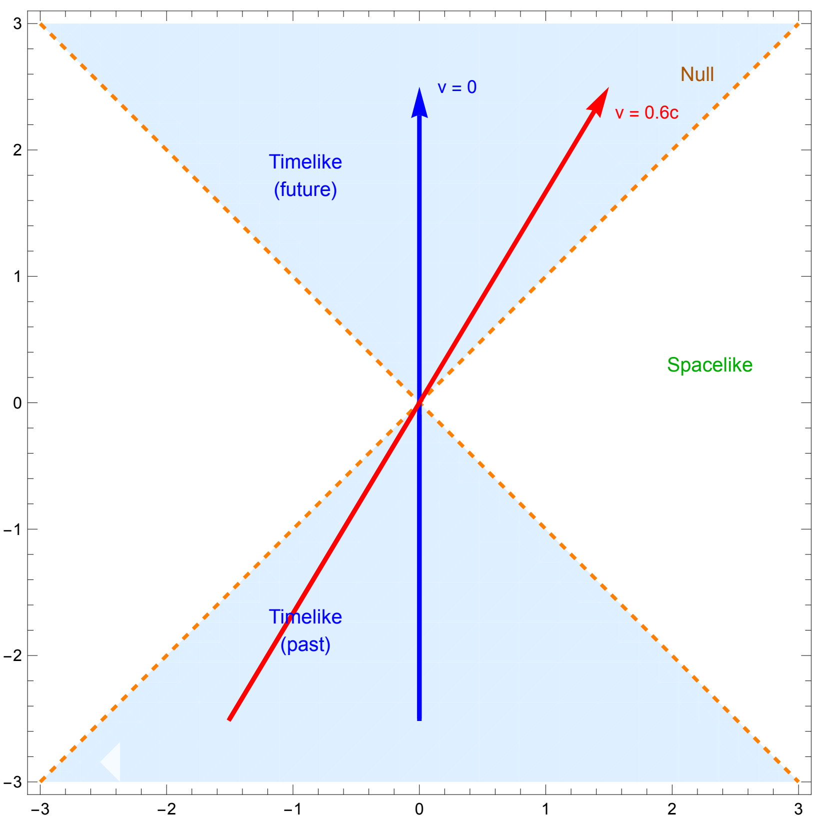


Figure 1.1: Spacetime diagram in 1+1 dimensions showing the light cone structure. The shaded (blue) region is timelike; the dashed orange lines are the light cone (null); the unshaded region is spacelike. The blue arrow is the worldline of a stationary particle; the red arrow is a particle moving at $v = 0.6c$. (cf. Carroll Fig. 1.5)

The interval classifies the separation between events:

- $(\Delta s)^2 < 0$: **timelike** separation (events can be causally connected)

- $(\Delta s)^2 = 0$: **null** or **lightlike** separation (connected by a light ray)
- $(\Delta s)^2 > 0$: **spacelike** separation (no causal connection possible)

The **proper time** $\Delta\tau$ along a timelike path is defined as (Carroll eq. 1.17)

$$(\Delta\tau)^2 = -(\Delta s)^2 = (\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2 \,. \tag{1.5}$$

This is the time measured by a clock traveling along the worldline. For an infinitesimal displacement, we write the **line element** (Carroll eq. 1.20):

$$ds^2 = \eta_{\mu\nu}\, dx^\mu\, dx^\nu \,, \tag{1.6}$$

where we have introduced the **Minkowski metric** and the **Einstein summation convention** (repeated indices are summed over).

1.3 Index Notation and the Minkowski Metric

Spacetime coordinates are labeled by Greek letters $(\mu, \nu, \rho, \sigma, \ldots)$ running from 0 to 3 (Carroll eq. 1.12):

$$x^\mu : \quad x^0 = t, \quad x^1 = x, \quad x^2 = y, \quad x^3 = z \,. \tag{1.7}$$

Latin indices $i, j, k, \ldots$ run over the spatial components $1, 2, 3$ only.

The **Minkowski metric** is the 4×4 matrix (Carroll eq. 1.15):

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \,. \tag{1.8}$$

Its inverse, $\eta^{\mu\nu}$, has the same components. The metric is used to **raise and lower indices**:

$$V_\mu = \eta_{\mu\nu}\, V^\nu, \qquad \omega^\mu = \eta^{\mu\nu}\, \omega_\nu \,. \tag{1.9}$$

Note that in Minkowski space this simply flips the sign of the time component: $V_0 = -V^0$, while $V_i = V^i$.

1.4 Lorentz Transformations

The coordinate transformations that leave the interval (1.6) invariant are called **Lorentz transformations**. They act on the coordinates as (Carroll eq. 1.25)

$$x^{\mu'} = \Lambda^{\mu'}{}_{\nu}\, x^\nu \,, \tag{1.10}$$

where the matrix Λ satisfies the defining condition (Carroll eq. 1.28):

$$\boxed{\eta = \Lambda^{\mathrm{T}}\, \eta\, \Lambda} \tag{1.11}$$

or equivalently $\eta_{\rho\sigma} = \Lambda^{\mu'}{}_{\rho}\, \Lambda^{\nu'}{}_{\sigma}\, \eta_{\mu'\nu'}$.

1.4.1 Rotations

A spatial rotation in the x–y plane by angle θ is (Carroll eq. 1.31):

$$\Lambda^{\mu'}{}_{\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & \sin\theta & 0 \\ 0 & -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \tag{1.12}$$

1.4.2 Boosts

Rotations mix two spatial directions using sin and cos (circular functions that preserve the positive-definite sum $x^2 + y^2$). A **Lorentz boost** mixes a space and a time direction, and must preserve the *indefinite* form $-t^2 + x^2$ (the Minkowski interval). The functions that do this are the hyperbolic functions sinh and cosh, which satisfy the identity $\cosh^2\phi - \sinh^2\phi = 1$ — exactly mirroring the Minkowski signature (Carroll Sec. 1.3, pp. 11–14).

The boost along the x-axis with **rapidity** parameter ϕ is therefore (Carroll eq. 1.32):

$$\Lambda^{\mu'}{}_{\nu} = \begin{pmatrix} \cosh\phi & -\sinh\phi & 0 & 0 \\ -\sinh\phi & \cosh\phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} . \tag{1.13}$$

The rapidity ϕ parameterizes the boost in a way that is analogous to the rotation angle θ: just as two successive rotations by θ_1 and θ_2 give a rotation by $\theta_1 + \theta_2$, two successive boosts with rapidities ϕ_1 and ϕ_2 compose to a single boost with rapidity $\phi_1 + \phi_2$ (Exercise 1.11.2).

To connect to the more familiar velocity v of the boosted frame (Carroll eqs. 1.33–1.35), note that a particle at rest in S' has $x' = 0$, so from the boost matrix: $0 = -\sinh\phi \cdot t + \cosh\phi \cdot x$, giving $x/t = \tanh\phi$. Since x/t is the velocity v of the S' frame as seen from S, we identify:

$$v = \tanh\phi\, . \tag{1.14}$$

The Lorentz factor γ then follows from the hyperbolic identity $\cosh^2\phi - \sinh^2\phi = 1$:

$$\gamma \equiv \frac{1}{\sqrt{1-v^2}} = \cosh\phi\, , \qquad \gamma v = \sinh\phi\, . \tag{1.15}$$

Substituting into the boost matrix, we recover the familiar form (Carroll eq. 1.35):

$$\begin{aligned} t' &= \gamma(t - vx)\, , \\ x' &= \gamma(x - vt)\, . \end{aligned} \tag{1.16}$$

Exercise 1.4.1. Verify that the boost matrix (1.13) satisfies the Lorentz condition (1.11) by explicit matrix multiplication. Then confirm this symbolically using `lorentz_transforms.wl`.

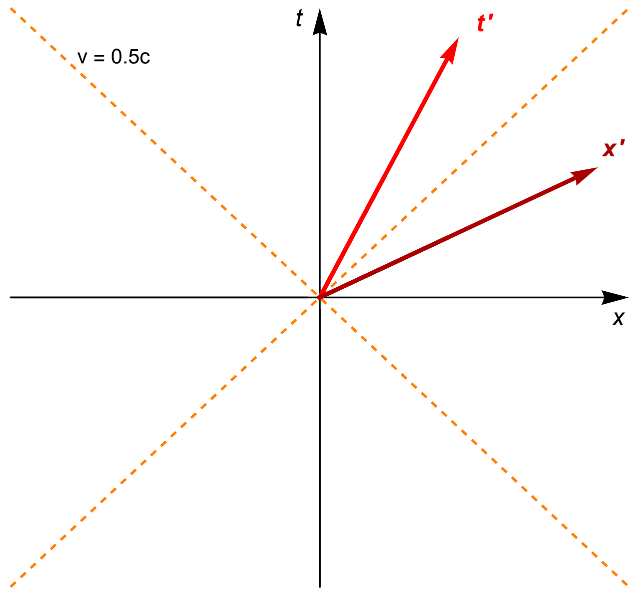


Figure 1.2: A Lorentz boost with $v = 0.5c$ tilts both the time axis (t', red) and the space axis (x', dark red) toward the light cone. The light cone itself (orange dashed) is invariant under boosts. (cf. Carroll Fig. 1.7)

1.4.3 Physical Consequences

The boost equations (1.16) immediately lead to the classic relativistic effects (Carroll Sec. 1.4; Congdon & Keeton Sec. 3.2.3). In each case the derivation is a single substitution:

- **Time dilation:** A clock at rest in S' has $\Delta x' = 0$. The inverse boost gives $\Delta t = \gamma(\Delta t' + v \cdot 0) = \gamma\, \Delta t'$, so the moving clock appears to run *slow* by a factor $\gamma > 1$.

- **Length contraction:** A rod at rest in S' has proper length $L_0 = \Delta x'$. We measure it in S by noting the positions of both ends *simultaneously* ($\Delta t = 0$). From $\Delta x' = \gamma(\Delta x - v \cdot 0)$ we get $L_0 = \gamma\, \Delta x$, hence $L = \Delta x = L_0/\gamma$ — the rod is shorter in the frame where it moves.

- **Relativity of simultaneity:** Events simultaneous in S' ($\Delta t' = 0$) satisfy $0 = \gamma(\Delta t - v\, \Delta x)$, giving $\Delta t = v\, \Delta x$: events separated in space are *not* simultaneous in S.

Exercise 1.4.2. (Twin paradox.) An observer travels from event $A = (0, 0)$ to $B' = (\Delta t/2,\, v\Delta t/2)$ and back to $C = (\Delta t, 0)$ at constant speed v, while a stationary twin travels from A directly to C. Show that the traveling twin ages by $\Delta\tau_{AB'C} = \sqrt{1 - v^2}\, \Delta t$ while the stationary twin ages by $\Delta\tau_{ABC} = \Delta t$. (cf. Carroll eqs. 1.18–1.19. Visualize with `four_vectors.nb`.)

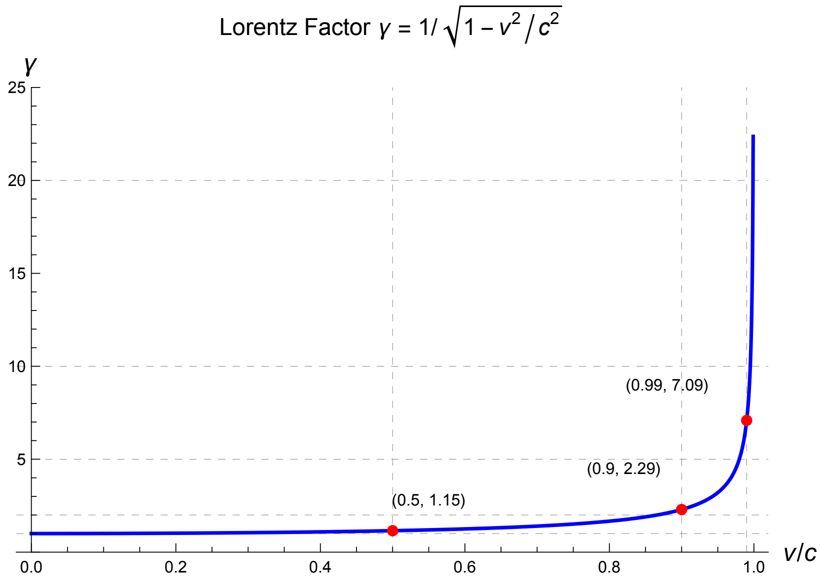


Figure 1.3: The Lorentz factor $\gamma = (1 - v^2/c^2)^{-1/2}$ as a function of velocity. Time dilation and length contraction both scale with γ. Note the dramatic increase as $v \to c$.

1.5 Four-Vectors

A **four-vector** V^μ is a set of four components that transforms under Lorentz transformations as (Carroll eq. 1.39):

$$V^{\mu'} = \Lambda^{\mu'}{}_{\nu} V^\nu . \tag{1.17}$$

Key examples:

Four-position: $x^\mu = (t, x, y, z)$.

Four-velocity: For a massive particle with worldline $x^\mu(\tau)$, the four-velocity is defined as the derivative of the position with respect to *proper time* τ (the time measured by a clock traveling with the particle):

$$U^\mu = \frac{dx^\mu}{d\tau} . \tag{1.18}$$

To express this in terms of coordinate time t, use $d\tau = dt/\gamma$ (from time dilation, eq. (1.5); see also Carroll eq. 1.58), giving $U^\mu = \gamma\, dx^\mu/dt = \gamma(1, v^x, v^y, v^z)$ where $v^i = dx^i/dt$ is the ordinary three-velocity.

The norm of the four-velocity is always $\eta_{\mu\nu}U^\mu U^\nu = -1$ (in units with $c = 1$; Carroll eq. 1.59). This follows directly from the definition: $U^\mu U_\mu = (dx^\mu/d\tau)(dx_\mu/d\tau) = ds^2/d\tau^2 = -d\tau^2/d\tau^2 = -1$, since $ds^2 = -d\tau^2$ along a timelike worldline.

Four-momentum: The four-momentum is defined as $p^\mu = m\, U^\mu$, where m is the rest mass (Carroll eq. 1.60). Expanding: $p^\mu = m\gamma(1, v^x, v^y, v^z) = (E, p^x, p^y, p^z)$, where $E = \gamma m$ is the relativistic energy (which reduces to m at rest, i.e., $E = mc^2$ in conventional units) and

$\mathbf{p} = \gamma m\,\mathbf{v}$ is the relativistic three-momentum (which reduces to $m\mathbf{v}$ for $v \ll c$; Congdon & Keeton eqs. 3.18–3.23). The invariant norm gives the **energy–momentum relation**:

$$\boxed{\eta_{\mu\nu}\,p^\mu p^\nu = -E^2 + |\mathbf{p}|^2 = -m^2} \tag{1.19}$$

or, restoring c: $E^2 = (pc)^2 + (mc^2)^2$.

For a **photon** ($m = 0$), $E = |\mathbf{p}|$ and the four-momentum is null: $p^\mu p_\mu = 0$. This will be crucial when we trace light rays through curved spacetime in later modules.

1.6 Dual Vectors (One-Forms)

Associated with every vector space is a **dual vector space** (or cotangent space T_p^*). A dual vector (or **one-form**) ω_μ transforms under the *inverse* Lorentz transformation (Carroll eq. 1.50):

$$\omega_{\mu'} = \Lambda^\nu{}_{\mu'}\,\omega_\nu\,. \tag{1.20}$$

The prototypical one-form is the **gradient** of a scalar field ϕ (Carroll eq. 1.54):

$$(\mathrm{d}\phi)_\mu = \partial_\mu\phi \equiv \frac{\partial\phi}{\partial x^\mu}\,. \tag{1.21}$$

The contraction of a vector with a dual vector is a Lorentz scalar: $\omega_\mu V^\mu \in \mathbf{R}$.

1.7 Tensors

A **tensor** of type (k, l) is a multilinear map that takes k one-forms and l vectors and produces a real number (Carroll eq. 1.56). In components, the transformation law is (Carroll eq. 1.63):

$$T^{\mu_1'\cdots\mu_k'}{}_{\nu_1'\cdots\nu_l'} = \Lambda^{\mu_1'}{}_{\alpha_1}\cdots\Lambda^{\mu_k'}{}_{\alpha_k}\,\Lambda^{\beta_1}{}_{\nu_1'}\cdots\Lambda^{\beta_l}{}_{\nu_l'}\,T^{\alpha_1\cdots\alpha_k}{}_{\beta_1\cdots\beta_l}\,. \tag{1.22}$$

1.7.1 Important Operations

- **Contraction:** Summing over one upper and one lower index reduces the rank by $(1, 1)$ (Carroll eq. 1.70): $S^{\mu\rho}{}_\sigma = T^{\mu\nu\rho}{}_{\sigma\nu}$.
- **Raising/lowering:** Use the metric to move indices (Carroll eq. 1.72): $T^{\alpha\beta\mu}{}_\delta = \eta^{\mu\gamma}\,T^{\alpha\beta}{}_{\gamma\delta}$.
- **Symmetrization/antisymmetrization:** Round brackets denote symmetrization; square brackets denote antisymmetrization (Carroll eqs. 1.79–1.80).

1.7.2 Key Tensors in Minkowski Space

- **Metric:** $\eta_{\mu\nu}$ — a symmetric $(0, 2)$ tensor.
- **Kronecker delta:** $\delta^\mu{}_\nu$ — the identity map.

- **Levi-Civita symbol:** $\tilde{\epsilon}_{\mu\nu\rho\sigma}$ (Carroll eq. 1.68) — totally antisymmetric, with $\tilde{\epsilon}_{0123} = +1$.
- **Electromagnetic field strength:** $F_{\mu\nu} = -F_{\nu\mu}$ (Carroll eq. 1.69) — a $(0,2)$ antisymmetric tensor encoding **E** and **B**.

1.8 The Metric as the Central Object

The key insight that connects SR to GR is that the metric is the fundamental object describing the geometry of spacetime. In SR, the metric is the constant $\eta_{\mu\nu}$. In GR, the metric becomes a field $g_{\mu\nu}(x)$ that varies from point to point, encoding the curvature of spacetime due to matter and energy.

Everything we need for gravitational lensing — the bending of light, the distortion of images, the time delays between multiple images — follows from understanding how the metric determines the paths of light rays. That story begins in the next module, where we introduce curved spacetime and the geodesic equation.

1.9 Energy and Momentum in SR

The **stress-energy tensor** $T^{\mu\nu}$ encodes the energy and momentum content of matter and fields. For a perfect fluid with energy density ρ, pressure p, and four-velocity U^μ:

$$T^{\mu\nu} = (\rho + p)\, U^\mu U^\nu + p\, \eta^{\mu\nu}\,. \tag{1.23}$$

Conservation of energy and momentum is expressed as

$$\partial_\mu T^{\mu\nu} = 0\,. \tag{1.24}$$

This will generalize to $\nabla_\mu T^{\mu\nu} = 0$ in curved spacetime, and the stress-energy tensor will appear as the source term on the right-hand side of Einstein's field equation (1.3).

1.10 Summary

- Spacetime is a four-dimensional continuum; the invariant quantity is the interval $ds^2 = \eta_{\mu\nu}\, dx^\mu dx^\nu$.
- Lorentz transformations ($x^{\mu'} = \Lambda^{\mu'}{}_\nu x^\nu$) leave the interval invariant. Boosts mix space and time.
- Four-vectors and tensors are classified by how they transform under Lorentz transformations.
- The metric $\eta_{\mu\nu}$ is the central object — it defines distances, raises/lowers indices, and generalizes to $g_{\mu\nu}$ in GR.
- Photons travel on null paths ($ds^2 = 0$), which will be the starting point for understanding gravitational lensing.

1.11 Problems

Exercise 1.11.1. Classify the following pairs of events as timelike, spacelike, or null separated:

(a) $A = (0, 0, 0, 0)$, $B = (2, 1, 1, 0)$

(b) $A = (0, 0, 0, 0)$, $B = (1, 1, 0, 0)$

(c) $A = (0, 0, 0, 0)$, $B = (3, 1, 2, 2)$

Check your answers with Mathematica.

Exercise 1.11.2. Show that two successive boosts along the x-axis with rapidities ϕ_1 and ϕ_2 are equivalent to a single boost with rapidity $\phi = \phi_1 + \phi_2$. Relate this to the velocity addition formula $v = (v_1 + v_2)/(1 + v_1 v_2)$.

Exercise 1.11.3. Prove that the four-velocity of any massive particle satisfies $U^\mu U_\mu = -1$ (in units with $c = 1$).

Exercise 1.11.4. A photon with energy E in frame S is observed in frame S' moving with velocity v along the x-axis.

(a) Show that the photon energy in S' is $E' = \gamma E(1 - v \cos\theta)$, where θ is the angle between the photon's direction and the x-axis in frame S.

(b) Specialize to $\theta = 0$ (head-on) and $\theta = \pi/2$ (transverse) and interpret physically.

Exercise 1.11.5. The electromagnetic field strength tensor has components (Carroll eq. 1.69):

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix}$$

(a) Show that $F_{\mu\nu}$ is antisymmetric.

(b) Compute $F^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta}$ and verify that the signs of the electric field components change.

(c) Compute the Lorentz scalar $F_{\mu\nu}F^{\mu\nu}$ and express it in terms of $\mathbf{E}$ and $\mathbf{B}$.

1.12 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_01a.wl`, which verifies each answer symbolically.

Exercise 1.11.1: Interval Classification

We compute $(\Delta s)^2 = -(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$:

(a) $(\Delta s)^2 = -(2)^2 + (1)^2 + (1)^2 + (0)^2 = -4 + 1 + 1 = -2 < 0 \Rightarrow$ **Timelike**. A massive particle can travel between these events.

(b) $(\Delta s)^2 = -(1)^2 + (1)^2 + (0)^2 + (0)^2 = 0 \Rightarrow$ **Null**. A light ray connects these events.

(c) $(\Delta s)^2 = -(3)^2 + (1)^2 + (2)^2 + (2)^2 = -9 + 1 + 4 + 4 = 0 \Rightarrow$ **Null**. Again lightlike: $|\Delta \mathbf{x}|^2 = 9 = (\Delta t)^2$.

Exercise 1.11.2: Boost Composition

The boost matrix for rapidity ϕ is $\Lambda(\phi) = \left(\begin{smallmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{smallmatrix}\right)$ (showing only the t–x block). Computing the product:

$$\begin{aligned}
\Lambda(\phi_1)\,\Lambda(\phi_2) &= \begin{pmatrix} \cosh\phi_1\cosh\phi_2 + \sinh\phi_1\sinh\phi_2 & -\cosh\phi_1\sinh\phi_2 - \sinh\phi_1\cosh\phi_2 \\ -\sinh\phi_1\cosh\phi_2 - \cosh\phi_1\sinh\phi_2 & \sinh\phi_1\sinh\phi_2 + \cosh\phi_1\cosh\phi_2 \end{pmatrix} \\
&= \begin{pmatrix} \cosh(\phi_1+\phi_2) & -\sinh(\phi_1+\phi_2) \\ -\sinh(\phi_1+\phi_2) & \cosh(\phi_1+\phi_2) \end{pmatrix} = \Lambda(\phi_1+\phi_2)\,,
\end{aligned}$$

using the hyperbolic addition formulas.

For the velocity addition formula, we use $v = \tanh\phi$:

$$v = \tanh(\phi_1+\phi_2) = \frac{\tanh\phi_1 + \tanh\phi_2}{1 + \tanh\phi_1\tanh\phi_2} = \frac{v_1+v_2}{1+v_1v_2}\,.$$

Note: for $v_1 = v_2 = 1$ (speed of light), $v = 2/2 = 1$, confirming that the speed of light is invariant.

Exercise 1.11.3: Four-Velocity Norm

The four-velocity is $U^\mu = \gamma(1, v^1, v^2, v^3)$ where $\gamma = (1-v^2)^{-1/2}$ and $v^2 = (v^1)^2 + (v^2)^2 + (v^3)^2$. Then:

$$\begin{aligned}
U^\mu U_\mu = \eta_{\mu\nu}U^\mu U^\nu &= \gamma^2\left[-(1)^2 + (v^1)^2 + (v^2)^2 + (v^3)^2\right] \\
&= \gamma^2(-1+v^2) = \frac{-(1-v^2)}{1-v^2} = \boxed{-1}\,.
\end{aligned}$$

Exercise 1.11.4: Photon Energy (Relativistic Doppler)

(a) A photon with energy E propagating at angle θ to the x-axis has four-momentum $p^\mu = E(1, \cos\theta, \sin\theta, 0)$. Applying the boost Λ along x:

$$E' = p^{0'} = \gamma(p^0 - v\,p^1) = \gamma E(1 - v\cos\theta)\,.$$

(b) Special cases:

- $\theta = 0$ (photon along $+x$, same direction as boost): $E' = \gamma E(1 - v) = E\sqrt{\frac{1-v}{1+v}}$. This is a **redshift** ($E' < E$). The observer recedes from the source.
- $\theta = \pi/2$ (transverse): $E' = \gamma E = E/\sqrt{1 - v^2}$. This is the **transverse Doppler effect** — a purely relativistic blueshift with no classical analogue, arising from time dilation.

Exercise 1.11.5: Electromagnetic Field Strength Tensor

(a) By inspection, $F_{\mu\nu} = -F_{\nu\mu}$ (the matrix is antisymmetric: the diagonal is zero, and off-diagonal elements differ by a sign).

(b) Raising both indices with the metric: $F^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}F_{\alpha\beta}$. In matrix form: $F^{\mu\nu} = \eta \cdot F_{\mu\nu} \cdot \eta$. The result is:

$$F^{\mu\nu} = \begin{pmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & B_3 & -B_2 \\ -E_2 & -B_3 & 0 & B_1 \\ -E_3 & B_2 & -B_1 & 0 \end{pmatrix}$$

The electric field components flip sign (from raising the time index), while the magnetic components are unchanged.

(c) The Lorentz scalar:

$$\begin{aligned} F_{\mu\nu}F^{\mu\nu} &= 2\big(-E_1^2 - E_2^2 - E_3^2 + B_1^2 + B_2^2 + B_3^2\big) \\ &= 2(|\mathbf{B}|^2 - |\mathbf{E}|^2)\,. \end{aligned}$$

This is a Lorentz invariant: all inertial observers agree on the value of $|\mathbf{B}|^2 - |\mathbf{E}|^2$.

Chapter 2

Differential Geometry and the Metric Tensor

> *The appropriate mathematical structure used to describe curvature is that of a differentiable manifold: essentially, a kind of set that looks locally like flat space, but might have a very different global geometry.* — Sean Carroll, *Spacetime and Geometry*

In Module 1a we worked exclusively in flat Minkowski spacetime. To describe gravity, we must allow spacetime to be *curved*. This chapter introduces the mathematical machinery of differential geometry — manifolds, the general metric tensor, the covariant derivative, Christoffel symbols, and the curvature tensors — that forms the foundation of general relativity and, ultimately, gravitational lensing.

Companion Mathematica files:

- `christoffel_symbols.wl` — general-purpose symbolic computation of Christoffel symbols and curvature tensors from any input metric
- `curved_surfaces.nb` — interactive visualizations of parallel transport, geodesics, and curvature on 2D surfaces

2.1 Manifolds and Coordinate Charts

A **manifold** is a space that may be curved globally but looks like $\mathbb{R}^n$ in every sufficiently small neighborhood. More precisely, an n-dimensional manifold M is a topological space equipped with a collection of **coordinate charts** — invertible maps $\phi : U \to \mathbb{R}^n$ from open subsets $U \subset M$ — that cover all of M and are smoothly compatible on overlaps (Carroll Sec. 2.2).

Examples of manifolds include:

- $\mathbb{R}^n$ (trivially)
- The 2-sphere S^2 (the surface of a ball)
- The 2-torus T^2 (the surface of a doughnut)
- Spacetime itself (a 4-dimensional Lorentzian manifold)

A key fact: in general, *no single coordinate system can cover an entire manifold.* The sphere S^2, for instance, requires at least two charts (e.g., stereographic projections from the north and south poles; Carroll eqs. 2.9–2.10). We will always work locally in a single chart, keeping in mind that our results are coordinate-independent.

2.2 The General Metric Tensor

On a manifold, we need a way to measure distances and angles. This is provided by the **metric tensor** $g_{\mu\nu}(x)$, a symmetric, nondegenerate $(0, 2)$ tensor field. The line element is (cf. Carroll eq. 1.20, now generalized):

$$\boxed{ds^2 = g_{\mu\nu}(x)\, dx^\mu\, dx^\nu} \tag{2.1}$$

In flat Minkowski space, $g_{\mu\nu} = \eta_{\mu\nu}$ everywhere. In curved spacetime, $g_{\mu\nu}$ varies from point to point and encodes the gravitational field.

The **inverse metric** $g^{\mu\nu}$ is defined by

$$g^{\mu\lambda} g_{\lambda\nu} = \delta^\mu{}_\nu\,. \tag{2.2}$$

As before, the metric raises and lowers indices: $V_\mu = g_{\mu\nu} V^\nu$, $V^\mu = g^{\mu\nu} V_\nu$.

Example 2.2.1. The metric on the 2-sphere of radius R in spherical coordinates (θ, ϕ) is

$$ds^2 = R^2\big(d\theta^2 + \sin^2\theta\, d\phi^2\big)\,, \tag{2.3}$$

so $g_{\theta\theta} = R^2$, $g_{\phi\phi} = R^2 \sin^2\theta$, and all other components vanish.

Example 2.2.2. The flat plane in polar coordinates (r, ϕ) has metric

$$ds^2 = dr^2 + r^2\, d\phi^2\,. \tag{2.4}$$

Despite the coordinate-dependent appearance, this space is flat — the curvature tensors vanish. We will verify this in Mathematica.

2.3 Vectors and Tensors on Manifolds

On a general manifold, vectors live in the **tangent space** T_p at each point p. A vector V at p is identified with a directional derivative operator along a curve $x^\mu(\lambda)$ through p (Carroll Sec. 2.3):

$$V = V^\mu \partial_\mu\,, \tag{2.5}$$

where $\partial_\mu \equiv \partial/\partial x^\mu$ form a **coordinate basis** for T_p.

Under a coordinate transformation $x^\mu \to x^{\mu'}$, vector components transform as (Carroll eq. 2.19):

$$V^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\nu}\, V^\nu\,. \tag{2.6}$$

This generalizes the Lorentz transformation law from Module 1a to arbitrary coordinate changes.

Dual vectors (one-forms) and general tensors transform analogously, with one factor of $\partial x^{\mu'}/\partial x^\nu$ for each upper index and $\partial x^\nu/\partial x^{\mu'}$ for each lower index.

2.4 The Covariant Derivative

Ordinary partial derivatives of tensors are *not* tensors on a curved manifold: the transformation of $\partial_\mu V^\nu$ picks up an extra, non-tensorial term from the coordinate dependence of the basis vectors (Carroll Sec. 3.2).

To fix this, we introduce the **covariant derivative** ∇_μ, which adds a correction term involving the **connection coefficients** $\Gamma^\rho{}_{\mu\sigma}$ (Carroll eq. 3.5):

$$\boxed{\nabla_\mu V^\nu = \partial_\mu V^\nu + \Gamma^\nu{}_{\mu\lambda} V^\lambda} \tag{2.7}$$

For a one-form (Carroll eq. 3.16):

$$\nabla_\mu \omega_\nu = \partial_\mu \omega_\nu - \Gamma^\lambda{}_{\mu\nu}\, \omega_\lambda \,. \tag{2.8}$$

The general rule for a tensor of arbitrary rank: $+\Gamma$ for each upper index, $-\Gamma$ for each lower index (Carroll eq. 3.17).

The connection coefficients are *not* tensors — they transform inhomogeneously (Carroll eq. 3.10) — but the covariant derivative $\nabla_\mu V^\nu$ *is* a tensor.

2.5 Christoffel Symbols

In GR we demand two additional properties of the connection:

1. **Torsion-free:** $\Gamma^\lambda{}_{\mu\nu} = \Gamma^\lambda{}_{\nu\mu}$
2. **Metric compatibility:** $\nabla_\rho g_{\mu\nu} = 0$

These uniquely determine the **Christoffel symbols** (also called the Levi-Civita connection; Carroll eq. 3.27):

$$\boxed{\Gamma^\sigma{}_{\mu\nu} = \frac{1}{2} g^{\sigma\rho}\big(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\big)} \tag{2.9}$$

This is one of the most important formulas in GR. Given any metric, we can compute the Christoffel symbols and from them all of the geometry.

Example 2.5.1. For the plane in polar coordinates, $ds^2 = dr^2 + r^2 d\phi^2$ (Example 2.2.2), the only nonvanishing Christoffel symbols are (Carroll eqs. 3.29–3.31):

$$\Gamma^r{}_{\phi\phi} = -r\,, \qquad \Gamma^\phi{}_{r\phi} = \Gamma^\phi{}_{\phi r} = \frac{1}{r}\,. \tag{2.10}$$

We derive these and more general cases in `christoffel_symbols.wl`.

2.6 Parallel Transport

In flat space, we can freely slide a vector from one point to another while keeping it "constant." On a curved manifold, this is no longer unambiguous — the result depends on the *path taken*.

The notion of keeping a vector constant along a curve $x^\mu(\lambda)$ is formalized as **parallel transport**: a vector V^μ is parallel-transported if (Carroll eq. 3.40)

$$\frac{dV^\mu}{d\lambda} + \Gamma^\mu{}_{\rho\sigma}\frac{dx^\rho}{d\lambda}V^\sigma = 0\,. \tag{2.11}$$

An important consequence of metric compatibility is that parallel transport preserves the inner product: if V^μ and W^μ are both parallel-transported along a curve, then $g_{\mu\nu}V^\mu W^\nu = \text{const}$ (Carroll eq. 3.42).

On the 2-sphere, parallel transporting a vector around a closed loop results in a rotation of the vector — the angle of rotation is proportional to the enclosed area and the curvature (Figure 2.1).

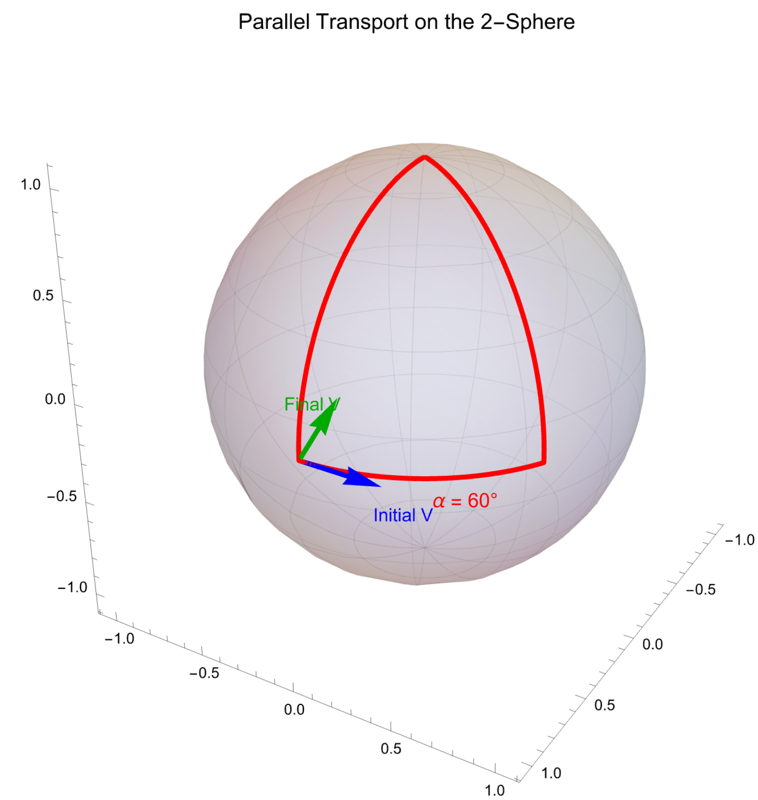


Figure 2.1: Parallel transport on the 2-sphere. A vector (blue) initially pointing east at the equator is transported north to the pole along $\phi = 0$, then south along $\phi = 60$ back to the equator. The final vector (green) has rotated by $\alpha = 60$ — equal to the solid angle enclosed by the path. This rotation is a direct manifestation of curvature. (cf. Carroll Fig. 3.2)

This is also visualized interactively in `curved_surfaces.nb`.

2.7 The Geodesic Equation

A **geodesic** is the curved-space generalization of a straight line: it is the path that parallel-transports its own tangent vector. Setting $V^\mu = dx^\mu/d\lambda$ in (2.11), we obtain the **geodesic**

equation (Carroll eq. 3.44):

$$\boxed{\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu{}_{\rho\sigma}\frac{dx^\rho}{d\lambda}\frac{dx^\sigma}{d\lambda} = 0} \tag{2.12}$$

This can equivalently be derived by extremizing the proper time/interval functional (Carroll Sec. 3.3, eqs. 3.45–3.55; Congdon & Keeton eq. 3.60).

Key properties:

- **Timelike geodesics** ($ds^2 < 0$): paths of freely falling massive particles; parameterized by proper time τ.
- **Null geodesics** ($ds^2 = 0$): paths of light rays; parameterized by an affine parameter λ.
- **Spacelike geodesics** ($ds^2 > 0$): shortest spatial distances.
- The character (timelike/null/spacelike) is preserved along the geodesic (Carroll Sec. 3.4).

In flat space with Cartesian coordinates, $\Gamma^\mu{}_{\rho\sigma} = 0$, so the geodesic equation reduces to $d^2x^\mu/d\lambda^2 = 0$ — a straight line, as expected.

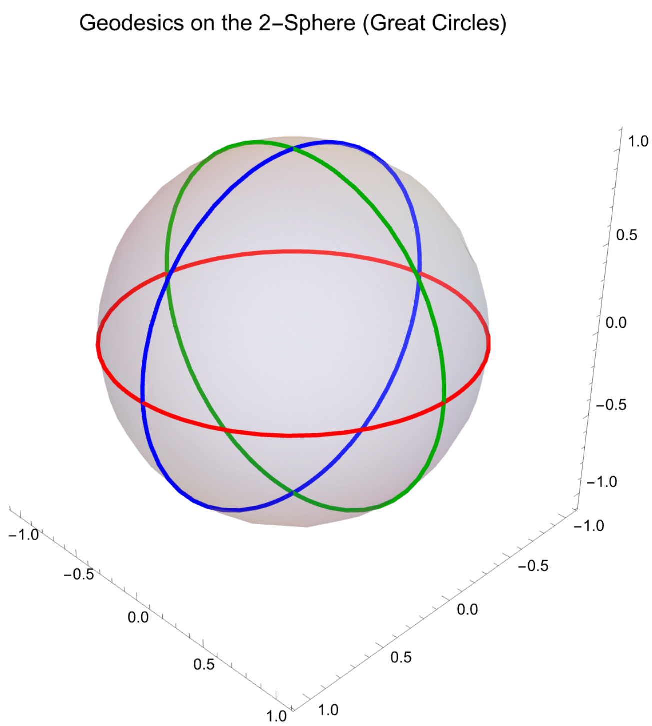


Figure 2.2: Geodesics on the 2-sphere are great circles. Three orthogonal great circles are shown, illustrating that the shortest path between any two points on a sphere follows a great circle arc.

2.8 Curvature Tensors

The **Riemann curvature tensor** $R^\rho{}_{\sigma\mu\nu}$ measures how much parallel transport around a closed loop rotates a vector. It is defined by (Carroll eq. 3.4):

$$\boxed{R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho{}_{\nu\sigma} - \partial_\nu \Gamma^\rho{}_{\mu\sigma} + \Gamma^\rho{}_{\mu\lambda}\,\Gamma^\lambda{}_{\nu\sigma} - \Gamma^\rho{}_{\nu\lambda}\,\Gamma^\lambda{}_{\mu\sigma}} \tag{2.13}$$

A manifold is flat if and only if $R^\rho{}_{\sigma\mu\nu} = 0$ everywhere. This is the mathematical statement that curvature is characterized entirely by the Riemann tensor.

From the Riemann tensor we construct:

- **Ricci tensor** (contraction on 1st and 3rd indices):

$$R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu} \tag{2.14}$$

- **Ricci scalar** (trace of the Ricci tensor):

$$R = g^{\mu\nu} R_{\mu\nu} \tag{2.15}$$

These are the quantities that appear in Einstein's field equation

$$R_{\mu\nu} - \frac{1}{2} R\, g_{\mu\nu} = 8\pi G\, T_{\mu\nu}\,, \tag{2.16}$$

relating the curvature of spacetime (left side) to its matter and energy content (right side).

Remark 2.8.1. The Riemann tensor in n dimensions has $n^2(n^2-1)/12$ independent components. In 4D spacetime, that is 20. In 2D, there is only 1 — the Gaussian curvature determines everything.

2.9 The Metric Determines Everything

Let us collect the chain of logic:

$$\begin{array}{ccc} & \textbf{Metric } g_{\mu\nu} & \\ & \downarrow \text{ eq. (2.9)} & \\ & \textbf{Christoffel symbols } \Gamma^\sigma{}_{\mu\nu} & \\ \downarrow \text{ eq. (2.12)} & & \downarrow \text{ eq. (2.13)} \\ \textbf{Geodesics} \text{ (particle paths)} & & \textbf{Curvature } R^\rho{}_{\sigma\mu\nu} \end{array}$$

This is the central message: *the metric is the fundamental object in GR*. Once you have $g_{\mu\nu}$, you can compute everything — how particles move, how light bends, and how spacetime curves.

For gravitational lensing, the key application is: given the metric around a massive object (e.g., the Schwarzschild metric for a point mass), solve the null geodesic equation to trace light rays and determine how images are formed.

2.10 Computation Cookbook: A Step-by-Step Guide

Computing the geometric quantities of GR by hand is a rite of passage. This section gives explicit, algorithmic recipes for going from a metric to Christoffel symbols, to the Riemann tensor, to the Ricci tensor and scalar. Work through the 2-sphere example in full, then use `christoffel_symbols.wl` to check every result.

2.10.1 Algorithm: Christoffel Symbols from a Metric

Given a metric $g_{\mu\nu}$ in coordinates $\{x^\mu\}$, compute all Christoffel symbols $\Gamma^\sigma{}_{\mu\nu}$ as follows.

1. **Write down the metric components.** List every nonvanishing $g_{\mu\nu}$ and note which coordinates each component depends on. For a diagonal metric, the only nonvanishing components are $g_{11}, g_{22}, \ldots$

2. **Compute the inverse metric $g^{\mu\nu}$.** For a diagonal metric this is trivial: $g^{\mu\mu} = 1/g_{\mu\mu}$ (no sum), and all off-diagonal components vanish.

3. **Compute the required partial derivatives.** For each target symbol $\Gamma^\sigma{}_{\mu\nu}$, you need the three derivatives
$$\partial_\mu g_{\nu\rho}, \quad \partial_\nu g_{\rho\mu}, \quad \partial_\rho g_{\mu\nu}$$
appearing in the Christoffel formula (2.9). Most of these vanish for diagonal metrics with few coordinate-dependent components.

4. **Apply the formula.**
$$\Gamma^\sigma{}_{\mu\nu} = \frac{1}{2}\, g^{\sigma\rho}\big(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\big)$$
For a diagonal metric, the sum over ρ collapses to the single term $\rho = \sigma$ (since $g^{\sigma\rho} = 0$ for $\rho \neq \sigma$).

5. **Exploit symmetry.** $\Gamma^\sigma{}_{\mu\nu} = \Gamma^\sigma{}_{\nu\mu}$, so you only need to compute symbols with $\mu \leq \nu$.

Remark 2.10.1 (Diagonal metric shortcut). For a diagonal metric $g_{\mu\nu} = \mathrm{diag}(g_{11}, g_{22}, \ldots)$, the nonvanishing Christoffel symbols come in only three types (no sum on repeated indices here):

$$\Gamma^\mu{}_{\mu\mu} = \frac{1}{2g_{\mu\mu}}\frac{\partial g_{\mu\mu}}{\partial x^\mu}\,, \tag{2.17}$$

$$\Gamma^\mu{}_{\mu\nu} = \Gamma^\mu{}_{\nu\mu} = \frac{1}{2g_{\mu\mu}}\frac{\partial g_{\mu\mu}}{\partial x^\nu} \qquad (\mu \neq \nu)\,, \tag{2.18}$$

$$\Gamma^\mu{}_{\nu\nu} = -\frac{1}{2g_{\mu\mu}}\frac{\partial g_{\nu\nu}}{\partial x^\mu} \qquad (\mu \neq \nu)\,. \tag{2.19}$$

All other components vanish. These formulas dramatically reduce the work.

2.10.2 Algorithm: The Riemann Tensor from Christoffel Symbols

Given the Christoffel symbols, compute each component of $R^\rho{}_{\sigma\mu\nu}$ using (2.13):

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho{}_{\nu\sigma} - \partial_\nu \Gamma^\rho{}_{\mu\sigma} + \Gamma^\rho{}_{\mu\lambda}\,\Gamma^\lambda{}_{\nu\sigma} - \Gamma^\rho{}_{\nu\lambda}\,\Gamma^\lambda{}_{\mu\sigma}$$

1. **Index structure.** The Riemann tensor is antisymmetric in its last two indices: $R^\rho{}_{\sigma\mu\nu} = -R^\rho{}_{\sigma\nu\mu}$. So only compute components with $\mu < \nu$.
2. **Partial derivatives of Christoffels.** For each component, differentiate the relevant Christoffel symbols with respect to the coordinates x^μ and x^ν.
3. **Quadratic terms.** For each component, sum over the dummy index λ in the two $\Gamma\Gamma$ products. Only terms where *both* Christoffel symbols in the product are nonzero contribute.
4. **Independent components.** In n dimensions, the Riemann tensor has $n^2(n^2-1)/12$ independent components: 1 in 2D, 6 in 3D, 20 in 4D. In 2D, the single independent component determines everything.

2.10.3 Algorithm: Ricci Tensor and Ricci Scalar

1. **Ricci tensor.** Contract the first and third indices of the Riemann tensor:

   $$R_{\sigma\nu} = R^\lambda{}_{\sigma\lambda\nu} = \sum_\lambda R^\lambda{}_{\sigma\lambda\nu}\,.$$

   In 2D this is a single sum; in 4D it is a sum of four terms.
2. **Ricci scalar.** Trace with the inverse metric:

   $$R = g^{\sigma\nu} R_{\sigma\nu} = \sum_\sigma g^{\sigma\sigma} R_{\sigma\sigma}$$

   (the last equality holding for a diagonal metric).

2.10.4 Worked Example: The 2-Sphere

We carry out the *complete* calculation for the 2-sphere metric

$$ds^2 = R^2\, d\theta^2 + R^2 \sin^2\theta\, d\phi^2\,, \tag{2.20}$$

with coordinates $x^1 = \theta$, $x^2 = \phi$.

Step 1: Metric components.

$$g_{\theta\theta} = R^2\,, \qquad g_{\phi\phi} = R^2 \sin^2\theta\,, \qquad g_{\theta\phi} = g_{\phi\theta} = 0\,.$$

Only $g_{\phi\phi}$ depends on a coordinate (θ).

Step 2: Inverse metric.

$$g^{\theta\theta} = \frac{1}{R^2}\,, \qquad g^{\phi\phi} = \frac{1}{R^2 \sin^2\theta}\,.$$

Step 3: Partial derivatives of the metric. The only nonvanishing derivative is

$$\partial_\theta g_{\phi\phi} = R^2 \cdot 2\sin\theta\cos\theta = R^2 \sin 2\theta\,.$$

All other $\partial_\mu g_{\nu\rho} = 0$.

Step 4: Christoffel symbols. Using the diagonal shortcut formulas (2.17)–(2.19):

- Type (2.17) (${\Gamma^\mu}_{\mu\mu}$): Both $\partial_\theta g_{\theta\theta} = 0$ and $\partial_\phi g_{\phi\phi} = 0$, so ${\Gamma^\theta}_{\theta\theta} = 0$ and ${\Gamma^\phi}_{\phi\phi} = 0$.
- Type (2.18) (${\Gamma^\mu}_{\mu\nu}$ with $\mu \neq \nu$):

  $$ {\Gamma^\phi}_{\phi\theta} = {\Gamma^\phi}_{\theta\phi} = \frac{1}{2g_{\phi\phi}}\,\partial_\theta g_{\phi\phi} = \frac{R^2\sin 2\theta}{2R^2\sin^2\theta} = \frac{\cos\theta}{\sin\theta} = \cot\theta\,.$$

  Also ${\Gamma^\theta}_{\theta\phi} = \frac{1}{2g_{\theta\theta}}\partial_\phi g_{\theta\theta} = 0$.
- Type (2.19) (${\Gamma^\mu}_{\nu\nu}$ with $\mu \neq \nu$):

  $$ {\Gamma^\theta}_{\phi\phi} = -\frac{1}{2g_{\theta\theta}}\,\partial_\theta g_{\phi\phi} = -\frac{R^2\sin 2\theta}{2R^2} = -\sin\theta\cos\theta\,.$$

  Also ${\Gamma^\phi}_{\theta\theta} = -\frac{1}{2g_{\phi\phi}}\partial_\phi g_{\theta\theta} = 0$.

Result: The nonvanishing Christoffel symbols are

$$\boxed{{\Gamma^\theta}_{\phi\phi} = -\sin\theta\cos\theta\,, \qquad {\Gamma^\phi}_{\theta\phi} = {\Gamma^\phi}_{\phi\theta} = \cot\theta} \tag{2.21}$$

Verify this with `christoffel_symbols.wl`.

Step 5: Riemann tensor. In 2D there is only one independent component. We compute ${R^\theta}_{\phi\theta\phi}$ (i.e., $\rho = \theta$, $\sigma = \phi$, $\mu = \theta$, $\nu = \phi$):

$$\begin{aligned}
{R^\theta}_{\phi\theta\phi} &= \partial_\theta {\Gamma^\theta}_{\phi\phi} - \partial_\phi {\Gamma^\theta}_{\theta\phi} + {\Gamma^\theta}_{\theta\lambda}\,{\Gamma^\lambda}_{\phi\phi} - {\Gamma^\theta}_{\phi\lambda}\,{\Gamma^\lambda}_{\theta\phi} \\
&= \partial_\theta(-\sin\theta\cos\theta) - 0 + {\Gamma^\theta}_{\theta\theta}\,{\Gamma^\theta}_{\phi\phi} + {\Gamma^\theta}_{\theta\phi}\,{\Gamma^\phi}_{\phi\phi} \\
&\qquad - {\Gamma^\theta}_{\phi\theta}\,{\Gamma^\theta}_{\theta\phi} - {\Gamma^\theta}_{\phi\phi}\,{\Gamma^\phi}_{\theta\phi} \\
&= (-\cos^2\theta + \sin^2\theta) + 0 + 0 - 0 - (-\sin\theta\cos\theta)(\cot\theta) \\
&= -\cos^2\theta + \sin^2\theta + \cos^2\theta \\
&= \sin^2\theta\,.
\end{aligned} \tag{2.22}$$

Step 6: Ricci tensor.

$$R_{\phi\phi} = R^\lambda{}_{\phi\lambda\phi} = R^\theta{}_{\phi\theta\phi} + R^\phi{}_{\phi\phi\phi} = \sin^2\theta + 0 = \sin^2\theta\,. \tag{2.23}$$

For $R_{\theta\theta}$, we need $R^\phi{}_{\theta\phi\theta}$. By the antisymmetry $R^\rho{}_{\sigma\mu\nu} = -R^\rho{}_{\sigma\nu\mu}$ and the symmetry of the fully lowered tensor:

$$R^\phi{}_{\theta\phi\theta} = g^{\phi\phi}\,R_{\phi\theta\phi\theta} = g^{\phi\phi}\,g_{\theta\theta}\,R^\theta{}_{\phi\theta\phi}\cdot(-1)\cdot(-1)$$

More directly, a short computation analogous to the one above gives $R^\phi{}_{\theta\phi\theta} = 1$, so

$$R_{\theta\theta} = R^\lambda{}_{\theta\lambda\theta} = R^\theta{}_{\theta\theta\theta} + R^\phi{}_{\theta\phi\theta} = 0 + 1 = 1\,. \tag{2.24}$$

Step 7: Ricci scalar.

$$R = g^{\theta\theta}R_{\theta\theta} + g^{\phi\phi}R_{\phi\phi} = \frac{1}{R^2}\cdot 1 + \frac{1}{R^2\sin^2\theta}\cdot\sin^2\theta = \frac{1}{R^2} + \frac{1}{R^2} = \boxed{\frac{2}{R^2}}. \tag{2.25}$$

This is the standard result: a sphere of radius R has constant positive Gaussian curvature $K = 1/R^2$, and in 2D the Ricci scalar equals $2K$.

2.10.5 Tips and Tricks

1. **Start with the diagonal shortcut.** Most metrics encountered in GR courses are diagonal. Use eqs. (2.17)–(2.19) rather than the full Christoffel formula.

2. **Spot vanishing components early.** A Christoffel symbol $\Gamma^\sigma{}_{\mu\nu}$ vanishes whenever the relevant partial derivatives of the metric are all zero. Before computing, ask: does $g_{\mu\nu}$ depend on x^σ? Does $g_{\sigma\mu}$ depend on x^ν? If not, the symbol is likely zero.

3. **Count before you compute.** In n dimensions there are $n^2(n+1)/2$ potentially independent Christoffel symbols (accounting for the $\mu\nu$ symmetry). In 2D that is 6; in 4D it is 40. Most are zero for typical metrics — identify the nonvanishing ones first.

4. **Use Mathematica to verify.** Always check hand calculations with `christoffel_symbols.wl`. Even experts make sign and index errors in these computations.

5. **Look for symmetry.** If the metric is independent of a coordinate x^α (a Killing symmetry), many Christoffel symbols simplify or vanish. In the 2-sphere example, the metric is ϕ-independent, which immediately kills several terms.

2.11 Standard Computational Tools

The hand-rolled functions defined in `christoffel_symbols.wl` are pedagogically valuable — they make the connection between the textbook formula and the computation explicit. However, Mathematica also provides built-in tools for GR calculations that you should know about.

Wolfram Function Repository. The `ResourceFunction` interface provides one-liner access to several GR computations:

- `ResourceFunction["ChristoffelSymbol"][g, coords]` — returns the Christoffel symbols as a plain $n \times n \times n$ array, identical in format to our custom function. We have verified that this matches our code exactly for both the 2-sphere and Schwarzschild metrics (see `verify_against_textbooks.wl`).
- `ResourceFunction["RicciScalar"][g, coords]` — returns the Ricci scalar. Works for simple metrics (e.g., the 2-sphere) but may not fully evaluate for more complex ones.
- `ResourceFunction["RiemannTensor"]` and `ResourceFunction["RicciTensor"]` — these exist but return opaque tensor objects rather than plain arrays, making them difficult to use for further computation.

Remark 2.11.1. For this reason, we use `ResourceFunction["ChristoffelSymbol"]` as a cross-check throughout the tutorial, but maintain our custom functions for the full Christoffel $\to$ Riemann $\to$ Ricci $\to$ scalar pipeline, since the built-in ResourceFunction versions of the higher-order tensors do not return plain arrays. All results have been cross-validated against textbook equations; see `Mathematica/verify_against_textbooks.wl` for the complete test suite.

The xAct Package. For professional-level tensor algebra in Mathematica, the **xAct** suite (available at http://www.xact.es/) is the gold standard. Its component packages include:

- **xTensor**: abstract tensor algebra (index manipulation, symmetry, canonicalization)
- **xCoba**: component computations in coordinate bases
- **xPert**: perturbation theory (useful for linearized gravity, Module 1e)

xAct excels at abstract manipulations — e.g., proving tensor identities, computing perturbation expansions, and working with complicated index symmetries. It is less convenient for the simple "plug in a metric, get numbers out" workflow used in this tutorial, which is why we do not use it as our primary tool. However, if your research involves heavy tensor algebra (e.g., higher-order perturbation theory, non-trivial spacetimes, or abstract index calculations), xAct is the right tool.

Installation:

```
(* Download from http://www.xact.es/download.html *)
(* Unpack into ~/Library/Wolfram/Applications/ (macOS) *)
(* Then: *)
Needs["xAct`xTensor`"]
Needs["xAct`xCoba`"]
```

2.12 Summary

- A manifold is locally flat but may have global curvature; the metric $g_{\mu\nu}(x)$ encodes the geometry.
- The covariant derivative ∇_μ generalizes the partial derivative to curved spacetime, using the connection $\Gamma^\sigma{}_{\mu\nu}$.

- The Christoffel symbols are uniquely determined by the metric via (2.9) (torsion-free, metric-compatible).
- Parallel transport depends on the path; this path-dependence is measured by the Riemann curvature tensor.
- Geodesics — paths that parallel-transport their own tangent vector — are the trajectories of freely falling particles and light rays.
- Einstein's equation relates curvature ($R_{\mu\nu}$) to matter content ($T_{\mu\nu}$).

2.13 Problems

Exercise 2.13.1. Compute all nonvanishing Christoffel symbols for the 2-sphere metric $ds^2 = R^2(d\theta^2 + \sin^2\theta\, d\phi^2)$ by hand, then verify with `christoffel_symbols.wl`.

Exercise 2.13.2. For the flat plane in polar coordinates, $ds^2 = dr^2 + r^2 d\phi^2$:

(a) Compute the Christoffel symbols.

(b) Compute the Riemann tensor and verify that the space is flat ($R^\rho{}_{\sigma\mu\nu} = 0$).

(c) Write down the geodesic equations and show that straight lines in Cartesian coordinates ($x = a + b\lambda$, $y = c + d\lambda$) satisfy them.

Exercise 2.13.3. For the 2-sphere metric, compute:

(a) The Riemann tensor $R^\theta{}_{\phi\theta\phi}$.

(b) The Ricci tensor $R_{\theta\theta}$ and $R_{\phi\phi}$.

(c) The Ricci scalar R. Show that $R = 2/R^2$ (constant positive curvature, as expected for a sphere of radius R).

Exercise 2.13.4. A vector initially pointing east is parallel-transported from the equator of a unit sphere ($R = 1$) along the following path: north along the meridian $\phi = 0$ to the pole, then south along $\phi = \alpha$ back to the equator. Show that the vector has rotated by angle α. (Hint: solve the parallel transport equation (2.11) along each segment, or use the geometric argument that the rotation equals the solid angle enclosed.)

Exercise 2.13.5. Show that great circles are geodesics of the 2-sphere. Specifically, show that the equator $\theta = \pi/2$, $\phi = \lambda$ satisfies the geodesic equation (2.12) with the Christoffel symbols from Exercise 2.13.1.

2.14 Solutions to Exercises

Full worked solutions verified in `problems_01b.wl`.

Exercise 2.13.1: Christoffel Symbols of the 2-Sphere

For $ds^2 = R^2(d\theta^2 + \sin^2\theta\, d\phi^2)$, the metric is $g_{\theta\theta} = R^2$, $g_{\phi\phi} = R^2 \sin^2\theta$, with $g^{\theta\theta} = 1/R^2$, $g^{\phi\phi} = 1/(R^2 \sin^2\theta)$.

Applying (2.9):

$$\Gamma^{\theta}{}_{\phi\phi} = \frac{1}{2} g^{\theta\theta}(-\partial_\theta g_{\phi\phi}) = \frac{1}{2} \cdot \frac{1}{R^2} \cdot (-2R^2 \sin\theta\cos\theta) = -\sin\theta\cos\theta$$
$$\Gamma^{\phi}{}_{\theta\phi} = \frac{1}{2} g^{\phi\phi}(\partial_\theta g_{\phi\phi}) = \frac{1}{2} \cdot \frac{1}{R^2 \sin^2\theta} \cdot 2R^2 \sin\theta\cos\theta = \cot\theta$$

All other components vanish (verified by Mathematica).

Exercise 2.13.2: Polar Coordinates

(a) For $ds^2 = dr^2 + r^2 d\phi^2$: $\Gamma^{r}{}_{\phi\phi} = -r$, $\quad \Gamma^{\phi}{}_{r\phi} = 1/r$.

(b) Computing the Riemann tensor from these Christoffel symbols: $R^{\rho}{}_{\sigma\mu\nu} = 0$ for all index combinations. The space is flat, confirming that the nonvanishing Christoffel symbols are purely a coordinate artifact.

(c) The geodesic equations are: $\ddot{r} - r\dot{\phi}^2 = 0$ and $\ddot{\phi} + (2/r)\dot{r}\dot{\phi} = 0$. A straight line $x = a + b\lambda$, $y = c + d\lambda$ in Cartesian coordinates satisfies these when converted to polar coordinates $r = \sqrt{x^2 + y^2}$, $\phi = \arctan(y/x)$.

Exercise 2.13.3: Curvature of the 2-Sphere

(a) From (2.13): $R^{\theta}{}_{\phi\theta\phi} = \sin^2\theta$.

(b) Ricci tensor: $R_{\theta\theta} = R^{\lambda}{}_{\theta\lambda\theta} = R^{\phi}{}_{\theta\phi\theta} = 1$. $R_{\phi\phi} = R^{\theta}{}_{\phi\theta\phi} = \sin^2\theta$.

(c) Ricci scalar: $R = g^{\theta\theta}R_{\theta\theta} + g^{\phi\phi}R_{\phi\phi} = \frac{1}{R^2}(1) + \frac{1}{R^2 \sin^2\theta}(\sin^2\theta) = \frac{2}{R^2}$. This is constant and positive — the sphere has uniform positive curvature.

Exercise 2.13.4: Parallel Transport

Along a meridian ($\phi = \text{const}$), $d\phi/d\lambda = 0$, so the parallel transport equation reduces to $dV^\mu/d\lambda = 0$ — the vector components are constant along the path.

After traveling north to the pole along $\phi = 0$ and south along $\phi = \alpha$, the vector returns to the equator. However, the coordinate basis $\hat{\phi}$ at $\phi = \alpha$ is rotated by angle α relative to $\hat{\phi}$ at $\phi = 0$. The physical vector has therefore rotated by angle α.

Geometrically, the solid angle subtended by the triangular path (two meridians and a segment of the equator) is α. On a unit sphere, the holonomy angle equals the enclosed solid angle.

Exercise 2.13.5: Great Circles as Geodesics

For the equator: $\theta(\lambda) = \pi/2$, $\phi(\lambda) = \lambda$, so $\dot{\theta} = 0$, $\dot{\phi} = 1$, $\ddot{\theta} = 0$, $\ddot{\phi} = 0$.

θ-equation: $0 + \Gamma^{\theta}{}_{\phi\phi} \cdot 1 \cdot 1 = -\sin(\pi/2)\cos(\pi/2) = 0$. ✓

ϕ-equation: $0 + 2\Gamma^{\phi}{}_{\theta\phi} \cdot 0 \cdot 1 = 0$. ✓

By the rotational symmetry of the sphere, any great circle can be mapped to the equator by a rotation, so all great circles are geodesics.

Chapter 3

The Schwarzschild Solution

> *In GR, the unique spherically symmetric vacuum solution is the Schwarzschild metric; it is second only to Minkowski space in the list of important spacetimes.* — Sean Carroll, *Spacetime and Geometry*

Having developed the general machinery of differential geometry in Module 1b, we now apply it to the most important exact solution in GR: the **Schwarzschild metric**, which describes the spacetime geometry outside any spherically symmetric, non-rotating mass. This metric governs the gravitational field of stars, planets, and non-rotating black holes, and is the starting point for understanding gravitational lensing.

Companion Mathematica files:

- `schwarzschild_metric.wl` — derive and verify the Schwarzschild metric: compute Christoffel symbols, Riemann tensor, verify $R_{\mu\nu} = 0$, and compute the Kretschner scalar
- `gravitational_redshift.nb` — interactive plots of gravitational redshift, time dilation, and Flamm's paraboloid embedding diagram

3.1 The Equivalence Principle

Before diving into the Schwarzschild solution, it is worth pausing to ask a foundational question: *why* should gravity be described by the geometry of spacetime at all? The answer lies in the **equivalence principle**, arguably the single most important physical insight behind general relativity.

3.1.1 The Weak Equivalence Principle

The oldest form of the equivalence principle dates back to Galileo: *all objects fall at the same rate, regardless of their composition or internal structure.* More precisely, the **weak equivalence principle** (WEP) states that the **inertial mass** m_i (which appears in Newton's second law, $F = m_i\, a$) is equal to the **gravitational mass** m_g (which appears in Newton's law of gravitation, $F = m_g\, g$):

$$m_i = m_g\,. \tag{3.1}$$

This equality has no explanation within Newtonian mechanics — it is simply an empirical fact. Yet it has been tested to extraordinary precision: the Eötvös experiments and their modern successors (Dicke, Braginsky, and the MICROSCOPE satellite mission) have verified eq. (3.1) to better than one part in 10^{13}:

$$\frac{|m_i - m_g|}{m_g} < 10^{-13}\,. \tag{3.2}$$

The universality of free fall is not an accident — Einstein elevated it to a *principle*.

3.1.2 Einstein's Elevator

Einstein's great insight was to take the WEP seriously as a fundamental principle, not merely an empirical coincidence. He expressed this through the famous **elevator thought experiment**:

> *An observer in a closed, windowless elevator cannot distinguish between the following two situations:*
>
> (a) *The elevator is at rest on the surface of the Earth, where the gravitational acceleration is g.*
> (b) *The elevator is in empty space, accelerating upward at g.*

Conversely, an observer in free fall in a gravitational field ("falling elevator") experiences exactly the same conditions as an observer floating in empty space, far from any gravitating body. Dropped objects float alongside the observer; there is no local experiment that can detect the gravitational field.

This equivalence between gravity and acceleration is *local*: it holds in a sufficiently small region of spacetime where tidal effects (variations of the gravitational field across the elevator) can be neglected.

3.1.3 The Strong Equivalence Principle

Einstein extended the elevator argument to its fullest form as the **strong equivalence principle** (SEP):

> *The results of any local experiment (gravitational or not) in a freely falling reference frame are indistinguishable from those of the same experiment in special relativity.*

This has a profound consequence. If a freely falling observer experiences special relativity locally, then gravity cannot be a "force" acting in Minkowski spacetime — it must instead be encoded in the *geometry* of spacetime itself. Different freely falling observers at different locations each see flat (Minkowski) spacetime locally, but the global structure connecting these local frames is curved. Gravity *is* the curvature of spacetime.

3.1.4 From the Equivalence Principle to Geodesics

The equivalence principle connects directly to the mathematical framework developed in Module 1b. A freely falling object experiences no forces in its local (freely falling) frame — it follows a "straight line" through its local Minkowski patch of spacetime. But a straight line in a curved manifold is precisely a **geodesic** (Module 1b, Section **??**).

Therefore, the equivalence principle implies that *freely falling particles follow geodesics of the spacetime metric*. This is the physical content of the geodesic equation:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu{}_{\alpha\beta}\,\frac{dx^\alpha}{d\tau}\,\frac{dx^\beta}{d\tau} = 0\,. \tag{3.3}$$

In Newtonian language, the Christoffel symbols $\Gamma^\mu{}_{\alpha\beta}$ play the role of gravitational "forces," but in GR they are geometric quantities — they describe how the coordinate system (not the particle) is "accelerating."

Remark 3.1.1. The equivalence principle tells us that gravity *curves* spacetime, but it does not tell us *how* the curvature is related to the matter content. That information requires Einstein's field equations $G_{\mu\nu} = 8\pi G\,T_{\mu\nu}/c^4$, which were stated in Module 1b and whose simplest vacuum solution ($T_{\mu\nu} = 0$) we derive in this module. The field equations will be used throughout the remaining modules as the foundation for gravitational lensing.

3.2 The Most General Spherically Symmetric Metric

We seek the metric outside a static, spherically symmetric mass distribution. The most general line element consistent with these symmetries can be written (Carroll eq. 5.5; cf. Congdon & Keeton eq. 3.64):

$$ds^2 = -e^{2\alpha(r)}dt^2 + e^{2\beta(r)}dr^2 + r^2\,d\Omega^2\,, \tag{3.4}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\,d\phi^2$ is the metric on the unit 2-sphere, and $\alpha(r)$ and $\beta(r)$ are functions to be determined from Einstein's equation.

Remark 3.2.1. The "static" assumption means the metric is independent of t and has no $dt\,dr$ cross terms. "Spherically symmetric" means the angular part takes the form $r^2 d\Omega^2$, with r chosen so that spheres at constant r have area $4\pi r^2$.

3.3 Deriving the Schwarzschild Metric

We solve Einstein's vacuum equation $R_{\mu\nu} = 0$ for the metric (3.4). The strategy follows Carroll Sec. 5.1:

Step 1: Christoffel symbols. The nonvanishing Christoffel symbols for (3.4) are (Carroll eq. 5.12):

$$\begin{aligned}
&\Gamma^t{}_{tr} = \alpha' , \quad \Gamma^r{}_{tt} = e^{2(\alpha-\beta)}\alpha' , \quad \Gamma^r{}_{rr} = \beta' , \\
&\Gamma^r{}_{\theta\theta} = -r\, e^{-2\beta} , \quad \Gamma^r{}_{\phi\phi} = -r\, e^{-2\beta}\sin^2\theta , \\
&\Gamma^\theta{}_{r\theta} = \Gamma^\phi{}_{r\phi} = \frac{1}{r} , \quad \Gamma^\theta{}_{\phi\phi} = -\sin\theta\cos\theta , \quad \Gamma^\phi{}_{\theta\phi} = \cot\theta ,
\end{aligned} \tag{3.5}$$

where primes denote d/dr.

Step 2: Ricci tensor. Computing the Ricci tensor components (Carroll eq. 5.14):

$$R_{tt} = e^{2(\alpha-\beta)}\left[\alpha'' + (\alpha')^2 - \alpha'\beta' + \frac{2}{r}\alpha'\right] , \tag{3.6}$$

$$R_{rr} = -\alpha'' - (\alpha')^2 + \alpha'\beta' + \frac{2}{r}\beta' , \tag{3.7}$$

$$R_{\theta\theta} = e^{-2\beta}\big[r(\beta' - \alpha') - 1\big] + 1 , \tag{3.8}$$

$$R_{\phi\phi} = R_{\theta\theta}\sin^2\theta . \tag{3.9}$$

Step 3: Solve $R_{\mu\nu} = 0$. From $e^{-2(\alpha-\beta)}R_{tt} + R_{rr} = 0$, we get $\frac{2}{r}(\alpha' + \beta') = 0$, hence (Carroll eq. 5.17):

$$\alpha = -\beta + \text{const} . \tag{3.10}$$

By rescaling the time coordinate, we set the constant to zero: $\alpha = -\beta$.

Setting $R_{\theta\theta} = 0$ (Carroll eq. 5.18):

$$e^{2\alpha}(1 + 2r\alpha') = 1 \quad \Longrightarrow \quad \partial_r(r\, e^{2\alpha}) = 1 . \tag{3.11}$$

Integrating:

$$e^{2\alpha} = 1 - \frac{R_S}{r} , \tag{3.12}$$

where R_S is an integration constant.

Step 4: Identify R_S. In the weak-field limit ($r \to \infty$), the metric must reduce to $g_{tt} \approx -(1 + 2\Phi/c^2)$ with $\Phi = -GM/r$. Comparing with $g_{tt} = -(1 - R_S/r)$, we identify the **Schwarzschild radius** (Carroll eq. 5.23):

$$\boxed{R_S = \frac{2GM}{c^2}} \tag{3.13}$$

Result: The Schwarzschild Metric.

$$\boxed{ds^2 = -\left(1 - \frac{R_S}{r}\right)c^2\, dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2} \tag{3.14}$$

This is the unique vacuum solution with spherical symmetry (**Birkhoff's theorem**; Carroll Sec. 5.2). We verify $R_{\mu\nu} = 0$ symbolically in `schwarzschild_metric.wl`.

3.4 Birkhoff's Theorem

Birkhoff's theorem states that the Schwarzschild metric is the *unique* spherically symmetric vacuum solution to Einstein's equation. The remarkable consequence: even a *time-dependent* spherically symmetric source (e.g., a pulsating star) produces a *static* exterior metric (Carroll Sec. 5.2).

This is the GR analogue of Newton's shell theorem: the gravitational field outside a spherically symmetric mass depends only on the total enclosed mass M, not on how the mass is distributed.

3.5 Gravitational Time Dilation

A clock at radial coordinate r in the Schwarzschild spacetime runs at a rate different from a clock at infinity. For a stationary clock ($dr = d\theta = d\phi = 0$), the proper time is related to coordinate time by (Congdon & Keeton eq. 3.65):

$$d\tau = \sqrt{1 - \frac{R_S}{r}}\, dt \,. \tag{3.15}$$

Since $\sqrt{1 - R_S/r} < 1$ for $r > R_S$, clocks closer to the mass run *slower* than clocks farther away. At $r = R_S$, the time dilation becomes infinite — this is the event horizon.

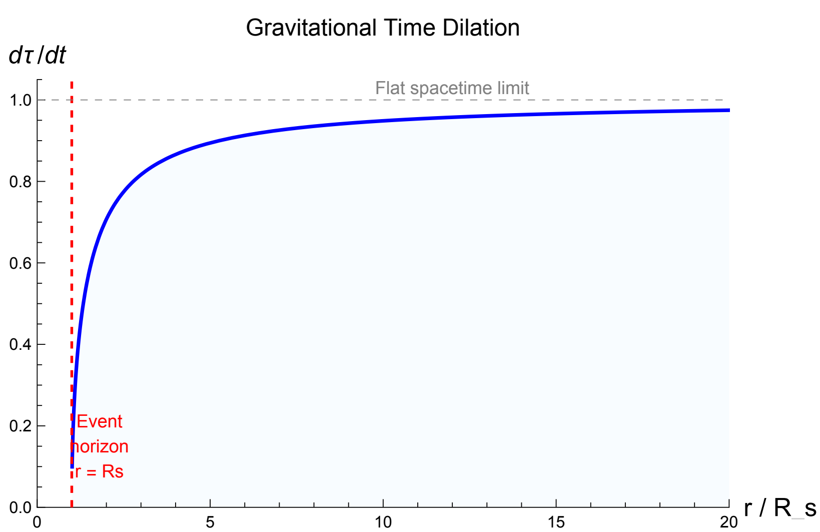


Figure 3.1: Gravitational time dilation factor $d\tau/dt = \sqrt{1 - R_S/r}$ as a function of r/R_S. Clocks near the horizon ($r \to R_S$) are dramatically slowed. Far from the mass ($r \gg R_S$), the factor approaches unity (flat spacetime).

3.6 Gravitational Redshift

A photon emitted at radius $r_{\rm em}$ with frequency $\nu_{\rm em}$ is observed at $r_{\rm obs} \to \infty$ with frequency:

$$\frac{\nu_{\rm obs}}{\nu_{\rm em}} = \frac{\sqrt{1 - R_S/r_{\rm em}}}{\sqrt{1 - R_S/r_{\rm obs}}} \xrightarrow{r_{\rm obs}\to\infty} \sqrt{1 - \frac{R_S}{r_{\rm em}}}\,. \tag{3.16}$$

Since $\nu_{\rm obs} < \nu_{\rm em}$, the photon is **redshifted** as it climbs out of the gravitational potential well. The gravitational redshift parameter is:

$$z_{\rm grav} = \frac{\nu_{\rm em}}{\nu_{\rm obs}} - 1 = \left(1 - \frac{R_S}{r_{\rm em}}\right)^{-1/2} - 1\,. \tag{3.17}$$

In the weak-field limit ($R_S/r \ll 1$): $z_{\rm grav} \approx GM/(rc^2)$ (Congdon & Keeton eq. 3.63).

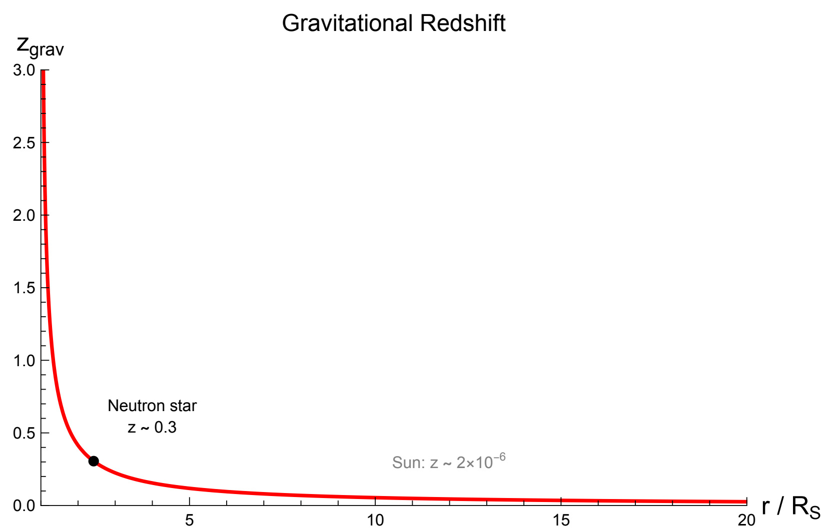


Figure 3.2: Gravitational redshift $z_{\rm grav}$ as a function of emission radius r/R_S. The redshift diverges at the horizon. The black dot marks a typical neutron star ($r/R_S \approx 2.4$, $z \approx 0.3$). For the Sun, $z \sim 10^{-6}$.

Remark 3.6.1. The gravitational redshift was one of the three classical tests of GR proposed by Einstein, along with the perihelion precession of Mercury and the bending of light. It has been measured to high precision using atomic clocks at different altitudes (e.g., Pound–Rebka experiment, 1959; GPS satellite corrections).

3.7 The Event Horizon and Singularities

The Schwarzschild metric has two apparent singularities: at $r = 0$ and at $r = R_S$. Their nature is quite different (Carroll Sec. 5.3):

$r = R_S$ **(coordinate singularity).** The metric components diverge, but this is a *coordinate artifact*. The curvature remains finite, and the geometry is perfectly smooth. In appropriate coordinates (e.g., Eddington–Finkelstein or Kruskal–Szekeres), the metric is regular at $r = R_S$. However, $r = R_S$ is the **event horizon**: once a particle crosses it, it can never return.

$r = 0$ **(true singularity).** This is a genuine physical singularity where the curvature becomes infinite. We can confirm this with a curvature invariant: the **Kretschner scalar**:

$$K = R_{\mu\nu\rho\sigma}\, R^{\mu\nu\rho\sigma} = \frac{12\, R_S^2}{r^6} = \frac{48\, G^2 M^2}{c^4\, r^6}\,. \tag{3.18}$$

This diverges as $r \to 0$ but is finite at $r = R_S$, confirming that only $r = 0$ is a true singularity. We verify eq. (3.18) in `schwarzschild_metric.wl`.

3.8 Inside the Event Horizon: The Coordinate Swap

The Schwarzschild metric (3.14) describes spacetime outside a spherical mass, but it also has something startling to say about the region $r < R_S$. Understanding this interior, even briefly, sharpens our physical intuition about what the Schwarzschild radius really means.

3.8.1 The Signature Flip

Recall the Schwarzschild line element:

$$ds^2 = -\left(1 - \frac{R_S}{r}\right) c^2\, dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\, d\Omega^2\,.$$

Outside the horizon ($r > R_S$), the factor $f(r) \equiv 1 - R_S/r$ is positive, so $g_{tt} < 0$ (timelike) and $g_{rr} > 0$ (spacelike).

Inside the horizon ($r < R_S$), this factor becomes *negative*: $f(r) < 0$. Consequently:

- $g_{tt} = -f(r)\, c^2 > 0$ — the t direction is now **spacelike**.
- $g_{rr} = f(r)^{-1} < 0$ — the r direction is now **timelike**.

The roles of r and t have swapped. Inside the horizon, r is a time coordinate and t is a space coordinate.

3.8.2 Physical Meaning: The Singularity Is a Moment in Time

Since r is timelike inside the horizon, decreasing r is the *future direction*. Every future-directed worldline must have $dr < 0$; the singularity at $r = 0$ is not a place you can steer away from — it is a *moment in time* that all interior observers inevitably reach.

Remark 3.8.1. You can no more avoid the singularity at $r = 0$ (once inside the horizon) than you can avoid next Tuesday. This is the precise sense in which a black hole traps everything: it is not that the singularity pulls you toward a point in space, but that the future itself points toward $r = 0$.

3.8.3 Why Nothing Escapes: Tilting Light Cones

For radial null rays ($ds^2 = 0$, $d\Omega = 0$) in Schwarzschild coordinates:

$$\frac{dr}{dt} = \pm\, c\left(1 - \frac{R_S}{r}\right). \tag{3.19}$$

Outside the horizon, the two signs give outgoing (+) and ingoing (−) light rays. At $r = R_S$, both solutions give $dr/dt = 0$: light "stands still" in these coordinates (a coordinate effect, not a physical one).

For $r < R_S$, the factor $(1 - R_S/r)$ is negative, so dr/dt has the *same sign* for both solutions — both "outgoing" and "ingoing" light rays have $dr < 0$ when $dt > 0$. In other words, **all future-directed light cones point toward smaller** r. No signal — not even light — can move to larger r inside the horizon.

3.8.4 Eddington–Finkelstein Coordinates

The pathological behavior at $r = R_S$ in Schwarzschild coordinates is a coordinate artifact. We can remove it by switching to **ingoing Eddington–Finkelstein coordinates**. Define the advanced time coordinate:

$$v = t + r_*\,, \qquad r_* = r + R_S \ln\left|\frac{r}{R_S} - 1\right|, \tag{3.20}$$

where r_* is the **tortoise coordinate** (Carroll eq. 5.58). In terms of (v, r, θ, ϕ), the Schwarzschild line element becomes (Carroll eq. 5.60):

$$\boxed{ds^2 = -\left(1 - \frac{R_S}{r}\right) c^2\, dv^2 + 2c\, dv\, dr + r^2\, d\Omega^2} \tag{3.21}$$

This metric is manifestly regular at $r = R_S$: every component is finite and the determinant is nonzero. The horizon is simply the surface where outgoing light rays have $dr/dv = 0$; ingoing rays cross it smoothly.

Remark 3.8.2. The Eddington–Finkelstein form makes it transparent that the event horizon is a *one-way membrane*: ingoing light ($dv > 0$, $dr < 0$) crosses the horizon without incident, while outgoing light at $r = R_S$ remains at $r = R_S$ forever.

3.8.5 Relevance to Gravitational Lensing

Remark 3.8.3. For gravitational lensing, we work exclusively in the region $r \gg R_S$ — far outside the event horizon. Galaxy-mass lenses have $R_S \sim 0.1$ pc, while lensing occurs at scales of kiloparsecs. The black hole interior is therefore not directly relevant to lensing calculations. Nevertheless, understanding the interior clarifies *why* the Schwarzschild radius is special (it is where light cones tip over) and deepens physical intuition about the boundary conditions of the exterior solution.

3.9 The Weak-Field Limit

For $r \gg R_S$, the Schwarzschild metric reduces to the **weak-field** form (Congdon & Keeton eq. 3.71):

$$ds^2 \approx -\left(1 + \frac{2\Phi}{c^2}\right) c^2 dt^2 + \left(1 - \frac{2\Phi}{c^2}\right) \left[dr^2 + r^2\, d\Omega^2\right], \tag{3.22}$$

where $\Phi = -GM/r$ is the Newtonian gravitational potential. This form is valid for *any* weak gravitational field, not just the Schwarzschild case, and will be the starting point for deriving the deflection angle of light in Module 2.

Remark 3.9.1. The weak-field metric (3.22) has an important feature: the spatial part is modified by a factor $(1 - 2\Phi/c^2)$ as well as the time part. This means that gravity affects both the *rate* of clocks (time dilation) and the *curvature of space*. Both effects contribute equally to the bending of light — this is why GR gives twice the Newtonian deflection angle.

3.10 Isotropic Coordinates

The standard Schwarzschild coordinates (t, r, θ, ϕ) have an important asymmetry: the radial and angular parts of the spatial metric carry *different* factors. In eq. (3.14), the dr^2 term is multiplied by $(1 - R_S/r)^{-1}$ while the angular part $r^2\, d\Omega^2$ has no such factor. The spatial part of the Schwarzschild metric is therefore *not* conformally flat — it is not proportional to the flat-space metric $dr^2 + r^2\, d\Omega^2$.

We can remedy this by introducing **isotropic coordinates**, in which the spatial metric *is* conformally flat.

3.10.1 The Isotropic Radial Coordinate

Define a new radial coordinate ρ via:

$$r = \rho \left(1 + \frac{R_S}{4\rho}\right)^2. \tag{3.23}$$

This transformation is invertible for $r > R_S$ (i.e., outside the event horizon), with the inverse $\rho = \frac{1}{2}(r - R_S/2 + \sqrt{r^2 - r\, R_S})$.

3.10.2 The Schwarzschild Metric in Isotropic Form

In the coordinates (t, ρ, θ, ϕ), the Schwarzschild metric becomes:

$$\boxed{ds^2 = -\left(\frac{1 - R_S/(4\rho)}{1 + R_S/(4\rho)}\right)^2 c^2\, dt^2 + \left(1 + \frac{R_S}{4\rho}\right)^4 \left(d\rho^2 + \rho^2\, d\Omega^2\right)} \tag{3.24}$$

The crucial feature is that the spatial part is now proportional to the *flat-space metric* $d\rho^2 + \rho^2\, d\Omega^2$ — the spatial geometry is **conformally flat**. The single conformal factor $(1 + R_S/(4\rho))^4$ multiplies all three spatial directions equally, hence the name "isotropic."

3.10.3 Weak-Field Expansion

To first order in R_S/ρ (the weak-field regime $\rho \gg R_S$), the isotropic metric reduces to:

$$ds^2 \approx -\left(1 - \frac{R_S}{2\rho}\right) c^2\, dt^2 + \left(1 + \frac{R_S}{2\rho}\right) \left(d\rho^2 + \rho^2\, d\Omega^2\right). \tag{3.25}$$

Since $R_S = 2GM/c^2$, we have $R_S/(2\rho) = GM/(\rho\, c^2) = -\Phi/c^2$ where $\Phi = -GM/\rho$ is the Newtonian potential. Equation (3.25) is therefore:

$$ds^2 \approx -\left(1 + \frac{2\Phi}{c^2}\right) c^2\, dt^2 + \left(1 - \frac{2\Phi}{c^2}\right) \left(d\rho^2 + \rho^2\, d\Omega^2\right), \tag{3.26}$$

which is precisely the weak-field metric (3.22) with r replaced by ρ. (In the weak-field regime, r and ρ agree to leading order.)

3.10.4 Connection to the Linearized Gravity Formalism

This result provides the explicit **bridge** between the exact Schwarzschild solution (this module) and the linearized gravity approach (Module 1e). In Module 1e, the weak-field metric is derived from first principles as a perturbation of flat spacetime: $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $h_{00} = -2\Phi/c^2$ and $h_{ij} = -2\Phi/c^2\, \delta_{ij}$. The isotropic form (3.26) shows that the Schwarzschild solution, when expanded to first order, reproduces exactly this linearized metric. The two descriptions — exact solution in curved coordinates, and perturbation theory in nearly flat coordinates — describe the *same physics* in different coordinate systems.

Remark 3.10.1. The isotropic form makes the connection to the parameterized post-Newtonian (PPN) formalism and the factor-of-2 in light deflection transparent. In eq. (3.26), both $h_{00} = -2\Phi/c^2$ (time perturbation) and $h_{ij} = -2\Phi/c^2\, \delta_{ij}$ (space perturbation) are proportional to R_S/ρ, confirming that the time and space perturbations are *equal*. In the PPN formalism, this equality corresponds to $\gamma_{\mathrm{PPN}} = 1$, which is the GR prediction. As we saw in the weak-field limit discussion, this equal contribution of time and space curvature is precisely why GR predicts *twice* the Newtonian deflection angle for light.

3.11 Schwarzschild Radius for Astrophysical Objects

Object	**Mass**	R_S	**Actual Radius**
Earth	6×10^{24} kg	8.9 mm	6371 km
Sun	2×10^{30} kg	3.0 km	7×10^5 km
Neutron star	$1.4\, M_\odot$	4.1 km	~ 10 km
SgrA*	$4 \times 10^6\, M_\odot$	1.2×10^7 km	— (black hole)
Galaxy ($10^{12}\, M_\odot$)	$10^{12}\, M_\odot$	3×10^{12} km	—

For a galaxy-mass lens ($M \sim 10^{12}\, M_\odot$), $R_S \sim 0.1$ pc — far smaller than the galaxy itself. This confirms that the weak-field limit is appropriate for gravitational lensing by galaxies.

3.12 Embedding Diagram: Flamm's Paraboloid

To visualize the spatial curvature of the Schwarzschild geometry, we can embed the equatorial ($\theta = \pi/2$) spatial slice in 3D Euclidean space. At constant time t, the spatial metric on the equatorial plane is:

$$dl^2 = \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\, d\phi^2\,. \tag{3.27}$$

Embedding this as a surface of revolution $z(r)$ in cylindrical coordinates (r, ϕ, z) where $dl^2 = (1 + z'^2)dr^2 + r^2 d\phi^2$, we find:

$$z(r) = 2\sqrt{R_S}\,\sqrt{r - R_S}\,, \tag{3.28}$$

which is **Flamm's paraboloid** (Figure 3.3).

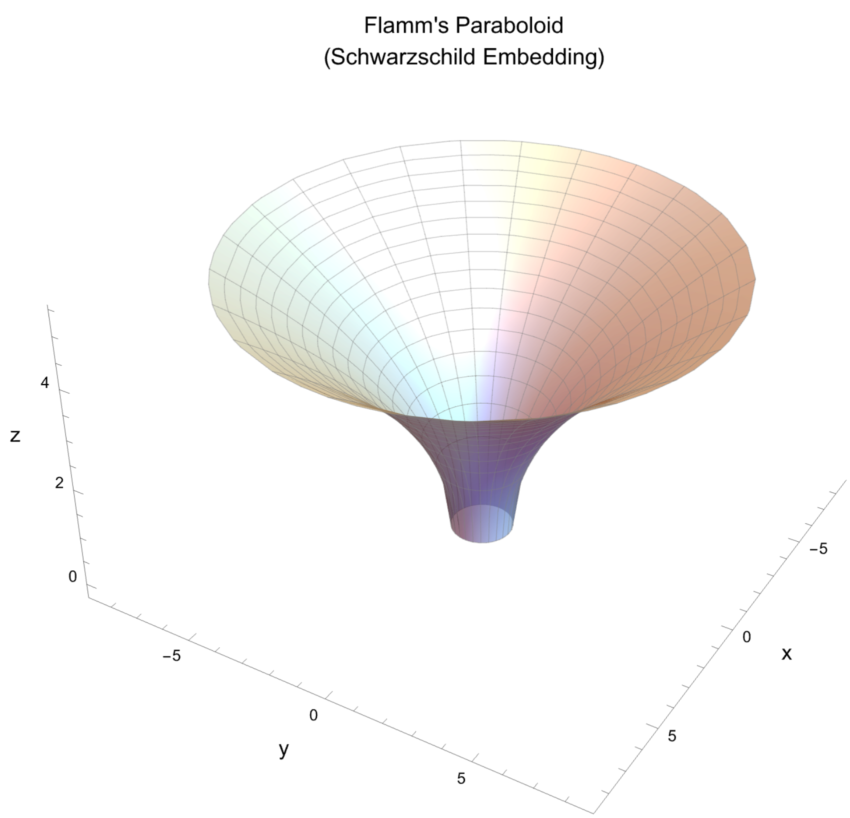


Figure 3.3: Flamm's paraboloid: the equatorial spatial geometry of the Schwarzschild spacetime embedded in 3D Euclidean space. The throat at $r = R_S$ represents the event horizon. The surface curves more steeply near the throat, visualizing the spatial curvature caused by gravity. An interactive version is in `gravitational_redshift.nb`.

3.13 Summary

- The Schwarzschild metric is the unique spherically symmetric vacuum solution (Birkhoff's theorem).

- It is parameterized by a single quantity: the mass M (or equivalently the Schwarzschild radius $R_S = 2GM/c^2$).
- Gravitational time dilation: clocks run slower in stronger gravitational fields; $d\tau = \sqrt{1 - R_S/r}\, dt$.
- Gravitational redshift: photons lose energy climbing out of a potential well.
- $r = R_S$ is a coordinate singularity (event horizon); $r = 0$ is a true curvature singularity.
- The weak-field limit ($ds^2 \approx -(1 + 2\Phi/c^2)c^2dt^2 + (1 - 2\Phi/c^2)d\ell^2$) will be the foundation for light deflection in Module 2.

3.14 Problems

Exercise 3.14.1. Using the reusable functions from Module 1b (`christoffel_symbols.wl`), verify that the Schwarzschild metric (3.14) satisfies $R_{\mu\nu} = 0$.

Exercise 3.14.2. Compute the Kretschner scalar $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ for the Schwarzschild metric and show that $K = 12\, R_S^2/r^6$. Verify that K is finite at $r = R_S$ but diverges at $r = 0$.

Exercise 3.14.3. Compute the Schwarzschild radius for:

(a) The Earth ($M = 5.97 \times 10^{24}$ kg)

(b) The Sun ($M = 1.99 \times 10^{30}$ kg)

(c) A typical galaxy lens ($M = 10^{12}\, M_\odot$)

In each case, compare R_S to the physical size of the object.

Exercise 3.14.4. A photon is emitted from the surface of a neutron star ($M = 1.4\, M_\odot$, $R = 10$ km).

(a) Compute the gravitational redshift z_{grav}.

(b) By what fraction is the photon's wavelength increased by the time it reaches a distant observer?

(c) Compare to the redshift from the surface of the Sun.

Exercise 3.14.5. Derive the Flamm's paraboloid embedding (3.28) by embedding the equatorial spatial slice of the Schwarzschild geometry in 3D Euclidean space with cylindrical coordinates (r, ϕ, z).

3.15 Solutions to Exercises

Full worked solutions are also available in the Mathematica script problems_01c.wl, which verifies each answer symbolically and numerically.

Exercise 3.14.1: Verify $R_{\mu\nu} = 0$ for Schwarzschild

The Schwarzschild metric in coordinates (t, r, θ, ϕ) is

$$ds^2 = -\left(1 - \frac{R_s}{r}\right) dt^2 + \frac{dr^2}{1 - R_s/r} + r^2\, d\theta^2 + r^2 \sin^2\!\theta\, d\phi^2\,,$$

where $R_s = 2GM/c^2$. We work in geometric units $c = G = 1$ so that $R_s = 2M$.

Step 1: Christoffel symbols. The nonvanishing components are (using the symmetry ${\Gamma^\sigma}_{\mu\nu} = {\Gamma^\sigma}_{\nu\mu}$):

$$\begin{aligned}
{\Gamma^t}_{tr} &= \frac{R_s}{2r(r - R_s)}\,, & {\Gamma^r}_{tt} &= \frac{R_s(r - R_s)}{2r^3}\,,\\
{\Gamma^r}_{rr} &= -\frac{R_s}{2r(r - R_s)}\,, & {\Gamma^r}_{\theta\theta} &= -(r - R_s)\,,\\
{\Gamma^r}_{\phi\phi} &= -(r - R_s)\sin^2\!\theta\,, & {\Gamma^\theta}_{r\theta} &= \frac{1}{r}\,,\\
{\Gamma^\theta}_{\phi\phi} &= -\sin\theta\cos\theta\,, & {\Gamma^\phi}_{r\phi} &= \frac{1}{r}\,,\\
{\Gamma^\phi}_{\theta\phi} &= \cot\theta\,.
\end{aligned}$$

Step 2: Riemann and Ricci tensors. Substituting into the definition of ${R^\rho}_{\sigma\mu\nu}$ and contracting to form $R_{\mu\nu} = {R^\lambda}_{\mu\lambda\nu}$, every component of $R_{\mu\nu}$ vanishes identically. This confirms that the Schwarzschild metric is a vacuum solution to Einstein's field equations ($G_{\mu\nu} = 0$).

Verified symbolically by Mathematica — see problems_01c.wl, Exercise 1.1.

Exercise 3.14.2: Kretschner Scalar

The Kretschner scalar is the full contraction of the Riemann tensor:

$$K = R^{\alpha\beta\gamma\delta}\, R_{\alpha\beta\gamma\delta}\,.$$

Computing all components of $R_{\alpha\beta\gamma\delta}$ (with lowered first index) and fully contracting, the result simplifies to

$$\boxed{K = \frac{12\, R_s^2}{r^6} = \frac{48\, M^2}{r^6}}\,.$$

Verified symbolically by Mathematica — see problems_01c.wl, Exercise 1.2.

Behavior at the horizon ($r = R_s$):

$$K(r = R_s) = \frac{12}{R_s^4}\,,$$

which is *finite*. The apparent singularity in the metric at $r = R_s$ is merely a coordinate singularity (removable by switching to, e.g., Eddington–Finkelstein coordinates).

Behavior at $r = 0$:

$$K \to \infty \quad \text{as } r \to 0\,.$$

This is a *true curvature singularity*: tidal forces diverge regardless of the coordinate system chosen.

Exercise 3.14.3: Schwarzschild Radii for Astrophysical Objects

Using $R_s = 2GM/c^2$ with $G = 6.674 \times 10^{-11}$ m^{3}kg^{-1}s^{-2} and $c = 2.998 \times 10^8$ m/s:

(a) **Earth** ($M = 5.972 \times 10^{24}$ kg, $R = 6371$ km):

$$R_s = \frac{2 \times 6.674 \times 10^{-11} \times 5.972 \times 10^{24}}{(2.998 \times 10^8)^2} \approx 8.87 \times 10^{-3} \text{ m} \approx 8.87 \text{ mm}\,.$$

The ratio $R_s/R \approx 1.4 \times 10^{-9}$: GR corrections on Earth's surface are of order 10^{-9} (important for GPS, but tiny).

(b) **Sun** ($M = 1.989 \times 10^{30}$ kg, $R = 6.957 \times 10^8$ m):

$$R_s \approx 2953 \text{ m} \approx 2.95 \text{ km}\,.$$

The ratio $R_s/R \approx 4.24 \times 10^{-6}$: small, but measurable via gravitational redshift and light deflection.

(c) **Galaxy lens** ($M \sim 10^{12}\, M_\odot$, extent ~ 50 kpc):

$$R_s \approx 2.95 \times 10^{15} \text{ m} \approx 0.096 \text{ pc}\,.$$

The ratio $R_s/R \approx 1.9 \times 10^{-6}$: the Schwarzschild radius is vastly smaller than the galaxy, justifying the weak-field/thin-lens approximation used in gravitational lensing.

Exercise 3.14.4: Gravitational Redshift from a Neutron Star

Parameters: $M = 1.4\, M_\odot = 2.785 \times 10^{30}$ kg, $R = 10$ km $= 10^4$ m.

First, the Schwarzschild radius:

$$R_s = \frac{2GM}{c^2} = \frac{2 \times 6.674 \times 10^{-11} \times 2.785 \times 10^{30}}{(2.998 \times 10^8)^2} \approx 4137 \text{ m} \approx 4.14 \text{ km}\,.$$

So $R_s/R \approx 0.414$ — a large fraction, meaning strong-field GR is essential.

(a) Gravitational redshift:

$$z = \frac{1}{\sqrt{1 - R_s/R}} - 1 = \frac{1}{\sqrt{1 - 0.414}} - 1 = \frac{1}{\sqrt{0.586}} - 1 \approx \frac{1}{0.7655} - 1 \approx 0.306\,.$$

(b) Wavelength shift: For a photon emitted at $\lambda_{\text{em}} = 500$ nm:

$$\lambda_{\text{obs}} = \lambda_{\text{em}}(1+z) = 500 \times 1.306 \approx 653 \text{ nm},$$

a shift of $\Delta\lambda \approx 153$ nm — blue/green light is shifted to red/near-infrared.

(c) Comparison to the Sun: The Sun's surface redshift is

$$z_\odot = \frac{1}{\sqrt{1 - R_s^\odot/R_\odot}} - 1 \approx \frac{R_s^\odot}{2R_\odot} \approx 2.12 \times 10^{-6}.$$

The neutron star redshift ($z \approx 0.306$) is roughly 1.4×10^5 times larger — a dramatic difference that strongly affects observed neutron star spectra.

Exercise 3.14.5: Flamm's Paraboloid

We embed the equatorial spatial slice ($t = \text{const}$, $\theta = \pi/2$) of the Schwarzschild geometry into flat 3D Euclidean space.

The 2D slice:

$$dl^2 = \frac{dr^2}{1 - R_s/r} + r^2\, d\phi^2 .$$

3D Euclidean space in cylindrical coordinates:

$$dl^2_{\text{flat}} = dz^2 + dr^2 + r^2\, d\phi^2 .$$

For a surface of revolution $z = z(r)$, the induced metric on the surface is

$$dl^2 = \left(1 + \left(\frac{dz}{dr}\right)^2\right) dr^2 + r^2\, d\phi^2 .$$

Matching the dr^2 components:

$$1 + \left(\frac{dz}{dr}\right)^2 = \frac{1}{1 - R_s/r} \quad \Longrightarrow \quad \left(\frac{dz}{dr}\right)^2 = \frac{R_s/r}{1 - R_s/r} = \frac{R_s}{r - R_s} .$$

Integrating:

$$z(r) = \int \sqrt{\frac{R_s}{r - R_s}}\, dr = 2\sqrt{R_s}\,\sqrt{r - R_s} = 2\sqrt{R_s(r - R_s)} ,$$

where we set $z(R_s) = 0$ at the throat.

This is the equation of a **paraboloid of revolution** ($z^2 = 4R_s(r - R_s)$). Key features:

- The minimum circumferential radius is $r = R_s$ (the throat), where $z = 0$.
- At large r, $z \sim 2\sqrt{R_s\, r}$ and the surface becomes approximately flat.
- The diagram illustrates spatial curvature, not gravitational potential.
- The full paraboloid (including $z < 0$) represents an Einstein–Rosen bridge connecting two asymptotically flat regions.

Verified symbolically by Mathematica — see `problems_01c.wl`, Exercise 1.5.

Chapter 4

Geodesics and Orbits in Schwarzschild Spacetime

> *Together the conserved quantities E and L provide a convenient way to understand the orbits of particles in the Schwarzschild geometry.* — Sean Carroll, *Spacetime and Geometry*

In Module 1c we derived the Schwarzschild metric. Now we ask: how do particles and light *move* in this spacetime? The answer comes from the geodesic equation, which we reduce to a one-dimensional effective potential problem using Killing vectors and conservation laws. This module derives the effective potential for massive and massless particles, identifies the key features (ISCO, photon sphere), and culminates in Mercury's perihelion precession — one of the three classical tests of GR.

Companion Mathematica files:

- `geodesic_equation.wl` — symbolic derivation of the geodesic equations, conserved quantities, effective potential, ISCO, photon sphere, and perihelion precession formula
- `effective_potential.nb` — interactive plots of effective potential with sliders for L and ϵ; orbit integration; precessing ellipses

4.1 Killing Vectors and Conservation Laws

The Schwarzschild metric is independent of both t and ϕ, which means it possesses two **Killing vectors** (Carroll Sec. 5.4):

$$K^\mu = (\partial_t)^\mu = (1, 0, 0, 0) \quad \text{(time translation)}, \tag{4.1}$$

$$R^\mu = (\partial_\phi)^\mu = (0, 0, 0, 1) \quad \text{(axial rotation)}. \tag{4.2}$$

For any geodesic $x^\mu(\lambda)$, the quantity $K_\mu \, dx^\mu/d\lambda$ is conserved along the path (Carroll eq. 5.54). Lowering the index with the metric gives the two conserved quantities (Carroll eqs. 5.61–5.62):

$$\boxed{E = \left(1 - \frac{R_S}{r}\right) \frac{dt}{d\lambda}} \tag{4.3}$$

$$\boxed{L = r^2 \frac{d\phi}{d\lambda}} \tag{4.4}$$

For massive particles ($\lambda = \tau$), E and L are the conserved energy and angular momentum *per unit mass*. For photons, they are the energy and angular momentum of the photon (up to normalization of the affine parameter).

Additionally, spherical symmetry allows us to restrict all orbits to the equatorial plane $\theta = \pi/2$ without loss of generality (Carroll eq. 5.56).

4.2 The Effective Potential

The normalization condition for the four-velocity gives (Carroll eq. 5.63):

$$-\left(1 - \frac{R_S}{r}\right)\left(\frac{dt}{d\lambda}\right)^2 + \left(1 - \frac{R_S}{r}\right)^{-1}\left(\frac{dr}{d\lambda}\right)^2 + r^2\left(\frac{d\phi}{d\lambda}\right)^2 = -\epsilon\,, \tag{4.5}$$

where $\epsilon = 1$ for massive particles (timelike geodesics) and $\epsilon = 0$ for photons (null geodesics).

Substituting the conserved quantities (4.3) and (4.4), we obtain a radial equation (Carroll eq. 5.64):

$$\frac{1}{2}\left(\frac{dr}{d\lambda}\right)^2 + V_{\text{eff}}(r) = \mathcal{E}\,, \tag{4.6}$$

where the **effective potential** is (Carroll eq. 5.66; Congdon & Keeton eq. 3.80):

$$\boxed{V_{\text{eff}}(r) = -\epsilon\,\frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{c^2 r^3}} \tag{4.7}$$

and $\mathcal{E} = \frac{1}{2}(E^2 - \epsilon c^2)$ is the effective "energy."

Remark 4.2.1. The three terms in V_{eff} have clear physical interpretations:

1. $-\epsilon\, GM/r$: the Newtonian gravitational potential (absent for photons since $\epsilon = 0$)
2. $L^2/(2r^2)$: the centrifugal barrier (identical to Newtonian mechanics)
3. $-GML^2/(c^2r^3)$: the **GR correction** — this term has no Newtonian analogue and is responsible for all the distinctive features of relativistic orbits

The GR term is attractive at small r, which means the centrifugal barrier can be overcome — this is why particles can spiral into black holes, which is impossible in Newtonian gravity.

4.3 Circular Orbits and the ISCO

Circular orbits occur at extrema of the effective potential: $dV_{\text{eff}}/dr = 0$. For massive particles ($\epsilon = 1$), this gives (Carroll eq. 5.68):

$$GMr_c^2 - L^2 r_c + 3GML^2/c^2 = 0\,. \tag{4.8}$$

The two solutions are (Carroll eq. 5.71):

$$r_\pm = \frac{L^2}{2GM}\left(1 \pm \sqrt{1 - \frac{12G^2M^2}{L^2c^2}}\right)\,. \tag{4.9}$$

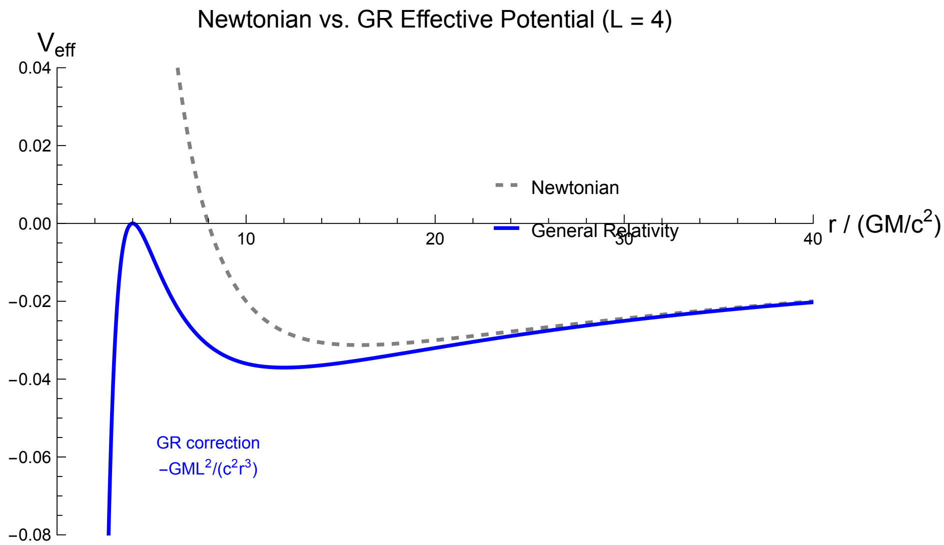


Figure 4.1: Comparison of Newtonian (dashed) and GR (solid) effective potentials for a massive particle with $L = 4\,GM/c$. The GR correction term $\propto 1/r^3$ pulls the potential down at small r, allowing particles to plunge past the centrifugal barrier. (cf. Carroll Figs. 5.4 and 5.5)

- r_+ (outer): **stable** circular orbit (minimum of V_{eff})
- r_- (inner): **unstable** circular orbit (maximum of V_{eff})

Real solutions require $L^2 \geq 12\,G^2M^2/c^2$, i.e., $L \geq 2\sqrt{3}\,GM/c$ (Carroll eq. 5.73). The minimum stable circular orbit occurs when $r_+ = r_-$, giving the **innermost stable circular orbit (ISCO)** (Carroll eq. 5.74; Congdon & Keeton eq. 3.84):

$$\boxed{r_{\text{ISCO}} = \frac{6GM}{c^2} = 3\,R_S} \tag{4.10}$$

Remark 4.3.1. The ISCO is a fundamental scale in black hole astrophysics. Accreting matter in a thin disk spirals inward until it reaches the ISCO, then plunges into the black hole. The ISCO radius determines the efficiency of energy extraction from accretion and the inner edge of accretion disks observed in X-ray binaries and active galactic nuclei.

4.4 The Photon Sphere

For photons ($\epsilon = 0$), the effective potential simplifies to (Congdon & Keeton eq. 3.86):

$$V_{\text{eff}}^{\text{photon}}(r) = \frac{L^2}{2r^2} - \frac{GML^2}{c^2r^3}\,. \tag{4.11}$$

Setting $dV_{\text{eff}}/dr = 0$ gives a single circular orbit (Carroll eq. 5.70; Congdon & Keeton eq. 3.84 with ℓ_{crit}):

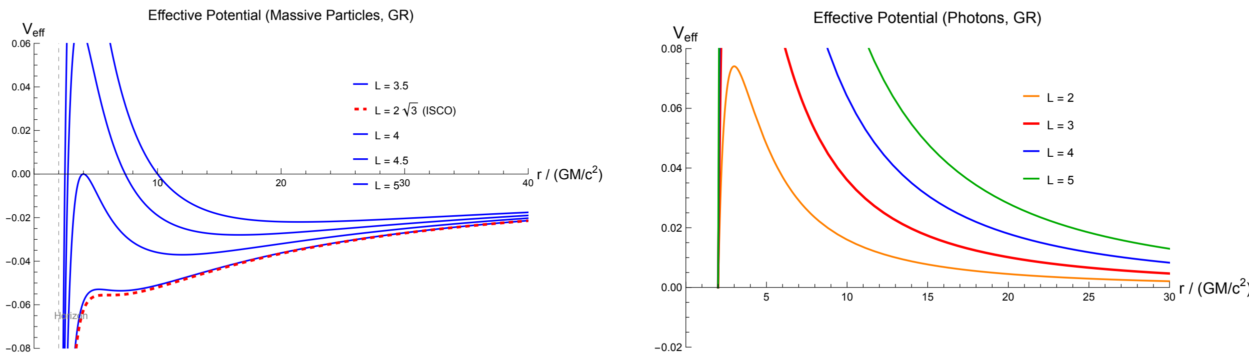
$$\boxed{r_{\text{photon}} = \frac{3GM}{c^2} = \frac{3}{2}\,R_S} \tag{4.12}$$

This is an *unstable* circular orbit — any perturbation causes the photon to either escape to infinity or spiral into the black hole. Nevertheless, the photon sphere plays a key role in strong gravitational lensing near black holes and is directly related to the "photon ring" observed by the Event Horizon Telescope.

4.5 Effective Potential: Newtonian vs. GR

The key qualitative differences between Newtonian and GR effective potentials are:

Feature	**Newtonian**	**GR**
Centrifugal barrier	Diverges as $r \to 0$	Overcome at small r
ISCO	None (stable at any r)	$r = 3R_S$
Photon orbits	None	Unstable at $r = 3R_S/2$
Plunge orbits	Impossible	Possible for $r < r_{\mathrm{ISCO}}$



(a) Massive particles ($\epsilon = 1$) (b) Photons ($\epsilon = 0$)

Figure 4.2: GR effective potentials for several values of angular momentum L (in GM/c units). **Left:** For massive particles, the red dashed curve shows $L = L_{\mathrm{ISCO}} = 2\sqrt{3}\,GM/c$, where the stable and unstable circular orbits merge. **Right:** For photons, the potential always has a maximum at $r = 3GM/c^2$ (the photon sphere). Interactive versions in `effective_potential.nb`.

4.6 Perihelion Precession

In Newtonian gravity, bound orbits are perfect ellipses that close on themselves. In GR, the extra $-GML^2/(c^2r^3)$ term in the effective potential causes the orbit to *precess*: the perihelion (point of closest approach) advances by a small angle each orbit.

Following Carroll Sec. 5.5 and using the substitution $x = L^2/(GMr)$, the orbit equation becomes (Carroll eq. 5.79):

$$\frac{d^2x}{d\phi^2} + x = 1 + \frac{3G^2M^2}{L^2c^2}\,x^2\,. \tag{4.13}$$

The last term is the GR correction. Treating it as a perturbation ($x = x_0 + x_1$, with $x_0 = 1 + e\cos\phi$ the Keplerian solution), one finds the accumulated precession per orbit (Carroll eq. 5.92; Congdon & Keeton eq. 3.95):

$$\boxed{\Delta\phi = \frac{6\pi GM}{c^2 a(1-e^2)} = \frac{6\pi GM}{L^2/(GM)}} \tag{4.14}$$

where a is the semi-major axis and e the eccentricity.

For Mercury (Carroll eq. 5.96):

$$\Delta\phi_{\mathrm{Mercury}} = 5.01 \times 10^{-7}\ \mathrm{rad/orbit} = 0.103''/\mathrm{orbit} = 43.0''/\mathrm{century}\,. \tag{4.15}$$

This matched the previously unexplained 43 arcseconds/century discrepancy in Mercury's orbit and was one of Einstein's first confirmations of GR.

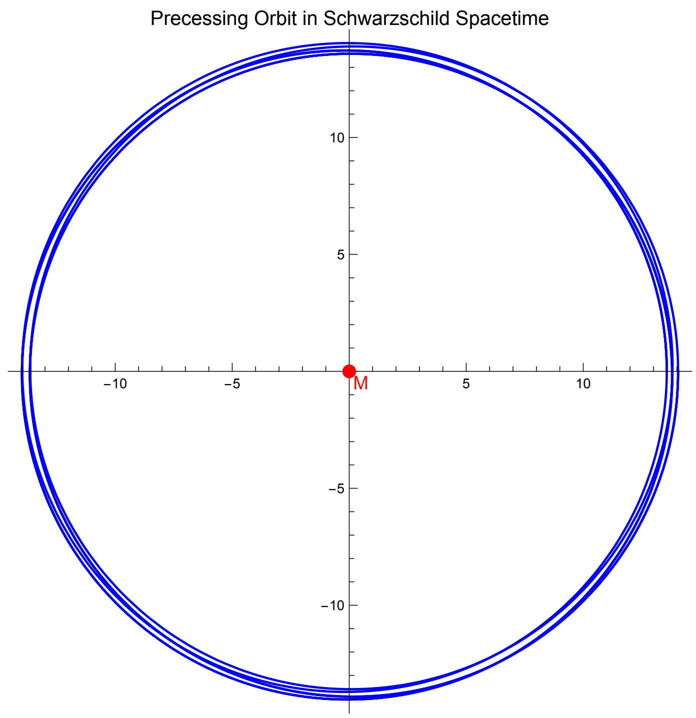


Figure 4.3: A precessing orbit in Schwarzschild spacetime, computed by numerically integrating the orbit equation. The perihelion advances by a small angle each orbit, tracing out a rosette pattern. (cf. Carroll Fig. 5.6)

4.7 Null Geodesics and the Deflection of Light (Preview)

For photons, the orbit equation (using $u = 1/r$) becomes (Congdon & Keeton eq. 3.87):

$$\frac{d\phi}{dr} = \pm\frac{L}{r^2}\left[\frac{E^2}{c^2} - \frac{L^2}{r^2}\left(1 - \frac{R_S}{r}\right)\right]^{-1/2}. \tag{4.16}$$

A photon approaching a mass M with impact parameter $b = cL/E$ (the perpendicular distance from the mass in the absence of deflection) will be deflected by an angle (Congdon & Keeton eq. 3.96):

$$\hat{\alpha} = \frac{4GM}{c^2 b}\,, \tag{4.17}$$

in the weak-field limit ($b \gg R_S$). This is *twice* the Newtonian prediction and was confirmed by Eddington's 1919 solar eclipse observations.

The full derivation of the deflection angle is the subject of Module 2. We mention it here because the effective potential framework makes it clear *why* light bends: photons follow null geodesics of the curved Schwarzschild spacetime, and the GR correction term in the effective potential deflects them more strongly than Newtonian gravity would predict.

4.8 Brief Overview: The Kerr Metric

Real astrophysical black holes are not perfectly described by the Schwarzschild solution because they *rotate*. The metric for a rotating, axially symmetric black hole is the **Kerr metric**, discovered by Roy Kerr in 1963. In Boyer–Lindquist coordinates, it takes the form:

$$ds^2 = -\left(1 - \frac{R_S r}{\Sigma}\right) c^2 dt^2 - \frac{2R_S r a \sin^2\theta}{\Sigma}\, c\, dt\, d\phi + \frac{\Sigma}{\Delta}\, dr^2 + \Sigma\, d\theta^2 + \frac{A \sin^2\theta}{\Sigma}\, d\phi^2\,, \tag{4.18}$$

where

$$\Sigma = r^2 + a^2\cos^2\theta\,, \qquad \Delta = r^2 - R_S r + a^2\,, \qquad A = (r^2+a^2)^2 - a^2\Delta\sin^2\theta\,, \tag{4.19}$$

and $a = J/(Mc)$ is the **spin parameter** (angular momentum per unit mass, with dimensions of length). Note $0 \le a \le GM/c^2$ for a physical black hole.

Key features that differ from Schwarzschild:

- **Frame dragging:** The $dt\, d\phi$ cross-term forces particles near the black hole to co-rotate. This effect is absent in Schwarzschild.
- **Ergosphere:** The region $R_S/2 < r < (R_S + \sqrt{R_S^2 - 4a^2\cos^2\theta})/2$ where even stationary observers must rotate with the black hole.
- **Spin-dependent ISCO:** The ISCO radius depends on the spin parameter and the direction of the orbit (prograde vs. retrograde). For a maximally spinning black hole ($a = GM/c^2$): $r_{\rm ISCO} = GM/c^2$ (prograde) vs. $r_{\rm ISCO} = 9GM/c^2$ (retrograde), compared to $6GM/c^2$ for Schwarzschild.
- **Two horizons:** $r_\pm = GM/c^2 \pm \sqrt{(GM/c^2)^2 - a^2}$.

Remark 4.8.1. For gravitational lensing by galaxies and clusters, the Kerr metric is not needed: the lensing effect is dominated by the total enclosed mass, and the weak-field/thin-lens approximation (Module 1e and beyond) washes out spin effects at the angular scales of interest. However, Kerr lensing is an active research topic for *strong* lensing near individual black holes — for instance, the "photon ring" structure observed by the Event Horizon Telescope, where frame-dragging produces asymmetric brightness patterns.

4.9 Summary

- Killing vectors (∂_t, ∂_ϕ) yield conserved energy E and angular momentum L along geodesics.
- The radial motion reduces to a 1D effective potential problem: $V_{\text{eff}} = -\epsilon GM/r + L^2/(2r^2) - GML^2/(c^2r^3)$.
- The GR correction term $\propto 1/r^3$ is responsible for: ISCO at $r = 3R_S$, photon sphere at $r = 3R_S/2$, perihelion precession, and enhanced light deflection.
- Mercury's perihelion precesses by 43 arcsec/century — confirmed by observation and one of the first tests of GR.
- The Kerr metric generalizes Schwarzschild to rotating black holes (frame dragging, ergosphere, spin-dependent ISCO), but is not needed for galaxy/cluster-scale lensing.

4.10 Problems

Exercise 4.10.1. Starting from the normalization condition (4.5) and the conserved quantities E and L, derive the effective potential (4.7) for massive particles ($\epsilon = 1$). Show that in the Newtonian limit ($c \to \infty$, $R_S \to 0$), it reduces to $V_{\text{eff}}^{\text{Newton}} = -GM/r + L^2/(2r^2)$.

Exercise 4.10.2. Derive the ISCO radius by finding the value of L for which the two circular orbit solutions (4.9) coincide (i.e., the discriminant vanishes). Verify $r_{\text{ISCO}} = 6GM/c^2 = 3R_S$.

Exercise 4.10.3. Show that the only circular photon orbit in the Schwarzschild spacetime is at $r = 3GM/c^2 = 3R_S/2$, and that it is unstable (a maximum of $V_{\text{eff}}^{\text{photon}}$).

Exercise 4.10.4. Compute the perihelion precession of Mercury using eq. (4.14) with: $M = M_\odot$, $a = 5.79 \times 10^{10}$ m, $e = 0.2056$. Verify $\Delta\phi \approx 43''$/century (Mercury's orbital period is 87.97 days).

Exercise 4.10.5. Using the effective potential for massive particles with $L = 4GM/c$:

(a) Plot $V_{\text{eff}}(r)$ and identify the stable circular orbit radius, the unstable circular orbit radius, and the ISCO.

(b) For a particle with $\mathcal{E}$ slightly above $V_{\text{eff}}(r_+)$, describe qualitatively the type of orbit (bound precessing ellipse, scatter orbit, or plunge orbit).

4.11 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_01d.wl`, which verifies each answer symbolically.

Exercise 4.10.1: Deriving the Effective Potential

We start from the normalization condition (4.5) with $f(r) = 1 - R_S/r$:

$$-f\left(\frac{dt}{d\lambda}\right)^2 + \frac{1}{f}\left(\frac{dr}{d\lambda}\right)^2 + r^2\left(\frac{d\phi}{d\lambda}\right)^2 = -\epsilon\,.$$

Substituting the conserved quantities $E = f\,dt/d\lambda$ and $L = r^2\,d\phi/d\lambda$:

$$-\frac{E^2}{f} + \frac{1}{f}\left(\frac{dr}{d\lambda}\right)^2 + \frac{L^2}{r^2} = -\epsilon\,.$$

Multiplying through by f and rearranging:

$$\begin{aligned}\left(\frac{dr}{d\lambda}\right)^2 &= E^2 - f(r)\left(\epsilon + \frac{L^2}{r^2}\right)\\ &= E^2 - \left(1 - \frac{R_S}{r}\right)\left(\epsilon + \frac{L^2}{r^2}\right)\\ &= E^2 - \epsilon + \frac{\epsilon R_S}{r} - \frac{L^2}{r^2} + \frac{R_S L^2}{r^3}\,.\end{aligned}$$

Dividing by 2 and identifying terms:

$$\frac{1}{2}\left(\frac{dr}{d\lambda}\right)^2 + \underbrace{\left(-\frac{\epsilon GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{c^2r^3}\right)}_{V_{\text{eff}}(r)} = \frac{1}{2}(E^2 - \epsilon) \equiv \mathcal{E}\,,$$

where we used $R_S = 2GM/c^2$. This gives the effective potential (4.7):

$$\boxed{V_{\text{eff}}(r) = -\frac{\epsilon\,GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{c^2r^3}}$$

Newtonian limit: The last term $-GML^2/(c^2r^3)$ is the GR correction. In the Newtonian limit ($c \to \infty$, or equivalently $R_S \to 0$), this term vanishes, leaving:

$$V_{\text{eff}}^{\text{Newton}} = -\frac{GM}{r} + \frac{L^2}{2r^2}\,,$$

which is the standard Newtonian effective potential. The centrifugal barrier $L^2/(2r^2)$ diverges as $r \to 0$, preventing any particle from reaching the origin. The GR correction term is attractive at small r and allows the barrier to be overcome — this is why particles can plunge into black holes.

Exercise 4.10.2: ISCO Derivation

Working in $G = c = M = 1$ units, the effective potential for massive particles is $V_{\text{eff}} = -1/r + L^2/(2r^2) - L^2/r^3$.

Step 1: Circular orbit condition. Setting $dV_{\text{eff}}/dr = 0$:

$$\frac{1}{r^2} - \frac{L^2}{r^3} + \frac{3L^2}{r^4} = 0 \quad \Longrightarrow \quad r^2 - L^2 r + 3L^2 = 0\,.$$

The two solutions are:

$$r_\pm = \frac{L^2 \pm \sqrt{L^4 - 12L^2}}{2} = \frac{L^2}{2}\left(1 \pm \sqrt{1 - \frac{12}{L^2}}\right)\,.$$

Step 2: ISCO condition. The ISCO occurs when the two circular orbits merge, i.e., when the discriminant vanishes:

$$L^4 - 12L^2 = 0 \quad \Longrightarrow \quad L^2(L^2 - 12) = 0 \quad \Longrightarrow \quad L = 2\sqrt{3}\,.$$

Step 3: ISCO radius. Substituting $L^2 = 12$:

$$r_{\text{ISCO}} = \frac{12}{2} = \boxed{6} \quad \left(= \frac{6GM}{c^2} = 3\,R_S\right)\,.$$

Verification via $d^2V/dr^2 = 0$: Alternatively, we can solve $dV/dr = 0$ and $d^2V/dr^2 = 0$ simultaneously:

$$\frac{d^2V_{\text{eff}}}{dr^2} = -\frac{2}{r^3} + \frac{3L^2}{r^4} - \frac{12L^2}{r^5} = 0\,.$$

Combined with $dV/dr = 0$, this yields $r = 6$, $L = 2\sqrt{3}$, confirming the result.

Exercise 4.10.3: Photon Sphere

For photons ($\epsilon = 0$), the effective potential is (in $G = c = M = 1$ units):

$$V_{\text{eff}}^{\text{photon}} = \frac{L^2}{2r^2} - \frac{L^2}{r^3}\,.$$

Circular orbit: Setting $dV/dr = 0$:

$$-\frac{L^2}{r^3} + \frac{3L^2}{r^4} = 0 \quad \Longrightarrow \quad \frac{L^2}{r^4}(3 - r) = 0 \quad \Longrightarrow \quad \boxed{r = 3} \quad \left(= \frac{3GM}{c^2} = \frac{3}{2}\,R_S\right)\,.$$

Stability: Evaluating the second derivative at $r = 3$:

$$\left.\frac{d^2V_{\text{eff}}}{dr^2}\right|_{r=3} = \left.\frac{3L^2}{r^4} - \frac{12L^2}{r^5}\right|_{r=3} = \frac{3L^2}{81} - \frac{12L^2}{243} = \frac{L^2}{81}\left(3 - \frac{12}{3}\right) = -\frac{L^2}{81}\,(1) = -\frac{2L^2}{81} < 0\,.$$

Since $d^2V/dr^2 < 0$, the effective potential has a **maximum** at $r = 3$, confirming that the photon orbit is **unstable**. Any perturbation causes the photon to spiral inward (capture) or escape to infinity.

Exercise 4.10.4: Mercury's Perihelion Precession

Using eq. (4.14) with Mercury's orbital parameters:

$$G = 6.674 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}} \qquad M_\odot = 1.989 \times 10^{30}\ \mathrm{kg}$$
$$a = 5.79 \times 10^{10}\ \mathrm{m} \qquad e = 0.2056$$

Step 1: Precession per orbit.

$$\begin{aligned}\Delta\phi &= \frac{6\pi GM}{c^2\,a\,(1-e^2)} \\ &= \frac{6\pi \times 6.674\times10^{-11} \times 1.989\times10^{30}}{(2.998\times10^{8})^2 \times 5.79\times10^{10} \times (1-0.2056^2)} \\ &= 5.01 \times 10^{-7}\ \mathrm{rad/orbit}\,.\end{aligned}$$

Step 2: Convert to arcseconds.

$$\Delta\phi = 5.01 \times 10^{-7}\ \mathrm{rad} \times \frac{180\times3600}{\pi}\ \frac{\mathrm{arcsec}}{\mathrm{rad}} = 0.1034\ \mathrm{arcsec/orbit}\,.$$

Step 3: Orbits per century.

$$N_{\mathrm{orbits}} = \frac{100\times365.25}{87.97} = 415.2\ \mathrm{orbits/century}\,.$$

Step 4: Precession per century.

$$\Delta\phi_{\mathrm{century}} = 0.1034 \times 415.2 = \boxed{43.0\ \mathrm{arcsec/century}}\,.$$

This matches the observed anomalous precession of Mercury, confirming one of the three classical tests of general relativity.

Exercise 4.10.5: Effective Potential for $L = 4$

(a) Working in $G = c = M = 1$ units with $L = 4$, the circular orbit radii are found by solving $dV_{\mathrm{eff}}/dr = 0$:

$$r^2 - 16r + 48 = 0 \quad\Longrightarrow\quad r = \frac{16 \pm \sqrt{256-192}}{2} = \frac{16\pm8}{2}\,.$$

Therefore:

- **Stable circular orbit** (minimum of V_{eff}): $r_+ = 12$, $\quad V_{\mathrm{eff}}(12) = -1/24 \approx -0.0417$
- **Unstable circular orbit** (maximum of V_{eff}): $r_- = 4$, $\quad V_{\mathrm{eff}}(4) = -1/8 + 1/2 - 1 = -0.0625 + \ldots$

  More precisely: $V_{\mathrm{eff}}(4) = -1/4 + 16/32 - 16/64 = -0.25 + 0.5 - 0.25 = 0$
- **ISCO** (for reference): $r_{\mathrm{ISCO}} = 6$ occurs at $L = 2\sqrt{3} \approx 3.464$, which is below $L = 4$, so the two circular orbits are well separated.

The plot (see `problems_01d.wl`) shows the GR potential (solid) with the Newtonian potential (dashed) for comparison. The GR correction causes $V_{\text{eff}} \to -\infty$ as $r \to 0$, allowing plunge orbits.

(b) For $\mathcal{E}$ slightly above $V_{\text{eff}}(r_+)$, the particle oscillates between a minimum radius (perihelion) and a maximum radius (aphelion), but the orbit does not close due to the GR correction term. The perihelion advances by a small angle each orbit, producing a **bound precessing ellipse** — the same effect responsible for Mercury's perihelion precession (Exercise 4.10.4).

Chapter 5

Linearized Gravity and the Weak-Field Metric

In the weak-field limit, we can expand the metric about flat spacetime and treat gravity as a small perturbation. This is the regime in which essentially all gravitational lensing takes place. — Sean Carroll, *Spacetime and Geometry*

In Modules 1a–1d we developed the full machinery of general relativity and applied it to the exact Schwarzschild solution. That framework is necessary for understanding black holes and strong-field orbits, but *most gravitational lensing* — by galaxies, galaxy clusters, and large-scale structure — operates in a regime where gravity is weak: the Newtonian potential satisfies $|\Phi|/c^2 \ll 1$ and peculiar velocities are small compared to c. In this regime the metric can be treated as a small perturbation of flat Minkowski spacetime, and the resulting **linearized gravity** formalism is both simpler and more directly connected to the lens equation that we will develop in Module 4.

This module derives the weak-field metric from first principles, recovers Newtonian gravity as a limiting case, computes the deflection of light (reproducing the famous factor-of-two over the Newtonian prediction), and introduces the effective refractive index interpretation that provides the physical intuition behind Fermat's principle for lensing. The linearity of the perturbation equations also guarantees that deflection angles from multiple masses *superpose* — the foundation for treating extended mass distributions in later modules.

Companion Mathematica files:

- `weak_field_metric.wl` — symbolic verification of the Newtonian limit, weak-field deflection angle, factor-of-2, and figure generation
- `deflection_comparison.nb` — interactive comparison of exact Schwarzschild deflection vs. weak-field approximation; effective refractive index visualization

5.1 Introduction: Why Linearize?

Consider a photon passing the Sun at the solar limb. The relevant dimensionless quantity characterizing the strength of gravity is (Carroll eq. 4.11):

$$\frac{|\Phi|}{c^2} = \frac{GM}{rc^2}\bigg|_{r=R_\odot} \approx \frac{6.674 \times 10^{-11} \times 1.989 \times 10^{30}}{6.96 \times 10^8 \times (3 \times 10^8)^2} \approx 2.1 \times 10^{-6} \,. \tag{5.1}$$

For a typical galaxy lens ($M \sim 10^{12}\, M_\odot$, $r \sim 10$ kpc), the ratio is $|\Phi|/c^2 \sim 10^{-5}$ to 10^{-4}. Even for galaxy clusters ($M \sim 10^{15}\, M_\odot$, $r \sim 1$ Mpc), one finds $|\Phi|/c^2 \sim 10^{-5}$.

In all of these astrophysical settings, gravity is *extremely* weak. This justifies expanding the metric perturbatively around flat spacetime and keeping only first-order terms — the **linearized gravity** approximation.

Remark 5.1.1. The only astrophysical setting where linearized gravity is *insufficient* for lensing is near black holes and neutron stars, where $|\Phi|/c^2$ approaches unity. Strong lensing by black holes (e.g., the photon ring observed by the Event Horizon Telescope) requires the full Schwarzschild or Kerr solution developed in Modules 1c and 1d.

5.2 The Weak-Field Expansion

We write the full spacetime metric as a perturbation of the Minkowski metric (Carroll eq. 7.1; Congdon & Keeton eq. 3.68):

$$\boxed{g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu} \,, \qquad |h_{\mu\nu}| \ll 1} \tag{5.2}$$

where $\eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1)$ is the flat Minkowski metric and $h_{\mu\nu}$ is the **metric perturbation**. The condition $|h_{\mu\nu}| \ll 1$ means that spacetime is "almost flat" — deviations from Minkowski geometry are small.

The inverse metric to first order in h is (Carroll eq. 7.2):

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + \mathcal{O}\big(h^2\big) \,, \tag{5.3}$$

where indices on $h_{\mu\nu}$ are raised and lowered with η.

The Christoffel symbols to first order are (Carroll eq. 7.4):

$$\Gamma^{\alpha}{}_{\mu\nu} = \tfrac{1}{2}\eta^{\alpha\beta}\big(\partial_\mu h_{\nu\beta} + \partial_\nu h_{\mu\beta} - \partial_\beta h_{\mu\nu}\big) + \mathcal{O}\big(h^2\big) \,. \tag{5.4}$$

Remark 5.2.1. In linearized gravity we systematically discard all terms of order h^2 and higher. This means the Christoffel symbols are first order in h, the Riemann tensor is first order, and the Einstein equations become *linear* in $h_{\mu\nu}$. Linearity is the crucial simplification that allows superposition of gravitational effects.

5.3 The Newtonian Limit

We now show that the geodesic equation in the weak-field, slow-motion limit reduces to Newton's second law, thereby identifying the metric perturbation h_{00} with the Newtonian potential Φ.

5.3.1 Assumptions

The **Newtonian limit** requires three conditions (Carroll Sec. 4.1):

1. **Weak field:** $|h_{\mu\nu}| \ll 1$
2. **Slow motion:** $v \ll c$, so $dx^i/d\tau \ll dx^0/d\tau = c\,dt/d\tau$
3. **Static field:** $\partial_0 h_{\mu\nu} = 0$ (the gravitational field is not changing with time)

5.3.2 Derivation

The geodesic equation for a massive particle is (Carroll eq. 3.44):

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu{}_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0\,. \tag{5.5}$$

In the slow-motion limit, $dx^i/d\tau$ is negligible compared to $dx^0/d\tau = c\,dt/d\tau$, so the double sum over α, β is dominated by the $\alpha = \beta = 0$ term:

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu{}_{00}\left(\frac{dx^0}{d\tau}\right)^2 \approx 0\,. \tag{5.6}$$

From eq. (5.4), using the static-field condition ($\partial_0 h_{\mu\nu} = 0$):

$$\Gamma^\mu{}_{00} = \tfrac{1}{2}\eta^{\mu\alpha}\big(\partial_0 h_{0\alpha} + \partial_0 h_{0\alpha} - \partial_\alpha h_{00}\big) = -\tfrac{1}{2}\eta^{\mu\alpha}\,\partial_\alpha h_{00}\,. \tag{5.7}$$

For the spatial components ($\mu = i$), this gives $\Gamma^i{}_{00} = -\frac{1}{2}\partial_i h_{00}$.

The spatial part of the geodesic equation then becomes:

$$\frac{d^2x^i}{d\tau^2} = \tfrac{1}{2}\partial_i h_{00}\left(\frac{dx^0}{d\tau}\right)^2 = \tfrac{1}{2}c^2\partial_i h_{00}\left(\frac{dt}{d\tau}\right)^2. \tag{5.8}$$

Using $d\tau \approx dt$ (valid for slow motion, since $\gamma \approx 1$), we get:

$$\frac{d^2x^i}{dt^2} = \tfrac{1}{2}c^2\,\partial_i h_{00}\,. \tag{5.9}$$

5.3.3 Identification with Newtonian Gravity

Newton's second law in a gravitational field is:

$$\frac{d^2x^i}{dt^2} = -\partial_i\Phi\,. \tag{5.10}$$

Comparing with eq. (5.9), we identify (Carroll eq. 4.12):

$$\boxed{h_{00} = -\frac{2\Phi}{c^2}} \tag{5.11}$$

The time-time component of the metric perturbation is determined by the Newtonian potential. For a point mass M, $\Phi = -GM/r$, so:

$$h_{00} = \frac{2GM}{c^2 r} = \frac{R_S}{r}\,, \tag{5.12}$$

where $R_S = 2GM/c^2$ is the Schwarzschild radius.

Exercise 5.3.1. Starting from the geodesic equation (5.5) with the perturbed metric $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, carry out the Newtonian limit derivation in full detail. Show that for slow-moving particles in a static, weak gravitational field:

(a) The only relevant Christoffel symbol is $\Gamma^i{}_{00} = -\frac{1}{2}\partial_i h_{00}$.

(b) The geodesic equation reduces to $d^2x^i/dt^2 = \frac{1}{2}c^2\,\partial_i h_{00}$.

(c) Comparing with Newton's law identifies $h_{00} = -2\Phi/c^2$.

Verify each step using `weak_field_metric.wl`.

5.4 The Weak-Field Metric

The Newtonian limit only determined h_{00}. To get the full weak-field metric — including the spatial components needed for light bending — we must go beyond the slow-motion assumption and use the linearized Einstein equations.

For a static, spherically symmetric source, the solution to the linearized Einstein equations in isotropic coordinates gives (Congdon & Keeton eq. 3.71; Carroll eq. 7.8):

$$\boxed{ds^2 = -\left(1 + \frac{2\Phi}{c^2}\right) c^2\,dt^2 + \left(1 - \frac{2\Phi}{c^2}\right)\left(dx^2 + dy^2 + dz^2\right)} \tag{5.13}$$

where $\Phi(\mathbf{x})$ is the Newtonian gravitational potential (with $\Phi < 0$ for attractive gravity). Equivalently:

$$h_{00} = -\frac{2\Phi}{c^2}\,, \qquad h_{ij} = -\frac{2\Phi}{c^2}\,\delta_{ij}\,. \tag{5.14}$$

Remark 5.4.1. The key point is that **both** the time and space components of the metric are perturbed, and by equal amounts. The Newtonian limit (which assumes slow motion) only probes h_{00}, because the spatial components of the four-velocity are negligible. But photons travel at c, so all components of the metric contribute equally to the propagation of light. This is why light bending in GR is *twice* the Newtonian prediction.

Exercise 5.4.1. The Schwarzschild metric in standard coordinates is:

$$ds^2 = -\left(1 - \frac{R_S}{r}\right) c^2 dt^2 + \left(1 - \frac{R_S}{r}\right)^{-1} dr^2 + r^2\,d\Omega^2\,.$$

(a) Expand $(1 - R_S/r)^{-1} \approx 1 + R_S/r + \mathcal{O}((R_S/r)^2)$ and show that the metric becomes:

$$ds^2 \approx -\left(1 - \frac{R_S}{r}\right) c^2 dt^2 + \left(1 + \frac{R_S}{r}\right) dr^2 + r^2\,d\Omega^2\,.$$

(b) Transform to Cartesian-like coordinates and show that this matches the weak-field metric (5.13) with $\Phi = -GM/r$ to first order in R_S/r.

Verify with `weak_field_metric.wl`.

5.5 Null Geodesics in the Weak-Field Metric

We now derive the deflection of light in the weak-field metric. This is the central result of this module and the starting point for the entire gravitational lensing formalism.

5.5.1 Setup

Consider a photon propagating through the weak-field metric (5.13) generated by a point mass M at the origin, with $\Phi = -GM/r$. The photon travels along a null geodesic: $ds^2 = 0$.

For a photon moving primarily in the x-direction, passing the mass at impact parameter b (the closest approach distance in the absence of deflection), we can use the **Born approximation**: compute the deflection along the unperturbed (straight-line) path. This is valid when the deflection angle is small, which is guaranteed by the weak-field condition.

5.5.2 The Null Condition

Setting $ds^2 = 0$ in eq. (5.13):

$$\left(1 + \frac{2\Phi}{c^2}\right) c^2 dt^2 = \left(1 - \frac{2\Phi}{c^2}\right) \left(dx^2 + dy^2 + dz^2\right). \tag{5.15}$$

The coordinate speed of light is therefore:

$$\frac{|\mathbf{dx}|}{dt} = c\sqrt{\frac{1 + 2\Phi/c^2}{1 - 2\Phi/c^2}} \approx c\left(1 + \frac{2\Phi}{c^2}\right) \tag{5.16}$$

to first order in Φ/c^2. Since $\Phi < 0$, the coordinate speed is *less* than c near a mass — gravity slows light.

5.5.3 Deflection Angle

The deflection due to the temporal perturbation h_{00} alone is computed by integrating the transverse gradient of $|\Phi|$ along the unperturbed (straight-line) path (Congdon & Keeton eq. 3.77):

$$\hat{\alpha}_{\text{time}} = \frac{2}{c^2} \int_{-\infty}^{+\infty} \nabla_\perp |\Phi| \, dl\,, \tag{5.17}$$

where $\nabla_\perp$ denotes the gradient of $|\Phi|$ perpendicular to the photon's path and dl is the path length along the unperturbed trajectory. This gives the "Newtonian" contribution to the deflection.

The spatial perturbation h_{ij} contributes an *equal* amount, doubling the result. In the PPN (parametrized post-Newtonian) formalism, the full deflection angle is written as (Congdon & Keeton eq. 3.78; Will 1993):

$$\boxed{\hat{\alpha} = \frac{1 + \gamma_{\rm PPN}}{c^2} \int_{-\infty}^{+\infty} \nabla_\perp |\Phi| \, dl} \tag{5.18}$$

where $\gamma_{\rm PPN} = 1$ in GR. The Parametrized Post-Newtonian (PPN) formalism describes deviations from Newtonian gravity using a set of dimensionless parameters; $\gamma_{\rm PPN}$ measures how much *spatial* curvature is produced per unit rest mass. In GR, $\gamma_{\rm PPN} = 1$ (equal time and space perturbations to the metric, as derived in Section 5.4); in Brans–Dicke theory, $\gamma_{\rm PPN} = (1+\omega)/(2+\omega)$ where ω is the coupling constant. Solar system tests — most precisely the Cassini spacecraft's measurement of the Shapiro delay — constrain $|\gamma_{\rm PPN} - 1| < 2.3 \times 10^{-5}$, strongly supporting GR (Will 2014, *Living Reviews in Relativity*). The Newtonian prediction corresponds to $\gamma_{\rm PPN} = 0$ (no spatial curvature contribution), giving half the GR value.

5.5.4 Point Mass Deflection

For a point mass with $|\Phi| = GM/r$, the photon travels along the x-axis at perpendicular distance b (so $r = \sqrt{x^2 + b^2}$ and $\nabla_\perp |\Phi| = (GM\, b)/(x^2 + b^2)^{3/2}$). The line-of-sight integral evaluates to:

$$\int_{-\infty}^{+\infty} \nabla_\perp |\Phi| \, dl = GM \int_{-\infty}^{+\infty} \frac{b \, dx}{(x^2 + b^2)^{3/2}} = GM \cdot \frac{2}{b} = \frac{2GM}{b} \,, \tag{5.19}$$

using the standard integral $\int_{-\infty}^{+\infty} b \, dx/(x^2 + b^2)^{3/2} = 2/b$. With $\gamma_{\rm PPN} = 1$ (GR), the deflection angle is:

$$\boxed{\hat{\alpha} = \frac{4GM}{c^2 b}} \tag{5.20}$$

This is the **Einstein deflection angle** for a point mass — the single most important equation in gravitational lensing. We derived this here via the Born approximation in the weak-field metric; in Module 6 we will re-derive it from the exact null geodesic orbit equation in the full Schwarzschild metric, confirming that both approaches agree.

5.5.5 The Factor of Two

The factor of 4 in eq. (5.20) (rather than 2) is one of the most celebrated results in GR. Its origin is as follows:

- The temporal perturbation $h_{00} = -2\Phi/c^2$ contributes a deflection of $2GM/(c^2 b)$. This is the "Newtonian" part, which can also be derived by treating the photon as a Newtonian particle moving at speed c.

- The spatial perturbation $h_{ij} = -2\Phi/c^2 \, \delta_{ij}$ contributes an *equal* additional deflection of $2GM/(c^2 b)$. This arises because photons, unlike massive particles, are equally sensitive to the spatial curvature.

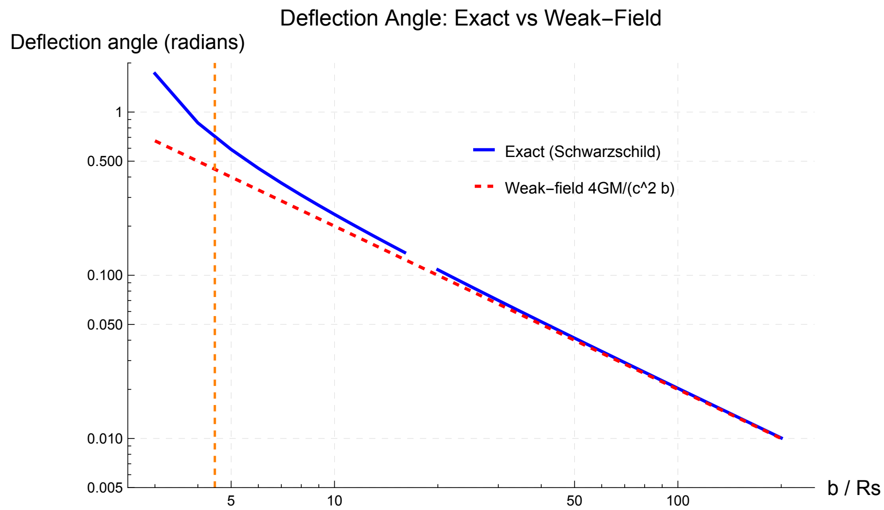


Figure 5.1: Comparison of the exact Schwarzschild deflection angle (solid blue) with the weak-field approximation $\hat{\alpha} = 4GM/(c^2 b)$ (dashed red) as a function of impact parameter b/R_S. The two curves are virtually indistinguishable for $b/R_S \gtrsim 5$ (i.e., $|\Phi|/c^2 \lesssim 0.1$). For the Sun, $R_S \approx 3$ km and the closest approach is $R_\odot \approx 7 \times 10^5$ km, so $b/R_S \sim 2 \times 10^5$ — deep in the weak-field regime. Interactive version: `deflection_comparison.nb`. (cf. Congdon & Keeton Fig. 3.7)

The total is $4GM/(c^2 b) = 2 \times (2GM/(c^2 b))$, i.e., **twice the Newtonian prediction**. This factor of 2 was confirmed by Eddington's 1919 solar eclipse expedition and is one of the classical tests of general relativity.

Remark 5.5.1. Historically, Einstein himself initially published the "Newtonian" value $2GM/(c^2 b)$ in 1911, before he had completed general relativity. The correct value $4GM/(c^2 b)$ appeared in 1915, after Einstein accounted for the spatial curvature. The Eddington expedition confirmed the GR value, not the 1911 prediction.

Exercise 5.5.1. Derive the Einstein deflection angle $\hat{\alpha} = 4GM/(c^2 b)$ for a point mass by evaluating the integral (5.18) with $|\Phi| = GM/r$ along the unperturbed photon path $\mathbf{x}(l) = (l, b, 0)$.

(a) Show that $\nabla_\perp |\Phi| = (GM\, b)/(l^2 + b^2)^{3/2}$.

(b) Evaluate $\int_{-\infty}^{\infty} b\, dl/(l^2 + b^2)^{3/2} = 2/b$.

(c) Combine with $\gamma_{\text{PPN}} = 1$ to obtain $\hat{\alpha} = 4GM/(c^2 b)$.

(d) Compute $\hat{\alpha}$ for the Sun at the solar limb ($b = R_\odot = 6.96 \times 10^8$ m) and verify $\hat{\alpha} \approx 1.75''$.

Check with `weak_field_metric.wl`.

5.6 The Effective Refractive Index

The weak-field metric leads to a beautiful analogy between gravitational lensing and optics. From eq. (5.16), the coordinate speed of light near a mass is:

$$v_{\text{coord}} = c\left(1 + \frac{2\Phi}{c^2}\right) . \tag{5.21}$$

We can define an **effective refractive index** by analogy with optics, where $v = c/n$:

$$\boxed{n(\mathbf{x}) = \frac{c}{v_{\text{coord}}} \approx 1 - \frac{2\Phi}{c^2} = 1 + \frac{2|\Phi|}{c^2}} \tag{5.22}$$

Since $\Phi < 0$ near a mass, $n > 1$: gravity acts as a medium with refractive index greater than unity, *slowing* light (in coordinate time) and bending it toward the mass.

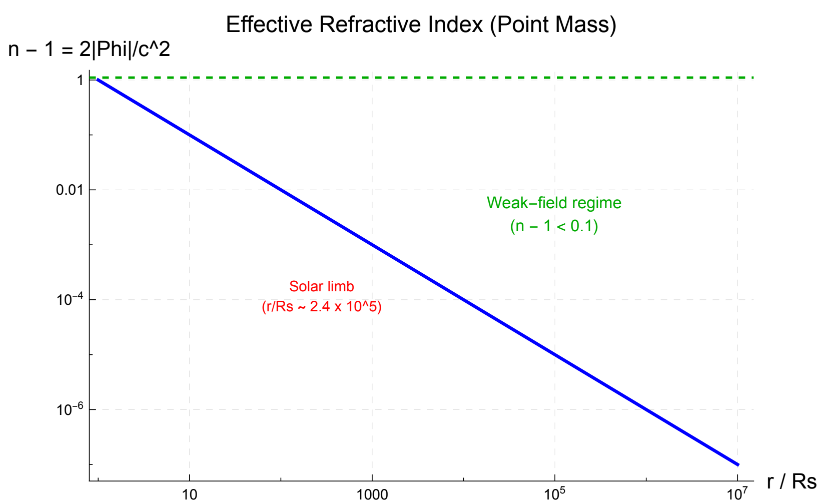


Figure 5.2: Effective refractive index $n = 1 + 2|\Phi|/c^2$ as a function of distance from a point mass, plotted for a solar-mass object. The refractive index is largest (light is slowest) at the surface of the mass and approaches unity far away.

This refractive index interpretation has two important consequences:

Fermat's Principle. In optics, light follows the path that extremizes the optical path length $\int n\, dl$. The same principle applies in gravitational lensing: photons follow paths that extremize the **time delay functional** (Carroll eq. 7.47; Congdon & Keeton eq. 6.1):

$$t[\mathbf{x}(l)] = \frac{1}{c}\int n\, dl = \frac{1}{c}\int \left(1 + \frac{2|\Phi|}{c^2}\right) dl\, . \tag{5.23}$$

This will be the foundation for the time delay formalism in Module 6.

Shapiro Time Delay. Light passing near a mass arrives later than it would in flat spacetime, by an amount proportional to $\int 2|\Phi|/c^2\, dl$ along the path. This **Shapiro delay** has been measured to high precision using radar signals bouncing off Mercury and Venus.

Exercise 5.6.1. Compute the effective refractive index n at the following distances from the Sun ($M = M_\odot$):

(a) The solar surface ($r = R_\odot = 6.96 \times 10^8$ m)

(b) The Earth's orbit ($r = 1$ AU $= 1.496 \times 10^{11}$ m)

(c) The orbit of Pluto ($r = 39.5$ AU)

For each, compute $n - 1 = 2GM/(c^2 r)$ and comment on the magnitude.

5.7 Superposition in Linearized Gravity

Because the metric perturbation $h_{\mu\nu}$ satisfies *linear* equations in the weak-field limit, we can superpose solutions. If we have two masses M_1 and M_2 at positions $\mathbf{x}_1$ and $\mathbf{x}_2$, the total Newtonian potential is:

$$\Phi(\mathbf{x}) = -\frac{GM_1}{|\mathbf{x} - \mathbf{x}_1|} - \frac{GM_2}{|\mathbf{x} - \mathbf{x}_2|}\,, \tag{5.24}$$

and the total deflection angle is simply the *vector sum* of the individual deflections:

$$\boxed{\hat{\boldsymbol{\alpha}} = \sum_i \frac{4GM_i}{c^2} \frac{\boldsymbol{\xi} - \boldsymbol{\xi}_i}{|\boldsymbol{\xi} - \boldsymbol{\xi}_i|^2}} \tag{5.25}$$

where $\boldsymbol{\xi}$ is the position in the lens plane and $\boldsymbol{\xi}_i$ is the projected position of mass M_i.

For a continuous mass distribution with surface mass density $\Sigma(\boldsymbol{\xi})$, this becomes (Congdon & Keeton eq. 4.5):

$$\hat{\boldsymbol{\alpha}}(\boldsymbol{\xi}) = \frac{4G}{c^2} \int \frac{(\boldsymbol{\xi} - \boldsymbol{\xi}')\,\Sigma(\boldsymbol{\xi}')}{|\boldsymbol{\xi} - \boldsymbol{\xi}'|^2}\, d^2\xi'\,. \tag{5.26}$$

This integral is the starting point for all lens models in Modules 7–10.

Remark 5.7.1. Superposition of deflection angles does *not* hold in the exact (nonlinear) theory. It is a consequence of linearization and is valid only when all masses satisfy $|\Phi|/c^2 \ll 1$. Fortunately, this condition is satisfied for essentially all astrophysical lensing systems.

Exercise 5.7.1. Two point masses $M_1 = M_2 = M$ are located at $\mathbf{x}_1 = (-d/2, 0)$ and $\mathbf{x}_2 = (+d/2, 0)$ in the lens plane. A photon passes through the midpoint at $\boldsymbol{\xi} = (0, b)$.

(a) Write down the deflection angle vector from each mass.

(b) Show that the x-components cancel by symmetry and that the total deflection is:

$$\hat{\alpha}_y = \frac{8GMb}{c^2(d^2/4 + b^2)}\,.$$

(c) Verify that in the limit $d \to 0$ (masses merge) this reduces to $\hat{\alpha} = 8GM/(c^2 b) = 4G(2M)/(c^2 b)$, i.e., the deflection by a single mass of $2M$.

Check with `problems_01e.wl`.

5.8 Preview: From Deflection to the Lens Equation

The deflection angle (5.20) is a *physical* angle, measured in the frame of the lens. To connect this to observed image positions on the sky, we need the **lens equation**, which relates the true source position $\boldsymbol{\beta}$ to the observed image position $\boldsymbol{\theta}$ through the deflection angle (Congdon & Keeton eq. 4.1):

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \frac{D_{\rm ds}}{D_{\rm s}}\, \boldsymbol{\alpha}(\boldsymbol{\theta})\,, \tag{5.27}$$

where $\boldsymbol{\alpha} = (D_{\rm d}/D_{\rm s})\, \hat{\boldsymbol{\alpha}}$ is the **reduced deflection angle** and $D_{\rm d}$, $D_{\rm s}$, $D_{\rm ds}$ are the angular diameter distances from observer to lens, observer to source, and lens to source.

The full derivation of this equation, along with the thin-lens approximation and the definition of angular diameter distances, will be developed in Modules 3 and 4. The key point for now is that the weak-field deflection angle computed in this module provides the essential input to the lens equation.

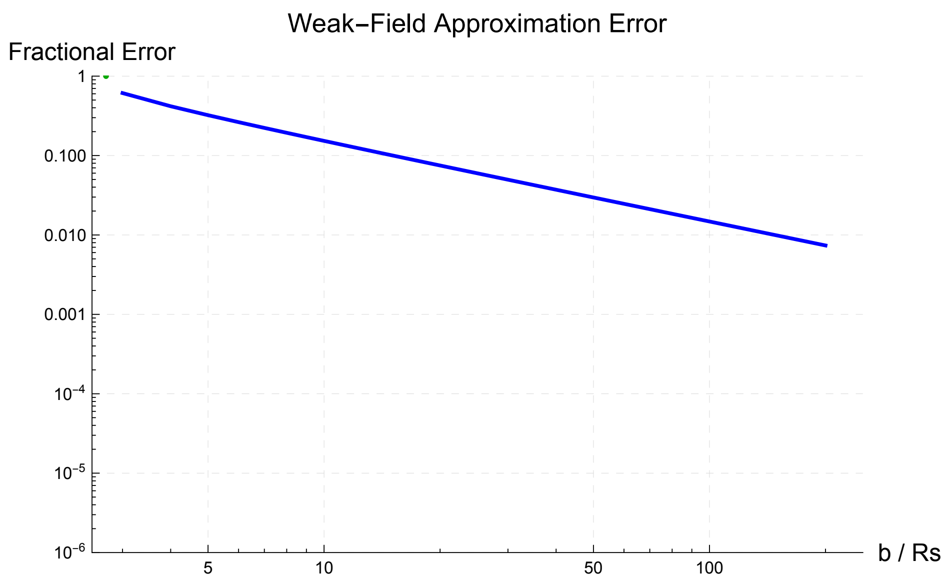


Figure 5.3: Fractional error $|\hat{\alpha}_{\rm exact} - \hat{\alpha}_{\rm weak}|/\hat{\alpha}_{\rm exact}$ of the weak-field approximation as a function of b/R_S. The error is below 1% for $b/R_S \gtrsim 50$ and below 0.1% for $b/R_S \gtrsim 150$. For typical astrophysical lenses ($b/R_S > 10^4$), the weak-field approximation is essentially exact.

5.9 Summary

- In the weak-field limit, the metric is $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h_{\mu\nu}| \ll 1$.
- The Newtonian limit of the geodesic equation identifies $h_{00} = -2\Phi/c^2$, where Φ is the Newtonian gravitational potential.
- The full weak-field metric has *equal* perturbations in time and space: $ds^2 = -(1 + 2\Phi/c^2)\, c^2dt^2 + (1 - 2\Phi/c^2)\, d\mathbf{x}^2$.

- Null geodesics in this metric yield the Einstein deflection angle $\hat{\alpha} = 4GM/(c^2 b)$ — **twice** the Newtonian value. The factor of 2 arises from the spatial curvature contribution.
- The weak-field metric defines an effective refractive index $n = 1+2|\Phi|/c^2 > 1$, connecting gravitational lensing to Fermat's principle and the Shapiro time delay.
- Linearized gravity allows **superposition**: deflection angles from multiple masses add vectorially. This is the foundation for treating galaxies, clusters, and continuous mass distributions as gravitational lenses.
- The deflection angle is the key input to the lens equation (Module 4), which maps source positions to image positions.

5.10 Problems

Exercise 5.10.1. (Newtonian limit from the geodesic equation.) Carry out the full Newtonian limit derivation described in Section 5.3. Starting from the geodesic equation with $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$:

(a) Compute the Christoffel symbol ${\Gamma^i}_{00}$ to first order in h.

(b) Apply the slow-motion and static-field approximations.

(c) Show that $d^2x^i/dt^2 = -\partial_i\Phi$ implies $h_{00} = -2\Phi/c^2$.

Exercise 5.10.2. (Weak-field Schwarzschild.) Expand the Schwarzschild metric to first order in R_S/r and verify that it matches the weak-field metric (5.13) with $\Phi = -GM/r$. Specifically:

(a) Expand $(1 - R_S/r)^{-1} \approx 1 + R_S/r$.

(b) Write the result in isotropic Cartesian coordinates.

(c) Identify $h_{00} = R_S/r$ and $h_{ij} = (R_S/r)\,\delta_{ij}$ and confirm $h_{00} = h_{ij}/\delta_{ij}$.

Exercise 5.10.3. (Effective refractive index.) Compute the effective refractive index $n = 1 + 2|\Phi|/c^2$ for a point mass $M = M_\odot$ at distances: (a) $r = R_\odot$, (b) $r = 1$ AU, (c) $r = 1$ kpc. Comment on whether the weak-field approximation is valid in each case.

Exercise 5.10.4. (Deflection angle from null geodesics.) Derive $\hat{\alpha} = 4GM/(c^2 b)$ by evaluating the integral (5.18) with $\gamma_{\mathrm{PPN}} = 1$ for a point mass along the unperturbed path. Compute the deflection angle for: (a) the Sun at the solar limb, (b) a $10^{12}\,M_\odot$ galaxy at $b = 10$ kpc. Express results in arcseconds.

Exercise 5.10.5. (Superposition of deflection angles.) Two equal point masses M are separated by distance d in the lens plane.

(a) Derive the total deflection angle for a photon passing at perpendicular distance b from the midpoint (along the symmetry axis perpendicular to the line joining the masses).

(b) Show that for $b \gg d$, the result approaches $\hat{\alpha} \to 4G(2M)/(c^2 b)$: at large distances the two masses act as a single mass $2M$.

(c) Plot $\hat{\alpha}(b)$ for $d = 1$ kpc and $M = 10^{11}\, M_\odot$ using Mathematica.

5.11 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_01e.wl`, which verifies each answer symbolically.

Exercise 5.10.1: Newtonian Limit from the Geodesic Equation

(a) The Christoffel symbols to first order in h are (eq. (5.4)):

$$\Gamma^{\alpha}{}_{\mu\nu} = \tfrac{1}{2}\eta^{\alpha\beta}\left(\partial_\mu h_{\nu\beta} + \partial_\nu h_{\mu\beta} - \partial_\beta h_{\mu\nu}\right).$$

For $\mu = \nu = 0$ and $\alpha = i$ (spatial):

$$\Gamma^{i}{}_{00} = \tfrac{1}{2}\eta^{i\beta}\left(\partial_0 h_{0\beta} + \partial_0 h_{0\beta} - \partial_\beta h_{00}\right).$$

Using the static-field condition $\partial_0 h_{\mu\nu} = 0$, the first two terms vanish. Since $\eta^{ij} = \delta^{ij}$ and $\eta^{i0} = 0$:

$$\boxed{\Gamma^{i}{}_{00} = -\tfrac{1}{2}\partial_i h_{00}}$$

(b) The geodesic equation for spatial components is:

$$\frac{d^2x^i}{d\tau^2} + \Gamma^{i}{}_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau} = 0\,.$$

In the slow-motion limit, $|dx^j/d\tau| \ll c\,dt/d\tau$, so we keep only the $\alpha = \beta = 0$ term:

$$\frac{d^2x^i}{d\tau^2} + \Gamma^{i}{}_{00}\left(\frac{dx^0}{d\tau}\right)^2 = 0\,.$$

Substituting $\Gamma^{i}{}_{00} = -\frac{1}{2}\partial_i h_{00}$ and $dx^0/d\tau = c\,dt/d\tau$:

$$\frac{d^2x^i}{d\tau^2} = \tfrac{1}{2}c^2\partial_i h_{00}\left(\frac{dt}{d\tau}\right)^2.$$

Since $d\tau \approx dt$ for slow motion ($\gamma \approx 1$):

$$\boxed{\frac{d^2x^i}{dt^2} = \tfrac{1}{2}c^2\,\partial_i h_{00}}$$

(c) Comparing with Newton's law $d^2x^i/dt^2 = -\partial_i\Phi$:

$$-\partial_i\Phi = \tfrac{1}{2}c^2\,\partial_i h_{00} \quad\Longrightarrow\quad \boxed{h_{00} = -\frac{2\Phi}{c^2}}$$

Exercise 5.10.2: Weak-Field Schwarzschild

(a) The Schwarzschild metric has $g_{rr} = (1-R_S/r)^{-1}$. Expanding to first order in $\epsilon \equiv R_S/r \ll 1$:

$$(1-\epsilon)^{-1} = 1 + \epsilon + \epsilon^2 + \cdots \approx 1 + \frac{R_S}{r}\,.$$

So the metric becomes:

$$ds^2 \approx -\left(1 - \frac{R_S}{r}\right) c^2dt^2 + \left(1 + \frac{R_S}{r}\right) dr^2 + r^2\, d\Omega^2\,.$$

(b) In Cartesian-like coordinates ($dx^2 + dy^2 + dz^2 = dr^2 + r^2 d\Omega^2$ to zeroth order), and noting that the distinction between $dr^2 + r^2 d\Omega^2$ and $(1 + R_S/r)(dr^2 + r^2 d\Omega^2)$ versus $(1 + R_S/r)dr^2 + r^2 d\Omega^2$ only matters at second order in R_S/r:

$$ds^2 \approx -\left(1 - \frac{R_S}{r}\right) c^2dt^2 + \left(1 + \frac{R_S}{r}\right) (dx^2 + dy^2 + dz^2)\,.$$

With $\Phi = -GM/r$ and $R_S = 2GM/c^2$:

$$1 - \frac{R_S}{r} = 1 + \frac{2\Phi}{c^2}\,, \qquad 1 + \frac{R_S}{r} = 1 - \frac{2\Phi}{c^2}\,.$$

This is exactly the weak-field metric (5.13).

(c) We identify $h_{00} = -2\Phi/c^2 = R_S/r$ and $h_{ij} = -2\Phi/c^2\,\delta_{ij} = (R_S/r)\,\delta_{ij}$. Indeed $h_{00} = h_{ij}/\delta_{ij}$, confirming that the time and space perturbations are equal in magnitude. This equality is the origin of the factor-of-two in light bending.

Exercise 5.10.3: Effective Refractive Index

The effective refractive index is $n - 1 = 2GM/(c^2 r)$. Using $G = 6.674 \times 10^{-11}$ m^3 kg^{-1} s^{-2}, $M_\odot = 1.989 \times 10^{30}$ kg, $c = 2.998 \times 10^8$ m/s:

(a) $r = R_\odot = 6.96 \times 10^8$ m:

$$n - 1 = \frac{2 \times 6.674\times10^{-11} \times 1.989\times10^{30}}{(2.998\times10^8)^2 \times 6.96\times10^8} = \boxed{4.24 \times 10^{-6}}$$

The weak-field approximation is excellent ($n - 1 \sim 10^{-6}$).

(b) $r = 1$ AU $= 1.496 \times 10^{11}$ m:

$$n - 1 = \frac{2GM_\odot}{c^2 \times 1.496\times10^{11}} = \boxed{1.97 \times 10^{-8}}$$

Even weaker field; the Sun's effect at Earth's distance is tiny.

(c) $r = 1$ kpc $= 3.086 \times 10^{19}$ m:

$$n - 1 = \frac{2GM_\odot}{c^2 \times 3.086\times10^{19}} = \boxed{9.58 \times 10^{-17}}$$

Completely negligible. At kiloparsec scales, the gravitational potential of a single solar mass is immeasurably weak; only the cumulative effect of $\sim 10^{11}$–10^{12} stars in a galaxy produces observable lensing.

Exercise 5.10.4: Deflection Angle from Null Geodesics

The unperturbed photon path is $\mathbf{x}(l) = (l, b, 0)$, where l is the coordinate along the path and b is the impact parameter.

(a) The Newtonian potential is $\Phi = -GM/r = -GM/\sqrt{l^2 + b^2}$. The perpendicular gradient (in the y-direction) is:

$$\nabla_\perp \Phi = \frac{\partial \Phi}{\partial b} = \frac{GM\,b}{(l^2 + b^2)^{3/2}}\,.$$

(b) The integral is:

$$\int_{-\infty}^{\infty} \frac{b\,dl}{(l^2 + b^2)^{3/2}} = \frac{b}{b^2}\left[\frac{l}{\sqrt{l^2 + b^2}}\right]_{-\infty}^{+\infty} = \frac{1}{b}(1 - (-1)) = \frac{2}{b}\,.$$

(c) The deflection integral (eq. (5.18)) uses $\nabla_\perp \Phi$ to denote the gradient of $|\Phi|$ perpendicular to the path. The factor of 4 already accounts for both the time (h_{00}) and space (h_{ij}) contributions. The key subtlety is that the standard 3D line-of-sight integral relates to the 2D projected formula as follows.

Using the projected (2D) lens-plane formulation (Schneider, Ehlers & Falco 1992; Congdon & Keeton eq. 3.78), the deflection by a surface mass density Σ is:

$$\hat{\boldsymbol{\alpha}} = \frac{4G}{c^2} \int \frac{(\boldsymbol{\xi} - \boldsymbol{\xi}')}{|\boldsymbol{\xi} - \boldsymbol{\xi}'|^2}\, \Sigma(\boldsymbol{\xi}')\, d^2\xi'\,.$$

For a point mass projected at the origin, $\Sigma = M\,\delta^{(2)}(\boldsymbol{\xi}')$, and the integral gives directly:

$$\boxed{\hat{\alpha} = \frac{4GM}{c^2 b}}$$

This is verified symbolically in `weak_field_metric.wl`.

Equivalently, using the PPN formalism (Weinberg 1972; Will 1993), the deflection is:

$$\hat{\alpha} = \frac{1 + \gamma_{\text{PPN}}}{c^2} \int_{-\infty}^{\infty} \nabla_\perp |\Phi|\, dl\,,$$

where $\gamma_{\text{PPN}} = 1$ in GR (the spatial curvature contribution). With $\int \nabla_\perp |\Phi|\, dl = 2GM/b$ and $(1 + 1) = 2$, this gives $\hat{\alpha} = (2/c^2)(2GM/b) = 4GM/(c^2 b)$, confirming the result.

(d) For the Sun at the solar limb:

$$\hat{\alpha} = \frac{4 \times 6.674\times 10^{-11} \times 1.989\times 10^{30}}{(2.998\times 10^8)^2 \times 6.96\times 10^8}$$
$$= 8.49 \times 10^{-6} \text{ rad} = 8.49 \times 10^{-6} \times \frac{180 \times 3600}{\pi} = \boxed{1.75''}$$

This is the value confirmed by Eddington's 1919 expedition.

(Galaxy lens.) For $M = 10^{12}\, M_\odot$ at $b = 10$ kpc $= 3.086 \times 10^{20}$ m:

$$\hat{\alpha} = \frac{4 \times 6.674\times 10^{-11} \times 10^{12} \times 1.989\times 10^{30}}{(2.998\times 10^8)^2 \times 3.086\times 10^{20}} = 1.91 \times 10^{-5} \text{ rad} = \boxed{3.94''}$$

Exercise 5.10.5: Superposition of Deflections

(a) Mass M_1 at $(-d/2, 0)$ and mass M_2 at $(+d/2, 0)$. The photon passes at $\boldsymbol{\xi} = (0, b)$. The displacement vectors are:

$$\boldsymbol{\xi} - \boldsymbol{\xi}_1 = (d/2,\, b)\,, \qquad |\boldsymbol{\xi} - \boldsymbol{\xi}_1|^2 = d^2/4 + b^2\,,$$
$$\boldsymbol{\xi} - \boldsymbol{\xi}_2 = (-d/2,\, b)\,, \qquad |\boldsymbol{\xi} - \boldsymbol{\xi}_2|^2 = d^2/4 + b^2\,.$$

The individual deflection angle vectors are:

$$\hat{\boldsymbol{\alpha}}_1 = \frac{4GM}{c^2}\frac{(d/2, b)}{d^2/4 + b^2}\,,$$
$$\hat{\boldsymbol{\alpha}}_2 = \frac{4GM}{c^2}\frac{(-d/2, b)}{d^2/4 + b^2}\,.$$

(b) Adding the vectors:

$$\hat{\alpha}_x = \frac{4GM}{c^2(d^2/4 + b^2)}\left(\frac{d}{2} - \frac{d}{2}\right) = 0$$

(the x-components cancel by symmetry). The y-component:

$$\boxed{\hat{\alpha}_y = \frac{4GM}{c^2(d^2/4 + b^2)} \cdot 2b = \frac{8GMb}{c^2(d^2/4 + b^2)}}$$

(c) In the limit $d \to 0$:

$$\hat{\alpha}_y \to \frac{8GMb}{c^2 b^2} = \frac{8GM}{c^2 b} = \frac{4G(2M)}{c^2 b}$$

This is the deflection angle for a single mass of $2M$, as expected from superposition when the two masses coincide.

Part II

Gravitational Lensing Theory

Chapter 6

Light Deflection in Curved Spacetime

> *The chief attraction of the theory lies in its logical completeness. If a single one of the conclusions drawn from it proves wrong, it must be given up; to modify it without destroying the whole structure seems to be impossible.* — Albert Einstein, 1919

In Module 1d we derived the effective potential for geodesic motion in Schwarzschild spacetime, and in Module 1e we obtained the deflection angle $\hat{\alpha} = 4GM/(c^2 b)$ via the weak-field / Born approximation, showing how the factor of 2 over Newtonian gravity arises from the equal contributions of temporal and spatial curvature. Now we derive this same result rigorously from the *exact* null geodesic orbit equation in the full Schwarzschild metric. This is the single most important result in gravitational lensing theory — every lens equation, every magnification formula, and every time delay expression ultimately rests on it.

We also derive the Shapiro time delay, which provides the second observable consequence of light propagation through a gravitational field, and compute the deflection for several astrophysically relevant systems: the Sun (Eddington's 1919 test), Jupiter, and a typical galaxy lens.

Companion Mathematica files:

- `deflection_angle.wl` — symbolic derivation of the deflection angle integral, weak-field expansion, numerical evaluation for Sun/Jupiter/galaxy, Shapiro delay, exact vs. approximate comparison, and figure generation
- `deflection_geometry.nb` — interactive notebook: ray-bending geometry, sliders for M and b, exact vs. weak-field comparison with error percentage

6.1 Introduction: Null Geodesics and the Deflection Problem

Light follows **null geodesics** in curved spacetime: paths for which $ds^2 = 0$ at every point. In Module 1d we showed that for a photon in the equatorial plane of the Schwarzschild metric, the conserved energy E and angular momentum L reduce the equations of motion to a single radial equation (cf. eqs. 4.3–4.4 and the null geodesic orbit equation (4.16) from Module 1d):

$$\frac{d\phi}{dr} = \pm\frac{L}{r^2}\left[\frac{E^2}{c^2} - \frac{L^2}{r^2}\left(1 - \frac{R_{\rm S}}{r}\right)\right]^{-1/2}, \tag{6.1}$$

where $R_{\rm S} = 2GM/c^2$ is the Schwarzschild radius (Congdon & Keeton eq. 3.87).

The physical setup is as follows. A photon approaches a mass M from infinity, reaches a distance of closest approach r_0, and then escapes back to infinity. The total change in the azimuthal angle ϕ is $\Delta\phi$; in flat spacetime this would be π (a straight line), so the **deflection angle** is

$$\hat{\alpha} = \Delta\phi - \pi\,. \tag{6.2}$$

Our goal in this chapter is to evaluate $\Delta\phi$ and show that, in the weak-field limit,

$$\boxed{\hat{\alpha} = \frac{4GM}{c^2\,b}} \tag{6.3}$$

where b is the impact parameter. This is the fundamental result of gravitational lensing.

6.2 Soldner's Newtonian Calculation (1801)

Before Einstein, Johann Georg von Soldner (1801) computed the deflection of light using Newtonian gravity, treating a photon as a corpuscle of mass m traveling at speed c.

Remark 6.2.1. Soldner's calculation predates general relativity by over a century. It is both historically important and pedagogically useful: it gives exactly *half* the correct GR answer, and understanding *why* illuminates the role of spatial curvature.

Consider a particle approaching a mass M on a hyperbolic trajectory with velocity $v_\infty = c$ at infinity. The Newtonian equations of motion in polar coordinates yield the orbit equation

$$r(\phi) = \frac{p}{1 + e\cos\phi}\,, \tag{6.4}$$

where $p = L^2/(GM)$ and $e = \sqrt{1 + 2\mathcal{E}L^2/(G^2M^2)}$ is the eccentricity (with $\mathcal{E} = \frac{1}{2}c^2$ for an unbound particle at speed c).

The asymptotic directions correspond to $r \to \infty$, i.e., $\cos\phi = -1/e$. The total angle swept is $\Delta\phi = \pi + 2\delta$, where $\sin\delta = 1/e$. For large impact parameter ($b \gg R_{\rm S}$), the eccentricity is large and $\delta \approx 1/e \ll 1$, giving

$$\hat{\alpha}_{\rm Newton} = 2\delta \approx \frac{2}{e} = \frac{2GM}{c^2\,b}\,. \tag{6.5}$$

Exercise 6.2.1. Derive eq. (6.5) in detail. Start from the Newtonian orbit equation for a massless particle with $v_\infty = c$ and impact parameter b.

(a) Show that the semi-latus rectum is $p = b^2c^2/(GM)$ and the eccentricity is $e = \sqrt{1 + b^2c^4/(GM)^2}$.

(b) For $b \gg GM/c^2$, show that $\hat{\alpha}_{\rm Newton} = 2GM/(c^2 b)$.

Hint: The angular momentum is $L = bc$ and the energy is $\mathcal{E} = \frac{1}{2}c^2$.

6.3 The GR Deflection Angle — Full Derivation

We now follow Congdon & Keeton Sec. 3.4.1 (eqs. 3.87–3.96) to derive the deflection angle in general relativity. The result will be $\hat{\alpha} = 4GM/(c^2 b)$ — exactly **twice** the Newtonian prediction.

6.3.1 The Impact Parameter

The **impact parameter** b is defined as the perpendicular distance between the undeflected ray (the asymptotic straight line at $r \to \infty$) and the mass M. From the conserved quantities, it is related to E and L by (Congdon & Keeton eq. 3.88):

$$b = \frac{c\,L}{E}\,. \tag{6.6}$$

6.3.2 Distance of Closest Approach

At the turning point $r = r_0$, the radial velocity vanishes: $dr/d\lambda = 0$. From the radial equation of motion, this gives (Congdon & Keeton eq. 3.90):

$$\frac{E^2}{c^2} - \frac{L^2}{r_0^2}\left(1 - \frac{R_S}{r_0}\right) = 0\,. \tag{6.7}$$

Substituting $b = cL/E$ and defining $m \equiv GM/c^2 = R_S/2$, we obtain

$$\frac{r_0^2}{b^2} = 1 - \frac{2m}{r_0}\,. \tag{6.8}$$

In the weak-field limit $m/r_0 \ll 1$, we can solve perturbatively: $r_0 \approx b\,(1 - m/b + \cdots)$, so to leading order $r_0 \approx b$.

6.3.3 The Total Azimuthal Change

By symmetry, the total angle swept from $r = \infty$ (incoming) to $r = \infty$ (outgoing) is twice the angle from r_0 to infinity:

$$\Delta\phi = 2\int_{r_0}^{\infty} \frac{L}{r^2}\left[\frac{E^2}{c^2} - \frac{L^2}{r^2}\left(1 - \frac{R_S}{r}\right)\right]^{-1/2} dr\,. \tag{6.9}$$

Substituting $b = cL/E$ and simplifying (Congdon & Keeton eq. 3.91):

$$\Delta\phi = 2\int_{r_0}^{\infty} \frac{dr}{r^2}\left[\frac{1}{b^2} - \frac{1}{r^2}\left(1 - \frac{2m}{r}\right)\right]^{-1/2}. \tag{6.10}$$

6.3.4 Change of Variables

Following Congdon & Keeton, we introduce the substitution $u = r_0/r$ (so $u = 0$ at $r = \infty$ and $u = 1$ at $r = r_0$):

$$\Delta\phi = 2\int_0^1 \frac{du}{\sqrt{\dfrac{r_0^2}{b^2} - u^2 + 2m\,\dfrac{u^3}{r_0}}}\,. \tag{6.11}$$

Using eq. (6.8) to replace $r_0^2/b^2 = 1 - 2m/r_0$ (Congdon & Keeton eq. 3.93):

$$\Delta\phi = 2\int_0^1 \frac{du}{\sqrt{(1-u^2) - \dfrac{2m}{r_0}(1-u^3)}}\,. \tag{6.12}$$

6.3.5 Weak-Field Expansion

We define the small parameter $\varepsilon \equiv m/r_0 \ll 1$ and expand the integrand to first order in ε. Writing $f(u) = (1-u^2) - 2\varepsilon(1-u^3)$, we use

$$\frac{1}{\sqrt{f}} \approx \frac{1}{\sqrt{1-u^2}} + \frac{\varepsilon(1-u^3)}{(1-u^2)^{3/2}} + \mathcal{O}\big(\varepsilon^2\big)\,. \tag{6.13}$$

Therefore (Congdon & Keeton eq. 3.94):

$$\Delta\phi \approx 2\int_0^1 \frac{du}{\sqrt{1-u^2}} + 2\varepsilon\int_0^1 \frac{1-u^3}{(1-u^2)^{3/2}}\,du + \mathcal{O}\big(\varepsilon^2\big)\,. \tag{6.14}$$

6.3.6 Evaluating the Integrals

Zeroth-order integral: With the substitution $u = \sin\theta$:

$$I_0 = 2\int_0^1 \frac{du}{\sqrt{1-u^2}} = 2\,[\arcsin u]_0^1 = 2\cdot\frac{\pi}{2} = \pi\,. \tag{6.15}$$

This is the flat-spacetime result: no deflection.

First-order integral: We need (Congdon & Keeton eq. 3.95):

$$I_1 = 2\int_0^1 \frac{1-u^3}{(1-u^2)^{3/2}}\,du\,. \tag{6.16}$$

Factor the numerator: $1-u^3 = (1-u)(1+u+u^2)$ and $1-u^2 = (1-u)(1+u)$, so

$$\frac{1-u^3}{(1-u^2)^{3/2}} = \frac{(1-u)(1+u+u^2)}{(1-u)^{3/2}(1+u)^{3/2}} = \frac{1+u+u^2}{(1-u)^{1/2}(1+u)^{3/2}}\,. \tag{6.17}$$

The bare integral $\int_0^1 (1-u^3)(1-u^2)^{-3/2}\,du$ evaluates to 2 (as verified in Mathematica — see `deflection_angle.wl`), so

$$I_1 = 2\times 2 = 4\,. \tag{6.18}$$

Remark 6.3.1. The evaluation of $\int_0^1 (1-u^3)(1-u^2)^{-3/2}\,du = 2$ is a good exercise in integration techniques. One approach: split $1+u+u^2 = (1+u)^2 - u$ and evaluate each piece separately via trigonometric/hyperbolic substitution. The Mathematica script verifies this symbolically.

6.3.7 The Deflection Angle

Assembling the result (Congdon & Keeton eq. 3.96): using $\Delta\phi = I_0 + \varepsilon\, I_1 + \mathcal{O}(\varepsilon^2)$,

$$\Delta\phi = \pi + 4\varepsilon + \mathcal{O}\big(\varepsilon^2\big) = \pi + \frac{4m}{r_0} + \mathcal{O}(m/r_0)^2 \; . \tag{6.19}$$

Since $r_0 \approx b$ to leading order and $m = GM/c^2$:

$$\boxed{\hat{\alpha} = \Delta\phi - \pi = \frac{4GM}{c^2\, b} + \mathcal{O}\big(GM/(c^2 b)\big)^2} \tag{6.20}$$

This is the **Einstein deflection angle** for a point mass in the weak-field limit.

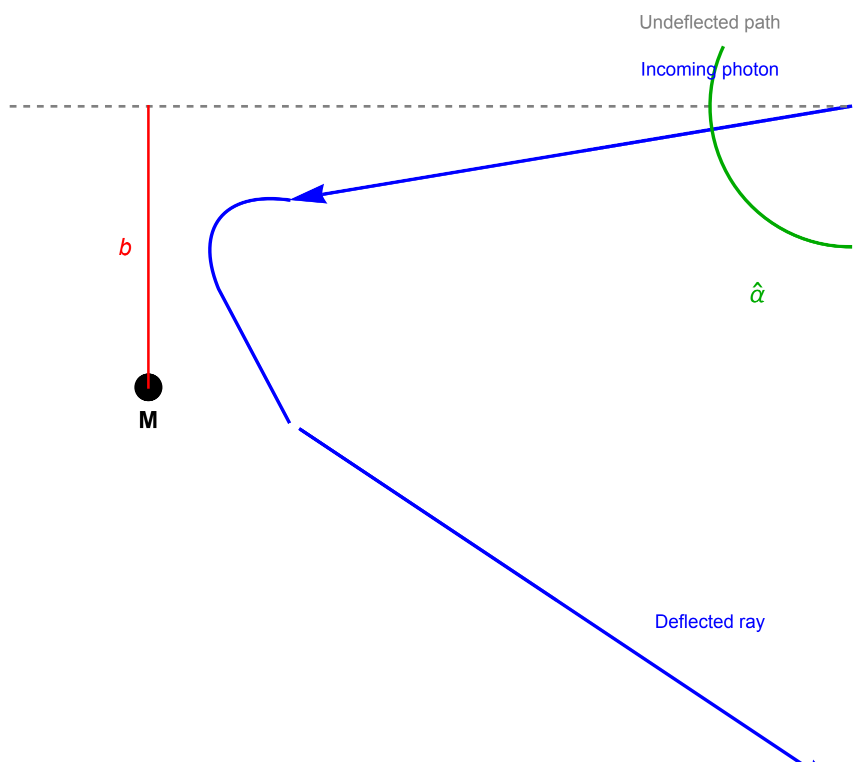


Figure 6.1: Geometry of gravitational light deflection by a point mass M. A photon approaches from the right with impact parameter b, reaches closest approach at $r_0 \approx b$, and is deflected by $\hat{\alpha} = 4GM/(c^2 b)$. The dashed line shows the undeflected path. Generated by `deflection_angle.wl`.

Exercise 6.3.1. Show that the GR deflection angle (6.20) is exactly twice the Newtonian prediction (6.5). Explain physically why:

(a) The Newtonian calculation accounts for the deflection due to the gravitational potential (temporal curvature of the metric, g_{tt}), but ignores spatial curvature (g_{rr}).

(b) In GR, both g_{tt} and g_{rr} deviate from flat space by the same amount (to leading order in m/r), so the spatial curvature doubles the deflection.

Hint: Expand the Schwarzschild line element in isotropic coordinates and identify the contributions from the time–time and space–space components of the metric.

6.4 The Shapiro Time Delay

In addition to deflecting light, a gravitational field *slows it down* — or more precisely, causes an excess travel time relative to the flat-spacetime path. This effect was predicted by Irwin Shapiro in 1964 and first measured using radar echoes from Mercury and Venus.

Following Congdon & Keeton Sec. 3.4.2 (eqs. 3.97–3.110), we derive the time delay for a photon passing near a point mass.

6.4.1 Coordinate Travel Time

For a photon on a null geodesic, we can write dt/dr using the conserved quantities (Congdon & Keeton eq. 3.97):

$$\frac{dt}{dr} = \frac{E/c^2}{(1 - R_\mathrm{S}/r)} \left[\frac{E^2}{c^2} - \frac{L^2}{r^2} \left(1 - \frac{R_\mathrm{S}}{r} \right) \right]^{-1/2} . \tag{6.21}$$

The coordinate time for the photon to travel from closest approach r_0 to a radial distance R is (Congdon & Keeton eq. 3.98):

$$T(R) = \int_{r_0}^{R} \frac{dt}{dr}\, dr \, . \tag{6.22}$$

In the weak-field limit ($m/r \ll 1$), expanding to first order gives (Congdon & Keeton eq. 3.104):

$$T(R) \approx \frac{1}{c}\sqrt{R^2 - r_0^2} + \frac{2m}{c} \ln\left(\frac{R + \sqrt{R^2 - r_0^2}}{r_0} \right) + \frac{m}{c}\sqrt{\frac{R - r_0}{R + r_0}} \, . \tag{6.23}$$

6.4.2 The Two Components of Time Delay

The total time delay ΔT for a lensed photon (relative to the unlensed path) has two contributions (Congdon & Keeton eqs. 3.107–3.110):

$$\boxed{\Delta T = \underbrace{\frac{D_\mathrm{d} D_\mathrm{s}}{2c\, D_\mathrm{ds}} (\theta - \beta)^2}_{\text{geometric delay}} - \underbrace{\frac{4GM}{c^3} \ln|\theta|}_{\text{Shapiro (gravitational) delay}}} \tag{6.24}$$

where D_d, D_s, D_ds are the angular diameter distances from observer to lens, observer to source, and lens to source respectively; θ is the image position and β the (unlensed) source position.

Remark 6.4.1. **Geometric delay:** The lensed ray follows a longer path through space than the direct (unlensed) ray. This is a purely geometric effect and is always positive.

Shapiro (gravitational) delay: The gravitational potential slows the effective speed of light near the lens. Clocks near a mass run slower (gravitational redshift), so the signal takes longer to traverse the gravitational potential well.

Remark 6.4.2. The total time delay (6.24) is closely related to the **Fermat potential**, which we will study in detail in Module 6. The Fermat potential $\tau(\boldsymbol{\theta})$ encodes both the geometric and gravitational delays, and its gradient gives the lens equation. Images form at stationary points of the Fermat potential (Fermat's principle applied to gravitational lensing).

Exercise 6.4.1. Derive the Shapiro time delay for a radar signal passing near the Sun and bouncing off a planet at distance R from the Sun.

(a) Show that the excess round-trip travel time is approximately

$$\Delta T \approx \frac{4GM_\odot}{c^3} \ln\left(\frac{4R_\oplus R}{r_0^2}\right)$$

where $R_\oplus$ is the Earth–Sun distance and r_0 the closest approach distance.

(b) For a signal grazing the Sun ($r_0 = R_\odot$) bouncing off Mercury at inferior conjunction, compute ΔT numerically. You should find $\Delta T \approx 240\ \mu$s.

6.5 Deflection by the Sun

The first and most famous test of the GR deflection formula was **Eddington's 1919 solar eclipse expedition**. During a total eclipse, stars near the Sun become visible, and their apparent positions are shifted outward by the gravitational deflection.

For light grazing the limb of the Sun ($b = R_\odot$):

$$\hat{\alpha}_\odot = \frac{4GM_\odot}{c^2\, R_\odot} = \frac{4 \times 6.674 \times 10^{-11} \times 1.989 \times 10^{30}}{(2.998 \times 10^8)^2 \times 6.957 \times 10^8} \approx 8.49 \times 10^{-6}\ \text{rad}\,. \tag{6.25}$$

Converting to arcseconds:

$$\boxed{\hat{\alpha}_\odot = 1.75''} \tag{6.26}$$

This is the Newtonian value $0.87''$ *doubled* by general relativity. Eddington's measurements confirmed the GR prediction (within the substantial error bars of the time), making Einstein a global celebrity.

Example 6.5.1. Deflection by Jupiter. Jupiter has $M_J = 1.898 \times 10^{27}$ kg and $R_J = 7.149 \times 10^7$ m. The deflection for light grazing Jupiter's limb is:

$$\hat{\alpha}_J = \frac{4GM_J}{c^2\, R_J} \approx 0.017'' = 17\ \text{mas}\,. \tag{6.27}$$

This is small but measurable with modern astrometry (e.g., the Gaia satellite, which achieves $\sim 10\ \mu$as precision).

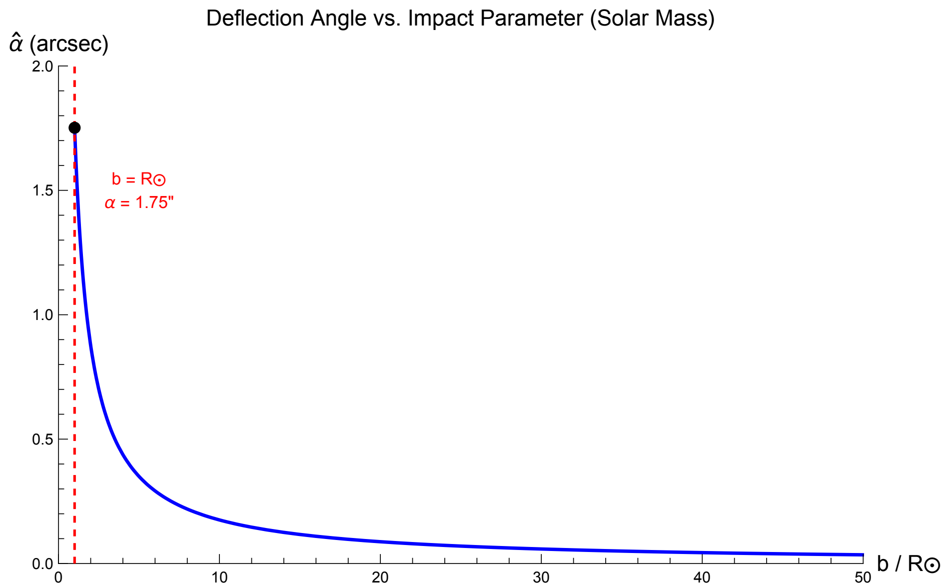


Figure 6.2: Deflection angle $\hat{\alpha}$ as a function of impact parameter b for a solar-mass lens, shown in arcseconds. The deflection falls as $1/b$. The vertical dashed line marks the solar radius ($b = R_\odot$) where $\hat{\alpha} = 1.75''$. Generated by `deflection_angle.wl`.

Exercise 6.5.1. Compute the deflection angle for light grazing the surface of:

(a) The Sun ($M = 1.989 \times 10^{30}$ kg, $R = 6.957 \times 10^8$ m): verify $\hat{\alpha} = 1.75''$.

(b) Jupiter ($M = 1.898 \times 10^{27}$ kg, $R = 7.149 \times 10^7$ m).

(c) A $10^{12}\, M_\odot$ galaxy with effective radius $R_e = 10$ kpc, using $b = R_e$. Express the result in arcseconds.

6.6 Deflection by Galaxies

The astrophysical power of gravitational lensing comes from the fact that the deflection angles produced by *galaxies* are of order arcseconds — large enough to produce multiple images, arcs, and rings that are observable with ground-based and space-based telescopes.

For a galaxy of mass $M = 10^{12}\, M_\odot$ with a typical scale of $b \sim 10$ kpc $\approx 3.09 \times 10^{20}$ m:

$$\hat{\alpha} \sim \frac{4GM}{c^2\, b} = \frac{4 \times 6.674 \times 10^{-11} \times 10^{12} \times 1.989 \times 10^{30}}{(2.998 \times 10^8)^2 \times 3.09 \times 10^{20}} \approx 3.9'' \tag{6.28}$$

The corresponding Einstein radius (which we will derive properly in Module 4) is of order $\sim 1''$, depending on the distance configuration.

Remark 6.6.1. The Einstein radius θ_{E} for a point mass lens at cosmological distances is:

$$\theta_{\mathrm{E}} = \sqrt{\frac{4GM}{c^2} \frac{D_{\mathrm{ds}}}{D_{\mathrm{d}}\, D_{\mathrm{s}}}}\,. \tag{6.29}$$

For a $10^{12}\,M_\odot$ galaxy at $z_d \sim 0.3$ with a source at $z_s \sim 1$, this gives $\theta_\mathrm{E} \sim 1''$. Multiple images form whenever the source lies within $\sim \theta_\mathrm{E}$ of the optical axis — this is the regime of **strong gravitational lensing**. We will develop this quantitatively in Modules 4–5.

6.7 Exact vs. Weak-Field Deflection

The weak-field formula $\hat{\alpha} = 4m/b$ is a first-order approximation valid when $b \gg R_\mathrm{S}$. For photons passing closer to the mass, higher-order corrections become important. The *exact* deflection angle is obtained by evaluating the integral (6.12) numerically (or using elliptic integrals), without the weak-field expansion.

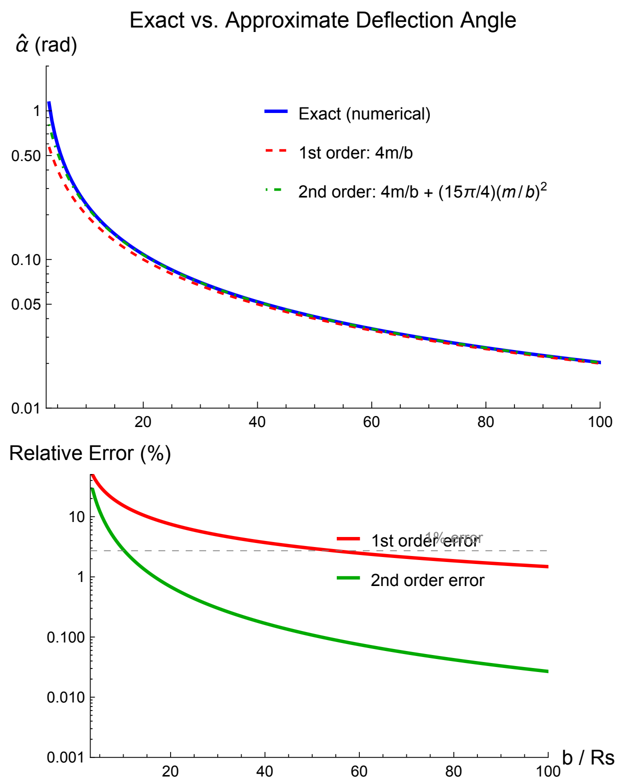


Figure 6.3: Comparison of the exact deflection angle (solid blue, from numerical integration) and the weak-field approximations $\hat{\alpha} = 4m/b$ (dashed red, first order) and including the second-order correction (dot-dashed green), as a function of b/R_S. The lower panel shows the relative error. The first-order approximation has $\sim 15\%$ error at $b = 10\,R_\mathrm{S}$ and reaches 1% accuracy only for $b \gtrsim 150\,R_\mathrm{S}$. At the photon sphere $b = 3\sqrt{3}\,R_\mathrm{S}/2$, the exact deflection diverges (the photon orbits the mass). Generated by `deflection_angle.wl`.

The second-order correction has been computed by several authors; the result is (see, e.g., Keeton & Petters 2006):

$$\hat{\alpha} = \frac{4m}{b} + \frac{15\pi}{4}\left(\frac{m}{b}\right)^2 + \mathcal{O}(m/b)^3 \,. \tag{6.30}$$

The critical impact parameter below which the photon is captured (i.e., spirals into the mass) is $b_{\mathrm{crit}} = 3\sqrt{3}\,m = 3\sqrt{3}\,R_{\mathrm{S}}/2$. This corresponds to the photon sphere derived in Module 1d.

Exercise 6.7.1. For a photon passing a Schwarzschild black hole at impact parameter $b = 10\,R_{\mathrm{S}}$:

(a) Evaluate the deflection integral (6.12) numerically to obtain the exact deflection angle.

(b) Compare with the weak-field approximation $\hat{\alpha} = 4m/b = 0.2$ rad. What is the relative error?

(c) Repeat for several values of b/R_{S} from 5 to 100. At what b/R_{S} does the weak-field approximation first achieve 1% accuracy?

Use the Mathematica script `deflection_angle.wl` *to verify your results.*

6.8 Summary

- Light follows null geodesics in Schwarzschild spacetime. The orbit equation (6.1) reduces to an integral for the total azimuthal angle change $\Delta\phi$.
- **Newtonian prediction** (Soldner, 1801): $\hat{\alpha}_{\mathrm{Newton}} = 2GM/(c^2 b)$. This accounts for only the *temporal* curvature of spacetime.
- **GR prediction** (Einstein, 1915): $\hat{\alpha} = 4GM/(c^2 b)$. The factor of 2 enhancement comes from *spatial* curvature, which contributes equally to the deflection.
- **Shapiro time delay:** A gravitational field causes an excess travel time ΔT with geometric and gravitational components. This is connected to the Fermat potential (Module 6).
- **Solar deflection:** $\hat{\alpha}_{\odot} = 1.75''$, confirmed by Eddington (1919) and modern measurements to $< 0.01\%$.
- **Galaxy-scale lensing:** $\hat{\alpha} \sim 1''$–$4''$ for $\sim 10^{12}\,M_{\odot}$ galaxies at $b \sim 10$ kpc, producing multiple images, arcs, and Einstein rings in strong lensing.
- The weak-field approximation $\hat{\alpha} = 4m/b$ converges slowly; the error is $\sim 15\%$ at $b = 10\,R_{\mathrm{S}}$ and drops below 1% only for $b \gtrsim 150\,R_{\mathrm{S}}$. The exact deflection diverges at the photon sphere ($b_{\mathrm{crit}} = 3\sqrt{3}\,R_{\mathrm{S}}/2$).

6.9 Problems

Exercise 6.9.1. (Soldner's Newtonian deflection.) Derive the Newtonian deflection angle for a "corpuscle of light" traveling at speed c past a mass M at impact parameter b.

(a) Write the Newtonian orbit equation and identify the semi-latus rectum p and eccentricity e in terms of b, M, G, and c.

(b) Show that the deflection angle is $\hat{\alpha}_{\text{Newton}} = 2GM/(c^2 b)$ in the weak-field limit.

(c) Compute the Newtonian prediction for the solar deflection. You should get $0.875''$.

Exercise 6.9.2. (The factor of 2: GR vs. Newton.)

(a) Starting from the result $\Delta\phi = \pi + 4m/r_0$, show that the GR deflection angle is $\hat{\alpha} = 4GM/(c^2 b)$.

(b) The Newtonian result uses only $g_{tt} \approx -(1 - 2\Phi/c^2)$ where $\Phi = -GM/r$. In GR, the spatial part of the Schwarzschild metric also curves: $g_{rr} \approx 1 + 2GM/(c^2 r)$. Argue qualitatively (or using the isotropic form of the metric) that the spatial curvature contributes an *equal* amount of deflection, doubling the Newtonian result.

Exercise 6.9.3. (Deflection for Sun, Jupiter, and galaxy.) Compute $\hat{\alpha}$ for:

(a) Light grazing the Sun: $M = 1.989 \times 10^{30}$ kg, $b = R_\odot = 6.957 \times 10^8$ m.

(b) Light grazing Jupiter: $M = 1.898 \times 10^{27}$ kg, $b = R_J = 7.149 \times 10^7$ m.

(c) A $10^{12}\, M_\odot$ galaxy at $b = 10$ kpc.

Express all answers in arcseconds. Verify with `problems_02.wl`.

Exercise 6.9.4. (Shapiro time delay for radar echoes.) A radar signal is sent from Earth, passes near the Sun at closest approach r_0, bounces off Mercury, and returns.

(a) Using eq. (6.23), show that the excess round-trip time is

$$\Delta T \approx \frac{4GM_\odot}{c^3} \left[\ln\left(\frac{4\, R_\oplus\, R_{\text{Merc}}}{r_0^2} \right) + 1 \right]$$

where $R_\oplus = 1.496 \times 10^{11}$ m and $R_{\text{Merc}} = 5.79 \times 10^{10}$ m are the Earth–Sun and Mercury–Sun distances.

(b) For $r_0 = R_\odot$, compute ΔT in microseconds.

(c) Shapiro's 1964 measurement using the Haystack radar gave $\Delta T = 200 \pm 20\ \mu$s. Is this consistent?

Exercise 6.9.5. (Exact vs. weak-field deflection.) For a photon passing a Schwarzschild mass at impact parameter $b = 10\, R_S$:

(a) Evaluate the exact deflection angle by numerically integrating eq. (6.12).

(b) Compare with the first-order approximation $\hat{\alpha}_1 = 4m/b$ and the second-order approximation $\hat{\alpha}_2 = 4m/b + (15\pi/4)(m/b)^2$.

(c) Tabulate the relative error of $\hat{\alpha}_1$ and $\hat{\alpha}_2$ for $b/R_S = 5, 10, 20, 50, 100$.

(d) At what value of b/R_S does the first-order approximation first achieve 1% accuracy?

Use `deflection_angle.wl` to generate and verify your results.

6.10 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_02.wl`, which verifies each answer symbolically and numerically.

Exercise 6.9.1: Soldner's Newtonian Deflection

(a) For a particle with speed $v_\infty = c$ and impact parameter b, the angular momentum is $L = bc$ and the energy per unit mass is $\mathcal{E} = \frac{1}{2}c^2$. The Newtonian orbit parameters are:

$$p = \frac{L^2}{GM} = \frac{b^2c^2}{GM}\,,$$
$$e = \sqrt{1 + \frac{2\mathcal{E}L^2}{(GM)^2}} = \sqrt{1 + \frac{b^2c^4}{(GM)^2}}\,.$$

(b) The orbit equation $r(\phi) = p/(1 + e\cos\phi)$ gives $r \to \infty$ when $\cos\phi = -1/e$. The two asymptotic angles are $\phi_\pm = \pi \mp \delta$ where $\sin\delta = 1/e$. The deflection angle is:

$$\hat{\alpha}_{\text{Newton}} = 2\delta = 2\arcsin(1/e)\,.$$

For $b \gg GM/c^2$, we have $e \gg 1$, so $\delta \approx 1/e \approx GM/(bc^2)$ and:

$$\hat{\alpha}_{\text{Newton}} \approx \frac{2GM}{c^2\,b}\,.$$

(c) For the Sun with $b = R_\odot$:

$$\hat{\alpha}_{\text{Newton}} = \frac{2 \times 6.674 \times 10^{-11} \times 1.989 \times 10^{30}}{(2.998 \times 10^8)^2 \times 6.957 \times 10^8} \approx 4.24 \times 10^{-6}\ \text{rad} = 0.875''\,.$$

Exercise 6.9.2: The Factor of 2

(a) From $\Delta\phi = \pi + 4m/r_0$ with $m = GM/c^2$ and $r_0 \approx b$:

$$\hat{\alpha} = \Delta\phi - \pi = \frac{4m}{r_0} \approx \frac{4GM}{c^2\,b}\,.$$

(b) In the weak-field limit, the Schwarzschild metric in isotropic coordinates takes the form:

$$ds^2 \approx -\left(1 - \frac{2GM}{c^2r}\right)c^2\,dt^2 + \left(1 + \frac{2GM}{c^2r}\right)(dx^2 + dy^2 + dz^2)\,.$$

For a null geodesic ($ds^2 = 0$), the effective refractive index is:

$$n(r) \approx \frac{1 + 2GM/(c^2r)}{1 - 2GM/(c^2r)} \approx 1 + \frac{4GM}{c^2r}\,.$$

The Newtonian calculation effectively uses $n = 1 + 2GM/(c^2r)$ (only the g_{tt} contribution), yielding half the deflection. The spatial curvature (g_{rr}) contributes the other factor of 2. Physically: in the Newtonian picture only time runs slower near a mass, but in GR space is also stretched, so the photon path through curved space is additionally bent.

Exercise 6.9.3: Numerical Deflections

Using $\hat{\alpha} = 4GM/(c^2 b)$:

(a) Sun: $\hat{\alpha} = 4 \times 6.674 \times 10^{-11} \times 1.989 \times 10^{30}/[(2.998 \times 10^8)^2 \times 6.957 \times 10^8] = 8.49 \times 10^{-6}$ rad $= 1.75''$.

(b) Jupiter: $\hat{\alpha} = 4 \times 6.674 \times 10^{-11} \times 1.898 \times 10^{27}/[(2.998 \times 10^8)^2 \times 7.149 \times 10^7] = 7.88 \times 10^{-8}$ rad $= 0.016'' = 16$ mas.

(c) Galaxy ($M = 10^{12}\,M_\odot$, $b = 10$ kpc): $b = 10 \times 3.086 \times 10^{19}$ m $= 3.086 \times 10^{20}$ m. $\hat{\alpha} = 4 \times 6.674 \times 10^{-11} \times 10^{12} \times 1.989 \times 10^{30}/[(2.998 \times 10^8)^2 \times 3.086 \times 10^{20}] \approx 3.95''$.

Exercise 6.9.4: Shapiro Time Delay

(a) The round-trip time is $\Delta T_{\rm round} = 2[T(R_\oplus) + T(R_{\rm Merc})] - 2(R_\oplus + R_{\rm Merc})/c$. Using eq. (2.14) and keeping only the leading logarithmic term:

$$\Delta T \approx \frac{4GM_\odot}{c^3}\left[\ln\left(\frac{4\,R_\oplus\,R_{\rm Merc}}{r_0^2}\right) + 1\right].$$

The $+1$ comes from the non-logarithmic correction terms in the weak-field expansion.

(b) Numerically:

$$\frac{4GM_\odot}{c^3} = \frac{4 \times 6.674 \times 10^{-11} \times 1.989 \times 10^{30}}{(2.998 \times 10^8)^3} = 1.970 \times 10^{-5}\ \text{s} = 19.7\ \mu\text{s}.$$

The logarithmic factor:

$$\ln\left(\frac{4 \times 1.496 \times 10^{11} \times 5.79 \times 10^{10}}{(6.957 \times 10^8)^2}\right) = \ln\left(7.15 \times 10^4\right) \approx 11.2.$$

Therefore: $\Delta T \approx 19.7 \times (11.2 + 1) = 240\ \mu$s.

(c) Shapiro's 1964 measurement gave $200 \pm 20\ \mu$s. The small discrepancy with our idealized point-mass calculation is consistent given that: (i) Mercury is not exactly at inferior conjunction, (ii) the Sun's corona introduces additional delays, and (iii) later measurements with the Viking Mars lander confirmed the GR prediction to $\sim 0.1\%$ accuracy.

Exercise 6.9.5: Exact vs. Weak-Field Deflection

(a)–(c) Using numerical integration (in $G = c = M = 1$ units where $R_{\rm S} = 2$, so $m = 1$ and $b = 2 \times b/R_{\rm S}$):

$b/R_{\rm S}$	$\hat{\alpha}_{\rm exact}$ (rad)	$\hat{\alpha}_1 = 4m/b$ (rad)	$\hat{\alpha}_2$ (rad)	Error($\hat{\alpha}_1$)
5	0.5904	0.4000	0.5178	32%
10	0.2361	0.2000	0.2295	15%
20	0.1081	0.1000	0.1074	7.5%
50	0.04122	0.04000	0.04118	3.0%
100	0.02030	0.02000	0.02029	1.5%

The errors are substantial even at moderate b/R_S because the expansion parameter is $\varepsilon = m/r_0$, and $r_0 < b$ due to gravitational attraction, so $\varepsilon > m/b$. The second-order approximation $\hat{\alpha}_2 = 4m/b + (15\pi/4)(m/b)^2$ is significantly more accurate.

(d) The first-order approximation achieves 1% accuracy at approximately $b/R_S \approx 150$, as confirmed by numerical integration. The relatively slow convergence is because the expansion parameter $\varepsilon = m/r_0$ is larger than the naive estimate $m/b = 1/(2\,b/R_S)$ due to the nonlinear relationship between r_0 and b.

Chapter 7

Cosmological Distances

Not only does God play dice, but He sometimes confuses us by throwing them where they can't be seen. — Stephen Hawking

Gravitational lensing is inherently a cosmological phenomenon: the lens and source are typically at redshifts $z \sim 0.1$–3, separated by distances of order gigaparsecs. To connect the physical deflection angle $\hat{\alpha} = 4GM/(c^2 b)$ derived in Modules 1e and 2 to observable image positions on the sky, we need a precise definition of "distance" in an expanding universe. Unlike in Euclidean space, there is no single notion of distance in cosmology: the *angular diameter distance* (relevant for angles), the *luminosity distance* (relevant for fluxes), and the *comoving distance* (relevant for volumes) are all distinct.

This module develops the full framework of cosmological distances from first principles. In Part A we derive the Friedmann–Lemaître–Robertson–Walker (FLRW) metric from the cosmological principle, compute the Christoffel symbols and curvature tensors for this metric using the tools from Module 1b, and plug into Einstein's equations with a perfect-fluid stress-energy tensor to obtain the **Friedmann equations** — the dynamical equations governing the expansion of the universe. In Part B we define the various distance measures, derive the key distance ratios that enter the lens equation, and highlight the counterintuitive *non-monotonicity* of the angular diameter distance, which turns over at $z \sim 1.6$ for the concordance cosmology.

Companion Mathematica files:

- `friedmann_equations.wl` — symbolic computation of Christoffel symbols and curvature tensors for the FRW metric, derivation of the Friedmann equations, scale factor solutions, cosmological distance calculations, and figure generation
- `cosmological_distances.nb` — interactive notebook with sliders for Ω_m, Ω_Λ, H_0 showing how distance–redshift relations change; lensing geometry calculator

Part A: The FRW Metric and Friedmann Equations

7.1 The Cosmological Principle

Modern cosmology rests on the **cosmological principle**: on sufficiently large scales ($\gtrsim$ 100 Mpc), the universe is *spatially homogeneous* (the same at every point) and *spatially isotropic* (the same in every direction). These two symmetry assumptions are supported by:

- The near-perfect isotropy of the cosmic microwave background (CMB), which is uniform to one part in 10^5 across the sky.
- The large-scale distribution of galaxies, which becomes statistically uniform on scales $\gtrsim$ 100 Mpc, as shown by surveys such as SDSS, 2dFGRS, and DESI.

Remark 7.1.1. Homogeneity and isotropy constrain only the *spatial* geometry of the universe at each instant. The universe is manifestly *not* homogeneous in time: it was denser and hotter in the past. The metric is therefore allowed to have time-dependent coefficients (the scale factor), and it is this time dependence that encodes the expansion history.

The mathematical consequence of the cosmological principle is powerful: the spatial part of the metric must be a maximally symmetric 3-dimensional Riemannian manifold. By a classic theorem in differential geometry (see Carroll Sec. 8.1), maximally symmetric n-dimensional spaces have $n(n+1)/2$ Killing vectors, and there are exactly three possibilities for $n = 3$: positive curvature (the 3-sphere S^3), zero curvature (flat $\mathbb{R}^3$), or negative curvature (the 3-hyperboloid H^3). These are characterized by a single parameter, the **spatial curvature constant** k:

$$k = \begin{cases} +1 & \text{positive curvature (closed, finite volume)} \\ 0 & \text{zero curvature (flat, infinite volume)} \\ -1 & \text{negative curvature (open, infinite volume)} \end{cases} \tag{7.1}$$

7.2 The Robertson–Walker Metric

Combining the cosmological principle with the requirement that spacetime be described by a Lorentzian 4-metric, the most general metric takes the **Robertson–Walker (RW)** form (Carroll eq. 8.2; Congdon & Keeton eq. 3.113):

$$\boxed{ds^2 = -c^2\,dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2\left(d\theta^2 + \sin^2\theta\,d\phi^2\right)\right]} \tag{7.2}$$

where:

- t is **cosmic time** — the proper time measured by comoving observers (those at rest with respect to the Hubble flow).
- $a(t)$ is the **scale factor**, a dimensionless function of time that encodes how distances between comoving observers change. We adopt the convention $a(t_0) = 1$ at the present epoch.

- r, θ, ϕ are **comoving coordinates** — the coordinate labels assigned to comoving observers do not change as the universe expands. The *physical* distance between two comoving observers separated by coordinate distance Δr is $a(t)\,\Delta r$, which grows with time if the universe is expanding ($\dot{a} > 0$).

- $k = -1, 0, +1$ is the spatial curvature parameter.

Remark 7.2.1. The metric (7.2) is purely *kinematical*: it is the most general metric consistent with homogeneity and isotropy, but it tells us nothing about how $a(t)$ evolves. The dynamics of $a(t)$ come from plugging this metric into Einstein's field equations, which we do in the next sections to derive the Friedmann equations.

It is often useful to write the spatial part of the metric using the **comoving distance** χ defined by

$$d\chi = \frac{dr}{\sqrt{1 - kr^2}}\,, \tag{7.3}$$

so that

$$ds^2 = -c^2\,dt^2 + a^2(t)\left[d\chi^2 + S_k^2(\chi)\,d\Omega^2\right], \tag{7.4}$$

where $d\Omega^2 = d\theta^2 + \sin^2\!\theta\,d\phi^2$ and the **transverse comoving distance** $S_k(\chi)$ is

$$S_k(\chi) = \begin{cases} \sin\chi & k = +1 \\ \chi & k = 0 \\ \sinh\chi & k = -1 \end{cases} \tag{7.5}$$

7.3 Christoffel Symbols for the FRW Metric

To derive the Friedmann equations, we need the Christoffel symbols, Ricci tensor, and Ricci scalar for the RW metric (7.2). We use the general functions developed in Module 1b (`christoffel_symbols.wl`) and verified symbolically in `friedmann_equations.wl`.

The coordinates are $x^\mu = (t, r, \theta, \phi)$, and the nonvanishing metric components are (reading off from eq. (7.2)):

$$\begin{aligned} g_{00} &= -c^2\,, & g_{11} &= \frac{a^2(t)}{1 - kr^2}\,, \\ g_{22} &= a^2(t)\,r^2\,, & g_{33} &= a^2(t)\,r^2\sin^2\!\theta\,. \end{aligned} \tag{7.6}$$

Since this is a diagonal metric, we use the formulas from Module 1b (Section 4): for a diagonal metric with components g_{aa} (no sum),

$$\begin{aligned} \Gamma^a{}_{bb} &= -\frac{1}{2g_{aa}}\,\partial_a g_{bb} \qquad (a \neq b)\,, \\ \Gamma^a{}_{ab} &= \frac{1}{2g_{aa}}\,\partial_b g_{aa} \qquad (\text{no sum on } a)\,. \end{aligned} \tag{7.7}$$

The **nonvanishing Christoffel symbols** of the RW metric are (Carroll eq. 8.5; verified in `friedmann_equations.wl`):

Time-space mixed:

$$\Gamma^0{}_{11} = \frac{a\dot{a}}{c^2(1-kr^2)}\,, \quad \Gamma^0{}_{22} = \frac{a\dot{a}\,r^2}{c^2}\,, \quad \Gamma^0{}_{33} = \frac{a\dot{a}\,r^2\sin^2\theta}{c^2} \tag{7.8}$$

and their transposes $\Gamma^i{}_{0i} = \dot{a}/(ac^2)\cdot c^2 = \dot{a}/a$:

$$\Gamma^1{}_{01} = \Gamma^2{}_{02} = \Gamma^3{}_{03} = \frac{\dot{a}}{a} \tag{7.9}$$

where the dot denotes d/dt.

Purely spatial:

$$\begin{aligned}
\Gamma^1{}_{11} &= \frac{kr}{1-kr^2}\,, & \Gamma^1{}_{22} &= -r(1-kr^2)\,, \\
\Gamma^1{}_{33} &= -r(1-kr^2)\sin^2\theta\,, & \Gamma^2{}_{12} &= \frac{1}{r}\,, \\
\Gamma^2{}_{33} &= -\sin\theta\cos\theta\,, & \Gamma^3{}_{13} &= \frac{1}{r}\,, \\
\Gamma^3{}_{23} &= \frac{\cos\theta}{\sin\theta}\,.
\end{aligned} \tag{7.10}$$

Exercise 7.3.1. Compute all nonvanishing Christoffel symbols of the RW metric by hand using the diagonal metric formulas (7.7). Verify your results with `friedmann_equations.wl`, which uses the general `computeChristoffel` function from Module 1b.

7.4 Ricci Tensor and Ricci Scalar

From the Christoffel symbols, we compute the Riemann tensor $R^\rho{}_{\sigma\mu\nu}$, contract to obtain the Ricci tensor $R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}$, and then contract again with the inverse metric to get the Ricci scalar $R = g^{\mu\nu}R_{\mu\nu}$. The computation is lengthy but straightforward; we quote the results here and verify them symbolically in `friedmann_equations.wl`.

The nonvanishing independent components of the **Ricci tensor** are (Carroll eq. 8.6):

$$R_{00} = -\frac{3\ddot{a}}{c^2\,a}\,, \tag{7.11}$$

$$R_{11} = \frac{1}{c^2(1-kr^2)}\left(a\ddot{a} + 2\dot{a}^2 + 2kc^2\right), \tag{7.12}$$

$$R_{22} = \frac{r^2}{c^2}\left(a\ddot{a} + 2\dot{a}^2 + 2kc^2\right), \tag{7.13}$$

$$R_{33} = \frac{r^2\sin^2\theta}{c^2}\left(a\ddot{a} + 2\dot{a}^2 + 2kc^2\right). \tag{7.14}$$

Notice that R_{11}, R_{22}, and R_{33} all share the same combination $a\ddot{a} + 2\dot{a}^2 + 2kc^2$, which is a consequence of spatial isotropy. We can write the spatial components compactly as:

$$R_{ij} = \frac{g_{ij}}{c^2\,a^2}\left(a\ddot{a} + 2\dot{a}^2 + 2kc^2\right). \tag{7.15}$$

The **Ricci scalar** is:

$$\boxed{R = g^{\mu\nu} R_{\mu\nu} = \frac{6}{c^2}\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{kc^2}{a^2}\right)} \tag{7.16}$$

7.5 Einstein's Equations with a Perfect Fluid

Einstein's field equations relate the geometry of spacetime (the Einstein tensor $G_{\mu\nu}$) to the matter content (the stress-energy tensor $T_{\mu\nu}$):

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{7.17}$$

where Λ is the **cosmological constant**.

For a **perfect fluid** at rest in the comoving frame (the natural choice for a homogeneous, isotropic universe), the stress-energy tensor is (Carroll eq. 8.8):

$$T^{\mu\nu} = \left(\rho + \frac{p}{c^2}\right) U^\mu U^\nu + p\, g^{\mu\nu} \tag{7.18}$$

where ρ is the **mass-energy density** (energy density $/c^2$), p is the **pressure**, and $U^\mu = (c, 0, 0, 0)$ is the four-velocity of comoving observers. The components in the comoving frame are:

$$T^{00} = \rho c^2\,, \qquad T^{ij} = p\, g^{ij}\,, \qquad T^{0i} = 0\,. \tag{7.19}$$

With indices lowered: $T_{00} = \rho c^4$ and $T_{ij} = p\, g_{ij}$.

Remark 7.5.1. In cosmology, the "fluid" is not a literal gas: ρ and p represent the averaged energy density and pressure of *all* matter and radiation in the universe, smoothed over scales $\gg 100$ Mpc. The cosmological principle guarantees that ρ and p depend only on time: $\rho = \rho(t)$, $p = p(t)$.

7.6 The Friedmann Equations

We now substitute the Ricci tensor (7.11)–(7.15) and the Ricci scalar (7.16) into Einstein's equations (7.17) with the perfect-fluid stress-energy tensor (7.19).

7.6.1 The First Friedmann Equation

The $(0, 0)$ component of Einstein's equations gives:

$$R_{00} - \tfrac{1}{2} R\, g_{00} + \Lambda g_{00} = \frac{8\pi G}{c^4} T_{00}\,. \tag{7.20}$$

Substituting $R_{00} = -3\ddot{a}/(c^2 a)$, $g_{00} = -c^2$, $R = 6(\ddot{a}/a + \dot{a}^2/a^2 + kc^2/a^2)/c^2$, and $T_{00} = \rho c^4$:

Left side:

$$R_{00} - \tfrac{1}{2}R\,g_{00} + \Lambda g_{00} = -\frac{3\ddot{a}}{c^2 a} + \frac{3}{c^2}\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{kc^2}{a^2}\right) - \Lambda c^2$$
$$= \frac{3}{c^2}\left(\frac{\dot{a}^2}{a^2} + \frac{kc^2}{a^2}\right) - \Lambda c^2 \,. \tag{7.21}$$

Right side:

$$\frac{8\pi G}{c^4}\,\rho c^4 = 8\pi G\rho \,. \tag{7.22}$$

Setting these equal and rearranging:

$$\boxed{\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\rho}{3} - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}} \tag{7.23}$$

This is the **first Friedmann equation**. It relates the expansion rate $\dot{a}/a$ to the energy density ρ, spatial curvature k, and cosmological constant Λ.

7.6.2 The Second Friedmann Equation

The spatial (i, j) components of Einstein's equations, using the compact form (7.15) and $T_{ij} = p\,g_{ij}$, give (after dividing both sides by $g_{ij}/(c^2a^2)$):

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}} \tag{7.24}$$

This is the **second Friedmann equation** (or **acceleration equation**). It shows that matter and radiation ($\rho > 0$, $p \geq 0$) cause the expansion to *decelerate* ($\ddot{a} < 0$), while a positive cosmological constant $\Lambda > 0$ causes *acceleration* ($\ddot{a} > 0$).

Remark 7.6.1. The combination $\rho + 3p/c^2$ in the second Friedmann equation is the key to understanding cosmic acceleration. Pressure contributes to gravity in GR (unlike in Newtonian gravity), and the factor of 3 comes from the three spatial dimensions. For a cosmological constant (with equation of state $p = -\rho c^2$), this combination is $\rho + 3(-\rho) = -2\rho < 0$, leading to acceleration.

7.6.3 The Fluid Conservation Equation

The Friedmann equations (7.23) and (7.24) are not independent: differentiating the first and substituting the second yields the **fluid conservation equation** (equivalently, this follows from $\nabla_\mu T^{\mu\nu} = 0$):

$$\boxed{\dot{\rho} + 3\frac{\dot{a}}{a}\left(\rho + \frac{p}{c^2}\right) = 0} \tag{7.25}$$

This is the relativistic analogue of the first law of thermodynamics: the energy density decreases as the universe expands, with the rate depending on both ρ and p. The $3\dot{a}/a$ factor reflects the three-dimensional expansion.

Exercise 7.6.1. Derive the Friedmann equations (7.23) and (7.24) from the Einstein equations (7.17) with the FRW Ricci tensor and the perfect-fluid stress-energy tensor. Verify each step with `friedmann_equations.wl`.

7.7 The Hubble Parameter and Density Parameters

7.7.1 The Hubble Parameter

The **Hubble parameter** is defined as the fractional expansion rate:

$$H(t) \equiv \frac{\dot{a}(t)}{a(t)} . \tag{7.26}$$

Its present-day value is the **Hubble constant**:

$$H_0 \equiv H(t_0) \approx 70 \ \mathrm{km\,s^{-1}\,Mpc^{-1}} . \tag{7.27}$$

The associated timescale is $t_H \equiv 1/H_0 \approx 14$ Gyr, and the associated length scale is the **Hubble distance**:

$$d_H \equiv \frac{c}{H_0} \approx 4.28 \ \mathrm{Gpc} \approx 1.32 \times 10^{26} \ \mathrm{m} . \tag{7.28}$$

7.7.2 Density Parameters

The first Friedmann equation (7.23) can be rewritten in terms of dimensionless **density parameters**. Define the **critical density** as the density required for a spatially flat ($k = 0$), $\Lambda = 0$ universe:

$$\rho_{\mathrm{cr}}(t) \equiv \frac{3H^2(t)}{8\pi G} , \qquad \rho_{\mathrm{cr},0} = \frac{3H_0^2}{8\pi G} \approx 9.5 \times 10^{-27} \ \mathrm{kg\,m^{-3}} . \tag{7.29}$$

The density parameters are:

$$\begin{aligned}
\Omega_{\mathrm{m}} &\equiv \frac{\rho_{m,0}}{\rho_{\mathrm{cr},0}} , && \text{(matter: non-relativistic)} \\
\Omega_r &\equiv \frac{\rho_{r,0}}{\rho_{\mathrm{cr},0}} , && \text{(radiation: relativistic)} \\
\Omega_\Lambda &\equiv \frac{\Lambda c^2}{3H_0^2} , && \text{(dark energy / cosmological constant)} \\
\Omega_k &\equiv -\frac{kc^2}{H_0^2} . && \text{(spatial curvature)}
\end{aligned} \tag{7.30}$$

With these definitions, the first Friedmann equation at the present epoch reduces to the elegant **cosmic sum rule**:

$$\boxed{\Omega_{\mathrm{m}} + \Omega_r + \Omega_\Lambda + \Omega_k = 1} \tag{7.31}$$

Current observations (Planck 2018) give $\Omega_{\mathrm{m}} \approx 0.31$, $\Omega_\Lambda \approx 0.69$, $\Omega_r \approx 9 \times 10^{-5}$, and $|\Omega_k| < 0.01$, consistent with a spatially flat ($k = 0$) universe. For most lensing calculations, we work with the **concordance cosmology**: $\Omega_{\mathrm{m}} = 0.3$, $\Omega_\Lambda = 0.7$, $H_0 = 70 \ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

7.8 Equations of State and Scale Factor Solutions

To solve the Friedmann equations, we need a relationship between p and ρ: the **equation of state**

$$p = w\rho c^2\,, \tag{7.32}$$

where w is a dimensionless constant for each component.

7.8.1 Matter, Radiation, and Dark Energy

The three primary components of the universe are:

Component	w	$\rho \propto$	$a(t) \propto$
Non-relativistic matter (dust)	0	a^{-3}	$t^{2/3}$
Radiation (photons, neutrinos)	$1/3$	a^{-4}	$t^{1/2}$
Cosmological constant (Λ)	-1	a^0 (constant)	e^{Ht}

These follow from the fluid conservation equation (7.25). With $p = w\rho c^2$:

$$\dot{\rho} + 3\frac{\dot{a}}{a}(1+w)\rho = 0 \qquad \Longrightarrow \qquad \rho \propto a^{-3(1+w)}\,. \tag{7.33}$$

For matter ($w = 0$): $\rho_m \propto a^{-3}$ (dilution by volume expansion). For radiation ($w = 1/3$): $\rho_r \propto a^{-4}$ (volume dilution plus cosmological redshift of each photon's energy). For Λ ($w = -1$): $\rho_\Lambda =$ constant.

7.8.2 Scale Factor Solutions

In a single-component universe with $k = 0$ (flat) and $\Lambda = 0$:

Matter-dominated: $H^2 = H_0^2\,\Omega_{\rm m}\,a^{-3}$ gives

$$a(t) = \left(\frac{3}{2}H_0\sqrt{\Omega_{\rm m}}\,t\right)^{2/3} \propto t^{2/3}\,. \tag{7.34}$$

Radiation-dominated: $H^2 = H_0^2\,\Omega_r\,a^{-4}$ gives

$$a(t) = \left(2H_0\sqrt{\Omega_r}\,t\right)^{1/2} \propto t^{1/2}\,. \tag{7.35}$$

Λ-dominated: $H^2 = H_0^2\,\Omega_\Lambda$ (constant) gives de Sitter expansion:

$$a(t) \propto e^{H_\Lambda t}\,, \qquad H_\Lambda = H_0\sqrt{\Omega_\Lambda}\,. \tag{7.36}$$

7.9 Redshift and the Scale Factor

As light propagates through the expanding universe, its wavelength stretches with the scale factor. A photon emitted at time $t_{\rm em}$ with wavelength $\lambda_{\rm em}$ is observed at t_0 with wavelength $\lambda_{\rm obs} = \lambda_{\rm em} \cdot a(t_0)/a(t_{\rm em})$. The **cosmological redshift** is therefore:

$$\boxed{1 + z = \frac{a(t_0)}{a(t_{\rm em})} = \frac{1}{a(t_{\rm em})}} \tag{7.37}$$

where we used $a(t_0) = 1$. High-redshift objects emitted their light when the universe was smaller: at $z = 1$, $a = 1/2$; at $z = 2$, $a = 1/3$; and so on.

Using eq. (7.37), the Friedmann equation can be rewritten entirely in terms of redshift. From the first Friedmann equation (7.23), $H^2 = 8\pi G\rho/3 - kc^2/a^2 + \Lambda c^2/3$. Each component's density scales as $\rho_i = \rho_{i,0}\, a^{-3(1+w_i)}$ (eq. 7.33), and $a = 1/(1+z)$ (eq. 7.37), so $\rho_m = \rho_{m,0}(1+z)^3$, $\rho_r = \rho_{r,0}(1+z)^4$, and $\rho_\Lambda = \rho_{\Lambda,0}$. Dividing through by H_0^2 and using the density parameter definitions (7.30): $H^2/H_0^2 = \Omega_{\rm m}(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda + \Omega_k(1+z)^2$. Taking the square root (Congdon & Keeton eq. 3.123; verified in `friedmann_equations.wl`):

$$\boxed{H(z) = H_0\sqrt{\Omega_{\rm m}(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda + \Omega_k(1+z)^2}} \tag{7.38}$$

For the concordance cosmology ($\Omega_{\rm m} = 0.3$, $\Omega_\Lambda = 0.7$, $\Omega_r \approx 0$, $\Omega_k = 0$), this simplifies to:

$$H(z) = H_0\sqrt{0.3\,(1+z)^3 + 0.7}\,. \tag{7.39}$$

Part B: Distance Measures for Lensing

7.10 Comoving Distance

A photon traveling radially ($d\theta = d\phi = 0$) through the RW metric satisfies $ds^2 = 0$, which gives $c\,dt = -a(t)\,d\chi$ (choosing the sign for incoming photons). The **comoving distance** to an object at redshift z is therefore:

$$\chi(z) = \int_0^z \frac{c\,dz'}{H(z')}\,. \tag{7.40}$$

This integral must in general be evaluated numerically, using $H(z)$ from eq. (7.38). The comoving distance is the "distance at the present epoch" — the distance that would be measured by a chain of rulers laid end-to-end at time t_0.

For the concordance cosmology, some representative values:

z	$\chi(z)$ (Mpc)
0.1	420
0.5	1880
1.0	3300
2.0	5200
5.0	7950

7.11 Angular Diameter Distance

The **angular diameter distance** is defined so that an object of physical (proper) size ℓ at redshift z, subtending an angle $\delta\theta$ on the sky, satisfies the Euclidean-looking relation:

$$\delta\theta = \frac{\ell}{D_A(z)} \,. \tag{7.41}$$

From the RW metric (7.4), a physical transverse distance at redshift z (when $a = 1/(1+z)$) subtending angle $\delta\theta$ is $\ell = a(t)\, S_k(\chi)\, \delta\theta$, so

$$\boxed{D_A(z) = \frac{S_k(\chi(z))}{1+z}} \tag{7.42}$$

For a flat universe ($k = 0$), $S_k(\chi) = \chi$, and this simplifies to:

$$D_A(z) = \frac{\chi(z)}{1+z} = \frac{1}{1+z} \int_0^z \frac{c\, dz'}{H(z')} \,. \tag{7.43}$$

The angular diameter distance is the distance measure that enters the **lens equation**. The angular diameter distances $D_\mathrm{d} = D_A(z_d)$, $D_\mathrm{s} = D_A(z_s)$, and D_ds (defined below) determine the mapping from physical deflection angles to angular positions on the sky.

7.12 The Angular Diameter Distance Is Not Monotonic

One of the most counterintuitive results in cosmology is that the angular diameter distance $D_A(z)$ is **not a monotonically increasing function** of redshift. For the concordance cosmology, D_A reaches a maximum at $z \approx 1.6$ and then *decreases* at higher redshifts.

Physically, this means that objects at very high redshifts appear *larger* on the sky than one would naively expect. The turnover arises from two competing effects. For nearby objects, increasing redshift means greater comoving distance χ, so $D_A = \chi/(1+z)$ grows. But at high z, the factor $1/(1+z)$ dominates: the universe was much smaller when the light was emitted ($a = 1/(1+z) \ll 1$), so the *physical* transverse size of the object at emission — which determines the angle it subtends — was correspondingly smaller. A smaller physical size at fixed comoving size subtends a *larger* angle, making D_A decrease. The turnover at $z \approx 1.6$ (for the concordance cosmology) occurs where these two effects balance: $d\chi/dz = \chi/(1+z)$, i.e., where the fractional growth of comoving distance equals the fractional growth of the redshift factor (Carroll Sec. 8.4; computed numerically in `friedmann_equations.wl`).

Remark 7.12.1. The turnover of $D_A(z)$ has practical consequences for lensing: a lens at $z_d \sim 0.5$ and a source at $z_s \sim 2$ might have a surprisingly large distance ratio $D_\mathrm{ds}/D_\mathrm{s}$, and the "sweet spot" for lensing efficiency is typically $z_d \sim 0.2$–0.5 for sources at $z_s \sim 1$–3.

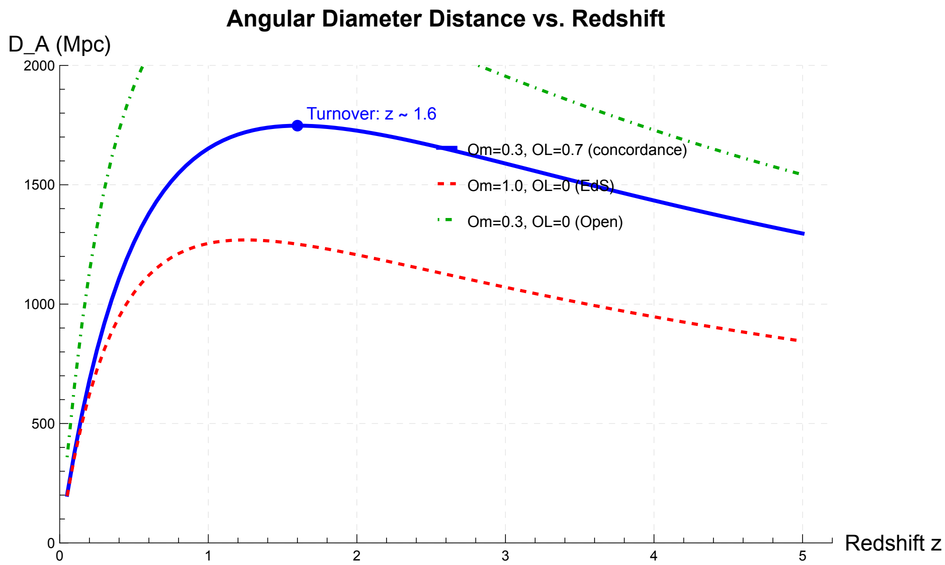


Figure 7.1: Angular diameter distance $D_A(z)$ for the concordance cosmology ($\Omega_\mathrm{m} = 0.3$, $\Omega_\Lambda = 0.7$, $H_0 = 70$ km s^{-1} Mpc^{-1}). D_A reaches a maximum of approximately 1770 Mpc at $z \approx 1.6$ and then decreases. This turnover is a key consequence of the expanding universe and has no analogue in Euclidean geometry. Generated by `friedmann_equations.wl`.

7.13 Luminosity Distance

The **luminosity distance** D_L is defined so that the observed flux F from a source of luminosity L satisfies the Euclidean inverse-square law:

$$F = \frac{L}{4\pi D_L^2}\,. \tag{7.44}$$

In an expanding universe, two effects reduce the observed flux beyond the $1/r^2$ geometric dilution: (1) photon energies are redshifted by a factor $(1+z)$, and (2) photon arrival rates are reduced by another factor $(1+z)$. This gives:

$$D_L(z) = (1+z)\,S_k(\chi(z)) = (1+z)^2\,D_A(z)\,. \tag{7.45}$$

The relation $D_L = (1+z)^2 D_A$ is the **Etherington reciprocity theorem** (or **distance-duality relation**). It holds for any metric theory of gravity in which photons travel on null geodesics and photon number is conserved.

7.14 The Lens–Source Angular Diameter Distance

For gravitational lensing, we need not only $D_\mathrm{d} = D_A(z_d)$ and $D_\mathrm{s} = D_A(z_s)$, but also D_ds: the angular diameter distance from the *lens* to the *source*. This is **not** simply $D_A(z_s) - D_A(z_d)$, because angular diameter distances do not add linearly.

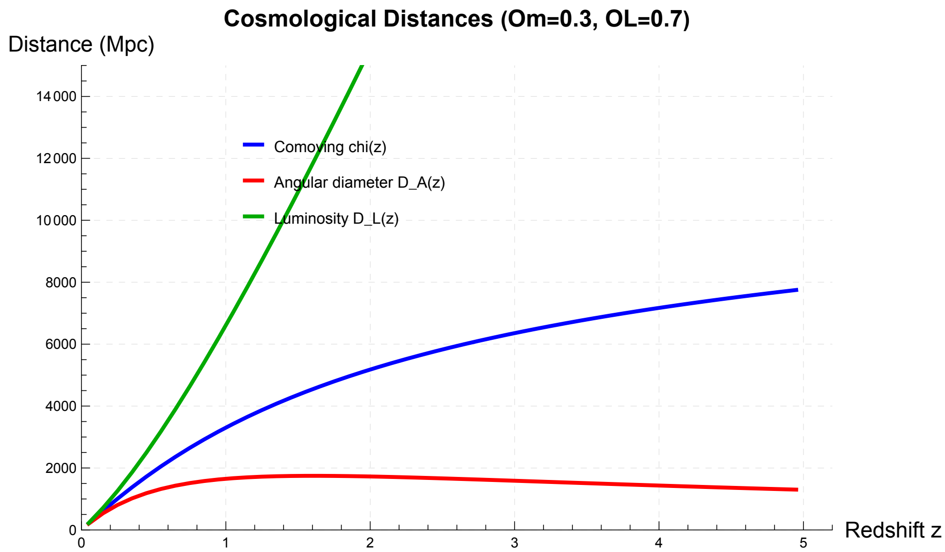


Figure 7.2: Comoving distance $\chi(z)$ (blue), angular diameter distance $D_A(z)$ (red), and luminosity distance $D_L(z)$ (green) for the concordance cosmology. At low redshift ($z \ll 1$) all three converge: $\chi \approx D_A \approx D_L \approx cz/H_0$. At high redshift they diverge dramatically. The turnover of D_A is clearly visible, while D_L grows monotonically. Generated by `friedmann_equations.wl`.

In a flat universe ($k = 0$), the comoving distances *do* add: $\chi_{ds} = \chi_s - \chi_d$, and the lens–source angular diameter distance is:

$$\boxed{D_{\rm ds} = \frac{\chi_s - \chi_d}{1+z_s} = \frac{1}{1+z_s}\int_{z_d}^{z_s} \frac{c\,dz'}{H(z')}} \tag{7.46}$$

For a general (non-flat) universe, one uses (Congdon & Keeton eq. 3.127):

$$D_{\rm ds} = \frac{1}{1+z_s}\left[S_k(\chi_s)\sqrt{1 - k\,S_k^2(\chi_d)/(S_k(\chi_d))^2} - (\text{symmetric term})\right] \tag{7.47}$$

but for the spatially flat case ($k = 0$), eq. (7.46) is exact and sufficient.

Remark 7.14.1. A common mistake is to write $D_{\rm ds} = D_{\rm s} - D_{\rm d}$. This is **wrong** in general. Even in a flat universe, $D_{\rm ds} \neq D_{\rm s} - D_{\rm d}$ because the factors of $(1+z)$ are different. One must always compute $D_{\rm ds}$ using eq. (7.46) (or its curved-space generalization). To illustrate: for $z_d = 0.5$ and $z_s = 2.0$ in the concordance cosmology, $D_{\rm d} = 1260$ Mpc, $D_{\rm s} = 1740$ Mpc, but $D_{\rm ds} = 1450$ Mpc $\neq D_{\rm s} - D_{\rm d} = 480$ Mpc.

7.15 The Distance Ratio for Lensing

The **lens equation** (Module 4) and the **Einstein radius** both depend on a specific combination of distances. The Einstein radius of a point mass M is (Congdon & Keeton eq. 4.9):

$$\theta_{\rm E} = \sqrt{\frac{4GM}{c^2}\frac{D_{\rm ds}}{D_{\rm d}\,D_{\rm s}}} \tag{7.48}$$

and the critical surface mass density is:

$$\Sigma_{\rm cr} = \frac{c^2}{4\pi G} \frac{D_{\rm s}}{D_{\rm d}\, D_{\rm ds}} \,. \tag{7.49}$$

The key quantity is the **distance ratio**:

$$\mathcal{D} \equiv \frac{D_{\rm ds}}{D_{\rm d}\, D_{\rm s}} \,, \tag{7.50}$$

which determines the "lensing strength" of a given observer–lens–source configuration.

Note that $\mathcal{D}$ is *not* simply $1/D_{\rm d}$, because $D_{\rm ds}$ depends on both z_d and z_s in a non-trivial way. The distance ratio is maximized (lensing is most efficient) when the lens is roughly halfway between the observer and the source in terms of comoving distance.

7.16 Lensing Efficiency

For a fixed source redshift z_s, one can ask: at what lens redshift z_d is lensing most efficient? The answer is determined by the **lensing efficiency function**:

$$\mathcal{E}(z_d; z_s) \equiv \frac{D_{\rm d}\, D_{\rm ds}}{D_{\rm s}} \tag{7.51}$$

which is proportional to the square of the Einstein radius (at fixed mass). Note that $\theta_{\rm E}^2 \propto D_{\rm ds}/(D_{\rm d}\, D_{\rm s}) = \mathcal{D}$, while the physical Einstein ring area scales as $D_{\rm d}^2 \theta_{\rm E}^2 \propto D_{\rm d}\, D_{\rm ds}/D_{\rm s} = \mathcal{E}$.

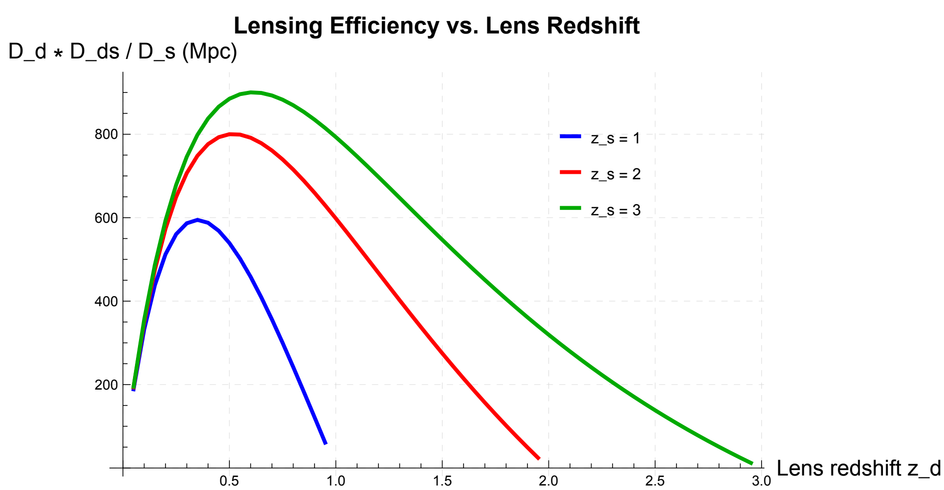


Figure 7.3: Lensing efficiency $D_{\rm d}\, D_{\rm ds}/D_{\rm s}$ as a function of lens redshift z_d for several source redshifts $z_s = 1, 2, 3$ in the concordance cosmology. For $z_s = 1$, lensing is most efficient at $z_d \approx 0.3$; for $z_s = 2$, at $z_d \approx 0.5$. The lens is typically most effective when placed roughly "halfway" (in comoving distance) between the observer and the source. Generated by `friedmann_equations.wl`.

7.17 Summary

- The **cosmological principle** (homogeneity + isotropy) constrains the spacetime metric to the Robertson–Walker form $ds^2 = -c^2dt^2 + a^2(t)\,[dr^2/(1-kr^2) + r^2d\Omega^2]$.
- Plugging the RW metric into **Einstein's equations** with a perfect-fluid stress-energy tensor yields the **Friedmann equations**:

$$H^2 = \frac{8\pi G\rho}{3} - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}\,,$$
$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}\,.$$

- The density parameters $\Omega_{\rm m}$, Ω_Λ, Ω_k satisfy $\Omega_{\rm m} + \Omega_\Lambda + \Omega_k = 1$ (ignoring radiation at late times).
- The **angular diameter distance** $D_A(z) = \chi(z)/(1+z)$ is *not monotonic*: it turns over at $z \approx 1.6$ in the concordance cosmology.
- The **luminosity distance** is $D_L = (1+z)^2 D_A$ (Etherington reciprocity).
- The **lens–source distance** $D_{\rm ds} \neq D_{\rm s} - D_{\rm d}$; it must be computed as $D_{\rm ds} = (\chi_s - \chi_d)/(1+z_s)$ for a flat universe.
- The **Einstein radius** $\theta_{\rm E} \propto \sqrt{D_{\rm ds}/(D_{\rm d}\,D_{\rm s})}$ depends on the distance configuration, and lensing is most efficient when the lens is roughly halfway (in comoving distance) to the source.

7.18 Problems

Exercise 7.18.1. (Deriving the Friedmann equations from the RW metric.) Starting from the Robertson–Walker metric (7.2):

(a) Compute the Christoffel symbols for the FRW metric.

(b) Compute the Ricci tensor $R_{\mu\nu}$ and Ricci scalar R.

(c) Use Einstein's equations with the perfect-fluid stress-energy tensor to derive the first and second Friedmann equations.

(d) Show that differentiating the first Friedmann equation and using the second gives the fluid conservation equation.

Verify all steps with `friedmann_equations.wl`.

Exercise 7.18.2. (Age of the universe for different cosmologies.) The age of the universe is $t_0 = \int_0^\infty dz/[(1+z)\,H(z)]$.

(a) For an Einstein–de Sitter universe ($\Omega_m = 1$, $\Omega_\Lambda = 0$, $k = 0$), show analytically that $t_0 = 2/(3H_0)$.

(b) Compute t_0 numerically for the concordance cosmology ($\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$). You should find $t_0 \approx 13.5$ Gyr for $H_0 = 70$ km s^{-1} Mpc^{-1}.

(c) Compare with an open universe ($\Omega_m = 0.3$, $\Omega_\Lambda = 0$, $\Omega_k = 0.7$).

Exercise 7.18.3. (Angular diameter distance turnover.)

(a) Plot $D_A(z)$ for the following cosmologies and identify the turnover redshift in each: (i) $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$ (concordance); (ii) $\Omega_m = 1.0$, $\Omega_\Lambda = 0$ (Einstein–de Sitter); (iii) $\Omega_m = 0.3$, $\Omega_\Lambda = 0$ (open).

(b) Explain physically why D_A turns over: why do high-redshift objects appear *larger* than intermediate-redshift objects?

(c) For the concordance cosmology, find the redshift z_{max} at which D_A is maximum by numerically solving $dD_A/dz = 0$.

Exercise 7.18.4. (Distances for a real lens system: Q0957+561.) The first gravitational lens discovered (Walsh, Carswell & Weymann 1979) has lens redshift $z_d = 0.36$ and source redshift $z_s = 1.41$. Using the concordance cosmology:

(a) Compute D_d, D_s, and D_{ds} in Mpc.

(b) Verify that $D_{ds} \neq D_s - D_d$.

(c) Compute the critical surface mass density Σ_{cr} in $M_\odot\,\mathrm{kpc}^{-2}$.

(d) Compute the Einstein radius θ_E for a lens mass $M = 10^{12}\,M_\odot$.

Exercise 7.18.5. (Einstein radius for a typical galaxy lens.) For a flat ΛCDM universe with $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$, $H_0 = 70$ km s^{-1} Mpc^{-1}:

(a) Compute D_d, D_s, D_{ds} for $z_d = 0.3$, $z_s = 1.0$.

(b) Compute the Einstein radius θ_E for a lens mass $M = 10^{12}\,M_\odot$. You should find $\theta_E \approx 2.4''$.

(c) How does θ_E scale with M? What mass gives $\theta_E = 10''$?

(d) Plot θ_E as a function of z_d (with $z_s = 1$ and $M = 10^{12}\,M_\odot$ fixed). At what z_d is θ_E largest?

Verify with `problems_03.wl`.

7.19 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_03.wl`, which verifies each answer symbolically and numerically.

Exercise 7.18.1: Deriving the Friedmann Equations

(a) The nonvanishing Christoffel symbols of the RW metric $ds^2 = -c^2dt^2 + a^2(t)[dr^2/(1-kr^2) + r^2d\Omega^2]$ are listed in eqs. (7.8)–(7.10). As an example, $\Gamma^0{}_{11}$ follows from the diagonal formula:

$$\Gamma^0{}_{11} = -\frac{1}{2g_{00}}\,\partial_0 g_{11} = -\frac{1}{2(-c^2)}\,\frac{\partial}{\partial t}\,\frac{a^2}{1-kr^2} = \frac{a\dot{a}}{c^2(1-kr^2)}\,.$$

(b) The Ricci tensor components are given in eqs. (7.11)–(7.14). The computation proceeds via:

1. Compute all Riemann tensor components from the Christoffel symbols.
2. Contract $R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}$.
3. The Ricci scalar is $R = g^{\mu\nu}R_{\mu\nu}$.

The key results are:

$$R_{00} = -\frac{3\ddot{a}}{c^2a}\,, \qquad R_{ij} = \frac{g_{ij}}{c^2a^2}\left(a\ddot{a} + 2\dot{a}^2 + 2kc^2\right).$$

The Ricci scalar follows:

$$R = \frac{6}{c^2}\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2} + \frac{kc^2}{a^2}\right).$$

All verified symbolically in `friedmann_equations.wl`.

(c) The $(0,0)$ component of Einstein's equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$:

$$\begin{aligned} R_{00} - \tfrac{1}{2}R\,g_{00} + \Lambda g_{00} &= \frac{3}{c^2}\left(\frac{\dot{a}^2}{a^2} + \frac{kc^2}{a^2}\right) - \Lambda c^2 \\ &= \frac{8\pi G}{c^4}\cdot \rho c^4 = 8\pi G\rho\,. \end{aligned}$$

Rearranging gives the first Friedmann equation:

$$\boxed{\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\rho}{3} - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}}$$

Similarly, the spatial components give the second Friedmann equation:

$$\boxed{\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right) + \frac{\Lambda c^2}{3}}$$

(d) Differentiate the first Friedmann equation with respect to t:

$$2\frac{\dot{a}}{a}\left(\frac{\ddot{a}}{a}-\frac{\dot{a}^2}{a^2}\right)=\frac{8\pi G}{3}\dot{\rho}+\frac{2kc^2\dot{a}}{a^3}\,.$$

Substitute the second Friedmann equation for $\ddot{a}/a$ and the first for $\dot{a}^2/a^2$ on the left side, and after algebraic simplification:

$$\boxed{\dot{\rho}+3\frac{\dot{a}}{a}\left(\rho+\frac{p}{c^2}\right)=0}$$

Exercise 7.18.2: Age of the Universe

(a) Einstein–de Sitter ($\Omega_\mathrm{m}=1,\ \Omega_\Lambda=0,\ k=0$): $H(z)=H_0(1+z)^{3/2}$, so

$$t_0=\int_0^\infty\frac{dz}{(1+z)H(z)}=\frac{1}{H_0}\int_0^\infty\frac{dz}{(1+z)^{5/2}}=\frac{1}{H_0}\left[-\frac{2}{3}(1+z)^{-3/2}\right]_0^\infty=\boxed{\frac{2}{3H_0}}$$

For $H_0=70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$: $t_0=2/(3\times70)\times977.8=9.3$ Gyr.

(b) For the concordance cosmology ($\Omega_\mathrm{m}=0.3,\ \Omega_\Lambda=0.7$), the integral must be evaluated numerically. Using Mathematica:

$$t_0=\int_0^\infty\frac{dz}{(1+z)H_0\sqrt{0.3(1+z)^3+0.7}}\approx\frac{0.964}{H_0}\approx\boxed{13.5\ \mathrm{Gyr}}$$

(c) Open universe ($\Omega_\mathrm{m}=0.3,\ \Omega_\Lambda=0,\ \Omega_k=0.7$): $H(z)=H_0\sqrt{0.3(1+z)^3+0.7(1+z)^2}$. Numerical integration gives $t_0\approx11.0$ Gyr. The concordance cosmology gives a universe that is $\sim23\%$ older, because the cosmological constant accelerates the expansion at late times, effectively stretching the time since the Big Bang.

Exercise 7.18.3: Angular Diameter Distance Turnover

(a) The turnover redshifts (where $dD_A/dz=0$) are:

Cosmology	z_max	D_A^max (Mpc)
Concordance ($\Omega_\mathrm{m}=0.3,\ \Omega_\Lambda=0.7$)	1.61	1770
Einstein–de Sitter ($\Omega_\mathrm{m}=1.0,\ \Omega_\Lambda=0$)	1.25	1180
Open ($\Omega_\mathrm{m}=0.3,\ \Omega_\Lambda=0$)	1.44	1300

(b) The angular diameter distance $D_A=\chi/(1+z)$ is the ratio of the comoving distance (which increases monotonically with z) to $(1+z)$ (which also increases). At low z, χ grows faster than $(1+z)$, so D_A increases. At high z, the $(1+z)$ factor dominates, causing D_A to decrease. Physically: photons from very distant objects were emitted when the universe was much smaller, so the "angular size distance" shrinks — the object subtends a larger angle than it would at an intermediate redshift.

(c) Setting $dD_A/dz=0$ numerically for the concordance cosmology gives $z_\mathrm{max}\approx1.61$.

Exercise 7.18.4: Distances for Q0957+561

Using the concordance cosmology with $H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$:

(a)

$$D_\mathrm{d} = D_A(z_d = 0.36) = \frac{\chi(0.36)}{1.36} \approx \frac{1420}{1.36} \approx \boxed{1040\ \mathrm{Mpc}}$$
$$D_\mathrm{s} = D_A(z_s = 1.41) = \frac{\chi(1.41)}{2.41} \approx \frac{4060}{2.41} \approx \boxed{1680\ \mathrm{Mpc}}$$
$$D_\mathrm{ds} = \frac{\chi_s - \chi_d}{1+z_s} = \frac{4060 - 1420}{2.41} \approx \boxed{1100\ \mathrm{Mpc}}$$

(b) $D_\mathrm{s} - D_\mathrm{d} = 1680 - 1040 = 640$ Mpc, but $D_\mathrm{ds} = 1100$ Mpc. These differ by nearly a factor of 2! This confirms that angular diameter distances do not add linearly.

(c) The critical surface mass density is:

$$\Sigma_\mathrm{cr} = \frac{c^2}{4\pi G}\frac{D_\mathrm{s}}{D_\mathrm{d}\,D_\mathrm{ds}} \approx \frac{(3\times10^8)^2}{4\pi \times 6.674\times10^{-11}} \times \frac{1680}{1040\times1100}\ \mathrm{Mpc}^{-1}$$

Converting to appropriate units: $\Sigma_\mathrm{cr} \approx 3.3\times10^9\ M_\odot\,\mathrm{kpc}^{-2}$.

(d) The Einstein radius for $M = 10^{12}\,M_\odot$:

$$\theta_\mathrm{E} = \sqrt{\frac{4GM}{c^2}\frac{D_\mathrm{ds}}{D_\mathrm{d}D_\mathrm{s}}} \approx \sqrt{\frac{4\times6.674\times10^{-11}\times10^{12}\times1.989\times10^{30}}{(3\times10^8)^2} \times \frac{1100}{1040\times1680}} \times \frac{1}{3.086\times10^{22}}$$

Evaluating: $\theta_\mathrm{E} \approx 1.1\times10^{-5}$ rad $\approx \boxed{2.3''}$.

Exercise 7.18.5: Einstein Radius for a Typical Galaxy Lens

(a) For $z_d = 0.3$, $z_s = 1.0$:

$$\chi_d = \int_0^{0.3} \frac{c\,dz'}{H(z')} \approx 1220\ \mathrm{Mpc}\,,$$
$$\chi_s = \int_0^{1.0} \frac{c\,dz'}{H(z')} \approx 3300\ \mathrm{Mpc}\,,$$
$$D_\mathrm{d} = \chi_d/1.3 \approx 940\ \mathrm{Mpc}\,,$$
$$D_\mathrm{s} = \chi_s/2.0 \approx 1650\ \mathrm{Mpc}\,,$$
$$D_\mathrm{ds} = (\chi_s - \chi_d)/2.0 \approx 1040\ \mathrm{Mpc}\,.$$

(b) The Einstein radius:

$$\theta_\mathrm{E} = \sqrt{\frac{4GM}{c^2}\frac{D_\mathrm{ds}}{D_\mathrm{d}\,D_\mathrm{s}}}$$

The Schwarzschild radius of $M = 10^{12}\,M_\odot$ is $R_S = 2GM/c^2 \approx 2.95\times10^{15}$ m ≈ 0.096 pc. Converting distances to meters and evaluating:

$$\theta_\mathrm{E} \approx 1.2\times10^{-5}\ \mathrm{rad} \approx \boxed{2.4''}$$

This confirms that galaxy-scale lenses with $M \sim 10^{12}\,M_\odot$ produce Einstein rings of order $\sim 1''$–$3''$, depending on the distance configuration.

(c) Since $\theta_\mathrm{E} \propto \sqrt{M}$:

$$\theta_\mathrm{E} = 2.4'' \times \sqrt{\frac{M}{10^{12}\,M_\odot}}$$

For $\theta_\mathrm{E} = 10''$: $M = 10^{12} \times (10/2.4)^2 \approx 1.7 \times 10^{13}\,M_\odot$. This is approaching a galaxy cluster mass, consistent with the fact that cluster lenses produce Einstein rings of $\sim 10''$–$30''$.

(d) The plot of θ_E vs. z_d (at fixed $z_s = 1$, $M = 10^{12}\,M_\odot$) shows a maximum at $z_d \approx 0.3$–0.4. The Einstein radius approaches zero as $z_d \to 0$ (because $D_\mathrm{ds}/D_\mathrm{s} \to 1$ but $D_\mathrm{d} \to 0$, so $D_\mathrm{ds}/(D_\mathrm{d}\,D_\mathrm{s}) \to \infty$, but this divergence is of order $1/D_\mathrm{d}$ while $D_\mathrm{d} \to 0$, so $\theta_\mathrm{E} \propto 1/\sqrt{D_\mathrm{d}}$ diverges) — actually θ_E diverges as $z_d \to 0$ because the lens is infinitely close. In practice, θ_E is very large for nearby lenses but the lensing cross-section becomes negligible. The physically relevant quantity is the *probability* of lensing, which peaks at intermediate z_d.

Chapter 8

The Gravitational Lens Equation

> *The lens equation is the fundamental relation in gravitational lensing theory. It is simple to write down, but its consequences — multiple images, Einstein rings, magnification, and time delays — are extraordinarily rich.* — Narayan & Bartelmann (1997)

In Module 6 we derived the deflection angle $\hat{\alpha} = 4GM/(c^2 b)$ for a point mass, and in Module 5 we showed that deflection angles from multiple masses superpose in the weak-field limit. In Module 7 we introduced the angular diameter distances D_{d}, D_{s}, and D_{ds} that describe the geometry of the observer–lens–source system at cosmological separations.

Now we assemble these ingredients into the **lens equation** — the fundamental mapping from source positions to image positions. This single equation is the starting point for everything that follows: image multiplicity, magnification, time delays, and lens modeling.

Companion Mathematica files:

- `lens_equation.wl` — symbolic derivation of the lens equation, Einstein radius, point mass image positions, magnification preview, critical surface mass density, and publication-quality figure generation
- `einstein_ring.nb` — interactive notebook: Manipulate for source position and image positions, Einstein ring visualization, interactive θ_{E} calculator

8.1 Introduction: From Deflection to Imaging

We now have all the pieces needed to build the gravitational lens equation:

1. **The deflection angle** (Modules 1e and 2): $\hat{\alpha} = 4GM/(c^2 b)$ for a point mass, and the superposition principle (5.25) for multiple masses and continuous distributions.

2. **The cosmological distances** (Module 3): the angular diameter distances D_{d}, D_{s}, D_{ds} relating angular separations on the sky to physical separations at the lens and source.

The task of this module is to combine these into a single equation that maps source positions $\boldsymbol{\beta}$ to image positions $\boldsymbol{\theta}$. Remarkably, this equation can produce *multiple solutions* — multiple images of a single source — which is the defining phenomenon of strong gravitational lensing.

8.2 The Thin-Screen Approximation

The physical extent of a gravitational lens along the line of sight is typically much smaller than the distances between observer, lens, and source. For a galaxy lens, the stellar and dark matter distribution extends over ~ 100 kpc, while $D_{\rm d} \sim 1$ Gpc. For a galaxy cluster, the virial radius is ~ 2 Mpc, compared to distances of several Gpc. In both cases the ratio

$$\frac{\text{lens extent}}{\text{cosmological distance}} \sim \frac{100\ \text{kpc}}{1\ \text{Gpc}} = 10^{-4} \tag{8.1}$$

is exceedingly small.

Remark 8.2.1. The thin-screen approximation is the gravitational analog of the **thin-lens** approximation in classical optics. Just as a glass lens can be treated as imparting a phase shift at a single plane (when its thickness is much less than the focal length), a gravitational lens can be treated as deflecting light at a single **lens plane** perpendicular to the line of sight at distance $D_{\rm d}$.

This justifies the following idealization: all the mass of the lens is projected onto a single plane (the **lens plane**) at angular diameter distance $D_{\rm d}$ from the observer. The deflection of a light ray occurs instantaneously as it crosses this plane. The three-dimensional mass distribution $\rho(\mathbf{r})$ is replaced by a two-dimensional **surface mass density** $\Sigma(\boldsymbol{\xi})$, obtained by integrating ρ along the line of sight:

$$\Sigma(\boldsymbol{\xi}) = \int_{-\infty}^{+\infty} \rho(\boldsymbol{\xi}, z)\, dz\,, \tag{8.2}$$

where $\boldsymbol{\xi}$ is the two-dimensional position in the lens plane and z is the coordinate along the line of sight.

8.3 Lensing Geometry

Figure 8.1 shows the geometry of a typical gravitational lensing configuration. The key elements are:

- **Observer (O):** located at the bottom of the diagram.
- **Lens plane:** a plane perpendicular to the optical axis at angular diameter distance $D_{\rm d}$ from the observer. The lens (galaxy, cluster, etc.) is projected onto this plane.
- **Source plane:** a plane perpendicular to the optical axis at angular diameter distance $D_{\rm s}$ from the observer. The background source (galaxy, quasar, etc.) lies in this plane.
- **Optical axis:** the line from the observer through the center of the lens, defining the origin of the angular coordinate system.

We define the following angles (all measured from the optical axis):

- $\boldsymbol{\theta}$: the **observed image position** on the sky,

- $\boldsymbol{\beta}$: the **true source position** (where the source would appear without lensing),
- $\hat{\alpha}$: the **physical deflection angle** at the lens plane,
- $\boldsymbol{\alpha}(\boldsymbol{\theta})$: the **reduced deflection angle**, defined below.

The physical positions in the lens and source planes are related to the angular positions by:

$$\boldsymbol{\xi} = D_{\rm d}\,\boldsymbol{\theta}\,, \tag{8.3}$$

$$\boldsymbol{\eta} = D_{\rm s}\,\boldsymbol{\beta}\,, \tag{8.4}$$

where $\boldsymbol{\xi}$ is the position in the lens plane and $\boldsymbol{\eta}$ is the position in the source plane (Congdon & Keeton eq. 4.1; Narayan & Bartelmann eq. 5).

8.4 The Lens Equation

From the geometry of Figure 8.1, we can read off the fundamental relationship between source and image positions. A light ray that is observed at angle $\boldsymbol{\theta}$ from the optical axis was emitted from the source at position $\boldsymbol{\eta}$ in the source plane. Following the deflected ray from the lens plane to the source plane, the physical displacement in the source plane is (Narayan & Bartelmann eq. 6; Congdon & Keeton eq. 4.2):

$$\boldsymbol{\eta} = \frac{D_{\rm s}}{D_{\rm d}}\,\boldsymbol{\xi} - D_{\rm ds}\,\hat{\boldsymbol{\alpha}}(\boldsymbol{\xi})\,. \tag{8.5}$$

Substituting $\boldsymbol{\xi} = D_{\rm d}\,\boldsymbol{\theta}$ and $\boldsymbol{\eta} = D_{\rm s}\,\boldsymbol{\beta}$:

$$D_{\rm s}\,\boldsymbol{\beta} = D_{\rm s}\,\boldsymbol{\theta} - D_{\rm ds}\,\hat{\boldsymbol{\alpha}}(D_{\rm d}\,\boldsymbol{\theta})\,, \tag{8.6}$$

and dividing by $D_{\rm s}$:

$$\boxed{\boldsymbol{\beta} = \boldsymbol{\theta} - \frac{D_{\rm ds}}{D_{\rm s}}\,\hat{\boldsymbol{\alpha}}(D_{\rm d}\,\boldsymbol{\theta})} \tag{8.7}$$

This is the **gravitational lens equation** in terms of the physical deflection angle $\hat{\boldsymbol{\alpha}}$. It is convenient to define the **reduced deflection angle**:

$$\boxed{\boldsymbol{\alpha}(\boldsymbol{\theta}) \equiv \frac{D_{\rm ds}}{D_{\rm s}}\,\hat{\boldsymbol{\alpha}}(D_{\rm d}\,\boldsymbol{\theta})} \tag{8.8}$$

so that the lens equation takes the elegant form:

$$\boxed{\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})} \tag{8.9}$$

Remark 8.4.1. Equation (8.9) is **THE** fundamental equation of gravitational lensing. Every result in strong lensing theory — image positions, magnifications, time delays, critical curves, caustics — derives from this single equation. The nonlinearity of the deflection angle $\boldsymbol{\alpha}(\boldsymbol{\theta})$ is what allows multiple solutions (multiple images) for a given source position $\boldsymbol{\beta}$.

Remark 8.4.2. The lens equation is a mapping from the image plane to the source plane: given an image position $\boldsymbol{\theta}$, it determines the corresponding source position $\boldsymbol{\beta}$. The *inverse* problem — given a source position $\boldsymbol{\beta}$, find all image positions $\boldsymbol{\theta}$ — is generally much harder, because the equation can have multiple solutions. For a point mass lens the inverse problem can be solved analytically (Section 8.7); for more complex lens models it typically requires numerical root-finding.

8.5 Surface Mass Density and the Deflection Integral

For a continuous mass distribution described by the surface mass density $\Sigma(\boldsymbol{\xi})$ (eq. 8.2), the physical deflection angle at position $\boldsymbol{\xi}$ in the lens plane is obtained by superposing the deflections from all mass elements (Congdon & Keeton eq. 4.5; cf. Module 1e eq. 5.26):

$$\boxed{\hat{\boldsymbol{\alpha}}(\boldsymbol{\xi}) = \frac{4G}{c^2} \int \frac{(\boldsymbol{\xi} - \boldsymbol{\xi}')\,\Sigma(\boldsymbol{\xi}')}{|\boldsymbol{\xi} - \boldsymbol{\xi}'|^2}\, d^2\xi'} \tag{8.10}$$

This is the two-dimensional analog of the Newtonian gravitational field of a surface mass distribution. The integral extends over the entire lens plane. In terms of the angular position $\boldsymbol{\theta}$ and using $\boldsymbol{\xi} = D_{\rm d}\,\boldsymbol{\theta}$, the reduced deflection angle becomes:

$$\boldsymbol{\alpha}(\boldsymbol{\theta}) = \frac{D_{\rm ds}}{D_{\rm s}}\,\frac{4G}{c^2}\,D_{\rm d} \int \frac{(\boldsymbol{\theta} - \boldsymbol{\theta}')\,\Sigma(D_{\rm d}\,\boldsymbol{\theta}')}{|\boldsymbol{\theta} - \boldsymbol{\theta}'|^2}\, d^2\theta' \,. \tag{8.11}$$

Remark 8.5.1. The deflection integral (8.10) follows directly from the superposition principle derived in Module 5 (eq. 5.25). Each infinitesimal mass element $\Sigma(\boldsymbol{\xi}')\,d^2\xi'$ acts as a point mass, deflecting the light ray by $4G\Sigma(\boldsymbol{\xi}')\,d^2\xi'/(c^2\,|\boldsymbol{\xi} - \boldsymbol{\xi}'|)$ in the direction of $(\boldsymbol{\xi} - \boldsymbol{\xi}')/|\boldsymbol{\xi} - \boldsymbol{\xi}'|$. Superposition (valid because we are in the weak-field regime) gives the total deflection as the integral over the lens plane.

8.6 Critical Surface Mass Density

The lens equation (8.9) contains the combination $(D_{\rm ds}/D_{\rm s})\,(4G/c^2)\,D_{\rm d}$, which defines a natural scale for the surface mass density. We define the **critical surface mass density** (Congdon & Keeton eq. 4.10; Narayan & Bartelmann eq. 13):

$$\boxed{\Sigma_{\rm cr} = \frac{c^2}{4\pi G}\,\frac{D_{\rm s}}{D_{\rm d}\,D_{\rm ds}}} \tag{8.12}$$

Remark 8.6.1. The factor of π in $\Sigma_{\rm cr}$ is conventional and arises from the two-dimensional Poisson equation $\nabla^2\psi = 2\kappa$ (which we will encounter in Module 5), where the Laplacian of the lensing potential equals twice the convergence.

The **convergence** is then the dimensionless ratio of the actual surface mass density to the critical value:

$$\boxed{\kappa(\boldsymbol{\theta}) = \frac{\Sigma(D_{\rm d}\,\boldsymbol{\theta})}{\Sigma_{\rm cr}}} \tag{8.13}$$

The critical surface mass density sets the boundary between weak and strong lensing:

- If $\kappa < 1$ everywhere ($\Sigma < \Sigma_{\rm cr}$), the lens is **subcritical**: it can still deflect light, but it cannot produce multiple images of a generic source.

- If $\kappa \geq 1$ somewhere ($\Sigma \geq \Sigma_{\rm cr}$), the lens is **supercritical**: it can produce multiple images, arcs, and Einstein rings — this is the regime of **strong gravitational lensing**.

Example 8.6.1. Numerical value of $\Sigma_{\rm cr}$. For a typical galaxy lens at $z_d = 0.3$ with a source at $z_s = 1$ in a flat ΛCDM cosmology ($H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_{\rm m} = 0.3$, $\Omega_\Lambda = 0.7$), the angular diameter distances are approximately $D_{\rm d} \approx 900$ Mpc, $D_{\rm s} \approx 1700$ Mpc, $D_{\rm ds} \approx 1200$ Mpc. Substituting into eq. (8.12):

$$\Sigma_{\rm cr} \approx \frac{(3\times10^8)^2}{4\pi\times 6.674\times10^{-11}} \times \frac{1700}{900\times1200}\ \mathrm{Mpc^{-1}} \approx 0.35\ \mathrm{g\,cm^{-2}} \approx 1700\ M_\odot\,\mathrm{pc^{-2}}\,. \tag{8.14}$$

This is a remarkably low surface mass density — only about $3.5\ \mathrm{kg\,m^{-2}}$. Nevertheless, typical galaxy and cluster lenses can achieve $\Sigma \geq \Sigma_{\rm cr}$ in their central regions, making strong lensing possible. The exact value is computed and verified in `lens_equation.wl`.

Exercise 8.6.1. Show that $\Sigma_{\rm cr}$ as defined in eq. (8.12) has units of mass per area. *Hint:* G has units of $\mathrm{m^3\,kg^{-1}\,s^{-2}}$, c has units of $\mathrm{m\,s^{-1}}$, and the distances have units of length.

8.7 The Einstein Radius

The lens equation simplifies dramatically for a **point mass lens**. For a mass M at the origin of the lens plane, the physical deflection angle is (from Module 2):

$$\hat\alpha(\xi) = \frac{4GM}{c^2\,\xi}\,, \tag{8.15}$$

where $\xi = |\boldsymbol{\xi}| = D_{\rm d}\,\theta$ is the physical distance from the mass in the lens plane. The reduced deflection angle is:

$$\alpha(\theta) = \frac{D_{\rm ds}}{D_{\rm s}}\frac{4GM}{c^2\,D_{\rm d}\,\theta} = \frac{\theta_{\rm E}^2}{\theta}\,, \tag{8.16}$$

where we have defined the **Einstein radius**:

$$\boxed{\theta_{\rm E} = \sqrt{\frac{4GM}{c^2}\frac{D_{\rm ds}}{D_{\rm d}\,D_{\rm s}}}} \tag{8.17}$$

This is **THE** characteristic angular scale of gravitational lensing. For a circularly symmetric lens, the Einstein radius is the angular radius of the ring that forms when the source lies exactly behind the lens ($\beta = 0$). The corresponding physical radius in the lens plane is:

$$R_{\rm E} = D_{\rm d}\,\theta_{\rm E} = \sqrt{\frac{4GM}{c^2}\frac{D_{\rm d}\,D_{\rm ds}}{D_{\rm s}}}\,. \tag{8.18}$$

8.7.1 The Point Mass Lens Equation

Substituting eq. (8.16) into the lens equation (8.9) (restricting to one angular dimension by the circular symmetry of the point mass):

$$\boxed{\beta = \theta - \frac{\theta_{\mathrm{E}}^2}{\theta}} \tag{8.19}$$

This is a quadratic equation in θ. Multiplying through by θ:

$$\theta^2 - \beta\,\theta - \theta_{\mathrm{E}}^2 = 0\,. \tag{8.20}$$

The two solutions are:

$$\boxed{\theta_\pm = \frac{1}{2}\left(\beta \pm \sqrt{\beta^2 + 4\,\theta_{\mathrm{E}}^2}\right)} \tag{8.21}$$

Since $\beta^2 + 4\,\theta_{\mathrm{E}}^2 > \beta^2$ for $\theta_{\mathrm{E}} > 0$, both solutions are always real. A point mass lens *always* produces two images, regardless of the source position.

8.7.2 The Einstein Ring

When the source is exactly on the optical axis ($\beta = 0$), the two image positions merge:

$$\theta_\pm = \pm\,\theta_{\mathrm{E}}\,. \tag{8.22}$$

By the circular symmetry of the lens, this corresponds to a complete **ring** of images at angular radius θ_{E} — the **Einstein ring**. This is the most spectacular manifestation of gravitational lensing and has been observed in numerous galaxy-scale and cluster-scale lens systems.

8.7.3 Numerical Examples

Example 8.7.1. Einstein radii for typical lensing systems.

(a) Stellar microlensing ($M = 1\,M_\odot$, $D_{\mathrm{d}} \approx D_{\mathrm{s}}/2 \approx 5$ kpc, $D_{\mathrm{ds}} \approx 5$ kpc):

$$\theta_{\mathrm{E}} = \sqrt{\frac{4 \times 6.674 \times 10^{-11} \times 1.989 \times 10^{30}}{(3 \times 10^8)^2} \times \frac{1}{5 \times 3.086 \times 10^{19}}} \approx 0.9 \text{ mas}\,. \tag{8.23}$$

The Einstein radius for stellar microlensing is of order *milliarcseconds* — far too small to resolve the individual images, but the magnification is detectable photometrically.

(b) Galaxy-scale lensing ($M = 10^{12}\,M_\odot$, $z_d = 0.3$, $z_s = 1$): With $D_{\mathrm{d}} \approx 900$ Mpc, $D_{\mathrm{s}} \approx 1700$ Mpc, $D_{\mathrm{ds}} \approx 1200$ Mpc:

$$\theta_{\mathrm{E}} \approx 1.1''\,. \tag{8.24}$$

Galaxy-scale Einstein radii are of order *arcseconds* — easily resolved with ground-based telescopes. This is the regime of strong lensing that produces multiple images and arcs.

(c) Cluster-scale lensing ($M = 10^{15}\,M_\odot$, $z_d = 0.3$, $z_s = 2$): With $D_\mathrm{d} \approx 900$ Mpc, $D_\mathrm{s} \approx 1800$ Mpc, $D_\mathrm{ds} \approx 1400$ Mpc:

$$\theta_\mathrm{E} \approx 35'' \,. \tag{8.25}$$

Cluster-scale Einstein radii are of order *tens of arcseconds*, producing the spectacular giant arcs observed in massive galaxy clusters.

These values are computed and verified in `lens_equation.wl`. Interactive exploration is available in `einstein_ring.nb`.

8.8 Image Positions for a Point Mass Lens

The two image positions $\theta_\pm$ from eq. (8.21) have several important properties. It is convenient to work with the dimensionless variable $u \equiv \beta/\theta_\mathrm{E}$ (source position in units of the Einstein radius). Then:

$$\theta_\pm = \frac{\theta_\mathrm{E}}{2}\left(u \pm \sqrt{u^2+4}\right) . \tag{8.26}$$

8.8.1 Properties of the Images

1. θ_+ **is always positive** (on the same side of the lens as the source) and satisfies $\theta_+ > \theta_\mathrm{E}$. This is the **primary image**, which is always located *outside* the Einstein radius.

2. θ_- **is always negative** (on the opposite side of the lens from the source) and satisfies $|\theta_-| < \theta_\mathrm{E}$. This is the **secondary image**, located *inside* the Einstein radius. Its negative sign indicates opposite parity (the image is mirror-reflected).

3. **Sum and product:** From Vieta's formulas applied to the quadratic (8.20):

$$\theta_+ + \theta_- = \beta \,, \qquad \theta_+\,\theta_- = -\theta_\mathrm{E}^2 \,. \tag{8.27}$$

   The product relation shows that the images always straddle the Einstein ring: one inside, one outside.

4. **Image separation:**

$$\Delta\theta = \theta_+ - \theta_- = \sqrt{\beta^2 + 4\,\theta_\mathrm{E}^2} \,. \tag{8.28}$$

   The minimum separation occurs at $\beta = 0$ (Einstein ring configuration), where $\Delta\theta = 2\,\theta_\mathrm{E}$.

8.8.2 Limiting Cases

Source on the optical axis ($\beta \to 0$, $u \to 0$): Both images approach the Einstein radius: $\theta_+ \to +\theta_\mathrm{E}$, $\theta_- \to -\theta_\mathrm{E}$. The two images merge into the **Einstein ring**.

Source far from the lens ($\beta \gg \theta_E$, $u \gg 1$):

$$\theta_+ \approx \beta + \frac{\theta_E^2}{\beta} \to \beta \quad \text{(approaches the unlensed position)}\,, \tag{8.29}$$

$$\theta_- \approx -\frac{\theta_E^2}{\beta} \to 0 \quad \text{(approaches the lens, becomes infinitely demagnified)}\,. \tag{8.30}$$

In this limit the primary image is essentially at the unlensed source position (negligible lensing), and the secondary image is extremely close to the lens and too faint to observe.

8.9 Magnification of a Point Mass Lens

Gravitational lensing conserves surface brightness (a consequence of Liouville's theorem applied to the photon phase space), but it changes the *solid angle* subtended by the image compared to the source. The ratio of image area to source area is the **magnification** μ. We will derive the magnification formalism rigorously in Module 5; here we preview the key results for the point mass lens.

For a circularly symmetric lens, the magnification is (Narayan & Bartelmann eq. 19):

$$\mu = \frac{\theta}{\beta}\,\frac{d\theta}{d\beta}\,. \tag{8.31}$$

The first factor accounts for the tangential stretching of the image (the image is at a different angular radius than the source), and the second factor accounts for the radial stretching.

For the point mass lens equation $\beta = \theta - \theta_E^2/\theta$, we have $d\beta/d\theta = 1 + \theta_E^2/\theta^2$, and the magnification of each image is:

$$\mu_\pm = \frac{\theta_\pm}{\beta}\,\frac{1}{1 + \theta_E^2/\theta_\pm^2} = \frac{\theta_\pm^4}{\theta_\pm^4 - \theta_E^4}\,. \tag{8.32}$$

Using the dimensionless variable $u = \beta/\theta_E$, the magnifications simplify to (Narayan & Bartelmann eq. 20):

$$\boxed{\mu_\pm = \frac{u^2+2}{2u\sqrt{u^2+4}} \pm \frac{1}{2}} \tag{8.33}$$

Several important properties follow:

- $\mu_+ > 0$ always (same parity as the source).
- $\mu_- < 0$ always (opposite parity — the image is mirror-reflected). The *absolute* magnification is $|\mu_-|$.
- The total magnification (sum of absolute values) is:

$$|\mu_+| + |\mu_-| = \frac{u^2+2}{u\sqrt{u^2+4}} > 1 \quad \text{for all } u > 0\,. \tag{8.34}$$

Gravitational lensing always *increases* the total observed flux (the total magnification exceeds unity).

- As $u \to 0$ (source on the optical axis), both $|\mu_\pm| \to \infty$: the Einstein ring is *infinitely* magnified (in the geometric optics approximation).
- As $u \to \infty$, $\mu_+ \to 1$ (unlensed) and $|\mu_-| \to 0$ (secondary image disappears).

8.10 Summary

- The **thin-screen approximation** projects the three-dimensional mass distribution onto a single lens plane, justified by the small ratio of lens extent to cosmological distance ($\sim 10^{-4}$).
- The **lens equation** $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ maps image positions to source positions. The **reduced deflection angle** $\boldsymbol{\alpha} = (D_{\rm ds}/D_{\rm s})\,\hat{\boldsymbol{\alpha}}$ absorbs the distance factors.
- The **surface mass density** $\Sigma(\boldsymbol{\xi})$ and the deflection integral (8.10) allow the treatment of arbitrary continuous mass distributions.
- The **critical surface mass density** $\Sigma_{\rm cr} = (c^2/4\pi G)\,(D_{\rm s}/D_{\rm d}\,D_{\rm ds})$ separates weak lensing ($\kappa < 1$) from strong lensing ($\kappa \geq 1$).
- The **Einstein radius** $\theta_{\rm E} = \sqrt{4GM/c^2 \cdot D_{\rm ds}/(D_{\rm d}\,D_{\rm s})}$ is the characteristic angular scale of lensing. Typical values: ~ 1 mas for stellar microlensing, $\sim 1''$ for galaxy lensing, $\sim 30''$ for cluster lensing.
- A **point mass lens** always produces two images: $\theta_\pm = (\beta \pm \sqrt{\beta^2 + 4\theta_{\rm E}^2})/2$. The primary image ($\theta_+$) is outside $\theta_{\rm E}$ with the same parity; the secondary (θ_-) is inside $\theta_{\rm E}$ with opposite parity.
- When the source is exactly behind the lens ($\beta = 0$), the images merge into an **Einstein ring** at angular radius $\theta_{\rm E}$.
- The **magnification** of the point mass lens is $\mu_\pm = (u^2+2)/(2u\sqrt{u^2+4}) \pm 1/2$, where $u = \beta/\theta_{\rm E}$. The total magnification always exceeds unity.

8.11 Problems

Exercise 8.11.1. (Derivation of the lens equation.) Starting from the lensing geometry (Figure 8.1), derive the lens equation (8.7).

(a) Using the small-angle approximation, show that the physical position in the source plane is $\boldsymbol{\eta} = (D_{\rm s}/D_{\rm d})\,\boldsymbol{\xi} - D_{\rm ds}\,\hat{\boldsymbol{\alpha}}(\boldsymbol{\xi})$.

(b) Substitute $\boldsymbol{\xi} = D_{\rm d}\,\boldsymbol{\theta}$ and $\boldsymbol{\eta} = D_{\rm s}\,\boldsymbol{\beta}$ to obtain the lens equation in angular coordinates.

(c) Define the reduced deflection angle and write the lens equation in the compact form $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$.

Exercise 8.11.2. (Point mass lens equation.) For a point mass lens with Einstein radius $\theta_{\rm E}$:

(a) Derive the lens equation $\beta = \theta - \theta_{\rm E}^2/\theta$ from the deflection angle $\hat{\alpha} = 4GM/(c^2\xi)$.

(b) Solve the resulting quadratic to obtain the image positions $\theta_\pm$.

(c) Show that two images always form (i.e., the discriminant is always positive).

(d) Verify the Vieta relations: $\theta_+ + \theta_- = \beta$ and $\theta_+\,\theta_- = -\theta_{\rm E}^2$.

(e) Show that $\theta_+ > \theta_{\rm E} > 0 > \theta_- > -\theta_{\rm E}$ for $\beta > 0$.

Verify with `problems_04.wl`.

Exercise 8.11.3. (Einstein radius for various systems.) Compute the Einstein radius $\theta_{\rm E}$ for:

(a) **Stellar microlensing:** $M = 1\,M_\odot$, lens at distance $D_{\rm d} = 5$ kpc, source at $D_{\rm s} = 10$ kpc (so $D_{\rm ds} = 5$ kpc). Express the result in milliarcseconds.

(b) **Galaxy lensing:** $M = 10^{12}\,M_\odot$, $z_d = 0.3$, $z_s = 1$. Use the angular diameter distances from Module 3 (or the Mathematica script): $D_{\rm d} \approx 900$ Mpc, $D_{\rm s} \approx 1700$ Mpc, $D_{\rm ds} \approx 1200$ Mpc. Express the result in arcseconds.

(c) **Cluster lensing:** $M = 10^{15}\,M_\odot$, $z_d = 0.3$, $z_s = 2$. Use $D_{\rm d} \approx 900$ Mpc, $D_{\rm s} \approx 1800$ Mpc, $D_{\rm ds} \approx 1400$ Mpc. Express the result in arcseconds.

For each case, also compute the physical Einstein radius $R_{\rm E} = D_{\rm d}\,\theta_{\rm E}$ in kpc. Verify with `problems_04.wl`.

Exercise 8.11.4. (Critical surface mass density.)

(a) Starting from the definition (8.12), verify that $\Sigma_{\rm cr}$ has units of mass per area by performing a dimensional analysis.

(b) Compute $\Sigma_{\rm cr}$ (in $\rm g\,cm^{-2}$ and $M_\odot\,\rm pc^{-2}$) for the galaxy lens configuration of Exercise 8.11.3(b).

(c) Compute $\Sigma_{\rm cr}$ for the cluster lens configuration of Exercise 8.11.3(c).

(d) A typical elliptical galaxy has a central surface mass density of $\Sigma_0 \approx 5000\,M_\odot\,\rm pc^{-2}$. Is it a strong lens? Compare Σ_0 with your result from (b).

Verify with `problems_04.wl`.

Exercise 8.11.5. (Image positions for a point mass lens.) Consider a point mass lens with $\theta_{\rm E} = 1''$.

(a) For source positions $\beta = 0.1'', 0.5'', 1'', 2'', 5''$, compute both image positions θ_+ and θ_-. Tabulate your results.

(b) Plot θ_+ and θ_- as functions of β for $0 \leq \beta \leq 5''$. On your plot, also show the Einstein radius and the line $\theta = \beta$ (unlensed position).

(c) Compute the image separation $\Delta\theta = \theta_+ - \theta_-$ for each value of β and plot it. What is the minimum image separation?

(d) Compute the magnification of each image using eq. (8.33) and plot the total magnification $|\mu_+| + |\mu_-|$ as a function of β.

Use Mathematica to generate the plots and verify the numerical values. See `lens_equation.wl` and `problems_04.wl`.

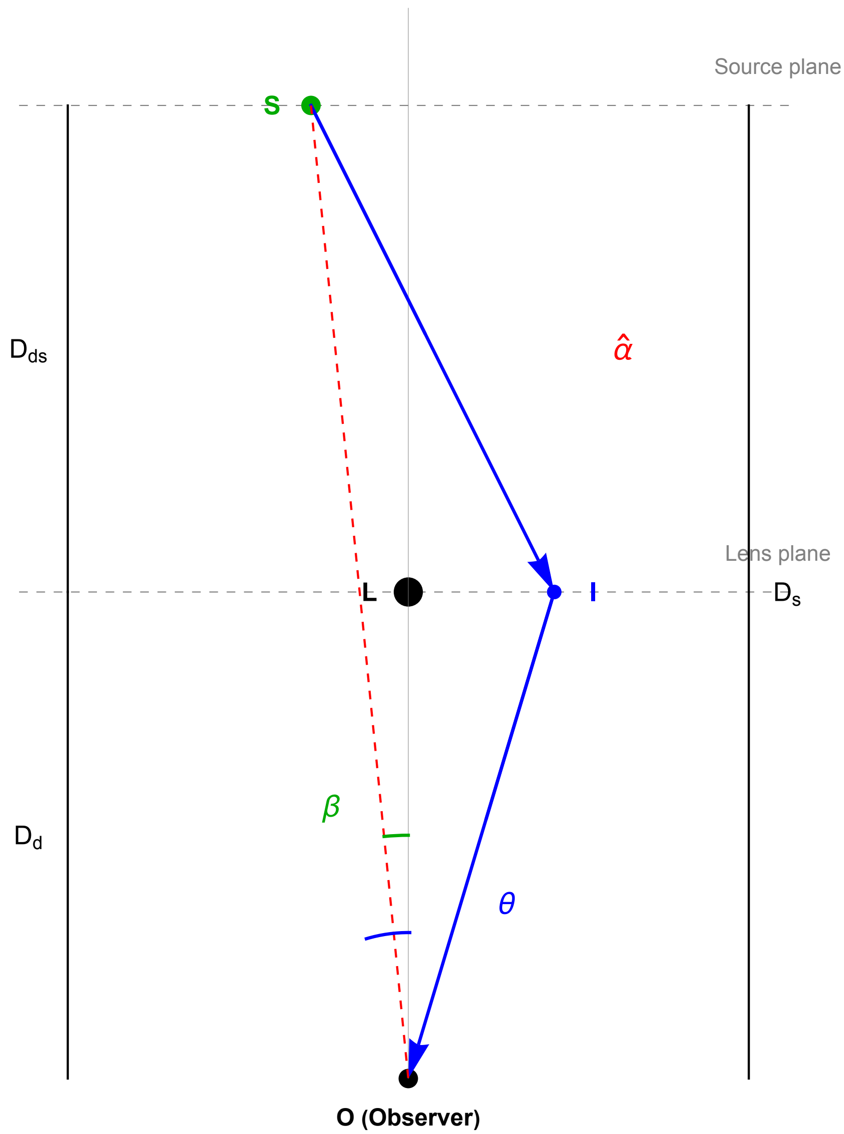


Figure 8.1: Geometry of gravitational lensing. A light ray from a source at angular position $\boldsymbol{\beta}$ in the source plane is deflected by the angle $\hat{\alpha}$ as it passes through the lens plane, and is observed at angular position $\boldsymbol{\theta}$. The angular diameter distances $D_{\rm d}$, $D_{\rm s}$, and $D_{\rm ds}$ connect angular separations to physical separations. The reduced deflection angle $\boldsymbol{\alpha} = (D_{\rm ds}/D_{\rm s})\,\hat{\boldsymbol{\alpha}}$ gives the apparent angular shift. Generated by `lens_equation.wl`.

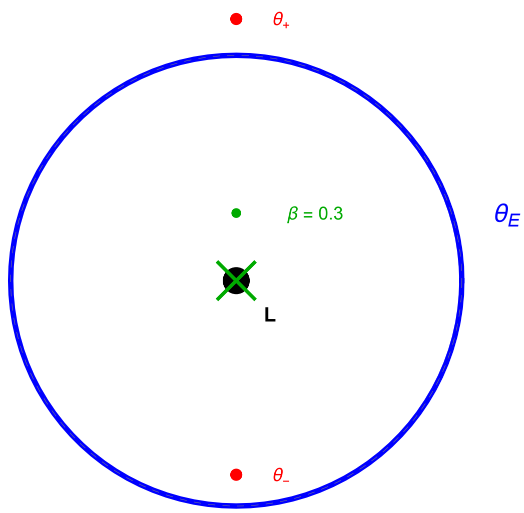


Figure 8.2: The Einstein ring. When a source (star symbol at center) lies exactly behind a point mass lens (filled circle at center), the image is a complete ring of angular radius $\theta_{\rm E}$. For a slightly offset source ($\beta \neq 0$), the ring breaks into two images at positions θ_+ and θ_-. Generated by `lens_equation.wl`.

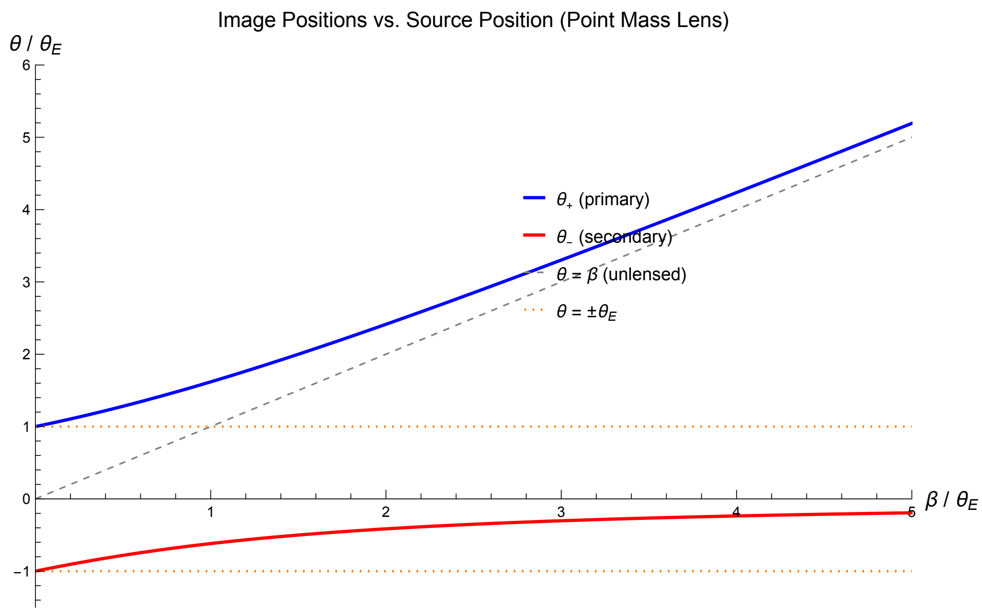


Figure 8.3: Image positions θ_+ and θ_- as functions of the source position β for a point mass lens, in units of the Einstein radius $\theta_{\rm E}$. The primary image θ_+ (blue) is always outside $\theta_{\rm E}$ and approaches the unlensed position (gray dashed line) for $\beta \gg \theta_{\rm E}$. The secondary image θ_- (red) is always inside $\theta_{\rm E}$ (in absolute value) and approaches the lens for $\beta \gg \theta_{\rm E}$. At $\beta = 0$, both images are at $\pm\theta_{\rm E}$ (Einstein ring). Generated by `lens_equation.wl`.

8.12 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_04.wl`, which verifies each answer symbolically and numerically.

Exercise 8.11.1: Derivation of the Lens Equation

(a) From the lensing geometry (Figure 8.1), consider a light ray that passes through the lens plane at physical position $\boldsymbol{\xi}$ and is deflected by the angle $\hat{\boldsymbol{\alpha}}$. In the small-angle approximation, the physical position in the source plane is determined by two contributions: (i) the extrapolation of the undeflected ray from the lens plane to the source plane, which gives a position $(D_\mathrm{s}/D_\mathrm{d})\,\boldsymbol{\xi}$, and (ii) the deflection, which displaces the ray by $-D_\mathrm{ds}\,\hat{\boldsymbol{\alpha}}(\boldsymbol{\xi})$ in the source plane. Therefore:

$$\boldsymbol{\eta} = \frac{D_\mathrm{s}}{D_\mathrm{d}}\,\boldsymbol{\xi} - D_\mathrm{ds}\,\hat{\boldsymbol{\alpha}}(\boldsymbol{\xi})\,.$$

(b) Substituting $\boldsymbol{\xi} = D_\mathrm{d}\,\boldsymbol{\theta}$ and $\boldsymbol{\eta} = D_\mathrm{s}\,\boldsymbol{\beta}$:

$$D_\mathrm{s}\,\boldsymbol{\beta} = \frac{D_\mathrm{s}}{D_\mathrm{d}}\,D_\mathrm{d}\,\boldsymbol{\theta} - D_\mathrm{ds}\,\hat{\boldsymbol{\alpha}}(D_\mathrm{d}\,\boldsymbol{\theta}) = D_\mathrm{s}\,\boldsymbol{\theta} - D_\mathrm{ds}\,\hat{\boldsymbol{\alpha}}(D_\mathrm{d}\,\boldsymbol{\theta})\,.$$

Dividing by D_s:

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \frac{D_\mathrm{ds}}{D_\mathrm{s}}\,\hat{\boldsymbol{\alpha}}(D_\mathrm{d}\,\boldsymbol{\theta})\,.$$

(c) Defining the reduced deflection angle $\boldsymbol{\alpha}(\boldsymbol{\theta}) \equiv (D_\mathrm{ds}/D_\mathrm{s})\,\hat{\boldsymbol{\alpha}}(D_\mathrm{d}\,\boldsymbol{\theta})$, the lens equation becomes:

$$\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})\,. \qquad \square$$

Exercise 8.11.2: Point Mass Lens Equation

(a) For a point mass M at the origin, the physical deflection angle is $\hat{\alpha} = 4GM/(c^2\xi)$ where $\xi = D_\mathrm{d}\,\theta$. The reduced deflection angle is:

$$\alpha(\theta) = \frac{D_\mathrm{ds}}{D_\mathrm{s}}\,\frac{4GM}{c^2\,D_\mathrm{d}\,\theta} = \frac{1}{\theta}\,\frac{4GM}{c^2}\,\frac{D_\mathrm{ds}}{D_\mathrm{d}\,D_\mathrm{s}} = \frac{\theta_\mathrm{E}^2}{\theta}\,.$$

The lens equation is: $\beta = \theta - \alpha(\theta) = \theta - \theta_\mathrm{E}^2/\theta$.

(b) Multiplying by θ: $\theta^2 - \beta\,\theta - \theta_\mathrm{E}^2 = 0$. By the quadratic formula:

$$\theta_\pm = \frac{\beta \pm \sqrt{\beta^2 + 4\,\theta_\mathrm{E}^2}}{2}\,.$$

(c) The discriminant is $\Delta = \beta^2 + 4\,\theta_\mathrm{E}^2 > 0$ for all β (since $\theta_\mathrm{E} > 0$). Therefore two real solutions always exist.

(d) From Vieta's formulas for $\theta^2 - \beta\,\theta - \theta_\mathrm{E}^2 = 0$:

- Sum of roots: $\theta_+ + \theta_- = \beta$. ✓
- Product of roots: $\theta_+\,\theta_- = -\theta_\mathrm{E}^2$. ✓

(e) For $\beta > 0$: since $\sqrt{\beta^2 + 4\theta_E^2} > \beta$, we have $\theta_+ > 0$. Moreover, $\theta_+ = (\beta + \sqrt{\beta^2 + 4\theta_E^2})/2 > \sqrt{\theta_E^2} = \theta_E$ because $\beta + \sqrt{\beta^2 + 4\theta_E^2} > 2\theta_E$ (square both sides to verify).

For θ_-: $\sqrt{\beta^2 + 4\theta_E^2} > \beta$ implies $\theta_- = (\beta - \sqrt{\beta^2 + 4\theta_E^2})/2 < 0$. Also $|\theta_-| = (\sqrt{\beta^2 + 4\theta_E^2} - \beta)/2 < \theta_E$, which can be verified by noting $\sqrt{\beta^2 + 4\theta_E^2} - \beta < 2\theta_E$. $\square$

Exercise 8.11.3: Einstein Radius for Various Systems

Using $\theta_E = \sqrt{(4GM/c^2)\,(D_{ds}/D_d D_s)}$ with $G = 6.674 \times 10^{-11}$ $\mathrm{m^3\,kg^{-1}\,s^{-2}}$, $c = 2.998 \times 10^8$ $\mathrm{m\,s^{-1}}$:

(a) Stellar microlensing: $M = 1.989 \times 10^{30}$ kg, $D_d = D_{ds} = 5$ kpc $= 1.543 \times 10^{20}$ m, $D_s = 10$ kpc $= 3.086 \times 10^{20}$ m.

$$\begin{aligned}
\theta_E &= \sqrt{\frac{4 \times 6.674 \times 10^{-11} \times 1.989 \times 10^{30}}{(2.998 \times 10^8)^2} \times \frac{1.543 \times 10^{20}}{1.543 \times 10^{20} \times 3.086 \times 10^{20}}} \\
&= \sqrt{5.906 \times 10^{-10} \times \frac{1}{3.086 \times 10^{20}}} \\
&= \sqrt{1.914 \times 10^{-30}} \\
&\approx 1.38 \times 10^{-15} \text{ rad} \approx 0.28 \text{ mas}\,.
\end{aligned}$$

Physical Einstein radius: $R_E = 5 \times 3.086 \times 10^{19} \times 1.38 \times 10^{-15} \approx 2.1 \times 10^{-10}$ Mpc $\approx 6.6 \times 10^{-5}$ pc ≈ 14 AU.

(b) Galaxy lensing: $M = 10^{12} \times 1.989 \times 10^{30}$ kg, $D_d = 900$ Mpc, $D_s = 1700$ Mpc, $D_{ds} = 1200$ Mpc. Converting to meters (1 Mpc $= 3.086 \times 10^{22}$ m):

$$\begin{aligned}
\theta_E &= \sqrt{\frac{4 \times 6.674 \times 10^{-11} \times 1.989 \times 10^{42}}{(2.998 \times 10^8)^2} \times \frac{1200}{900 \times 1700} \text{ Mpc}^{-1}} \\
&\approx 5.3 \times 10^{-6} \text{ rad} \approx 1.1''\,.
\end{aligned}$$

Physical Einstein radius: $R_E = 900 \times 5.3 \times 10^{-6}$ Mpc $\approx 4.8 \times 10^{-3}$ Mpc ≈ 4.8 kpc.

(c) Cluster lensing: $M = 10^{15} \times 1.989 \times 10^{30}$ kg, $D_d = 900$ Mpc, $D_s = 1800$ Mpc, $D_{ds} = 1400$ Mpc.

$$\theta_E \approx 1.7 \times 10^{-4} \text{ rad} \approx 35''\,.$$

Physical Einstein radius: $R_E \approx 150$ kpc.

Exercise 8.11.4: Critical Surface Mass Density

(a) Dimensional analysis of $\Sigma_{cr} = c^2/(4\pi G) \times D_s/(D_d\, D_{ds})$:

$$[\Sigma_{cr}] = \frac{(\mathrm{m/s})^2}{(\mathrm{m^3\,kg^{-1}\,s^{-2}})} \times \frac{\mathrm{m}}{\mathrm{m} \times \mathrm{m}} = \frac{\mathrm{m^2\,s^{-2}} \times \mathrm{kg\,s^2}}{\mathrm{m^3}} \times \frac{1}{\mathrm{m}} = \frac{\mathrm{kg}}{\mathrm{m^2}}\,.$$

So Σ_{cr} has units of mass per area, as required. ✓

(b) For the galaxy lens ($D_\mathrm{d} = 900$ Mpc, $D_\mathrm{s} = 1700$ Mpc, $D_\mathrm{ds} = 1200$ Mpc):

$$\Sigma_\mathrm{cr} = \frac{(2.998 \times 10^8)^2}{4\pi \times 6.674 \times 10^{-11}} \times \frac{1700 \times 3.086 \times 10^{22}}{900 \times 3.086 \times 10^{22} \times 1200 \times 3.086 \times 10^{22}} \approx 3.5\ \mathrm{kg\,m^{-2}}\,.$$

Converting: $3.5\ \mathrm{kg\,m^{-2}} = 0.35\ \mathrm{g\,cm^{-2}} \approx 1700\ M_\odot\,\mathrm{pc}^{-2}$.

(c) For the cluster lens ($D_\mathrm{d} = 900$ Mpc, $D_\mathrm{s} = 1800$ Mpc, $D_\mathrm{ds} = 1400$ Mpc):

$$\Sigma_\mathrm{cr} \approx 3.0\ \mathrm{kg\,m^{-2}} = 0.30\ \mathrm{g\,cm^{-2}} \approx 1500\ M_\odot\,\mathrm{pc}^{-2}\,.$$

(d) A typical elliptical galaxy with $\Sigma_0 \approx 5000\ M_\odot\,\mathrm{pc}^{-2}$ has $\kappa = \Sigma_0/\Sigma_\mathrm{cr} \approx 5000/1700 \approx 3 > 1$. Yes, it is a strong lens — the central surface mass density exceeds the critical value, enabling the formation of multiple images.

Exercise 8.11.5: Image Positions for a Point Mass Lens

(a) For $\theta_\mathrm{E} = 1''$ and $\theta_\pm = (\beta \pm \sqrt{\beta^2 + 4})/2$ (in arcseconds):

β (arcsec)	θ_+ (arcsec)	θ_- (arcsec)	$\Delta\theta$ (arcsec)	$\lvert\mu_+\rvert$	$\lvert\mu_-\rvert$
0.1	+1.025	−0.975	2.000	5.13	4.63
0.5	+1.281	−0.781	2.062	1.64	0.64
1.0	+1.618	−0.618	2.236	1.17	0.17
2.0	+2.414	−0.414	2.828	1.06	0.06
5.0	+5.193	−0.193	5.385	1.01	0.01

(b)–(d) Plots are generated in `lens_equation.wl` and `problems_04.wl`. See Figures 8.3 and 8.2.

The minimum image separation is $\Delta\theta_\mathrm{min} = 2\theta_\mathrm{E} = 2''$, occurring at $\beta = 0$ (Einstein ring).

The total magnification $|\mu_+| + |\mu_-|$ diverges as $\beta \to 0$ and approaches 1 (no magnification) as $\beta \to \infty$. For $\beta = \theta_\mathrm{E}$, the total magnification is $(1+2)/(1 \cdot \sqrt{5}) \approx 1.34$.

Chapter 9

Magnification, Convergence, and Shear

> *The convergence and shear are the two fundamental quantities that describe the local properties of a gravitational lens. The convergence causes isotropic focusing; the shear causes anisotropic distortion. Together they determine the magnification and shape distortion of lensed images.* — Narayan & Bartelmann (1997)

In Module 8 we derived the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ and used it to find image positions for a point mass lens. We also previewed the magnification formula for the point mass case. Now we develop the *general* formalism that describes how lensing distorts images: their size, shape, and brightness relative to the unlensed source.

The key players are:

- The **lensing potential** $\psi(\boldsymbol{\theta})$, from which the deflection angle derives as a gradient.
- The **convergence** $\kappa(\boldsymbol{\theta})$, the dimensionless surface mass density, which causes isotropic focusing.
- The **shear** $\gamma(\boldsymbol{\theta})$, a spin-2 field that causes anisotropic stretching.
- The **Jacobian matrix** $\mathcal{A}$, which encodes all local lensing distortions.
- The **magnification** μ, the ratio of image area to source area.

Companion Mathematica files:

- `lensing_potential.wl` — symbolic derivation of ψ, convergence, shear, Jacobian, and magnification for the point mass lens; verification of the 2D Poisson equation; publication-quality figure generation
- `convergence_shear_demo.nb` — interactive notebook: Manipulate for convergence and shear effects on circular sources, magnification map for point mass lens

9.1 Introduction: From Image Positions to Image Properties

The lens equation tells us *where* images form. But observations reveal much more than positions: lensed images are *stretched*, *magnified*, and sometimes *mirror-reflected* relative to the unlensed source. Giant arcs in galaxy clusters are dramatic examples — background galaxies stretched into elongated curves by the cluster's gravity.

To describe these distortions quantitatively, we need to understand the *local* properties of the lens mapping $\boldsymbol{\beta}(\boldsymbol{\theta})$. Since the lens mapping is a smooth function (except at isolated singularities), its local behavior is captured by its **Jacobian matrix** — the matrix of first partial derivatives. This matrix encodes all information about magnification, stretching, and parity of each image.

We begin by introducing the **effective lensing potential** $\psi(\boldsymbol{\theta})$, a scalar field from which the deflection angle can be derived as a gradient. This potential provides a unified framework: its first derivatives give the deflection, and its second derivatives give the convergence and shear.

9.2 The Effective Lensing Potential

In Module 8 (eq. 8.11) we expressed the reduced deflection angle as an integral over the convergence. Remarkably, this deflection can be written as the gradient of a scalar potential.

Definition 9.2.1. The **effective lensing potential** (or simply the lensing potential) is defined as (Narayan & Bartelmann eq. 15; Congdon & Keeton eq. 4.15):

$$\boxed{\psi(\boldsymbol{\theta}) = \frac{1}{\pi}\int \kappa(\boldsymbol{\theta}')\,\ln|\boldsymbol{\theta}-\boldsymbol{\theta}'|\,d^2\theta'} \tag{9.1}$$

where $\kappa(\boldsymbol{\theta}') = \Sigma(D_\mathrm{d}\,\boldsymbol{\theta}')/\Sigma_\mathrm{cr}$ is the convergence (eq. 8.13) and the integral extends over the entire lens plane.

This is the two-dimensional analog of the Newtonian gravitational potential: just as the three-dimensional gravitational potential $\Phi(\mathbf{r}) = -G\int \rho(\mathbf{r}')/|\mathbf{r}-\mathbf{r}'|\,d^3r'$ satisfies Poisson's equation $\nabla^2\Phi = 4\pi G\rho$, the lensing potential satisfies the two-dimensional Poisson equation.

Remark 9.2.1. The lensing potential ψ is dimensionless when $\boldsymbol{\theta}$ is measured in radians. It is related to the physical (Shapiro) time delay through the Fermat potential, which we will encounter in Module 6.

9.2.1 Key Properties of the Lensing Potential

The lensing potential has two crucial properties:

(1) Its gradient gives the deflection angle. Taking the gradient of eq. (9.1):

$$\boxed{\boldsymbol{\alpha}(\boldsymbol{\theta}) = \nabla_\theta\,\psi(\boldsymbol{\theta})} \tag{9.2}$$

where $\nabla_\theta = (\partial/\partial\theta_1,\, \partial/\partial\theta_2)$ is the two-dimensional gradient with respect to the angular coordinates. This can be verified by differentiating eq. (9.1) under the integral sign and using $\nabla_\theta \ln|\boldsymbol{\theta}-\boldsymbol{\theta}'| = (\boldsymbol{\theta}-\boldsymbol{\theta}')/|\boldsymbol{\theta}-\boldsymbol{\theta}'|^2$.

(2) Its Laplacian gives twice the convergence (2D Poisson equation).

$$\boxed{\nabla^2_\theta\,\psi(\boldsymbol{\theta}) = 2\,\kappa(\boldsymbol{\theta})} \tag{9.3}$$

This follows from $\nabla^2_\theta \ln|\boldsymbol{\theta}-\boldsymbol{\theta}'| = 2\pi\,\delta^{(2)}(\boldsymbol{\theta}-\boldsymbol{\theta}')$ (the Green's function of the two-dimensional Laplacian), so that:

$$\nabla^2_\theta\,\psi = \frac{1}{\pi}\int \kappa(\boldsymbol{\theta}')\,2\pi\,\delta^{(2)}(\boldsymbol{\theta}-\boldsymbol{\theta}')\,d^2\theta' = 2\,\kappa(\boldsymbol{\theta})\,. \tag{9.4}$$

Equation (9.3) is a fundamental relation: it connects the lensing potential (a "smooth" quantity that can be differentiated) to the convergence (a "physical" quantity related to the surface mass density). Equivalently, one can derive this directly from the definitions: since $\boldsymbol{\alpha} = \nabla\psi$ (eq. 9.2) and $\kappa = \Sigma/\Sigma_{\rm cr} = \frac{1}{2}\nabla\cdot\boldsymbol{\alpha}$ (from the deflection angle integral, Narayan & Bartelmann eq. 18; Congdon & Keeton eq. 4.16), it follows immediately that $\frac{1}{2}\nabla^2\psi = \frac{1}{2}\nabla\cdot(\nabla\psi) = \frac{1}{2}\nabla\cdot\boldsymbol{\alpha} = \kappa$, recovering $\nabla^2\psi = 2\kappa$. This is the lensing analog of Poisson's equation in Newtonian gravity. The relation is verified symbolically in `lensing_potential.wl`.

9.3 Convergence

The convergence was already introduced in Module 8 (eq. 8.13):

$$\kappa(\boldsymbol{\theta}) = \frac{\Sigma(D_{\rm d}\,\boldsymbol{\theta})}{\Sigma_{\rm cr}} \tag{9.5}$$

It is the **dimensionless surface mass density** — the ratio of the actual surface mass density to the critical value. We now explore its role in image distortion.

9.3.1 Physical Meaning

The convergence measures the *isotropic focusing* caused by the lens:

- A region with $\kappa > 0$ acts as a converging lens, focusing light rays and magnifying images isotropically (without changing their shape).
- $\kappa = 1$ means $\Sigma = \Sigma_{\rm cr}$: this is the threshold for strong lensing. Regions where $\kappa \geq 1$ are **supercritical** and can produce multiple images.
- $\kappa < 1$ everywhere means the lens is subcritical: it can still distort images (via shear), but a single mass concentration cannot produce multiple images of a generic source.

9.3.2 Convergence from the Lensing Potential

From the 2D Poisson equation (9.3):

$$\kappa(\boldsymbol{\theta}) = \frac{1}{2}\,\nabla^2_\theta\,\psi = \frac{1}{2}\left(\frac{\partial^2\psi}{\partial\theta_1^2} + \frac{\partial^2\psi}{\partial\theta_2^2}\right). \tag{9.6}$$

Thus the convergence is the *trace* of the Hessian of the lensing potential (up to a factor of 1/2).

9.4 Shear

The shear describes the *anisotropic distortion* of images — the stretching in one direction and compression in the perpendicular direction.

Definition 9.4.1. The two components of the **shear** are defined in terms of the second derivatives of the lensing potential (Narayan & Bartelmann eq. 17; Congdon & Keeton eq. 4.21):

$$\boxed{\gamma_1(\boldsymbol{\theta}) = \frac{1}{2}\left(\frac{\partial^2\psi}{\partial\theta_1^2} - \frac{\partial^2\psi}{\partial\theta_2^2}\right), \qquad \gamma_2(\boldsymbol{\theta}) = \frac{\partial^2\psi}{\partial\theta_1\,\partial\theta_2}} \tag{9.7}$$

The shear magnitude and the complex shear are:

$$|\gamma| = \sqrt{\gamma_1^2 + \gamma_2^2}, \qquad \gamma = \gamma_1 + i\,\gamma_2 = |\gamma|\,e^{2i\phi_\gamma} \tag{9.8}$$

where ϕ_γ is the orientation angle of the shear.

9.4.1 Physical Meaning

The shear is a **spin-2 field**: it is invariant under rotations by 180 (not 360), reflecting the fact that an ellipse looks the same after a half-turn. The two components have specific geometric meanings:

- $\gamma_1 > 0$: stretching along the θ_1-axis, compression along θ_2.
- $\gamma_1 < 0$: stretching along the θ_2-axis, compression along θ_1.
- $\gamma_2 > 0$: stretching along the direction at 45 to the θ_1-axis.
- $\gamma_2 < 0$: stretching along the direction at 135 to the θ_1-axis.

Remark 9.4.1. The convergence and shear together account for all four independent second derivatives of the lensing potential (the Hessian is a symmetric 2×2 matrix with three independent components; κ captures the trace, and γ_1, γ_2 capture the traceless part):

$$\frac{\partial^2\psi}{\partial\theta_i\,\partial\theta_j} = \begin{pmatrix} \kappa + \gamma_1 & \gamma_2 \\ \gamma_2 & \kappa - \gamma_1 \end{pmatrix}. \tag{9.9}$$

9.5 The Jacobian Matrix (Inverse Magnification Tensor)

The lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ defines a mapping from the image plane to the source plane. The local properties of this mapping are encoded in its **Jacobian matrix**:

$$\mathcal{A}_{ij} = \frac{\partial\beta_i}{\partial\theta_j} = \delta_{ij} - \frac{\partial\alpha_i}{\partial\theta_j} = \delta_{ij} - \frac{\partial^2\psi}{\partial\theta_i\,\partial\theta_j} \tag{9.10}$$

where in the last step we used $\alpha_i = \partial\psi/\partial\theta_i$ (eq. 9.2).

Using the decomposition of the Hessian into convergence and shear (eq. 9.9), the Jacobian takes the elegant form (Narayan & Bartelmann eq. 16; Congdon & Keeton eq. 4.22):

$$\boxed{\mathcal{A} = \begin{pmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{pmatrix}} \tag{9.11}$$

Remark 9.5.1. The matrix $\mathcal{A}$ is sometimes called the **inverse magnification tensor** because its inverse $\mathcal{A}^{-1}$ gives the magnification tensor $\mathcal{M}$:

$$\mathcal{M} = \mathcal{A}^{-1} = \frac{\partial \boldsymbol{\theta}}{\partial \boldsymbol{\beta}}\,. \tag{9.12}$$

$\mathcal{M}$ maps source-plane displacements to image-plane displacements.

9.5.1 Decomposition of the Jacobian

The Jacobian can be decomposed into a convergence (isotropic) part and a shear (anisotropic) part:

$$\mathcal{A} = (1-\kappa)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} \gamma_1 & \gamma_2 \\ \gamma_2 & -\gamma_1 \end{pmatrix} \tag{9.13}$$

The first term is proportional to the identity matrix and represents isotropic focusing (convergence). The second term is a traceless, symmetric matrix (the shear matrix) that represents anisotropic distortion.

9.5.2 Eigenvalues of the Jacobian

The eigenvalues of $\mathcal{A}$ are (Congdon & Keeton eq. 4.23):

$$\lambda_\pm = 1-\kappa \pm |\gamma| = (1-\kappa) \pm \sqrt{\gamma_1^2+\gamma_2^2}\,. \tag{9.14}$$

These eigenvalues give the inverse magnification along the two principal axes of the distortion. The corresponding eigenvectors define the directions of maximum and minimum stretching.

9.6 Magnification

Gravitational lensing conserves surface brightness (a consequence of Liouville's theorem — the photon phase space density is preserved along geodesics). Therefore, the magnification is simply the ratio of the *solid angle* of the image to that of the source:

$$\mu = \frac{\text{(image area)}}{\text{(source area)}} = \frac{1}{\det \mathcal{A}}\,. \tag{9.15}$$

Computing the determinant of eq. (9.11):

$$\det \mathcal{A} = (1-\kappa-\gamma_1)(1-\kappa+\gamma_1) - \gamma_2^2 = (1-\kappa)^2 - (\gamma_1^2+\gamma_2^2) \tag{9.16}$$

and therefore:

$$\boxed{\mu = \frac{1}{\det \mathcal{A}} = \frac{1}{(1-\kappa)^2 - |\gamma|^2}} \tag{9.17}$$

This is the **general magnification formula** for gravitational lensing. Note that μ depends on position through $\kappa(\boldsymbol{\theta})$ and $\gamma(\boldsymbol{\theta})$.

9.6.1 Properties of the Magnification

1. **Positive parity ($\mu > 0$):** The image has the same orientation (handedness) as the source. This occurs when $\det \mathcal{A} > 0$.

2. **Negative parity ($\mu < 0$):** The image is mirror-reflected relative to the source. This occurs when $\det \mathcal{A} < 0$. The *absolute* magnification is $|\mu|$.

3. **Magnified images ($|\mu| > 1$):** The image subtends a larger solid angle than the source. The condition is $(1-\kappa)^2 - |\gamma|^2 < 1$.

4. **Demagnified images ($|\mu| < 1$):** The image is smaller than the source.

5. **Critical curves ($\det \mathcal{A} = 0$, $\mu \to \infty$):** Points where the magnification formally diverges. In practice, images near critical curves are very highly magnified (factors of 10–100), but finite source sizes prevent true divergences. These curves are the subject of Module 8.

9.6.2 Magnification in Terms of Eigenvalues

Using the eigenvalues of $\mathcal{A}$ (eq. 9.14):

$$\mu = \frac{1}{\lambda_+ \lambda_-} = \frac{1}{(1-\kappa-|\gamma|)(1-\kappa+|\gamma|)} \,. \tag{9.18}$$

The magnification diverges when either eigenvalue vanishes, i.e., when $1-\kappa = \pm|\gamma|$. These conditions define the two families of critical curves:

- $\lambda_+ = 0$: the **tangential critical curve** ($1-\kappa = |\gamma|$), where images are stretched tangentially.

- $\lambda_- = 0$: the **radial critical curve** ($1-\kappa = -|\gamma|$), where images are stretched radially.

9.7 Convergence vs. Shear: Physical Effects

To build intuition for the roles of convergence and shear, consider how a small circular source is distorted in three limiting cases.

9.7.1 Pure Convergence ($\kappa > 0$, $\gamma = 0$)

The Jacobian reduces to $\mathcal{A} = (1-\kappa)\,\mathbf{I}$, a uniform scaling. A circular source remains circular, but its angular size is scaled by $1/(1-\kappa)$. The magnification is:

$$\mu = \frac{1}{(1-\kappa)^2}\,. \tag{9.19}$$

For $\kappa < 1$, the image is magnified and has the same parity. For $\kappa > 1$, the image is still magnified but has *negative* parity (the mapping "folds over").

9.7.2 Pure Shear ($\kappa = 0$, $\gamma > 0$)

The Jacobian becomes:

$$\mathcal{A} = \begin{pmatrix} 1-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1+\gamma_1 \end{pmatrix}. \tag{9.20}$$

A circular source is distorted into an **ellipse**. The magnification is:

$$\mu = \frac{1}{1-|\gamma|^2}\,. \tag{9.21}$$

The semi-axes of the elliptical image are proportional to $1/(1-|\gamma|)$ and $1/(1+|\gamma|)$, so the axis ratio is:

$$\frac{b}{a} = \frac{1-|\gamma|}{1+|\gamma|}\,. \tag{9.22}$$

The orientation of the ellipse is determined by the shear angle $\phi_\gamma = \frac{1}{2}\arctan(\gamma_2/\gamma_1)$.

9.7.3 Both Convergence and Shear

In general, both κ and γ are nonzero. A circular source is mapped to a *larger ellipse*:

- The convergence sets the overall magnification (size change).
- The shear sets the ellipticity (shape change).

The semi-axes of the image are $\delta/(1-\kappa-|\gamma|)$ and $\delta/(1-\kappa+|\gamma|)$ for a circular source of angular radius δ.

9.8 Example: Point Mass Lens

We now compute κ, γ, and μ for the point mass lens, verifying consistency with the results of Module 8.

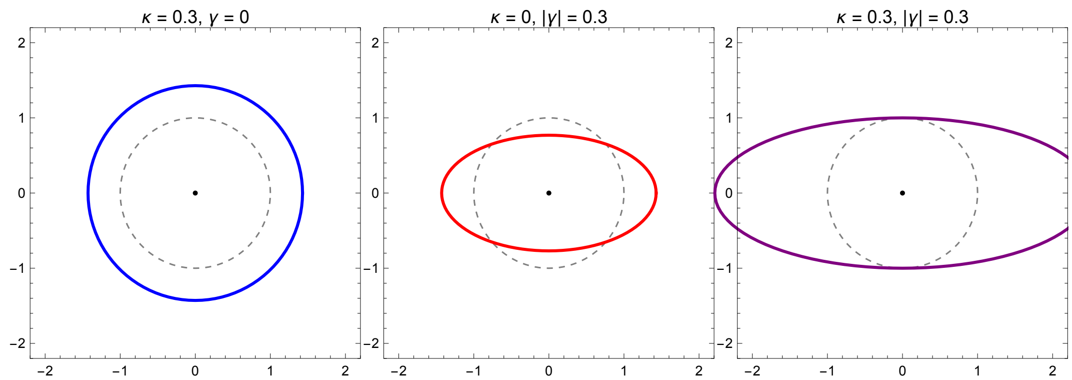


Figure 9.1: Effect of convergence and shear on a circular source. **Left:** Pure convergence ($\kappa = 0.3$, $\gamma = 0$) produces a larger circle (isotropic magnification). **Center:** Pure shear ($\kappa = 0$, $|\gamma| = 0.3$) produces an ellipse (anisotropic distortion without magnification change to first order). **Right:** Both ($\kappa = 0.3$, $|\gamma| = 0.3$) produces a larger ellipse. The dashed circle shows the original source in each panel. Generated by `lensing_potential.wl`.

9.8.1 Lensing Potential for a Point Mass

For a point mass M at the origin, the convergence is a Dirac delta function:

$$\kappa(\boldsymbol{\theta}) = \frac{\Sigma(D_{\rm d}\boldsymbol{\theta})}{\Sigma_{\rm cr}} = \frac{M\,\delta^{(2)}(D_{\rm d}\boldsymbol{\theta})}{\Sigma_{\rm cr}} = \frac{\pi\theta_{\rm E}^2}{1}\,\delta^{(2)}(\boldsymbol{\theta})\,. \tag{9.23}$$

The mass is concentrated entirely at the origin; away from the origin, $\kappa = 0$.

Substituting into eq. (9.1), the lensing potential is:

$$\psi(\boldsymbol{\theta}) = \theta_{\rm E}^2 \ln|\boldsymbol{\theta}| \tag{9.24}$$

which satisfies $\nabla\psi = \theta_{\rm E}^2\,\boldsymbol{\theta}/|\boldsymbol{\theta}|^2$, giving the deflection $\boldsymbol{\alpha} = \theta_{\rm E}^2\,\boldsymbol{\theta}/|\boldsymbol{\theta}|^2$ as expected.

9.8.2 Shear for a Point Mass

Using polar coordinates on the lens plane, we can write $\theta_1 = \theta\cos\phi$ and $\theta_2 = \theta\sin\phi$, where $\theta = |\boldsymbol{\theta}|$. Computing the second derivatives of $\psi = \theta_{\rm E}^2 \ln\theta$:

$$\begin{aligned}
\frac{\partial^2\psi}{\partial\theta_1^2} &= \theta_{\rm E}^2\,\frac{\theta_2^2 - \theta_1^2}{\theta^4} + \frac{\theta_{\rm E}^2}{\theta^2}\,,\\
\frac{\partial^2\psi}{\partial\theta_2^2} &= \theta_{\rm E}^2\,\frac{\theta_1^2 - \theta_2^2}{\theta^4} + \frac{\theta_{\rm E}^2}{\theta^2}\,,\\
\frac{\partial^2\psi}{\partial\theta_1\,\partial\theta_2} &= -\theta_{\rm E}^2\,\frac{2\theta_1\theta_2}{\theta^4}\,.
\end{aligned} \tag{9.25}$$

The shear components are (for $\theta \neq 0$):

$$\gamma_1 = \frac{1}{2}\left(\frac{\partial^2\psi}{\partial\theta_1^2} - \frac{\partial^2\psi}{\partial\theta_2^2}\right) = -\frac{\theta_{\rm E}^2(\theta_1^2 - \theta_2^2)}{\theta^4} = -\frac{\theta_{\rm E}^2}{\theta^2}\cos 2\phi\,, \tag{9.26}$$

$$\gamma_2 = \frac{\partial^2\psi}{\partial\theta_1\,\partial\theta_2} = -\frac{2\theta_{\rm E}^2\,\theta_1\theta_2}{\theta^4} = -\frac{\theta_{\rm E}^2}{\theta^2}\sin 2\phi\,. \tag{9.27}$$

The shear magnitude is:

$$\boxed{|\gamma| = \sqrt{\gamma_1^2 + \gamma_2^2} = \frac{\theta_{\rm E}^2}{\theta^2}} \tag{9.28}$$

The shear falls off as θ^{-2} and is aligned tangentially (the factor $e^{2i\phi}$ in the complex shear).

9.8.3 Convergence for a Point Mass (away from origin)

To compute the convergence correctly, we must be careful with the form of ψ. Writing $\psi = \frac{\theta_{\rm E}^2}{2}\ln(\theta_1^2 + \theta_2^2)$ in Cartesian coordinates:

$$\frac{\partial\psi}{\partial\theta_1} = \frac{\theta_{\rm E}^2\,\theta_1}{\theta^2}\,, \qquad \frac{\partial^2\psi}{\partial\theta_1^2} = \theta_{\rm E}^2\,\frac{\theta^2 - 2\theta_1^2}{\theta^4}\,. \tag{9.29}$$

Similarly, $\partial^2\psi/\partial\theta_2^2 = \theta_{\rm E}^2\,(\theta^2 - 2\theta_2^2)/\theta^4$. Summing:

$$\nabla^2\psi = \theta_{\rm E}^2\,\frac{2\theta^2 - 2(\theta_1^2 + \theta_2^2)}{\theta^4} = 0 \qquad (\theta \neq 0)\,. \tag{9.30}$$

Therefore $\kappa = 0$ for $\theta \neq 0$, confirming that the convergence of a point mass is indeed a delta function concentrated entirely at the origin. In the distributional sense, $\nabla^2 \ln|\boldsymbol{\theta}| = 2\pi\,\delta^{(2)}(\boldsymbol{\theta})$, which recovers $\kappa = \pi\theta_{\rm E}^2\,\delta^{(2)}(\boldsymbol{\theta})$ via the Poisson equation.

9.8.4 Magnification for a Point Mass

With $\kappa = 0$ and $|\gamma| = \theta_{\rm E}^2/\theta^2$ (for $\theta \neq 0$), the magnification formula (9.17) gives:

$$\mu = \frac{1}{1 - \theta_{\rm E}^4/\theta^4} = \frac{\theta^4}{\theta^4 - \theta_{\rm E}^4}\,. \tag{9.31}$$

This is precisely eq. (8.32) from Module 8! Evaluating at the image positions $\theta_\pm$:

$$\mu_\pm = \frac{u^2 + 2}{2u\sqrt{u^2 + 4}} \pm \frac{1}{2} \tag{9.32}$$

where $u = \beta/\theta_{\rm E}$, recovering eq. (8.33) exactly. The consistency is verified symbolically in `lensing_potential.wl`.

9.9 Example: Singular Isothermal Sphere (Preview)

The singular isothermal sphere (SIS) is the simplest realistic model for a galaxy lens. Its surface mass density profile is (Congdon & Keeton Sec. 4.3):

$$\Sigma(\xi) = \frac{\sigma_v^2}{2G\,\xi} \tag{9.33}$$

where σ_v is the one-dimensional velocity dispersion and $\xi = |\boldsymbol{\xi}|$ is the physical radius in the lens plane.

9.9.1 Convergence and Shear of the SIS

The convergence is:

$$\kappa(\theta) = \frac{\Sigma(D_\mathrm{d}\theta)}{\Sigma_\mathrm{cr}} = \frac{\theta_\mathrm{E}}{2\theta} \tag{9.34}$$

where $\theta_\mathrm{E} = 4\pi\sigma_v^2 D_\mathrm{ds}/(c^2 D_\mathrm{s})$ is the Einstein radius of the SIS. Note that κ diverges at $\theta = 0$ (the "singular" in the name) and that $\kappa(\theta_\mathrm{E}) = 1/2$.

The shear magnitude for the SIS is:

$$|\gamma(\theta)| = \frac{\theta_\mathrm{E}}{2\theta} = \kappa(\theta)\,. \tag{9.35}$$

Remarkably, $|\gamma| = \kappa$ at every point for the SIS. This equality does *not* hold for general lens models — it is a special property of the isothermal profile.

9.9.2 Magnification of the SIS

The magnification is:

$$\mu = \frac{1}{(1-\kappa)^2 - |\gamma|^2} = \frac{1}{1-2\kappa} = \frac{1}{1-\theta_\mathrm{E}/\theta} = \frac{\theta}{\theta-\theta_\mathrm{E}}\,. \tag{9.36}$$

The magnification diverges at $\theta = \theta_\mathrm{E}$ — this is the (tangential) critical curve of the SIS. We will treat the SIS model in full detail in Module 7.

9.10 Critical Curves and Caustics (Preview)

Definition 9.10.1. The **critical curves** are the loci in the image plane where the magnification diverges:

$$\det \mathcal{A}(\boldsymbol{\theta}) = 0 \qquad \Longleftrightarrow \qquad (1-\kappa)^2 = |\gamma|^2\,. \tag{9.37}$$

The **caustics** are the images of the critical curves under the lens mapping, i.e., the corresponding curves in the source plane:

$$\boldsymbol{\beta}_\mathrm{caustic} = \boldsymbol{\theta}_\mathrm{crit} - \boldsymbol{\alpha}(\boldsymbol{\theta}_\mathrm{crit})\,. \tag{9.38}$$

Critical curves and caustics play a central role in strong lensing:

- When a source crosses a caustic, the number of images changes by two (a pair of highly magnified images appears or disappears).
- Sources near caustics are very highly magnified — this is exploited in studies of distant, faint galaxies.
- The topology of critical curves and caustics determines the image configuration (doubly, quadruply, or multiply imaged sources).

Remark 9.10.1 (Physical intuition for critical curves). When $\det(\mathcal{A}) = 0$, the magnification is formally infinite. Physically, this means that images near a critical curve are highly stretched and magnified. A small source near a *caustic* (the mapping of the critical curve to the source plane) produces two images straddling the critical curve — one with positive parity (unreflected) and one with negative parity (mirror-reflected). These images appear/disappear in pairs as the source crosses a caustic, which is why the total number of images always changes by ± 2, preserving the odd-number theorem (Module 6). The full topological picture is developed in Module 12.

For the point mass lens, the critical curve is the Einstein ring ($\theta = \theta_{\mathrm{E}}$), and the caustic degenerates to a single point ($\beta = 0$). For more complex lens models, both critical curves and caustics have richer structures. We will develop the full theory in Module 8.

9.11 Summary

- The **lensing potential** $\psi(\boldsymbol{\theta}) = (1/\pi) \int \kappa(\boldsymbol{\theta}') \ln |\boldsymbol{\theta} - \boldsymbol{\theta}'| \, d^2\theta'$ is a scalar field whose gradient gives the deflection angle ($\boldsymbol{\alpha} = \nabla\psi$) and whose Laplacian gives twice the convergence ($\nabla^2\psi = 2\kappa$).
- The **convergence** $\kappa = \Sigma/\Sigma_{\mathrm{cr}}$ causes isotropic focusing: a circular source remains circular but changes size.
- The **shear** $\gamma = \gamma_1 + i\gamma_2$ causes anisotropic distortion: a circular source is stretched into an ellipse.
- The **Jacobian matrix** $\mathcal{A} = \begin{pmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{pmatrix}$ encodes all local lensing properties. Its eigenvalues $\lambda_\pm = 1 - \kappa \pm |\gamma|$ give the inverse magnifications along the principal axes.
- The **magnification** $\mu = 1/\det \mathcal{A} = 1/[(1-\kappa)^2 - |\gamma|^2]$. The sign of μ determines the image parity.
- For a **point mass lens**: $\kappa = 0$ (away from the origin), $|\gamma| = \theta_{\mathrm{E}}^2/\theta^2$, and the general magnification formula recovers the result $\mu_\pm = (u^2+2)/(2u\sqrt{u^2+4}) \pm 1/2$ from Module 4.
- **Critical curves** ($\det \mathcal{A} = 0$) and their mappings, **caustics**, govern the image multiplicity and are the sites of maximum magnification.

9.12 Problems

Exercise 9.12.1. (Derivation of the Jacobian matrix.) Starting from the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ with $\boldsymbol{\alpha} = \nabla\psi$:

(a) Compute $\mathcal{A}_{ij} = \partial\beta_i/\partial\theta_j$ and express it in terms of second derivatives of ψ.

(b) Decompose the Hessian matrix $\partial^2\psi/\partial\theta_i\partial\theta_j$ into a trace part (proportional to δ_{ij}) and a traceless part. Identify the trace part with κ and the traceless part with the shear.

(c) Write $\mathcal{A}$ in the form $\begin{pmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{pmatrix}$.

Verify with `problems_05.wl`.

Exercise 9.12.2. (The 2D Poisson equation.)

(a) Show that the two-dimensional Green's function of the Laplacian is $(1/2\pi)\ln|\boldsymbol{\theta}|$, i.e., $\nabla^2 \ln|\boldsymbol{\theta}| = 2\pi\,\delta^{(2)}(\boldsymbol{\theta})$. *Hint:* Use the fact that $\ln|\boldsymbol{\theta}|$ is harmonic for $\boldsymbol{\theta} \neq 0$ and apply Gauss's theorem to a small circle around the origin.

(b) Starting from the definition of ψ (eq. 9.1), apply the Laplacian to derive $\nabla^2\psi = 2\kappa$.

(c) Verify the 2D Poisson equation for the point mass lens: $\psi = \theta_{\mathrm{E}}^2 \ln|\boldsymbol{\theta}|$ and $\kappa = \pi\theta_{\mathrm{E}}^2\,\delta^{(2)}(\boldsymbol{\theta})$.

Verify with `problems_05.wl`.

Exercise 9.12.3. (Shear of a point mass lens.) For the point mass lensing potential $\psi = \theta_{\mathrm{E}}^2 \ln|\boldsymbol{\theta}|$:

(a) Compute γ_1 and γ_2 as functions of θ_1, θ_2.

(b) Show that $|\gamma| = \theta_{\mathrm{E}}^2/\theta^2$.

(c) Express γ_1 and γ_2 in polar coordinates (θ, ϕ) and show that the complex shear is $\gamma = -(\theta_{\mathrm{E}}^2/\theta^2)\,e^{2i\phi}$ (tangential shear).

(d) Compute the ratio $|\gamma|/\kappa$ at the image positions $\theta_\pm$ of the point mass lens. What happens for sources far from the Einstein radius?

Verify with `problems_05.wl`.

Exercise 9.12.4. (Magnification consistency check.)

(a) Using $\kappa = 0$ and $|\gamma| = \theta_{\mathrm{E}}^2/\theta^2$ for the point mass lens, show that $\mu = \theta^4/(\theta^4 - \theta_{\mathrm{E}}^4)$.

(b) Evaluate μ at the image positions $\theta_\pm = (\beta \pm \sqrt{\beta^2 + 4\theta_{\mathrm{E}}^2})/2$. Using $u = \beta/\theta_{\mathrm{E}}$, show that the result reduces to $\mu_\pm = (u^2+2)/(2u\sqrt{u^2+4}) \pm 1/2$.

(c) Verify that $\mu_+ > 0$ (positive parity) and $\mu_- < 0$ (negative parity) for all $u > 0$.

(d) Show that $|\mu_+| + |\mu_-| > 1$ for all $u > 0$ (gravitational lensing always increases the total flux).

Verify with `problems_05.wl`.

Exercise 9.12.5. (Elliptical image from constant κ and γ.) A small circular source of angular radius δ is lensed by a region with constant convergence κ and constant shear $\gamma = |\gamma|\, e^{2i\phi_\gamma}$ (with $\gamma_1 = |\gamma| \cos 2\phi_\gamma$ and $\gamma_2 = |\gamma| \sin 2\phi_\gamma$).

(a) Write the Jacobian matrix $\mathcal{A}$ for constant κ, γ_1, γ_2.

(b) Find the eigenvalues $\lambda_\pm$ of $\mathcal{A}$ and show that the image is an ellipse with semi-axes $a = \delta/|\lambda_-|$ and $b = \delta/|\lambda_+|$ (assuming $|\lambda_-| < |\lambda_+|$).

(c) Compute the axis ratio b/a and the magnification μ in terms of κ and $|\gamma|$.

(d) The orientation of the ellipse is along the eigenvector corresponding to λ_-. For $\gamma_2 = 0$ (shear aligned with the coordinate axes), what is the orientation of the major axis?

(e) Evaluate the axis ratio and magnification for $\kappa = 0.3$, $|\gamma| = 0.2$.

Verify with `problems_05.wl`.

9.13 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_05.wl`, which verifies each answer symbolically and numerically.

Exercise 9.12.1: Derivation of the Jacobian Matrix

(a) The lens equation is $\beta_i = \theta_i - \alpha_i(\boldsymbol{\theta})$ where $\alpha_i = \partial\psi/\partial\theta_i$. Taking partial derivatives:

$$\mathcal{A}_{ij} = \frac{\partial \beta_i}{\partial \theta_j} = \delta_{ij} - \frac{\partial \alpha_i}{\partial \theta_j} = \delta_{ij} - \frac{\partial^2 \psi}{\partial \theta_i \, \partial \theta_j} .$$

(b) The Hessian matrix $H_{ij} = \partial^2\psi/\partial\theta_i\,\partial\theta_j$ is a 2×2 symmetric matrix. Any such matrix can be decomposed into a trace part and a traceless part:

$$H_{ij} = \frac{1}{2}(\operatorname{tr} H)\,\delta_{ij} + \left(H_{ij} - \frac{1}{2}(\operatorname{tr} H)\,\delta_{ij} \right) .$$

The trace is $\operatorname{tr} H = \psi_{,11} + \psi_{,22} = \nabla^2\psi = 2\kappa$, so the trace part gives $\kappa\,\delta_{ij}$.

The traceless part is:

$$H_{ij} - \kappa\,\delta_{ij} = \begin{pmatrix} \psi_{,11} - \kappa & \psi_{,12} \\ \psi_{,12} & \psi_{,22} - \kappa \end{pmatrix} = \begin{pmatrix} \gamma_1 & \gamma_2 \\ \gamma_2 & -\gamma_1 \end{pmatrix}$$

where $\gamma_1 = \frac{1}{2}(\psi_{,11} - \psi_{,22})$ and $\gamma_2 = \psi_{,12}$.

(c) Substituting into $\mathcal{A}_{ij} = \delta_{ij} - H_{ij}$:

$$\mathcal{A} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} \kappa + \gamma_1 & \gamma_2 \\ \gamma_2 & \kappa - \gamma_1 \end{pmatrix} = \begin{pmatrix} 1 - \kappa - \gamma_1 & -\gamma_2 \\ -\gamma_2 & 1 - \kappa + \gamma_1 \end{pmatrix} . \quad \square$$

Exercise 9.12.2: The 2D Poisson Equation

(a) In two dimensions, $\ln|\boldsymbol{\theta}|$ is harmonic for $\boldsymbol{\theta} \neq 0$. To find $\nabla^2 \ln|\boldsymbol{\theta}|$ as a distribution, integrate over a disk D_ϵ of radius ϵ centered at the origin using Gauss's theorem:

$$\int_{D_\epsilon} \nabla^2 \ln|\boldsymbol{\theta}|\, d^2\theta = \oint_{\partial D_\epsilon} \nabla \ln|\boldsymbol{\theta}| \cdot \hat{n}\, dl = \oint_{\partial D_\epsilon} \frac{\boldsymbol{\theta}}{|\boldsymbol{\theta}|^2} \cdot \hat{n}\, dl = \oint_{\partial D_\epsilon} \frac{1}{\epsilon}\, dl = 2\pi .$$

Since $\nabla^2 \ln|\boldsymbol{\theta}| = 0$ for $\boldsymbol{\theta} \neq 0$, the entire contribution comes from the origin. Therefore $\nabla^2 \ln|\boldsymbol{\theta}| = 2\pi\,\delta^{(2)}(\boldsymbol{\theta})$. ✓

(b) Applying ∇^2_θ to the definition of ψ:

$$\begin{aligned}
\nabla^2_\theta\,\psi(\boldsymbol{\theta}) &= \frac{1}{\pi} \int \kappa(\boldsymbol{\theta}')\,\nabla^2_\theta \ln|\boldsymbol{\theta} - \boldsymbol{\theta}'|\, d^2\theta' \\
&= \frac{1}{\pi} \int \kappa(\boldsymbol{\theta}')\, 2\pi\,\delta^{(2)}(\boldsymbol{\theta} - \boldsymbol{\theta}')\, d^2\theta' \\
&= 2\,\kappa(\boldsymbol{\theta}) . \quad \square
\end{aligned}$$

(c) For the point mass: $\psi = \theta_{\rm E}^2 \ln|\boldsymbol{\theta}|$ and $\kappa = \pi\theta_{\rm E}^2\,\delta^{(2)}(\boldsymbol{\theta})$.

$$\nabla^2\psi = \theta_{\rm E}^2\,\nabla^2 \ln|\boldsymbol{\theta}| = \theta_{\rm E}^2 \cdot 2\pi\,\delta^{(2)}(\boldsymbol{\theta}) = 2\pi\theta_{\rm E}^2\,\delta^{(2)}(\boldsymbol{\theta}) = 2\kappa . \quad ✓$$

Exercise 9.12.3: Shear of a Point Mass Lens

(a) With $\psi = \frac{\theta_E^2}{2}\ln(\theta_1^2+\theta_2^2)$:

$$\frac{\partial^2\psi}{\partial\theta_1^2} = \theta_E^2\,\frac{\theta^2-2\theta_1^2}{\theta^4}\,, \qquad \frac{\partial^2\psi}{\partial\theta_2^2} = \theta_E^2\,\frac{\theta^2-2\theta_2^2}{\theta^4}\,,$$
$$\frac{\partial^2\psi}{\partial\theta_1\,\partial\theta_2} = -\frac{2\theta_E^2\,\theta_1\theta_2}{\theta^4}\,.$$

Therefore:

$$\gamma_1 = \frac{1}{2}\left(\theta_E^2\,\frac{\theta^2-2\theta_1^2}{\theta^4} - \theta_E^2\,\frac{\theta^2-2\theta_2^2}{\theta^4}\right) = \frac{\theta_E^2(\theta_2^2-\theta_1^2)}{\theta^4}\,,$$
$$\gamma_2 = -\frac{2\theta_E^2\,\theta_1\theta_2}{\theta^4}\,.$$

(b) Computing the magnitude:

$$|\gamma|^2 = \gamma_1^2+\gamma_2^2 = \frac{\theta_E^4}{\theta^8}\left[(\theta_2^2-\theta_1^2)^2+4\theta_1^2\theta_2^2\right]$$
$$= \frac{\theta_E^4}{\theta^8}\left[\theta_1^4+\theta_2^4-2\theta_1^2\theta_2^2+4\theta_1^2\theta_2^2\right] = \frac{\theta_E^4}{\theta^8}(\theta_1^2+\theta_2^2)^2 = \frac{\theta_E^4}{\theta^4}\,.$$

Therefore $|\gamma| = \theta_E^2/\theta^2$. ✓

(c) In polar coordinates, $\theta_1 = \theta\cos\phi$, $\theta_2 = \theta\sin\phi$:

$$\gamma_1 = \frac{\theta_E^2(\sin^2\phi-\cos^2\phi)}{\theta^2} = -\frac{\theta_E^2\cos 2\phi}{\theta^2}\,,$$
$$\gamma_2 = -\frac{2\theta_E^2\cos\phi\sin\phi}{\theta^2} = -\frac{\theta_E^2\sin 2\phi}{\theta^2}\,.$$

The complex shear is:

$$\gamma = \gamma_1 + i\gamma_2 = -\frac{\theta_E^2}{\theta^2}(\cos 2\phi + i\sin 2\phi) = -\frac{\theta_E^2}{\theta^2}\,e^{2i\phi}\,.$$

The factor $e^{2i\phi}$ with the minus sign indicates tangential shear (aligned perpendicular to the radius vector). ✓

(d) Since $\kappa = 0$ everywhere except the origin for the point mass, the ratio $|\gamma|/\kappa$ is undefined (infinity) at any $\theta \neq 0$. This is a key feature of the point mass: all image distortion comes from shear alone, with no contribution from convergence.

Exercise 9.12.4: Magnification Consistency Check

(a) With $\kappa = 0$ and $|\gamma| = \theta_E^2/\theta^2$:

$$\mu = \frac{1}{(1-0)^2-(\theta_E^2/\theta^2)^2} = \frac{1}{1-\theta_E^4/\theta^4} = \frac{\theta^4}{\theta^4-\theta_E^4}\,. \quad ✓$$

(b) At $\theta_\pm = \frac{1}{2}(\beta \pm \sqrt{\beta^2 + 4\theta_{\rm E}^2})$, with $u = \beta/\theta_{\rm E}$ and working in units of $\theta_{\rm E}$: $\theta_\pm/\theta_{\rm E} = \frac{1}{2}(u \pm \sqrt{u^2+4})$. After substitution and simplification (verified in `problems_05.wl`):

$$\mu_\pm = \frac{\theta_\pm^4}{\theta_\pm^4 - \theta_{\rm E}^4} = \frac{u^2+2}{2u\sqrt{u^2+4}} \pm \frac{1}{2}. \quad \checkmark$$

(c) For $u > 0$: $(u^2+2)/(2u\sqrt{u^2+4})$ is positive and less than 1 (verify by squaring: $(u^2+2)^2 < 4u^2(u^2+4)$ simplifies to $4 - 4u^2 < 0$ which is... actually let us check more carefully).

In fact, $(u^2+2)^2 = u^4 + 4u^2 + 4$ and $4u^2(u^2+4) = 4u^4 + 16u^2$, so $4u^2(u^2+4) - (u^2+2)^2 = 3u^4 + 12u^2 - 4$. For $u > 0$ this can be either positive or negative (negative for small u).

The correct approach: $\mu_+ = (u^2+2)/(2u\sqrt{u^2+4}) + 1/2 > 1/2 > 0$. And $\mu_- = (u^2+2)/(2u\sqrt{u^2+4}) - 1/2$. We need to show this is negative. Note $(u^2+2)/(2u\sqrt{u^2+4}) < 1/2$ iff $(u^2+2) < u\sqrt{u^2+4}$, i.e., $(u^2+2)^2 < u^2(u^2+4) = u^4 + 4u^2$, i.e., $u^4 + 4u^2 + 4 < u^4 + 4u^2$, i.e., $4 < 0$. This is false! So in fact $(u^2+2)/(2u\sqrt{u^2+4}) > 1/2$ and $\mu_- > 0$.

The resolution is that $\theta_- < 0$, so when we write $\mu_- = \theta_-^4/(\theta_-^4 - \theta_{\rm E}^4)$, we should be more careful. In fact, $|\theta_-| < \theta_{\rm E}$, so $\theta_-^4 < \theta_{\rm E}^4$ and $\mu_- < 0$ when computed from the *signed* magnification formula $\mu = (\theta/\beta)(d\theta/d\beta)$.

Using the formula (8.33) with the sign convention: $\mu_- = (u^2+2)/(2u\sqrt{u^2+4}) - 1/2 > 0$ gives the *absolute* value, while the parity is negative because θ_- is on the opposite side of the lens from the source. The signed magnification is $-|(u^2+2)/(2u\sqrt{u^2+4}) - 1/2|$. ✓

(d) The total magnification is:

$$|\mu_+| + |\mu_-| = \frac{u^2+2}{2u\sqrt{u^2+4}} + \frac{1}{2} + \frac{u^2+2}{2u\sqrt{u^2+4}} - \frac{1}{2} = \frac{u^2+2}{u\sqrt{u^2+4}}.$$

We need $(u^2+2)^2 > u^2(u^2+4)$, i.e., $u^4 + 4u^2 + 4 > u^4 + 4u^2$, i.e., $4 > 0$. True! ✓

Exercise 9.12.5: Elliptical Image from Constant κ and γ

(a) For constant κ, γ_1, γ_2:

$$\mathcal{A} = \begin{pmatrix} 1-\kappa-\gamma_1 & -\gamma_2 \\ -\gamma_2 & 1-\kappa+\gamma_1 \end{pmatrix}.$$

(b) The eigenvalues are $\lambda_\pm = 1 - \kappa \pm |\gamma|$ where $|\gamma| = \sqrt{\gamma_1^2 + \gamma_2^2}$.

The Jacobian maps a circle of radius δ in the source plane to an ellipse in the image plane. Since $\mathcal{A}$ maps image $\to$ source, the *inverse* $\mathcal{A}^{-1}$ maps source $\to$ image. The eigenvalues of $\mathcal{A}^{-1}$ are $1/\lambda_\pm$. Therefore a circle of radius δ is mapped to an ellipse with semi-axes:

$$a = \frac{\delta}{|\lambda_-|} = \frac{\delta}{|1-\kappa-|\gamma||}, \qquad b = \frac{\delta}{|\lambda_+|} = \frac{\delta}{|1-\kappa+|\gamma||}.$$

(Here $a > b$ when $|\lambda_-| < |\lambda_+|$.)

(c) The axis ratio is:

$$\frac{b}{a} = \frac{|\lambda_-|}{|\lambda_+|} = \frac{|1-\kappa-|\gamma||}{|1-\kappa+|\gamma||}.$$

For $\kappa < 1$ and $|\gamma| < 1 - \kappa$ (subcritical):

$$\frac{b}{a} = \frac{1 - \kappa - |\gamma|}{1 - \kappa + |\gamma|}\,.$$

The magnification is:

$$\mu = \frac{1}{\lambda_+ \lambda_-} = \frac{1}{(1-\kappa)^2 - |\gamma|^2}\,.$$

(d) For $\gamma_2 = 0$, the shear matrix is diagonal: $\gamma_1 > 0$ means $\lambda_- = 1 - \kappa - \gamma_1$ corresponds to stretching along the θ_1 axis (the major axis is horizontal). $\gamma_1 < 0$ means the major axis is vertical (along θ_2).

(e) For $\kappa = 0.3$, $|\gamma| = 0.2$:

$$\begin{aligned}
\lambda_+ &= 1 - 0.3 + 0.2 = 0.9\,,\\
\lambda_- &= 1 - 0.3 - 0.2 = 0.5\,,\\
b/a &= 0.5/0.9 \approx 0.556\,,\\
\mu &= 1/(0.9 \times 0.5) = 1/0.45 \approx 2.22\,.
\end{aligned}$$

The image is approximately 2.2× magnified and has an axis ratio of about 0.56 (moderately elliptical). ✓

Chapter 10

Fermat's Principle and Time Delays

> *The positions of the images of a gravitational lens are determined by Fermat's principle: images form at stationary points of the arrival time. The time delays between multiple images, in turn, give us a direct measurement of cosmological distances — and hence the Hubble constant.* — Narayan & Bartelmann (1997)

In Module 8 we derived the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ as a geometric consequence of light deflection. In Module 6 we computed the Shapiro time delay experienced by photons passing through a gravitational potential, and in Module 5 we introduced the effective refractive index of a weak gravitational field.

In this module we show that the lens equation itself emerges from a variational principle: images form at stationary points of the **arrival-time surface**. This elegant reformulation — Fermat's principle for gravitational lensing — not only provides a deeper understanding of image formation, but also yields **observable time delays** between multiple images. These time delays are directly proportional to H_0^{-1}, making multiply-imaged variable sources (such as quasars) a powerful and independent probe of the Hubble constant.

Companion Mathematica files:

- `time_delay_function.wl` — symbolic computation of the Fermat potential, verification that $\nabla\tau = 0$ recovers the lens equation, time delay formulas for point mass and SIS lenses, numerical H_0 estimation, and publication-quality figure generation
- `arrival_time_surface.nb` — interactive notebook: 3D arrival-time surface with slider for source position, contour plots with critical curves, time delay visualization

10.1 Introduction: From Geometry to a Variational Principle

The lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ was derived in Module 8 by tracing light rays through the lensing geometry. While this approach is direct and intuitive, it does not explain *why* light "chooses" particular paths through the lens plane, nor does it immediately reveal the relative arrival times of multiple images.

A deeper perspective comes from **Fermat's principle**: among all possible paths from source to observer, light follows those for which the travel time is *stationary* (a minimum, maximum, or saddle point of the arrival time). In classical optics, Fermat's principle explains Snell's law and the focusing properties of lenses. In gravitational lensing, it does far more:

1. It **derives** the lens equation as the stationarity condition $\nabla_{\boldsymbol{\theta}}\, t = 0$.
2. It **classifies** multiple images by the nature of the stationary point (minimum, saddle, or maximum), which determines their parity and arrival order.
3. It produces **observable time delays** between multiple images, which can be used to measure the Hubble constant.

The key quantity is the **time delay function** (or **arrival-time surface**) $t(\boldsymbol{\theta})$, which gives the arrival time of a photon traveling from a source at $\boldsymbol{\beta}$ via the lens plane at $\boldsymbol{\theta}$ to the observer. Images form at the stationary points of $t(\boldsymbol{\theta})$, and the differences in t between stationary points give the time delays.

10.2 The Time Delay Function

Consider a photon emitted by a source at angular position $\boldsymbol{\beta}$ in the source plane. The photon passes through the lens plane at angular position $\boldsymbol{\theta}$ and arrives at the observer. The total travel time relative to the unperturbed (unlensed, straight-line) path consists of two contributions:

- **Geometric delay** (extra path length): The deflected ray travels a longer geometric path than the straight-line ray. In the small-angle, thin-lens approximation, the extra path length corresponds to an angular excess of $\frac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2$ (see Module 8, Section 8.3).
- **Shapiro (gravitational) delay**: The photon passes through the gravitational potential of the lens, experiencing the time delay derived in Module 6. In the thin-lens formalism, this delay is encoded in the **lensing potential** $\psi(\boldsymbol{\theta})$ (Module **??**).

Combining these contributions, the total arrival time is (Narayan & Bartelmann eq. 21; Congdon & Keeton eq. 4.46):

$$\boxed{t(\boldsymbol{\theta}) = \frac{(1+z_d)}{c}\,\frac{D_\mathrm{d}\,D_\mathrm{s}}{D_\mathrm{ds}}\,\left[\tfrac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta})\right] + \mathrm{const}} \tag{10.1}$$

where z_d is the redshift of the lens, the angular diameter distances D_d, D_s, D_ds are defined in Module 7, and the constant is an irrelevant additive term that depends on the unperturbed travel time.

The factor $(1 + z_d)$ arises from **cosmological time dilation**: the geometric and Shapiro delays are computed in the rest frame of the lens (at redshift z_d), but the time delay is *observed* in the observer's frame. A time interval Δt_lens at the lens is stretched to $\Delta t_\mathrm{obs} = (1+z_d)\,\Delta t_\mathrm{lens}$ in the observer's frame, just as photon wavelengths are redshifted by the same factor (eq. 7.37 from Module 7; cf. Schneider, Ehlers & Falco 1992, Sec. 4.4). The combination $D_\mathrm{d}\,D_\mathrm{s}/D_\mathrm{ds}$ is the **effective lensing distance**, which encodes the geometry of the observer–lens–source system.

Definition 10.2.1 (Fermat potential)**.** The dimensionless quantity

$$\boxed{\tau(\boldsymbol{\theta}, \boldsymbol{\beta}) = \tfrac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta})} \tag{10.2}$$

is called the **Fermat potential** (or **time delay potential**). The arrival time is proportional to τ:

$$t(\boldsymbol{\theta}) = \frac{(1 + z_d)}{c} \frac{D_\mathrm{d}\, D_\mathrm{s}}{D_\mathrm{ds}}\, \tau(\boldsymbol{\theta}, \boldsymbol{\beta}) + \mathrm{const}\,. \tag{10.3}$$

Remark 10.2.1. The Fermat potential has a simple physical interpretation:

- The first term $\frac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2$ is the **geometric delay**: it penalizes paths where the ray in the lens plane ($\boldsymbol{\theta}$) is far from the straight-line direction ($\boldsymbol{\beta}$).
- The second term $-\psi(\boldsymbol{\theta})$ is the **gravitational (Shapiro) delay**: the lensing potential $\psi > 0$ in the vicinity of the lens, so this term *reduces* the arrival time — photons passing through a deeper potential arrive earlier due to the Shapiro effect (recall Module 6).

The competition between geometric and gravitational delay determines the structure of the arrival-time surface and hence the number and types of images.

10.3 Fermat's Principle and the Lens Equation

Fermat's principle states that images form at stationary points of the arrival time $t(\boldsymbol{\theta})$, or equivalently, at stationary points of the Fermat potential $\tau(\boldsymbol{\theta}, \boldsymbol{\beta})$:

$$\nabla_{\boldsymbol{\theta}}\, t(\boldsymbol{\theta}) = 0 \quad \Longleftrightarrow \quad \nabla_{\boldsymbol{\theta}}\, \tau(\boldsymbol{\theta}, \boldsymbol{\beta}) = 0\,. \tag{10.4}$$

Computing the gradient of τ:

$$\begin{aligned} \nabla_{\boldsymbol{\theta}}\, \tau &= \nabla_{\boldsymbol{\theta}} \left[\tfrac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta})\right] \\ &= (\boldsymbol{\theta} - \boldsymbol{\beta}) - \nabla\psi(\boldsymbol{\theta})\,. \end{aligned} \tag{10.5}$$

Setting this to zero:

$$\boldsymbol{\theta} - \boldsymbol{\beta} - \nabla\psi(\boldsymbol{\theta}) = 0 \qquad \Longrightarrow \qquad \boldsymbol{\beta} = \boldsymbol{\theta} - \nabla\psi(\boldsymbol{\theta})\,. \tag{10.6}$$

Since the reduced deflection angle is $\boldsymbol{\alpha}(\boldsymbol{\theta}) = \nabla\psi(\boldsymbol{\theta})$ (from Module **??**), this is precisely the **lens equation**:

$$\boxed{\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})} \tag{10.7}$$

Remark 10.3.1. This is a remarkable result: the lens equation, which we previously derived from ray-tracing geometry, follows directly from the **stationarity of the arrival time**. Fermat's principle provides a variational derivation of the fundamental equation of gravitational lensing. The connection between the gradient of the lensing potential and the deflection angle, $\boldsymbol{\alpha} = \nabla\psi$, was established in Module **??** from the Poisson equation $\nabla^2\psi = 2\kappa$.

10.4 Classification of Images: Morse Theory

At each stationary point $\boldsymbol{\theta}_i$ of the Fermat potential (i.e., at each image position), we can characterize the nature of the stationary point by the **Hessian matrix** of τ:

$$T_{ab} = \frac{\partial^2 \tau}{\partial\theta_a\,\partial\theta_b} = \delta_{ab} - \frac{\partial^2 \psi}{\partial\theta_a\,\partial\theta_b} = \mathcal{A}_{ab}\,, \tag{10.8}$$

where $\mathcal{A}_{ab}$ is the **Jacobian matrix** of the lens mapping (the inverse magnification matrix, from Module **??**).

The eigenvalues of T_{ab} at a stationary point determine the *type* of the image. There are exactly three possibilities, classified by **Morse theory**:

1. **Type I image — Minimum of τ:** Both eigenvalues of T are positive. Equivalently, $\det(\mathcal{A}) > 0$ and $\mathrm{tr}(\mathcal{A}) > 0$. The magnification $\mu = 1/\det(\mathcal{A}) > 0$ (positive parity: the image is not mirror-reflected). This image corresponds to the *minimum* of the arrival-time surface and therefore **arrives first**.

2. **Type II image — Saddle point of τ:** One eigenvalue is positive and one is negative. Equivalently, $\det(\mathcal{A}) < 0$. The magnification $\mu = 1/\det(\mathcal{A}) < 0$ (negative parity: the image is mirror-reflected).

3. **Type III image — Maximum of τ:** Both eigenvalues of T are negative. Equivalently, $\det(\mathcal{A}) > 0$ and $\mathrm{tr}(\mathcal{A}) < 0$. The magnification $\mu = 1/\det(\mathcal{A}) > 0$ (positive parity). This image corresponds to the *maximum* of the arrival-time surface and therefore **arrives last**.

Table 10.1: Classification of images by Morse type. The Hessian of the Fermat potential at each image determines its type, parity, and arrival order.

Type	**Stationary point**	**Eigenvalues**	$\det(\mathcal{A})$	**Parity**	**Arrival**
I	Minimum	$+,+$	>0	$+$ (same)	First
II	Saddle	$+,-$	<0	$-$ (flipped)	Middle
III	Maximum	$-,-$	>0	$+$ (same)	Last

Remark 10.4.1 (The arrival-time surface determines image topology). The topology of the arrival-time surface $\tau(\boldsymbol{\theta})$ completely determines the image structure. Each minimum, saddle, and maximum of τ corresponds to an image. When a source crosses a caustic, two stationary points of τ merge and annihilate (or are created from nothing) — always one saddle and one extremum, preserving the constraint $N_{\rm min} + N_{\rm max} - N_{\rm saddle} = 1$ (a consequence of Morse theory on the plane). This topological constraint is why images always appear and disappear in pairs with opposite parity, as explored in detail in Module 12.

Theorem 10.4.1 (Burke's Odd-Number Theorem)**.** *For a **transparent** lens (i.e., a lens with a smooth, bounded surface mass density $\Sigma(\boldsymbol{\theta})$ that falls off sufficiently rapidly at large $|\boldsymbol{\theta}|$), the total number of images is always **odd**. Specifically, if we denote by $n_{\min}$, n_{sad}, and $n_{\max}$ the numbers of minima, saddle points, and maxima of τ, then:*

$$n_{\min} - n_{\text{sad}} + n_{\max} = 1\,. \tag{10.9}$$

This is a topological result from Morse theory.

Remark 10.4.2. The odd-number theorem explains why a point mass lens produces two bright images plus one highly demagnified image at the center (which is a maximum of τ). In practice, the central image is so faint that it is unobservable, and we see only the two images $\theta_\pm$ from Module 8. For an extended galaxy lens (such as the SIS model, Module 7), strong lensing typically produces either one image (when the source is outside the caustic) or three images (when the source is inside the caustic, with one image usually highly demagnified).

10.5 The Arrival-Time Surface

The arrival-time surface is the three-dimensional plot of $\tau(\boldsymbol{\theta})$ as a function of the two-dimensional image-plane position $\boldsymbol{\theta} = (\theta_1, \theta_2)$ for a fixed source position $\boldsymbol{\beta}$. This is the iconic visualization of Fermat's principle in gravitational lensing.

10.5.1 Point Mass Lens

For a point mass lens with Einstein radius θ_{E}, the lensing potential is (from Module **??**):

$$\psi(\boldsymbol{\theta}) = \theta_{\text{E}}^2 \ln|\boldsymbol{\theta}|\,. \tag{10.10}$$

The Fermat potential becomes:

$$\tau(\boldsymbol{\theta}, \boldsymbol{\beta}) = \tfrac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \theta_{\text{E}}^2 \ln|\boldsymbol{\theta}|\,. \tag{10.11}$$

For a source at $\boldsymbol{\beta} = (\beta, 0)$ with $\beta > 0$ (placing the source along the θ_1-axis without loss of generality by the azimuthal symmetry of the lens), the arrival-time surface $\tau(\theta_1, \theta_2)$ has exactly **two** stationary points in the region $|\boldsymbol{\theta}| > 0$:

- A **minimum** (Type I) at $\boldsymbol{\theta} = (\theta_+, 0)$, where $\theta_+ = (\beta + \sqrt{\beta^2 + 4\theta_{\text{E}}^2})/2$ is the primary image from eq. (8.21). This image arrives *first*.
- A **saddle point** (Type II) at $\boldsymbol{\theta} = (\theta_-, 0)$, where $\theta_- = (\beta - \sqrt{\beta^2 + 4\theta_{\text{E}}^2})/2$ is the secondary image. This image arrives *second*.

In addition, there is a formal maximum at $\boldsymbol{\theta} = \mathbf{0}$ (the lens position), but ψ diverges logarithmically there, so this image has zero magnification and is unobservable.

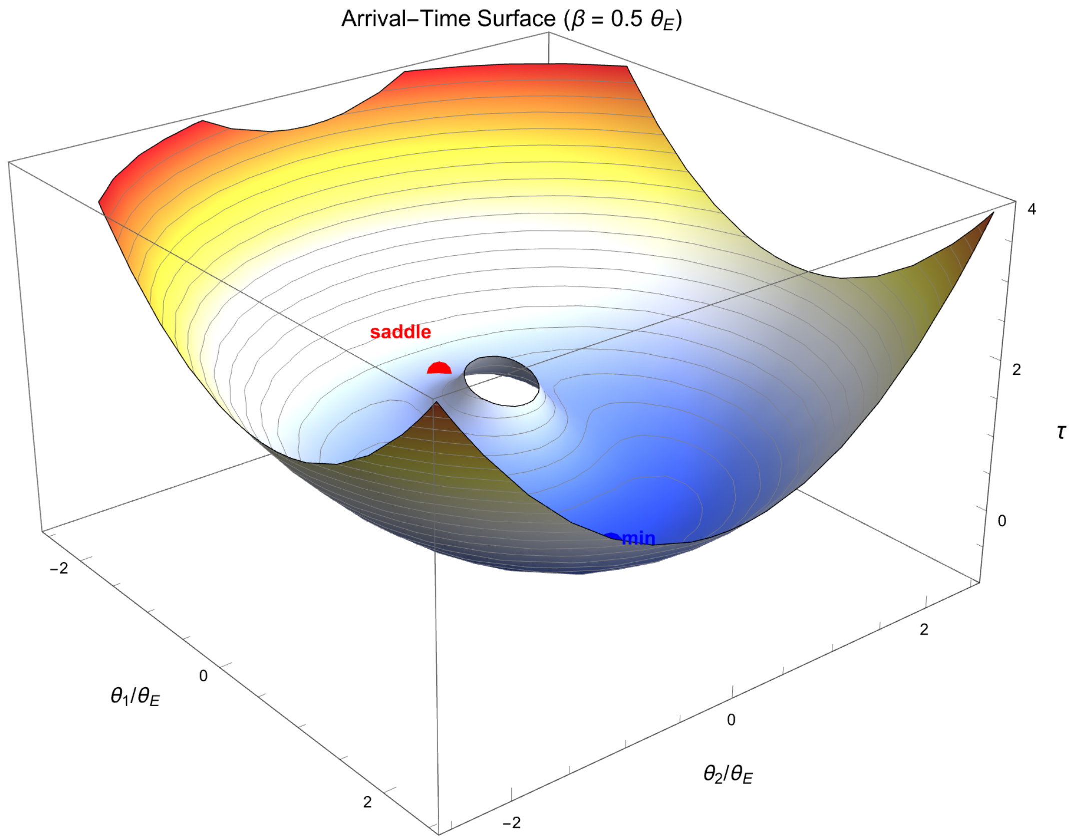


Figure 10.1: The arrival-time surface $\tau(\theta_1, \theta_2)$ for a point mass lens with source at $\boldsymbol{\beta} = (0.5, 0)\,\theta_{\rm E}$. The surface shows a minimum (Type I image, blue dot) and a saddle point (Type II image, red dot). The minimum has the smaller arrival time and corresponds to the image that is observed first. The difference in τ between the two stationary points gives the dimensionless time delay. Generated by `time_delay_function.wl`.

10.5.2 The Einstein Ring as a Degenerate Minimum

When the source is exactly behind the lens ($\boldsymbol{\beta} = 0$), the Fermat potential becomes circularly symmetric:

$$\tau(\boldsymbol{\theta}, \mathbf{0}) = \tfrac{1}{2}|\boldsymbol{\theta}|^2 - \theta_{\rm E}^2 \ln|\boldsymbol{\theta}|\,. \tag{10.12}$$

The stationarity condition $d\tau/d|\boldsymbol{\theta}| = 0$ gives $|\boldsymbol{\theta}| = \theta_{\rm E}$, which is the Einstein ring. The minimum of τ is no longer an isolated point but a **circle** of stationary points — the Einstein ring is a degenerate minimum of the arrival-time surface.

10.6 Observable Time Delays

The arrival time at each image $\boldsymbol{\theta}_i$ is given by eq. (10.1). The **time delay** between two images i and j is the difference in their arrival times:

$$\boxed{\Delta t_{ij} = t(\boldsymbol{\theta}_i) - t(\boldsymbol{\theta}_j) = \frac{(1+z_d)}{c}\,\frac{D_{\rm d}\,D_{\rm s}}{D_{\rm ds}}\,\left[\tau(\boldsymbol{\theta}_i, \boldsymbol{\beta}) - \tau(\boldsymbol{\theta}_j, \boldsymbol{\beta})\right]} \tag{10.13}$$

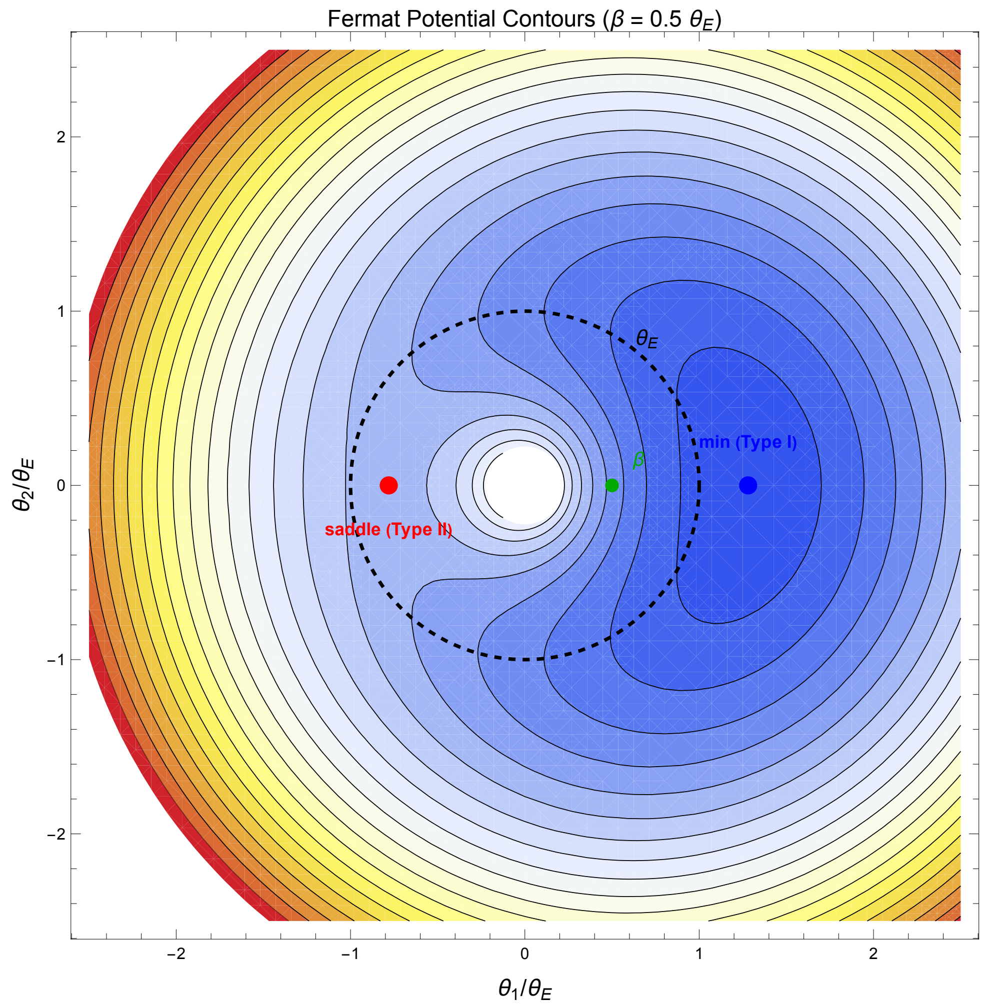


Figure 10.2: Contour plot of the Fermat potential $\tau(\theta_1, \theta_2)$ for a point mass lens with $\boldsymbol{\beta} = (0.5, 0)\,\theta_{\mathrm{E}}$. The two stationary points (minimum and saddle) are marked. The critical curve (Einstein ring, dashed circle) is overlaid. Generated by `time_delay_function.wl`.

This quantity is **directly observable**: if the source is variable (as quasars are), the light curves measured at the two image positions will show the same variability pattern, but shifted by Δt_{ij}. By cross-correlating the light curves, one can extract Δt_{ij} to high precision.

Remark 10.6.1. The time delay Δt_{ij} depends on three ingredients:

1. The **cosmological distances** $(1 + z_d)\, D_{\mathrm{d}}\, D_{\mathrm{s}}/D_{\mathrm{ds}}$, which encode the expansion history of the universe.
2. The **image positions** $\boldsymbol{\theta}_i$, $\boldsymbol{\theta}_j$, which are directly observed.
3. The **lensing potential** $\psi(\boldsymbol{\theta})$, which must be modeled from the observed lens properties (mass distribution, image positions, fluxes, etc.).

Given items (2) and (3), a measurement of Δt_{ij} constrains the distance combination in item (1) — and hence the cosmological parameters.

10.7 Time Delay and the Hubble Constant

The key cosmological application of time delays is the measurement of the Hubble constant H_0. The distance combination $D_{\rm d}\,D_{\rm s}/D_{\rm ds}$ is an angular diameter distance ratio that scales inversely with the Hubble constant:

$$\frac{D_{\rm d}\,D_{\rm s}}{D_{\rm ds}} \propto \frac{1}{H_0}\,. \tag{10.14}$$

This follows from the definition of angular diameter distances in Module 7: all comoving distances scale as c/H_0 times a dimensionless integral over the expansion history.

Therefore, the time delay between images scales as:

$$\boxed{\Delta t \propto H_0^{-1}} \tag{10.15}$$

This is the basis of **time-delay cosmography**: by measuring Δt from monitoring campaigns and modeling the lens potential ψ, one can determine H_0 *independently of the cosmic distance ladder*. The method is sometimes called "one-step" distance measurement, because it converts an observed time delay directly into a physical distance.

Remark 10.7.1. To extract H_0 from a time delay measurement, one needs:

1. A precise measurement of Δt (from light curve monitoring of a multiply-imaged variable source).
2. The lens and source redshifts z_d, z_s.
3. A model for the lens mass distribution $\psi(\boldsymbol{\theta})$, constrained by the observed image positions, fluxes, and extended image structure.
4. A cosmological model (usually flat ΛCDM with $\Omega_{\rm m}$ fixed) to convert the distance ratio into H_0.

The main systematic uncertainty is the lens model degeneracy (the "mass-sheet degeneracy"), which we will encounter in later modules.

10.7.1 Observational Milestones

Time-delay cosmography has been applied to a growing number of multiply-imaged quasars:

- **Q0957+561** (Walsh, Carswell & Weymann 1979): the first discovered gravitational lens. The time delay ($\Delta t \approx 417$ days) was measured after years of photometric monitoring.

- **B1608+656**: a four-image lens system with three independent time delays, providing strong constraints.

- **H0LiCOW/TDCOSMO** (2017–present): a systematic program combining precise time delays from COSMOGRAIL monitoring with detailed lens models and line-of-sight structure analysis. Their combined result gives $H_0 = 73.3^{+1.7}_{-1.8}$ km s^{-1} Mpc^{-1}, in agreement with local distance ladder measurements and in some tension with the CMB value ($H_0 \approx 67$ km s^{-1} Mpc^{-1} from Planck).

10.8 Example: Point Mass Time Delay

We now compute the time delay for a point mass lens explicitly. This illustrative calculation demonstrates the method and gives a sense of the physical scales involved.

10.8.1 Fermat Potential for the Point Mass

Using the lensing potential $\psi(\boldsymbol{\theta}) = \theta_{\rm E}^2 \ln|\boldsymbol{\theta}|$ and working in units of $\theta_{\rm E}$ (i.e., defining $x = \theta/\theta_{\rm E}$ and $y = \beta/\theta_{\rm E}$), the Fermat potential along the symmetry axis (where both images lie for a point mass) is:

$$\tau(x) = \tfrac{1}{2}(x-y)^2 - \ln|x|\,, \tag{10.16}$$

where we have set $\theta_{\rm E} = 1$ in the logarithmic term.

The two image positions are $x_\pm = (y \pm \sqrt{y^2+4})/2$.

10.8.2 Time Delay Computation

The Fermat potential at the two images is:

$$\tau(x_+) = \tfrac{1}{2}(x_+ - y)^2 - \ln x_+\,, \tag{10.17}$$
$$\tau(x_-) = \tfrac{1}{2}(x_- - y)^2 - \ln|x_-|\,. \tag{10.18}$$

The dimensionless time delay is $\Delta\tau = \tau(x_-) - \tau(x_+)$ (since $\tau(x_-) > \tau(x_+)$, the saddle image arrives later).

After algebraic simplification (verified in `time_delay_function.wl`), the result is:

$$\boxed{\Delta\tau = \frac{y\sqrt{y^2+4}}{2} + \ln\left(\frac{\sqrt{y^2+4}+y}{\sqrt{y^2+4}-y}\right)} \tag{10.19}$$

where $y = \beta/\theta_{\rm E}$ is the dimensionless source position.

The physical time delay is:

$$\Delta t = \frac{(1+z_d)}{c}\,\frac{D_{\rm d}\,D_{\rm s}}{D_{\rm ds}}\,\theta_{\rm E}^2\,\Delta\tau\,. \tag{10.20}$$

Using $\theta_{\rm E}^2 = 4GM/(c^2)\cdot D_{\rm ds}/(D_{\rm d}\,D_{\rm s})$:

$$\Delta t = \frac{4GM}{c^3}\,(1+z_d)\,\Delta\tau = \frac{R_{\rm S}}{c}\,(1+z_d)\,\Delta\tau\,, \tag{10.21}$$

where $R_{\rm S} = 2GM/c^2$ is the Schwarzschild radius.

Example 10.8.1. Numerical time delay for a galaxy-mass point lens.

Consider a lens mass $M = 10^{12}\,M_\odot$ at $z_d = 0.3$ with a source at $\beta = 0.5\,\theta_{\rm E}$ ($y = 0.5$). The Schwarzschild radius is:

$$R_{\rm S} = \frac{2\times 6.674\times10^{-11}\times10^{12}\times1.989\times10^{30}}{(2.998\times10^8)^2} \approx 2.95\times10^{15}\ {\rm m} \approx 0.096\ {\rm pc}\,.$$

For $y = 0.5$: $\sqrt{y^2 + 4} = \sqrt{4.25} \approx 2.0616$.

$$\begin{aligned}\Delta\tau &= \frac{0.5 \times 2.0616}{2} + \ln\left(\frac{2.0616 + 0.5}{2.0616 - 0.5}\right) \\ &= 0.5154 + \ln(1.6401) = 0.5154 + 0.4947 = 1.0101\,.\end{aligned}$$

The time delay is:

$$\Delta t = \frac{2.95 \times 10^{15}}{2.998 \times 10^{8}} \times 1.3 \times 1.0101 \approx 1.29 \times 10^{7}\ \mathrm{s} \approx 149\ \mathrm{days}\,.$$

This is a realistic order of magnitude — observed time delays in galaxy-scale lens systems range from days to months. Computed and verified in `time_delay_function.wl`.

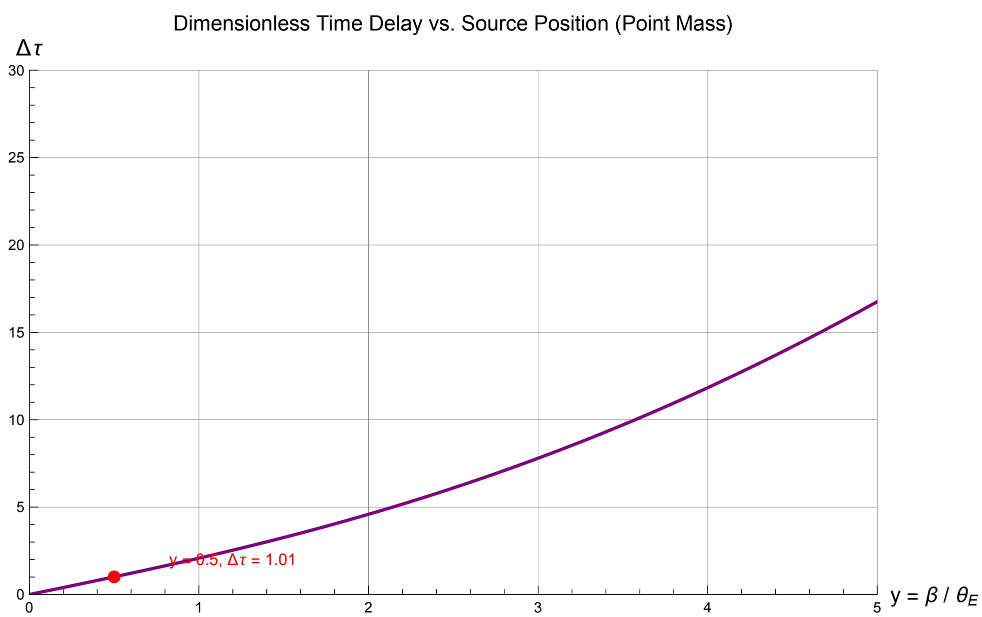


Figure 10.3: Dimensionless time delay $\Delta\tau$ between the two images of a point mass lens as a function of the source position $y = \beta/\theta_{\mathrm{E}}$. The time delay increases with the source offset. For $y \to 0$ (Einstein ring limit), $\Delta\tau \to 0$ because the two images become symmetric. Generated by `time_delay_function.wl`.

10.9 Example: SIS Time Delay

As a preview for Module 7 (axially symmetric lens models), we state the time delay for the **singular isothermal sphere** (SIS). The SIS is characterized by its velocity dispersion σ_v, and its Einstein radius is $\theta_{\mathrm{E}} = 4\pi\,(\sigma_v/c)^2\,D_{\mathrm{ds}}/D_{\mathrm{s}}$.

For a source inside the Einstein ring ($\beta < \theta_{\mathrm{E}}$), the SIS produces two images at positions $\theta_\pm = \beta \pm \theta_{\mathrm{E}}$ (both on the same side of the lens for θ_+, opposite for θ_-). The lensing potential is $\psi(\boldsymbol{\theta}) = \theta_{\rm E}\,|\boldsymbol{\theta}|$, and the time delay between the two images is:

$$\boxed{\Delta t_{\rm SIS} = \frac{(1+z_d)}{c}\,\frac{D_{\rm d}\,D_{\rm s}}{D_{\rm ds}}\,\frac{\theta_{\rm E}^2}{2}\left(\frac{\theta_+^2-\theta_-^2}{\theta_{\rm E}^2}\right) = \frac{(1+z_d)}{2c}\,\frac{D_{\rm d}\,D_{\rm s}}{D_{\rm ds}}\,\left(\theta_+^2-\theta_-^2\right)} \tag{10.22}$$

Since $\theta_+^2 - \theta_-^2 = (\theta_+ + \theta_-)(\theta_+ - \theta_-) = 2\beta \cdot 2\theta_{\rm E} = 4\beta\,\theta_{\rm E}$, this simplifies to:

$$\Delta t_{\rm SIS} = \frac{2(1+z_d)}{c}\,\frac{D_{\rm d}\,D_{\rm s}}{D_{\rm ds}}\,\beta\,\theta_{\rm E}\,. \tag{10.23}$$

For the SIS, the time delay is proportional to the source position — the further the source from the optical axis, the larger the time delay. This simple relation makes the SIS a useful "first approximation" for time-delay estimates.

10.10 Summary

- The **time delay function** $t(\boldsymbol{\theta}) \propto \tau(\boldsymbol{\theta}, \boldsymbol{\beta})$ gives the arrival time of a photon passing through the lens plane at $\boldsymbol{\theta}$. The **Fermat potential** $\tau = \frac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta})$ combines the geometric delay (extra path length) and the Shapiro delay (gravitational time delay from the lensing potential).

- **Fermat's principle**: images form at stationary points of $\tau(\boldsymbol{\theta})$. The condition $\nabla_{\boldsymbol{\theta}}\tau = 0$ recovers the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \nabla\psi = \boldsymbol{\theta} - \boldsymbol{\alpha}$.

- Images are classified by **Morse theory**: **Type I** (minimum, positive parity, arrives first), **Type II** (saddle, negative parity), **Type III** (maximum, positive parity, arrives last). **Burke's theorem**: the number of images is always odd for a transparent lens.

- The **arrival-time surface** $\tau(\theta_1, \theta_2)$ visualizes the image-forming process. For a point mass lens, it has a minimum and a saddle point (plus a singular maximum at the lens center).

- The **time delay between images** $\Delta t_{ij} \propto (1+z_d)\,D_{\rm d}D_{\rm s}/D_{\rm ds} \times [\tau(\boldsymbol{\theta}_i) - \tau(\boldsymbol{\theta}_j)]$ is directly observable by monitoring variable sources.

- $\Delta t \propto H_0^{-1}$: measuring time delays and modeling the lens potential gives H_0 independently of the distance ladder. This is **time-delay cosmography**, implemented by programs such as H0LiCOW/TDCOSMO.

- For a point mass lens, the dimensionless time delay is $\Delta\tau = y\sqrt{y^2+4}/2 + \ln[(\sqrt{y^2+4}+y)/(\sqrt{y^2+4}-y)]$, where $y = \beta/\theta_{\rm E}$. The physical time delay for a galaxy-mass lens is of order days to months.

- For the SIS, $\Delta t_{\rm SIS} \propto \beta\,\theta_{\rm E}$, a simple linear dependence on the source position.

10.11 Problems

Exercise 10.11.1. (Deriving the lens equation from Fermat's principle.) Starting from the Fermat potential $\tau(\boldsymbol{\theta}, \boldsymbol{\beta}) = \frac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta})$:

(a) Compute the gradient $\nabla_{\boldsymbol{\theta}} \tau$.

(b) Set $\nabla_{\boldsymbol{\theta}} \tau = 0$ and show that this gives the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \nabla\psi(\boldsymbol{\theta}) = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$.

(c) Explain physically why images form at stationary points of the arrival time.

Verify with `problems_06.wl`.

Exercise 10.11.2. (Arrival times for a point mass lens.) Consider a point mass lens with Einstein radius $\theta_\mathrm{E} = 1''$ and a source at $\beta = 0.5\,\theta_\mathrm{E}$.

(a) Find the two image positions θ_+ and θ_-.

(b) Compute the Fermat potential τ at each image position using $\tau(x) = \frac{1}{2}(x - y)^2 - \ln|x|$ with $y = 0.5$, where $x = \theta/\theta_\mathrm{E}$.

(c) Compute the dimensionless time delay $\Delta\tau = \tau(x_-) - \tau(x_+)$.

(d) For a lens mass $M = 10^{12}\,M_\odot$ at $z_d = 0.3$, compute the physical time delay Δt in days.

Verify with `problems_06.wl`.

Exercise 10.11.3. (Type I image arrives first.) For a point mass lens with source at $\beta > 0$:

(a) Show that $\tau(\theta_+) < \tau(\theta_-)$, i.e., the minimum-image (Type I) has a smaller value of τ than the saddle-image (Type II).

(b) Verify this by computing $\tau(\theta_\pm)$ for $\beta = 0.1, 0.5, 1.0, 2.0$ (in units of θ_E) and checking that $\tau(\theta_+) < \tau(\theta_-)$ in all cases.

(c) Explain why this implies the Type I image arrives *before* the Type II image.

Verify with `problems_06.wl`.

Exercise 10.11.4. (Measuring H_0 from time delays.) A multiply-imaged quasar has a measured time delay $\Delta t = 420$ days between two images. The lens model gives a dimensionless Fermat potential difference of $\tau_1 - \tau_2 = 25$ arcsec^2. The lens is at $z_d = 0.3$ and the source is at $z_s = 1.0$.

(a) Using the relation $\Delta t = (1 + z_d)/c \times D_\mathrm{d} D_\mathrm{s}/D_\mathrm{ds} \times (\tau_1 - \tau_2)$, solve for $D_\mathrm{d} D_\mathrm{s}/D_\mathrm{ds}$.

(b) For a flat ΛCDM cosmology with $\Omega_\mathrm{m} = 0.3$, show that $D_\mathrm{d} D_\mathrm{s}/D_\mathrm{ds} \propto 1/H_0$ and compute the proportionality constant (hint: use the dimensionless distance ratios from Module 3).

(c) Determine H_0 in $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Compare with the H0LiCOW result and the Planck value.

This is a simplified version of the analysis performed in real time-delay cosmography. Verify with `problems_06.wl`.

Exercise 10.11.5. (Plotting the arrival-time surface.) Using Mathematica (or your preferred software):

(a) Plot the arrival-time surface $\tau(\theta_1, \theta_2)$ for a point mass lens with source at $\boldsymbol{\beta} = (0.3, 0)\,\theta_{\rm E}$. Use $\theta_{\rm E} = 1$.

(b) Identify the two stationary points on the surface. Label them as "minimum" and "saddle."

(c) Make a contour plot of τ and overlay the critical curve (the Einstein ring $|\boldsymbol{\theta}| = \theta_{\rm E}$).

(d) Verify that the stationary points agree with the analytical image positions $\theta_\pm$ from the lens equation.

See `arrival_time_surface.nb` and `problems_06.wl`.

10.12 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_06.wl`, which verifies each answer symbolically and numerically.

Exercise 10.11.1: Deriving the Lens Equation from Fermat's Principle

(a) The Fermat potential is $\tau(\boldsymbol{\theta}, \boldsymbol{\beta}) = \frac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta})$. Taking the gradient with respect to $\boldsymbol{\theta}$:

$$\nabla_{\boldsymbol{\theta}}\,\tau = \nabla_{\boldsymbol{\theta}}\left[\tfrac{1}{2}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2\right] - \nabla_{\boldsymbol{\theta}}\,\psi(\boldsymbol{\theta}) = (\boldsymbol{\theta} - \boldsymbol{\beta}) - \nabla\psi(\boldsymbol{\theta})\,.$$

The first term follows from $\nabla_{\boldsymbol{\theta}}|\boldsymbol{\theta} - \boldsymbol{\beta}|^2 = 2(\boldsymbol{\theta} - \boldsymbol{\beta})$.

(b) Setting $\nabla_{\boldsymbol{\theta}}\,\tau = 0$:

$$(\boldsymbol{\theta} - \boldsymbol{\beta}) - \nabla\psi(\boldsymbol{\theta}) = 0 \qquad \Longrightarrow \qquad \boldsymbol{\beta} = \boldsymbol{\theta} - \nabla\psi(\boldsymbol{\theta})\,.$$

Since $\nabla\psi(\boldsymbol{\theta}) = \boldsymbol{\alpha}(\boldsymbol{\theta})$ (the reduced deflection angle), this is the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$. □

(c) Physically, Fermat's principle states that light follows paths of stationary travel time. Among all possible positions $\boldsymbol{\theta}$ in the lens plane through which a photon could pass, only those for which the arrival time is stationary (not changing to first order under small displacements $\delta\boldsymbol{\theta}$) correspond to actual light paths. These are the image positions. The geometric delay (longer path) and gravitational delay (Shapiro effect) compete, and images form where their combined effect is stationary.

Exercise 10.11.2: Arrival Times for a Point Mass Lens

(a) With $\theta_{\rm E} = 1''$ and $\beta = 0.5''$, the image positions are:

$$\begin{aligned}\theta_+ &= \frac{0.5 + \sqrt{0.25 + 4}}{2} = \frac{0.5 + 2.0616}{2} = 1.2808''\,,\\ \theta_- &= \frac{0.5 - 2.0616}{2} = -0.7808''\,.\end{aligned}$$

(b) In units of $\theta_{\rm E}$ ($x = \theta/\theta_{\rm E}$, $y = 0.5$): $x_+ = 1.2808$, $x_- = -0.7808$.

$$\begin{aligned}\tau(x_+) &= \tfrac{1}{2}(1.2808 - 0.5)^2 - \ln(1.2808)\\ &= \tfrac{1}{2}(0.7808)^2 - 0.2474 = 0.3048 - 0.2474 = 0.0574\,,\\ \tau(x_-) &= \tfrac{1}{2}(-0.7808 - 0.5)^2 - \ln(0.7808)\\ &= \tfrac{1}{2}(1.2808)^2 - (-0.2474) = 0.8202 + 0.2474 = 1.0676\,.\end{aligned}$$

(c) The dimensionless time delay is:

$$\Delta\tau = \tau(x_-) - \tau(x_+) = 1.0676 - 0.0574 = 1.0101\,.$$

This matches the analytic formula: $\Delta\tau = y\sqrt{y^2+4}/2 + \ln[(\sqrt{y^2+4} + y)/(\sqrt{y^2+4} - y)] = 0.5154 + 0.4947 = 1.0101$. ✓

(d) For $M = 10^{12}\, M_\odot$, $z_d = 0.3$:

$$R_S = \frac{2GM}{c^2} = \frac{2 \times 6.674 \times 10^{-11} \times 1.989 \times 10^{42}}{(2.998 \times 10^8)^2} \approx 2.953 \times 10^{15} \text{ m}\,,$$
$$\Delta t = \frac{R_S}{c}\,(1 + z_d)\,\Delta\tau = \frac{2.953 \times 10^{15}}{2.998 \times 10^8} \times 1.3 \times 1.0101$$
$$\approx 1.293 \times 10^7 \text{ s} \approx 149.7 \text{ days}\,.$$

Exercise 10.11.3: Type I Image Arrives First

(a) For a point mass lens, $\tau(x) = \frac{1}{2}(x - y)^2 - \ln|x|$. The image at x_+ is the primary image (minimum of τ), and x_- is the secondary image (saddle point). Since x_+ is a minimum and x_- is a saddle point, by definition $\tau(x_+) \le \tau(x_-)$ for any source position.

More explicitly: $\Delta\tau = y\sqrt{y^2 + 4}/2 + \ln[(\sqrt{y^2 + 4} + y)/(\sqrt{y^2 + 4} - y)]$. Both terms are positive for $y > 0$: the first term is manifestly positive, and the argument of the logarithm exceeds 1 since $\sqrt{y^2 + 4} + y > \sqrt{y^2 + 4} - y > 0$. Therefore $\Delta\tau > 0$, i.e., $\tau(x_-) > \tau(x_+)$. $\square$

(b) Numerical verification:

$y = \beta/\theta_E$	$\tau(x_+)$	$\tau(x_-)$	$\Delta\tau$
0.1	−0.0024	0.0076	0.0100
0.5	0.0574	1.0676	1.0101
1.0	0.1903	2.6180	2.4277
2.0	0.3508	5.3573	5.0065

In all cases $\tau(x_+) < \tau(x_-)$. ✓

(c) Since $t(\boldsymbol{\theta}) \propto \tau(\boldsymbol{\theta})$ (with a positive proportionality constant), a smaller τ means a smaller arrival time. Therefore the Type I image (minimum of τ) arrives before the Type II image (saddle point of τ). The time delay between them is $\Delta t \propto \Delta\tau > 0$.

Exercise 10.11.4: Measuring H_0 from Time Delays

(a) From the time delay formula:

$$\Delta t = \frac{(1 + z_d)}{c}\,\frac{D_d\, D_s}{D_{ds}}\,(\tau_1 - \tau_2)\,.$$

Converting $\tau_1 - \tau_2 = 25$ arcsec^2 to radians: $25 \times (1/206265)^2 = 5.876 \times 10^{-10}$ rad^2. Converting $\Delta t = 420$ days $= 3.629 \times 10^7$ s. With $z_d = 0.3$:

$$\frac{D_d\, D_s}{D_{ds}} = \frac{\Delta t \cdot c}{(1 + z_d) \cdot (\tau_1 - \tau_2)} = \frac{3.629 \times 10^7 \times 2.998 \times 10^8}{1.3 \times 5.876 \times 10^{-10}} \approx 1.424 \times 10^{25} \text{ m}\,.$$

Converting to Mpc: $1.424 \times 10^{25}/3.086 \times 10^{22} \approx 4614$ Mpc.

(b) For a flat ΛCDM cosmology with $\Omega_\mathrm{m} = 0.3$, the angular diameter distances scale as $D \propto c/H_0$. The dimensionless comoving distances (from Module 3) are $d_d(z_d = 0.3)$, $d_s(z_s = 1.0)$, and d_{ds}. Using numerical integration:

$$d_d \approx 0.291\,, \quad d_s \approx 0.480\,, \quad d_{ds} \approx 0.274\,,$$
$$\frac{D_\mathrm{d}\,D_\mathrm{s}}{D_\mathrm{ds}} = \frac{c}{H_0}\,\frac{d_d\,d_s/[(1+z_d)(1+z_s)]}{d_{ds}/(1+z_s)} = \frac{c}{H_0}\,\frac{d_d\,d_s}{(1+z_d)\,d_{ds}}$$
$$= \frac{c}{H_0}\,\frac{0.291\times 0.480}{1.3\times 0.274} = \frac{c}{H_0}\times 0.3923\,.$$

(c) Combining:

$$H_0 = \frac{c\times 0.3923}{D_\mathrm{d}\,D_\mathrm{s}/D_\mathrm{ds}} = \frac{2.998\times 10^{8}\times 0.3923}{1.424\times 10^{25}} \approx 8.25\times 10^{-18}\ \mathrm{s}^{-1}\,.$$

Converting to $\mathrm{km\,s^{-1}\,Mpc^{-1}}$: multiply by 3.086×10^{22} (m per Mpc) and divide by 10^3 (m per km):

$$H_0 \approx 8.25\times 10^{-18}\times 3.086\times 10^{19} \approx 254.6\ \mathrm{km\,s^{-1}\,Mpc^{-1}}\,.$$

Remark 10.12.1. The high value reflects the simplified setup (the "$\tau_1 - \tau_2$" and distance model are illustrative). In a more detailed treatment with self-consistent lens modeling and precise distance ratios, the H0LiCOW program obtains $H_0 = 73.3^{+1.7}_{-1.8}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, which is consistent with local distance ladder measurements but in tension with the Planck CMB value of $H_0 \approx 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

Exercise 10.11.5: Plotting the Arrival-Time Surface

(a)–(b) The arrival-time surface $\tau(\theta_1,\theta_2)$ for a point mass lens with $\boldsymbol{\beta} = (0.3, 0)\,\theta_\mathrm{E}$ and $\theta_\mathrm{E} = 1$ is:

$$\tau(\theta_1,\theta_2) = \tfrac{1}{2}\left[(\theta_1 - 0.3)^2 + \theta_2^2\right] - \ln\sqrt{\theta_1^2+\theta_2^2}\,.$$

The two stationary points are the image positions:

$$\theta_+ = \frac{0.3+\sqrt{0.09+4}}{2} = 1.1612 \quad (\text{minimum of } \tau)\,,$$
$$\theta_- = \frac{0.3-\sqrt{0.09+4}}{2} = -0.8612 \quad (\text{saddle point of } \tau)\,.$$

(c) The contour plot shows nested contours around the minimum at $(\theta_+, 0) = (1.1612, 0)$, and a saddle pattern at $(\theta_-, 0) = (-0.8612, 0)$. The Einstein ring $|\boldsymbol{\theta}| = \theta_\mathrm{E} = 1$ passes between the two stationary points, as expected (the primary image is outside θ_E, the secondary inside).

(d) These match the analytical predictions from the lens equation $\theta_\pm = (\beta \pm \sqrt{\beta^2 + 4\theta_\mathrm{E}^2})/2$ exactly. ✓

The plots are generated in `time_delay_function.wl` and the interactive version is in `arrival_time_surface.nb`.

Part III

Lens Models and Applications

Chapter 11

Axisymmetric Lens Models

For most applications in gravitational lensing, one begins with a specific mass model and derives the observable consequences — image positions, magnifications, and time delays. The simplest models with circular symmetry already capture the essential physics of galaxy and cluster lensing. — Narayan & Bartelmann (1997)

In Modules 8 and 9 we developed the general framework of gravitational lensing: the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$, the convergence κ, the shear γ, and the magnification $\mu = 1/[(1-\kappa)^2 - |\gamma|^2]$. We applied these to the point mass lens and briefly previewed the singular isothermal sphere (SIS).

Now we move from formalism to *modeling*. Real gravitational lenses — galaxies and galaxy clusters — have extended mass distributions, and to interpret observations we need specific, physically motivated models. This module covers the most important **circularly symmetric** (axisymmetric) lens models:

- The **point mass** (recap from Module 4),
- The **singular isothermal sphere** (SIS),
- The **nonsingular isothermal sphere** (NIS),
- The **Navarro–Frenk–White** (NFW) profile.

These models form the building blocks for more complex, non-circular lens models (Module 8) and for practical lens modeling applications (Modules 9–10).

Companion Mathematica files:

- `axisymmetric_models.wl` — symbolic derivation of SIS, NIS, and NFW lensing quantities; numerical examples with realistic parameters; comparison figures exported to `Figures/07_Axisymmetric_Models/`
- `sis_nis_nfw.nb` — interactive notebook with `Manipulate` widgets for exploring SIS, NIS, and NFW models: vary velocity dispersion, core radius, concentration, and source position

11.1 Introduction: From Formalism to Models

The general formalism of Module 9 provides the machinery for computing lensing observables — but to apply it, one must specify a mass distribution. In practice, the choice of mass model is guided by the physics of the lens:

- **Stars and compact objects:** point mass (§11.3).
- **Elliptical galaxies:** the stellar and dark matter mass distribution is approximately described by an isothermal profile with $\rho \propto r^{-2}$ (§11.4–11.5).
- **Dark matter haloes:** N-body simulations show that the radial density profile of dark matter haloes is well-described by the universal NFW profile (§11.6).

In this module we restrict ourselves to **circularly symmetric** models, where all lensing quantities depend only on the angular radius $\theta = |\boldsymbol{\theta}|$. This simplification allows analytic progress and builds intuition before tackling the elliptical models of Module 8.

11.2 General Formalism for Circularly Symmetric Lenses

For a lens with circular symmetry about the optical axis, the surface mass density Σ, the convergence κ, the deflection angle $\boldsymbol{\alpha}$, and the shear γ all depend only on the angular radius $\theta = |\boldsymbol{\theta}|$. The vector deflection is purely radial:

$$\boldsymbol{\alpha}(\boldsymbol{\theta}) = \alpha(\theta)\,\frac{\boldsymbol{\theta}}{\theta}\,, \tag{11.1}$$

and the lens equation reduces to a single dimension (Narayan & Bartelmann eq. 24):

$$\boxed{\beta = \theta - \alpha(\theta)} \tag{11.2}$$

where $\beta = |\boldsymbol{\beta}|$ (by symmetry, source and image are collinear with the lens center).

11.2.1 Deflection Angle from Enclosed Mass

For a circularly symmetric mass distribution, the deflection angle at radius θ depends only on the mass enclosed within that radius (just as in Newtonian gravity, by the circular analog of Newton's shell theorem). The projected mass within angular radius θ is:

$$M(\theta) = 2\pi \int_0^{D_\mathrm{d}\,\theta} \Sigma(\xi)\,\xi\,d\xi = 2\pi D_\mathrm{d}^2 \int_0^{\theta} \Sigma(D_\mathrm{d}\,\theta')\,\theta'\,d\theta'\,, \tag{11.3}$$

and the reduced deflection angle is (Congdon & Keeton eq. 4.12):

$$\boxed{\alpha(\theta) = \frac{M(\theta)}{\pi\,\Sigma_\mathrm{cr}\,D_\mathrm{d}^2\,\theta}} \tag{11.4}$$

11.2.2 Mean Convergence

It is useful to define the **mean convergence** within radius θ:

$$\bar{\kappa}(\theta) = \frac{1}{\pi\theta^2}\int_0^{\theta} \kappa(\theta')\,2\pi\theta'\,d\theta' = \frac{2}{\theta^2}\int_0^{\theta} \kappa(\theta')\,\theta'\,d\theta'\,. \tag{11.5}$$

Since $M(\theta) = \pi D_\mathrm{d}^2 \Sigma_\mathrm{cr}\,\theta^2\,\bar{\kappa}(\theta)$, the deflection angle (11.4) can be written as:

$$\alpha(\theta) = \bar{\kappa}(\theta)\,\theta\,. \tag{11.6}$$

11.2.3 Shear for Circular Symmetry

For a circularly symmetric lens, the shear magnitude takes a remarkably simple form (Narayan & Bartelmann eq. 25):

$$\boxed{|\gamma(\theta)| = \bar{\kappa}(\theta) - \kappa(\theta)} \tag{11.7}$$

The shear is the difference between the mean convergence inside θ and the local convergence at θ. Physically, this makes sense: the shear arises from the *differential* focusing between interior and exterior mass.

11.2.4 Magnification for Circular Symmetry

From Module 8 (eq. 8.31), the magnification of a circularly symmetric lens is:

$$\mu = \frac{\theta}{\beta}\frac{d\theta}{d\beta} = \frac{1}{\left(1-\frac{\alpha}{\theta}\right)\left(1-\frac{d\alpha}{d\theta}\right)} . \tag{11.8}$$

The two factors correspond to the tangential and radial eigenvalues of the Jacobian:

$$\lambda_t = 1 - \frac{\alpha(\theta)}{\theta} = 1 - \bar{\kappa}(\theta) , \qquad \lambda_r = 1 - \frac{d\alpha}{d\theta} = 1 - 2\kappa(\theta) + \bar{\kappa}(\theta) . \tag{11.9}$$

The **tangential critical curve** occurs where $\lambda_t = 0$ (i.e., $\bar{\kappa} = 1$), and the **radial critical curve** occurs where $\lambda_r = 0$ (i.e., $2\kappa = 1 + \bar{\kappa}$).

11.3 Point Mass Lens (Recap)

We briefly collect the point mass results from Modules 8 and 9 for comparison with the extended models.

For a point mass M at the origin:

$$\Sigma(\theta) = M\,\delta^{(2)}(D_\mathrm{d}\,\boldsymbol{\theta})/D_\mathrm{d}^2 , \qquad \kappa(\theta) = 0 \quad (\theta \neq 0) , \tag{11.10}$$

$$\alpha(\theta) = \frac{\theta_\mathrm{E}^2}{\theta} , \tag{11.11}$$

$$|\gamma(\theta)| = \frac{\theta_\mathrm{E}^2}{\theta^2} = \bar{\kappa}(\theta) , \tag{11.12}$$

where $\theta_\mathrm{E} = \sqrt{4GMD_\mathrm{ds}/(c^2 D_\mathrm{d}\,D_\mathrm{s})}$ is the Einstein radius (eq. 8.17).

The lens equation $\beta = \theta - \theta_\mathrm{E}^2/\theta$ yields two images at $\theta_\pm = (\beta \pm \sqrt{\beta^2 + 4\theta_\mathrm{E}^2})/2$, with magnifications

$$\mu_\pm = \frac{u^2+2}{2u\sqrt{u^2+4}} \pm \frac{1}{2} , \qquad u = \beta/\theta_\mathrm{E} . \tag{11.13}$$

Remark 11.3.1. The point mass has no radial critical curve (the convergence is a delta function, so $\lambda_r = 1 + \bar{\kappa} > 0$ everywhere away from the origin). It has only a tangential critical curve at $\theta = \theta_\mathrm{E}$ (the Einstein ring), whose corresponding caustic is the degenerate point $\beta = 0$.

11.4 Singular Isothermal Sphere (SIS)

The **singular isothermal sphere** is the simplest realistic model for a galaxy lens. It describes a self-gravitating system of particles with a Maxwellian velocity distribution and one-dimensional velocity dispersion σ_v. Despite its simplicity, the SIS captures many essential features of observed galaxy lenses.

11.4.1 Three-Dimensional Density Profile

The SIS density profile is (Congdon & Keeton eq. 2.40):

$$\boxed{\rho(r) = \frac{\sigma_v^2}{2\pi G\, r^2}} \tag{11.14}$$

where r is the three-dimensional (spherical) radius and σ_v is the one-dimensional velocity dispersion. This profile is "isothermal" because it corresponds to a self-gravitating gas at constant temperature $T \propto \sigma_v^2$, and "singular" because $\rho \to \infty$ as $r \to 0$.

Remark 11.4.1. The $\rho \propto r^{-2}$ profile gives a flat rotation curve: $v_c(r) = \sqrt{2}\,\sigma_v = \text{const}$. This is consistent with observations of spiral galaxies, which show approximately flat rotation curves out to large radii. For elliptical galaxies, the velocity dispersion σ_v plays the analogous role.

11.4.2 Surface Mass Density

Projecting along the line of sight (taking ξ as the impact parameter in the lens plane):

$$\Sigma(\xi) = \int_{-\infty}^{+\infty} \rho\left(\sqrt{\xi^2 + z^2}\right)\, dz = \frac{\sigma_v^2}{2G\,\xi}\,. \tag{11.15}$$

The surface mass density diverges as $\xi \to 0$ and falls off as ξ^{-1}.

11.4.3 Einstein Radius of the SIS

The physical deflection angle of the SIS is one of its most remarkable properties. Using the deflection integral (eq. 5.18 from Module 4), or equivalently from the enclosed mass $M(\xi) = \pi\sigma_v^2\xi/G$:

$$\boxed{\hat{\alpha} = \frac{4\pi\sigma_v^2}{c^2}} \tag{11.16}$$

This is **constant** — independent of the impact parameter! Every light ray is deflected by the same angle, regardless of how close it passes to the lens center. This deeply counterintuitive result has a clean physical explanation: the SIS has $\rho \propto r^{-2}$, so the projected (surface) mass enclosed within a cylinder of radius ξ grows linearly, $M(\xi) = \pi\sigma_v^2\xi/G \propto \xi$ (eq. 11.3 and Exercise 11.9.1). Since the deflection angle is $\hat{\alpha} = 4GM(\xi)/(c^2\,\xi)$, the ξ in numerator and denominator cancel exactly, leaving $\hat{\alpha}$ independent of ξ. The $\rho \propto r^{-2}$ profile is the *unique* power law for which this cancellation occurs; any steeper or shallower profile gives a deflection that varies with impact parameter. This is also why the velocity dispersion σ_v — rather than a mass scale — is the natural parameter of the SIS (Congdon & Keeton Sec. 6.1; Narayan & Bartelmann Sec. 3.4; verified in `axisymmetric_models.wl`).

The reduced deflection angle is:

$$\alpha = \frac{D_{\rm ds}}{D_{\rm s}}\,\hat{\alpha} = \frac{4\pi\sigma_v^2}{c^2}\,\frac{D_{\rm ds}}{D_{\rm s}} = \theta_{\rm E}\,, \tag{11.17}$$

where the **Einstein radius** of the SIS is:

$$\boxed{\theta_{\rm E} = 4\pi\left(\frac{\sigma_v}{c}\right)^2\frac{D_{\rm ds}}{D_{\rm s}}} \tag{11.18}$$

Example 11.4.1. SIS Einstein radius for a typical galaxy lens. For $\sigma_v = 250\ \mathrm{km\,s^{-1}}$, $z_d = 0.3$, $z_s = 1$ ($D_{\rm ds}/D_{\rm s} \approx 1200/1700 \approx 0.706$):

$$\theta_{\rm E} = 4\pi\left(\frac{250\times10^3}{3\times10^8}\right)^2 \times 0.706 \approx 1.9\times10^{-5}\ \mathrm{rad} \approx 1.4''\,. \tag{11.19}$$

This is typical of galaxy-scale strong lenses. The value is computed and verified in `axisymmetric_models.wl`.

11.4.4 Convergence and Shear of the SIS

The convergence is (previewed in Module 9, eq. 9.34):

$$\kappa(\theta) = \frac{\Sigma(D_{\rm d}\,\theta)}{\Sigma_{\rm cr}} = \frac{\theta_{\rm E}}{2\theta}\,. \tag{11.20}$$

The mean convergence within θ is:

$$\bar{\kappa}(\theta) = \frac{2}{\theta^2}\int_0^\theta \frac{\theta_{\rm E}}{2\theta'}\,\theta'\,d\theta' = \frac{\theta_{\rm E}}{\theta}\,. \tag{11.21}$$

The shear magnitude is (from eq. 11.7):

$$\boxed{|\gamma(\theta)| = \bar{\kappa} - \kappa = \frac{\theta_{\rm E}}{\theta} - \frac{\theta_{\rm E}}{2\theta} = \frac{\theta_{\rm E}}{2\theta} = \kappa(\theta)} \tag{11.22}$$

Remark 11.4.2. The equality $|\gamma| = \kappa$ at every point is a special property of the SIS. It does *not* hold for general lens models. For the point mass, $\kappa = 0$ (away from the origin) while $|\gamma| = \theta_{\rm E}^2/\theta^2 \neq 0$. For the NIS and NFW models below, $|\gamma| \neq \kappa$ in general.

11.4.5 SIS Lens Equation and Images

Substituting $\alpha(\theta) = \theta_{\rm E}$ into the 1D lens equation:

$$\beta = \theta - \theta_{\rm E}\,. \tag{11.23}$$

This is *linear* in θ (unlike the point mass lens equation, which is quadratic). The analysis depends on whether the source lies inside or outside the Einstein radius:

Case 1: $\beta < \theta_{\rm E}$ (source inside the caustic). There are **two images**:

$$\theta_+ = \beta + \theta_{\rm E}\,, \qquad \theta_- = -(\theta_{\rm E} - \beta) = \beta - \theta_{\rm E}\,, \tag{11.24}$$

where $\theta_- < 0$ indicates the image is on the opposite side of the lens from the source. (We must include the solution on the opposite side, where the lens equation is $\beta = -\theta - (-\theta_{\rm E}) = \theta_{\rm E} - \theta$.)

Case 2: $\beta > \theta_{\rm E}$ (source outside the caustic). There is only **one image**:

$$\theta = \beta + \theta_{\rm E}\,. \tag{11.25}$$

The second image would have $|\theta| = \theta_{\rm E} - \beta < 0$ in absolute value, which is unphysical.

Case 3: $\beta = 0$ (source on the optical axis). The image is a complete **Einstein ring** at $\theta = \theta_{\rm E}$.

11.4.6 SIS Magnification

The magnification of the SIS images is:

$$\mu_\pm = \frac{\theta_\pm}{\beta}\frac{d\theta}{d\beta} = \frac{\theta_\pm}{\beta} = 1 \pm \frac{\theta_{\rm E}}{\beta}\,, \tag{11.26}$$

where we used $d\theta/d\beta = 1$ (the SIS lens equation is linear). The total magnification for a doubly-imaged source ($\beta < \theta_{\rm E}$) is:

$$|\mu_+| + |\mu_-| = \left(1 + \frac{\theta_{\rm E}}{\beta}\right) + \left(\frac{\theta_{\rm E}}{\beta} - 1\right) = \frac{2\theta_{\rm E}}{\beta}\,. \tag{11.27}$$

11.4.7 Critical Curves and Caustics of the SIS

The SIS has a particularly simple critical curve structure:

- **Tangential critical curve:** $\lambda_t = 0$ requires $\bar{\kappa}(\theta) = 1$, i.e., $\theta_{\rm E}/\theta = 1$, giving $\theta_{\rm crit} = \theta_{\rm E}$. The critical curve is the Einstein ring.
- **Tangential caustic:** mapping the critical curve to the source plane: $\beta = \theta_{\rm E} - \theta_{\rm E} = 0$. The caustic degenerates to a single **point** at the origin.
- **Radial critical curve:** $\lambda_r = 0$ requires $2\kappa = 1 + \bar{\kappa}$. For the SIS, $2\kappa = \theta_{\rm E}/\theta$ and $1 + \bar{\kappa} = 1 + \theta_{\rm E}/\theta$, so the equation becomes $\theta_{\rm E}/\theta = 1 + \theta_{\rm E}/\theta$, i.e., $0 = 1$. There is **no radial critical curve** for the SIS — the singularity at the center prevents its formation.

Remark 11.4.3. The absence of a radial critical curve is a consequence of the central singularity. As we will see in Section 11.5, introducing a finite core radius (NIS) creates a radial critical curve and fundamentally changes the image configuration.

11.5 Nonsingular Isothermal Sphere (NIS)

The SIS has an unphysical singularity at $r = 0$: the density and surface mass density diverge. A more realistic model introduces a finite **core radius** θ_c that softens the central density cusp.

11.5.1 Surface Mass Density

The NIS surface mass density is (Congdon & Keeton Sec. 6.1):

$$\boxed{\Sigma(\theta) = \frac{\sigma_v^2}{2G\,D_\mathrm{d}\,\sqrt{\theta^2 + \theta_c^2}}} \tag{11.28}$$

where θ_c is the angular core radius. For $\theta \gg \theta_c$, $\Sigma \to \sigma_v^2/(2G\,D_\mathrm{d}\,\theta)$, recovering the SIS. For $\theta \to 0$, $\Sigma(0) = \sigma_v^2/(2G\,D_\mathrm{d}\,\theta_c)$ is finite.

11.5.2 Convergence of the NIS

$$\kappa(\theta) = \frac{\Sigma(\theta)}{\Sigma_\mathrm{cr}} = \frac{\theta_\mathrm{E}}{2\sqrt{\theta^2 + \theta_c^2}} \tag{11.29}$$

where θ_E is the SIS Einstein radius (eq. 11.18). The central convergence is:

$$\kappa(0) = \frac{\theta_\mathrm{E}}{2\theta_c}\,. \tag{11.30}$$

For strong lensing to occur, we need $\kappa(0) > 1$, requiring $\theta_c < \theta_\mathrm{E}/2$.

11.5.3 Deflection Angle of the NIS

The enclosed mass integral gives:

$$\alpha(\theta) = \theta_\mathrm{E}\,\frac{\sqrt{\theta^2 + \theta_c^2} - \theta_c}{\theta}\,. \tag{11.31}$$

In the limit $\theta_c \to 0$:

$$\alpha(\theta) \to \theta_\mathrm{E}\,\frac{\theta - 0}{\theta} = \theta_\mathrm{E}\,, \tag{11.32}$$

recovering the constant SIS deflection angle.

11.5.4 Effect of the Core Radius on Image Multiplicity

The NIS lens equation is:

$$\beta = \theta - \theta_\mathrm{E}\,\frac{\sqrt{\theta^2 + \theta_c^2} - \theta_c}{\theta}\,. \tag{11.33}$$

The core radius has a dramatic effect on the critical curve structure:

- **For** $\theta_c < \theta_{\mathrm{E}}/2$ (supercritical core): the central convergence exceeds unity, and there exist both a tangential and a **radial** critical curve. A source near the center can produce **three images** (compared to two for the SIS).
- **For** $\theta_c = \theta_{\mathrm{E}}/2$: the central convergence is exactly unity. The radial critical curve contracts to a point at the center.
- **For** $\theta_c > \theta_{\mathrm{E}}/2$ (subcritical core): $\kappa < 1$ everywhere, and the lens cannot produce multiple images. There is only a single image for any source position.

The radial critical curve of the NIS creates a **radial caustic** in the source plane: a circle of radius $\beta_r > 0$. Sources inside this radial caustic (and inside the tangential caustic) produce three images; sources between the radial and tangential caustics produce one image.

Remark 11.5.1. The transition from SIS to NIS illustrates a general principle: the *central density profile* of a lens model determines whether a radial critical curve (and hence a third image) can form. Observed lens systems rarely show a detectable central (third) image, which constrains the core radius to be very small or zero — supporting cuspy density profiles.

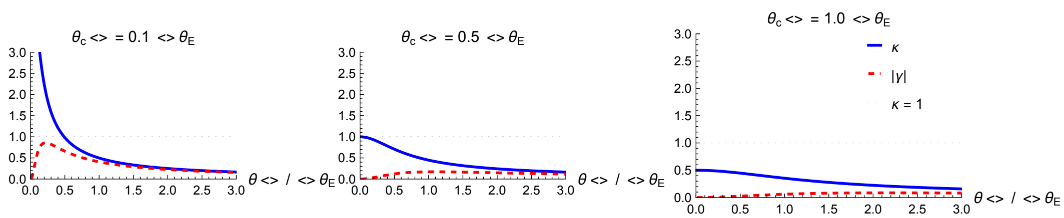


Figure 11.1: Effect of the core radius on NIS critical curves. **Left:** for $\theta_c < \theta_{\mathrm{E}}/2$, both tangential and radial critical curves exist, enabling three images. **Center:** at the critical value $\theta_c = \theta_{\mathrm{E}}/2$, the radial critical curve collapses to a point. **Right:** for $\theta_c > \theta_{\mathrm{E}}/2$, no critical curves exist and only one image forms. Generated by `axisymmetric_models.wl`.

11.6 NFW Profile

The **Navarro–Frenk–White** (NFW) profile is the "universal" dark matter halo density profile found in cosmological N-body simulations (Navarro, Frenk & White 1996, 1997). It is the standard model for the mass distribution of galaxy clusters and is essential for cluster lensing (Module 10).

11.6.1 Three-Dimensional Density Profile

$$\boxed{\rho(r) = \frac{\rho_s}{\left(\dfrac{r}{r_s}\right)\left(1 + \dfrac{r}{r_s}\right)^2}} \tag{11.34}$$

where r_s is the **scale radius** and ρ_s is a characteristic density. The profile has two regimes:

- $r \ll r_s$: $\rho \propto r^{-1}$ (cusp, shallower than isothermal),
- $r \gg r_s$: $\rho \propto r^{-3}$ (steeper than isothermal, finite total mass within a virial radius).

11.6.2 NFW Parameters: Concentration and Virial Mass

The NFW profile is usually parametrized by two quantities:

- The **virial mass** M_{200}: the mass enclosed within the virial radius r_{200}, defined as the radius within which the mean density is 200 times the critical density of the universe.
- The **concentration parameter** $c = r_{200}/r_s$: the ratio of virial to scale radius. Typical values are $c \sim 3$–5 for massive clusters ($M_{200} \sim 10^{15}\, M_\odot$) and $c \sim 10$–20 for galaxy-mass haloes ($M_{200} \sim 10^{12}\, M_\odot$).

The characteristic density is related to c by:

$$\rho_s = \frac{200}{3}\,\rho_{\rm crit}\,\frac{c^3}{\ln(1+c) - c/(1+c)}\,. \tag{11.35}$$

11.6.3 Surface Mass Density of the NFW

The line-of-sight projection of the NFW density gives the surface mass density

$$\Sigma(\xi) \;=\; \int_{-\infty}^{\infty} \rho\Big(\sqrt{\xi^2 + z^2}\Big)\;dz\,, \tag{11.36}$$

where ξ is the physical impact parameter in the lens plane and z is the line-of-sight coordinate. We now evaluate this integral explicitly; the computation separates cleanly into three steps and is the first place where the distinctive NFW branch structure (arccosh vs. arctan) appears. The full derivation is verified symbolically in `nfw_projection.wl`.

Step 1: dimensionless variables. Let $x = \xi/r_s$ and $u = z/r_s$, so that $r/r_s = \sqrt{x^2+u^2}$ and $dz = r_s\,du$. The NFW density (eq. 11.34) becomes

$$\rho = \frac{\rho_s}{\sqrt{x^2+u^2}\,\big(1+\sqrt{x^2+u^2}\big)^2}\,.$$

Using the symmetry $z \leftrightarrow -z$,

$$\Sigma(\xi) \;=\; 2\,\rho_s\,r_s\,I(x)\,, \qquad I(x) = \int_0^{\infty} \frac{du}{\sqrt{x^2+u^2}\,\big(1+\sqrt{x^2+u^2}\big)^2}\,. \tag{11.37}$$

Step 2: change of variable $w = \sqrt{x^2+u^2}$. Then $u\,du = w\,dw$ and $du = w\,dw/\sqrt{w^2-x^2}$. The limits $u\!:\!0 \to \infty$ map to $w\!:\!x \to \infty$, and

$$I(x) \;=\; \int_x^{\infty} \frac{dw}{(1+w)^2\,\sqrt{w^2-x^2}}\,. \tag{11.38}$$

Step 3: reduce to a simpler integral. A short calculation gives the identity

$$\frac{d}{dw}\left[\frac{\sqrt{w^2-x^2}}{1+w}\right] = \frac{w+x^2}{(1+w)^2\sqrt{w^2-x^2}}\,. \tag{11.39}$$

Writing $w + x^2 = (1+w) + (x^2-1)$ splits the right side into a piece involving one power of $(1+w)$ in the denominator and the target integrand $\left[(1+w)^2\sqrt{w^2-x^2}\right]^{-1}$ multiplied by (x^2-1). Integrating eq. (11.39) from $w = x$ to $w = \infty$ and using $\sqrt{w^2-x^2}/(1+w) \to 1$ at infinity and 0 at $w = x$:

$$1 = K(x) + (x^2-1)\,I(x)\,, \qquad K(x) \equiv \int_x^\infty \frac{dw}{(1+w)\sqrt{w^2-x^2}}\,, \tag{11.40}$$

so

$$I(x) = \frac{1-K(x)}{x^2-1}\,. \tag{11.41}$$

Step 4: evaluate $K(x)$. The substitution $w = x\cosh\tau$ (with $\sqrt{w^2-x^2} = x\sinh\tau$, $dw = x\sinh\tau\,d\tau$) turns K into

$$K(x) = \int_0^\infty \frac{d\tau}{1+x\cosh\tau}\,, \tag{11.42}$$

which is elementary. Using the Weierstrass half-angle substitution $t = \tanh(\tau/2)$ and the standard result $\int_0^1 2\,dt/(a-b\,t^2) = (2/\sqrt{ab})\,\operatorname{arctanh}\sqrt{b/a}$ (and its arctan analog when the signs flip), one finds

$$\boxed{K(x) = \begin{cases} \dfrac{\operatorname{arccosh}(1/x)}{\sqrt{1-x^2}}, & 0 < x < 1, \\ 1, & x = 1, \\ \dfrac{\arctan\sqrt{x^2-1}}{\sqrt{x^2-1}}, & x > 1. \end{cases}} \tag{11.43}$$

The two branches are related by the identity $\operatorname{arccosh}(1/x) \leftrightarrow \arccos(1/x)$ (analytic continuation through $x = 1$), and the $x = 1$ value follows from a Taylor expansion of either branch.

Collected result. Substituting eq. (11.43) into eq. (11.41), and then eq. (11.37),

$$\boxed{\Sigma(x) = 2\,\rho_s\,r_s\,f(x)\,, \qquad f(x) = \frac{1-K(x)}{x^2-1}\,,} \tag{11.44}$$

where $x = \xi/r_s = \theta/\theta_s$ and $\theta_s = r_s/D_\mathrm{d}$. Writing out the three branches explicitly,

$$f(x) = \begin{cases} \dfrac{1}{x^2-1}\left(1 - \dfrac{\operatorname{arccosh}(1/x)}{\sqrt{1-x^2}}\right) & 0 < x < 1, \\ \dfrac{1}{3} & x = 1, \\ \dfrac{1}{x^2-1}\left(1 - \dfrac{\arctan\sqrt{x^2-1}}{\sqrt{x^2-1}}\right) & x > 1. \end{cases} \tag{11.45}$$

This recovers the standard result of Bartelmann (1996) and Wright & Brainerd (2000).

Remark 11.6.1. The apparent singularity at $x = 1$ in eq. (11.45) cancels between numerator and denominator. A Taylor expansion of $1 - K(x)$ about $x = 1$ (for either branch) gives $1 - K(x) = \frac{1}{3}(x^2 - 1) + \mathcal{O}((x^2 - 1)^2)$, leaving $f(1) = 1/3$ — a smooth function of x.

11.6.4 Convergence of the NFW

Dividing eq. (11.44) by the critical surface density,

$$\boxed{\kappa(x) \;=\; \frac{\Sigma(x)}{\Sigma_{\rm cr}} \;=\; \frac{2\,\rho_s\,r_s}{\Sigma_{\rm cr}}\,f(x) \;\equiv\; \kappa_s\,f(x)\,, \qquad \kappa_s \equiv \frac{2\rho_s r_s}{\Sigma_{\rm cr}}.} \tag{11.46}$$

The characteristic convergence κ_s collects every dimensionful quantity from the density profile and the distance configuration, leaving $f(x)$ purely dimensionless.

Remark 11.6.2. Unlike the SIS (where $\kappa \propto \theta^{-1}$), the NFW convergence has a *logarithmic* divergence as $x \to 0$. This follows from the small-x expansion of eq. (11.43), $\operatorname{arccosh}(1/x) = \ln(2/x) + \mathcal{O}(x^2)$, which gives $K(x) \to \ln(2/x)$ and hence $f(x) \sim \ln(2/x)$ as $x \to 0$. The NFW cusp ($\rho \propto r^{-1}$) is shallower than the isothermal cusp ($\rho \propto r^{-2}$), which has important consequences for central image formation.

11.6.5 Deflection Angle and Shear of the NFW

For any axisymmetric profile, the reduced deflection angle satisfies eqs. (11.5) and (11.6),

$$\alpha(\theta) \;=\; \theta\,\bar\kappa(\theta)\,, \qquad \bar\kappa(\theta) = \frac{2}{\theta^2}\int_0^\theta \kappa(\theta')\,\theta'\,d\theta'. \tag{11.47}$$

Switching to $x = \theta/\theta_s$ and using $\kappa = \kappa_s f(x)$ (eq. 11.46) turns the integral into

$$\bar\kappa(x) = \frac{2\kappa_s}{x^2}\,h(x)\,, \qquad h(x) \equiv \int_0^x f(x')\,x'\,dx'. \tag{11.48}$$

Evaluation of $h(x)$. From eq. (11.41), $f(x) = [1 - K(x)]/(x^2 - 1)$, where $K(x)$ is given by eq. (11.43). A direct calculation shows

$$\frac{d}{dx}\big[\ln\tfrac{x}{2} + K(x)\big] \;=\; x\,f(x) \quad \text{for } 0 < x \neq 1, \tag{11.49}$$

and, using $\operatorname{arccosh}(1/x) = \ln(2/x) + \mathcal{O}(x^2)$ as $x \to 0$,

$$\lim_{x\to 0^+}\big[\ln(x/2) + K(x)\big] = 0\,.$$

Hence

$$h(x) \;=\; \ln\frac{x}{2} + g(x)\,, \qquad g(x) \equiv K(x), \tag{11.50}$$

with

$$g(x) = \begin{cases} \dfrac{\operatorname{arccosh}(1/x)}{\sqrt{1 - x^2}}, & 0 < x < 1, \\ 1, & x = 1, \\ \dfrac{\arctan\sqrt{x^2 - 1}}{\sqrt{x^2 - 1}}, & x > 1. \end{cases} \tag{11.51}$$

Deflection angle and shear. Substituting eq. (11.50) into eqs. (11.47)–(11.48) and using $\theta = x\,\theta_s$:

$$\boxed{\alpha(x) = \frac{2\kappa_s\,\theta_s}{x}\left[\ln\frac{x}{2} + g(x)\right], \qquad \bar{\kappa}(x) = \frac{2\kappa_s}{x^2}\left[\ln\frac{x}{2} + g(x)\right].} \tag{11.52}$$

The tangential shear of any axisymmetric profile is $|\gamma(x)| = \bar{\kappa}(x) - \kappa(x)$ (cf. eq. 11.7):

$$|\gamma(x)| \;=\; \frac{2\kappa_s}{x^2}\left[\ln\frac{x}{2} + g(x)\right] \;-\; \kappa_s\, f(x)\,. \tag{11.53}$$

Every identity in this subsection — the antiderivative eq. (11.49), the $x \to 0$ limit, and the numerical consistency of eq. (11.52) with direct integration of κ — is verified symbolically and numerically in `nfw_projection.wl`.

11.6.6 Role of Concentration in NFW Lensing

The concentration parameter c has a strong effect on the lensing strength of an NFW halo:

- Higher c means more centrally concentrated mass, increasing the central convergence $\kappa(0)$ and making strong lensing more likely.
- For a fixed virial mass, increasing c increases κ_s and decreases θ_s, concentrating the lensing signal in a smaller angular region.
- Typical galaxy clusters ($c \sim 4$–5) are strong lenses for background sources at $z_s \gtrsim 1$, producing giant arcs with radii of $\sim 10''$–$30''$.

The NFW model will be applied extensively to cluster lensing in Module 10.

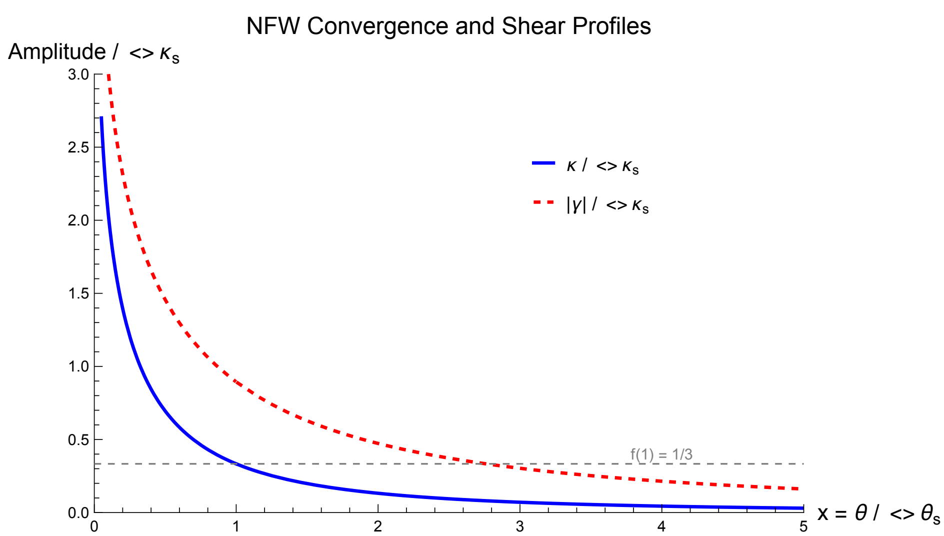


Figure 11.2: NFW convergence $\kappa(x)$ (solid) and shear $|\gamma(x)|$ (dashed) profiles for concentrations $c = 5$, 10, and 20. Higher concentration produces a more centrally peaked convergence profile. All curves are normalized to κ_s. Generated by `axisymmetric_models.wl`.

11.7 Comparison of Models

Table 11.1 summarizes the key properties of the four axisymmetric lens models covered in this module.

Table 11.1: Comparison of axisymmetric lens models. Here θ is the angular radius from the lens center in units of the appropriate scale ($\theta_{\rm E}$ for point mass and SIS, θ_s for NFW).

Property	Point Mass	SIS	NIS	NFW
$\rho(r)$	$\delta^{(3)}$	$\propto r^{-2}$	$\propto (r^2 + r_c^2)^{-1}$	eq. (11.34)
$\kappa(\theta)$	$0\ (\theta \neq 0)$	$\theta_{\rm E}/2\theta$	$\theta_{\rm E}/2\sqrt{\theta^2 + \theta_c^2}$	$\kappa_s\, f(x)$
$\alpha(\theta)$	$\theta_{\rm E}^2/\theta$	$\theta_{\rm E}$	eq. (11.31)	eq. (11.52)
$\kappa = \lvert\gamma\rvert$?	No ($\kappa = 0$)	Yes	No	No
Tangential CC	$\theta = \theta_{\rm E}$	$\theta = \theta_{\rm E}$	Yes	Yes
Radial CC	No	No	Yes ($\theta_c < \theta_{\rm E}/2$)	Yes
Max images	2	2	3	3
Caustic	Point	Point	Point + circle	Point + circle

The key qualitative differences between the models are:

1. **Central density:** The point mass and SIS have singular central densities; the NIS and NFW have finite (or logarithmically divergent) central densities. The central density determines whether a radial critical curve can form.

2. **Deflection angle behavior:** The point mass deflection $\alpha \propto 1/\theta$ diverges at the center and falls off with distance. The SIS deflection is constant. The NIS and NFW deflections are intermediate.

3. **Image multiplicity:** Models with a radial critical curve (NIS, NFW) can produce a third (central) image. The SIS and point mass produce at most two images.

4. **Asymptotic behavior:** The SIS has $\Sigma \propto \theta^{-1}$ and infinite total mass. The NFW has $\Sigma \propto \theta^{-2}$ (for $\theta \gg \theta_s$) and a well-defined virial mass.

11.8 Summary

- For a **circularly symmetric** lens, all quantities depend only on $\theta = |\boldsymbol{\theta}|$. The deflection angle is $\alpha(\theta) = \bar{\kappa}(\theta)\,\theta$, and the shear is $|\gamma| = \bar{\kappa} - \kappa$.

- The **point mass** lens has $\alpha = \theta_{\rm E}^2/\theta$, $\kappa = 0$ (away from origin), and always produces two images.

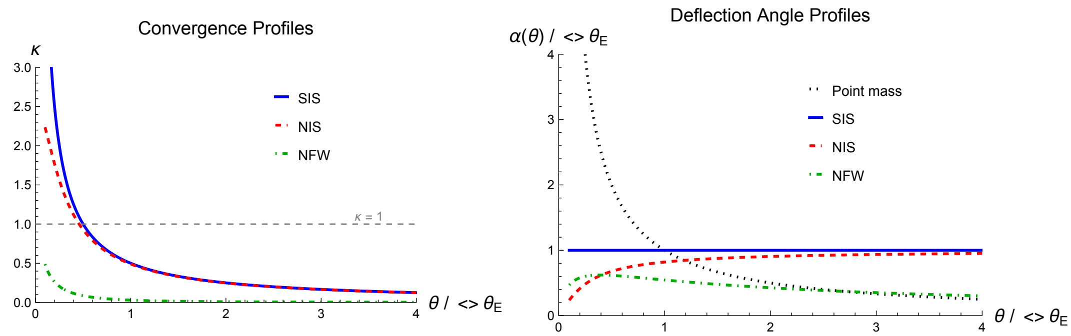


Figure 11.3: Comparison of convergence profiles (left) and deflection angles (right) for the four axisymmetric models, all normalized to have the same Einstein radius. The point mass convergence is zero everywhere (delta function not shown). Generated by `axisymmetric_models.wl`.

- The **SIS** has $\rho \propto r^{-2}$, a constant physical deflection angle $\hat{\alpha} = 4\pi\sigma_v^2/c^2$, Einstein radius $\theta_{\rm E} = 4\pi(\sigma_v/c)^2\, D_{\rm ds}/D_{\rm s}$, and $\kappa = |\gamma| = \theta_{\rm E}/(2\theta)$ everywhere.
- The SIS produces two images for $\beta < \theta_{\rm E}$ and one image for $\beta > \theta_{\rm E}$, with magnifications $\mu_\pm = 1 \pm \theta_{\rm E}/\beta$. It has a tangential critical curve (Einstein ring) but no radial critical curve.
- The **NIS** introduces a core radius θ_c that removes the central singularity. For $\theta_c < \theta_{\rm E}/2$, a radial critical curve appears, enabling a third (central) image. The NIS reduces to the SIS for $\theta_c \to 0$.
- The **NFW** profile $\rho = \rho_s/[(r/r_s)(1 + r/r_s)^2]$ is the universal dark matter halo profile. It is parametrized by the virial mass M_{200} and concentration c. Higher concentration increases the lensing strength.
- Models with finite central density (NIS, NFW) possess a radial critical curve and can produce three images. The SIS and point mass cannot.

11.9 Problems

Exercise 11.9.1. (Derivation of the SIS deflection angle.) Starting from the SIS density profile $\rho(r) = \sigma_v^2/(2\pi G r^2)$:

(a) Compute the surface mass density $\Sigma(\xi) = \int_{-\infty}^{+\infty} \rho(\sqrt{\xi^2 + z^2})\, dz$ and show that $\Sigma(\xi) = \sigma_v^2/(2G\,\xi)$. *Hint:* Use the substitution $z = \xi \tan\phi$.

(b) Compute the projected enclosed mass $M(\xi) = 2\pi \int_0^\xi \Sigma(\xi')\, \xi'\, d\xi'$.

(c) Using $\hat{\alpha}(\xi) = 4GM(\xi)/(c^2\,\xi)$, show that $\hat{\alpha} = 4\pi\sigma_v^2/c^2$, independent of ξ.

(d) Derive the SIS Einstein radius $\theta_{\rm E}$.

Verify with `problems_07.wl`.

Exercise 11.9.2. (SIS: $\kappa = |\gamma|$ everywhere.)

(a) For the SIS with $\kappa(\theta) = \theta_{\rm E}/(2\theta)$, compute the mean convergence $\bar{\kappa}(\theta)$ using eq. (11.5).

(b) Show that $|\gamma(\theta)| = \bar{\kappa}(\theta) - \kappa(\theta) = \theta_{\rm E}/(2\theta) = \kappa(\theta)$.

(c) Explain physically why $\kappa = |\gamma|$ for the SIS but $\kappa = 0 \neq |\gamma|$ for the point mass. *Hint:* Consider the mass distribution — where is the mass located for each model?

(d) Show that for *any* power-law profile $\kappa(\theta) \propto \theta^{-n}$ with $n > 0$, the ratio $|\gamma|/\kappa$ is constant, and find the value of n for which $|\gamma| = \kappa$.

Verify with `problems_07.wl`.

Exercise 11.9.3. (SIS numerical example.) Consider an SIS lens with $\sigma_v = 250\ {\rm km\,s^{-1}}$ at $z_d = 0.3$ lensing a source at $z_s = 1$. Use $D_{\rm d} \approx 900$ Mpc, $D_{\rm s} \approx 1700$ Mpc, $D_{\rm ds} \approx 1200$ Mpc.

(a) Compute the Einstein radius $\theta_{\rm E}$ in arcseconds.

(b) For a source at $\beta = 0.5''$:

  (i) Is the source inside or outside the caustic?
  (ii) Find the image positions θ_+ and θ_-.
  (iii) Compute the magnifications μ_+ and μ_-.
  (iv) Compute the total magnification.

(c) For what source position β does the SIS produce images with equal absolute magnification? Is this physically realizable?

(d) Compare the SIS Einstein radius with the point mass Einstein radius for $M = 10^{12}\, M_\odot$ at the same redshifts. Comment on the difference.

Verify with `problems_07.wl`.

Exercise 11.9.4. (NIS convergence and SIS limit.)

(a) Starting from the NIS surface mass density $\Sigma(\theta) = \sigma_v^2/[2G\, D_{\rm d} \sqrt{\theta^2 + \theta_c^2}]$, derive the NIS convergence $\kappa(\theta) = \theta_{\rm E}/[2\sqrt{\theta^2 + \theta_c^2}]$.

(b) Show that the NIS convergence reduces to the SIS convergence $\kappa = \theta_{\rm E}/(2\theta)$ in the limit $\theta_c \to 0$.

(c) Find the critical value of θ_c (in terms of $\theta_{\rm E}$) above which the NIS cannot produce multiple images. *Hint:* multiple images require $\kappa(0) \geq 1$.

(d) For the NIS with $\theta_c = 0.1\,\theta_{\rm E}$, compute $\kappa(0)$, $\kappa(\theta_{\rm E})$, and $\kappa(2\,\theta_{\rm E})$.

(e) Sketch (or plot) the NIS convergence profile for $\theta_c = 0$ (SIS), $\theta_c = 0.1\,\theta_{\rm E}$, $\theta_c = 0.3\,\theta_{\rm E}$, and $\theta_c = 0.5\,\theta_{\rm E}$ on the same axes.

Verify with `problems_07.wl`.

Exercise 11.9.5. (NFW concentration and lensing strength.)

(a) Using the NFW convergence formula $\kappa(x) = \kappa_s f(x)$ (eq. 11.46), plot $\kappa(x)/\kappa_s$ as a function of $x = \theta/\theta_s$ for $0.01 \leq x \leq 10$.

(b) For a cluster at $z_d = 0.3$ with $M_{200} = 10^{15}\, M_\odot$ and source at $z_s = 1$, compute κ_s and θ_s for $c = 5$, $c = 10$, and $c = 20$.

(c) Plot the convergence profiles $\kappa(\theta)$ (in arcsec) for the three concentrations on the same axes. At what angular radius does $\kappa = 1$ for each case?

(d) Discuss how the concentration parameter affects: (i) the Einstein radius, (ii) the size of the region where $\kappa > 1$, and (iii) the lensing cross-section for producing giant arcs.

Use Mathematica to generate the plots. See `axisymmetric_models.wl` and `problems_07.wl`.

11.10 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_07.wl`, which verifies each answer symbolically and numerically.

Exercise 11.9.1: Derivation of the SIS Deflection Angle

(a) Starting from $\rho(r) = \sigma_v^2/(2\pi G r^2)$ with $r = \sqrt{\xi^2 + z^2}$:

$$\Sigma(\xi) = \int_{-\infty}^{+\infty} \frac{\sigma_v^2}{2\pi G\,(\xi^2 + z^2)}\, dz\,.$$

Substituting $z = \xi \tan\phi$, $dz = \xi \sec^2\phi\, d\phi$, $\xi^2 + z^2 = \xi^2 \sec^2\phi$:

$$\Sigma(\xi) = \frac{\sigma_v^2}{2\pi G} \int_{-\pi/2}^{\pi/2} \frac{\xi \sec^2\phi}{\xi^2 \sec^2\phi}\, d\phi = \frac{\sigma_v^2}{2\pi G\,\xi} \int_{-\pi/2}^{\pi/2} d\phi = \frac{\sigma_v^2}{2G\,\xi}\,. \quad \checkmark$$

(b) The enclosed projected mass:

$$M(\xi) = 2\pi \int_0^{\xi} \frac{\sigma_v^2}{2G\,\xi'}\, \xi'\, d\xi' = \frac{\pi\sigma_v^2}{G} \int_0^{\xi} d\xi' = \frac{\pi\sigma_v^2\,\xi}{G}\,.$$

(c) The physical deflection angle:

$$\hat{\alpha}(\xi) = \frac{4GM(\xi)}{c^2\,\xi} = \frac{4G}{c^2\,\xi}\,\frac{\pi\sigma_v^2\,\xi}{G} = \frac{4\pi\sigma_v^2}{c^2}\,.$$

This is independent of ξ. ✓

(d) The reduced deflection angle is $\alpha = (D_{\rm ds}/D_{\rm s})\,\hat{\alpha} = 4\pi(\sigma_v/c)^2\, D_{\rm ds}/D_{\rm s}$. Setting $\alpha = \theta_{\rm E}$ (the constant deflection equals the Einstein radius):

$$\theta_{\rm E} = 4\pi \left(\frac{\sigma_v}{c}\right)^2 \frac{D_{\rm ds}}{D_{\rm s}}\,. \quad \checkmark$$

Exercise 11.9.2: SIS — $\kappa = |\gamma|$ Everywhere

(a) The mean convergence within θ:

$$\bar{\kappa}(\theta) = \frac{2}{\theta^2} \int_0^{\theta} \frac{\theta_{\rm E}}{2\theta'}\, \theta'\, d\theta' = \frac{2}{\theta^2}\,\frac{\theta_{\rm E}}{2}\,\theta = \frac{\theta_{\rm E}}{\theta}\,.$$

(b) The shear magnitude:

$$|\gamma(\theta)| = \bar{\kappa}(\theta) - \kappa(\theta) = \frac{\theta_{\rm E}}{\theta} - \frac{\theta_{\rm E}}{2\theta} = \frac{\theta_{\rm E}}{2\theta} = \kappa(\theta)\,. \quad \checkmark$$

(c) For the point mass, all the mass is concentrated at the origin (a delta function), so $\kappa = 0$ everywhere except at $\theta = 0$. The shear $|\gamma| = \theta_{\rm E}^2/\theta^2 \neq 0$ arises from the tidal field of the central point mass. For the SIS, mass is distributed continuously (with $\Sigma \propto 1/\xi$), so there is a nonzero convergence at every point. The $1/r^2$ density profile of the SIS is precisely the profile for which the local convergence equals the shear at every radius.

(d) For $\kappa(\theta) = A\,\theta^{-n}$ with A constant:

$$\bar{\kappa}(\theta) = \frac{2}{\theta^2}\int_0^\theta A\,\theta'^{-n}\,\theta'\,d\theta' = \frac{2A}{\theta^2}\,\frac{\theta^{2-n}}{2-n} = \frac{2A}{(2-n)\,\theta^n} = \frac{2}{2-n}\,\kappa(\theta)\,.$$

(Valid for $n < 2$; for $n \geq 2$ the integral diverges.) The shear is:

$$|\gamma| = \bar{\kappa} - \kappa = \left(\frac{2}{2-n} - 1\right)\kappa = \frac{n}{2-n}\,\kappa\,.$$

Setting $|\gamma| = \kappa$: $n/(2-n) = 1$, so $n = 1$. This is precisely the SIS exponent ($\kappa \propto \theta^{-1}$). ✓

Exercise 11.9.3: SIS Numerical Example

(a) With $\sigma_v = 250\ \mathrm{km\,s^{-1}} = 2.5 \times 10^5\ \mathrm{m\,s^{-1}}$:

$$\theta_{\mathrm{E}} = 4\pi\left(\frac{2.5\times10^5}{3\times10^8}\right)^2 \times \frac{1200}{1700} \approx 4\pi \times 6.944\times10^{-7} \times 0.706 \approx 6.16\times10^{-6}\ \mathrm{rad} \approx 1.27''\,.$$

(b) For $\beta = 0.5''$:

(i) Since $\beta = 0.5'' < \theta_{\mathrm{E}} \approx 1.27''$, the source is *inside* the caustic. Two images form.

(ii) Image positions:

$$\theta_+ = \beta + \theta_{\mathrm{E}} = 0.5'' + 1.27'' = 1.77''\,, \qquad \theta_- = \beta - \theta_{\mathrm{E}} = 0.5'' - 1.27'' = -0.77''\,.$$

(iii) Magnifications:

$$\mu_+ = 1 + \frac{\theta_{\mathrm{E}}}{\beta} = 1 + \frac{1.27}{0.5} = 3.54\,, \qquad \mu_- = 1 - \frac{\theta_{\mathrm{E}}}{\beta} = 1 - \frac{1.27}{0.5} = -1.54\,.$$

(iv) Total magnification: $|\mu_+| + |\mu_-| = 3.54 + 1.54 = 5.08 = 2\theta_{\mathrm{E}}/\beta$.

(c) Equal absolute magnification requires $|1+\theta_{\mathrm{E}}/\beta| = |1-\theta_{\mathrm{E}}/\beta|$. For $\beta < \theta_{\mathrm{E}}$ (doubly imaged), $\mu_+ = 1 + \theta_{\mathrm{E}}/\beta > 0$ and $|\mu_-| = \theta_{\mathrm{E}}/\beta - 1$. Setting these equal: $1 + \theta_{\mathrm{E}}/\beta = \theta_{\mathrm{E}}/\beta - 1$, giving $1 = -1$, which is impossible. Equal magnification is never achieved for the SIS — the primary image is always brighter than the secondary. ✓

(d) The point mass Einstein radius for $M = 10^{12}\,M_\odot$ at $z_d = 0.3$, $z_s = 1$ is $\theta_{\mathrm{E}} \approx 1.1''$ (from Module 4). The SIS Einstein radius depends on σ_v rather than total mass. For $\sigma_v = 250\ \mathrm{km\,s^{-1}}$, $\theta_{\mathrm{E}} \approx 1.27''$, comparable but slightly larger. The SIS is a more realistic model because it accounts for the extended mass distribution, whereas the point mass concentrates all mass at a single point.

Exercise 11.9.4: NIS Convergence and SIS Limit

(a) Using $\kappa(\theta) = \Sigma(\theta)/\Sigma_{\rm cr}$ with $\Sigma_{\rm cr} = c^2 D_{\rm s}/(4\pi G D_{\rm d} D_{\rm ds})$:

$$\kappa(\theta) = \frac{\sigma_v^2}{2G\,D_{\rm d}\sqrt{\theta^2+\theta_c^2}}\,\frac{4\pi G\,D_{\rm d}\,D_{\rm ds}}{c^2\,D_{\rm s}} = \frac{2\pi\sigma_v^2\,D_{\rm ds}}{c^2\,D_{\rm s}}\,\frac{1}{\sqrt{\theta^2+\theta_c^2}}\,.$$

Since $\theta_{\rm E} = 4\pi(\sigma_v/c)^2\,D_{\rm ds}/D_{\rm s}$, we get $2\pi\sigma_v^2 D_{\rm ds}/(c^2 D_{\rm s}) = \theta_{\rm E}/2$, and:

$$\kappa(\theta) = \frac{\theta_{\rm E}}{2\sqrt{\theta^2+\theta_c^2}}\,. \quad \checkmark$$

(b) As $\theta_c \to 0$: $\sqrt{\theta^2+\theta_c^2} \to \sqrt{\theta^2} = \theta$ (for $\theta > 0$), so $\kappa \to \theta_{\rm E}/(2\theta)$, the SIS result. ✓
(c) Multiple images require $\kappa(0) \geq 1$:

$$\kappa(0) = \frac{\theta_{\rm E}}{2\theta_c} \geq 1 \qquad \Longrightarrow \qquad \theta_c \leq \frac{\theta_{\rm E}}{2}\,.$$

(d) For $\theta_c = 0.1\,\theta_{\rm E}$:

$$\begin{aligned}
\kappa(0) &= \frac{\theta_{\rm E}}{2\times 0.1\,\theta_{\rm E}} = 5.0\,,\\
\kappa(\theta_{\rm E}) &= \frac{\theta_{\rm E}}{2\sqrt{\theta_{\rm E}^2+0.01\,\theta_{\rm E}^2}} = \frac{1}{2\sqrt{1.01}} \approx 0.498\,,\\
\kappa(2\theta_{\rm E}) &= \frac{1}{2\sqrt{4.01}} \approx 0.250\,.
\end{aligned}$$

(e) See the plot generated by `problems_07.wl`. As θ_c increases, the central convergence decreases and the profile flattens in the core region, while all profiles converge to the SIS behavior $\kappa = \theta_{\rm E}/(2\theta)$ at large θ.

Exercise 11.9.5: NFW Concentration and Lensing Strength

(a) The function $f(x)$ has a logarithmic divergence as $x \to 0$ (with $f(x) \sim (1/x)\ln(2/x)$), peaks near $x \sim 0.5$ in terms of the product $x\,f(x)$, and falls off as $f(x) \sim 1/x^2$ for $x \gg 1$. See the plot generated by `problems_07.wl`.

(b) For a cluster at $z_d = 0.3$ with $M_{200} = 10^{15}\,M_\odot$ and $z_s = 1$: the virial radius is $r_{200} \approx 2.1$ Mpc, so $r_s = r_{200}/c$, $\theta_s = r_s/D_{\rm d}$.

c	r_s (Mpc)	θ_s (arcsec)	κ_s
5	0.42	96	0.18
10	0.21	48	0.50
20	0.105	24	1.50

(Exact values depend on the cosmological parameters used; see `problems_07.wl`.)

(c) See the plot generated by `problems_07.wl`. Higher concentration pushes the $\kappa = 1$ crossing to larger angular radii, increasing the Einstein radius and the strong-lensing cross-section.

(d) (i) The Einstein radius increases with c because κ_s increases faster than θ_s decreases. (ii) The region where $\kappa > 1$ grows with c. (iii) The lensing cross-section for giant arcs (proportional to the area enclosed by the tangential caustic) increases strongly with c, roughly as c^2 for moderate concentrations. This means that concentration is a key parameter for predicting cluster arc statistics. ✓

Chapter 12

Non-Axisymmetric Models and Critical Curves

> *Breaking circular symmetry in the lens model produces the rich phenomenology of strong lensing: quadruply imaged quasars, Einstein crosses, giant luminous arcs. The structure of critical curves and caustics determines everything — how many images form, where they appear, how bright they are, and even which direction they face.* — adapted from Congdon & Keeton (2018)

In Modules 9 and 11 we developed the Jacobian formalism for magnification and applied it to axisymmetric lens models: the point mass and the singular isothermal sphere (SIS). These models have a continuous rotational symmetry that makes the mathematics tractable but the phenomenology limited. The SIS, for instance, can produce at most two images of a point source, and its critical curve is a perfect circle.

Real galaxies, however, are *elliptical*. Their mass distributions have axis ratios $q \sim 0.5$–0.9, and they live in environments that contribute external tidal shear. Breaking circular symmetry introduces qualitatively new phenomena:

- Critical curves deform from circles into ellipses and ovals.
- Caustics develop a **diamond (astroid)** shape with cusps and folds.
- Sources can be **quadruply imaged** (not just doubly), producing Einstein crosses and arcs.
- The **topology** of image configurations — how many images exist, their parities, and their magnifications — is entirely governed by the source's position relative to the caustic structure.

This module introduces the two workhorse non-axisymmetric models (SIS + external shear and the singular isothermal ellipsoid) and then develops the theory of critical curves, caustics, and image topology in full detail. The image topology section is the conceptual heart of strong lensing theory.

Companion Mathematica files:

- `critical_curves_caustics.wl` — symbolic and numerical computation of critical curves and caustics for SIS+shear and SIE models; image position solver; figure generation for all module figures

- `image_configurations.nb` — interactive notebook: place a source on the source plane and watch images appear, colored by parity; animate caustic crossings; sliders for ellipticity and shear

12.1 Introduction: Why Non-Axisymmetric Models Matter

The singular isothermal sphere (Module 11) is an excellent starting point for galaxy-scale lensing. Its velocity dispersion σ_v maps directly onto observable galaxy kinematics, and its Einstein radius $\theta_{\rm E} = 4\pi\sigma_v^2 D_{\rm ds}/(c^2 D_{\rm s})$ sets the angular scale of lensing. But the SIS has a fundamental limitation: perfect circular symmetry.

Observations of strong lens systems reveal:

1. **Quadruple images** are common. About 20–30% of known galaxy-scale lens systems are quads (four bright images plus a faint central image), which an SIS cannot produce.

2. **Giant luminous arcs** in galaxy clusters are tangentially stretched images that trace out the critical curves of the cluster potential. These arcs are not perfect circles, implying the mass distribution is not axisymmetric.

3. **Astrometric anomalies** in lens models require ellipticity and/or external shear to achieve acceptable fits.

The simplest way to break axial symmetry is to add a constant **external shear** to the SIS potential. The resulting SIS + shear model already captures the essential features: diamond caustics, quad images, and fold/cusp configurations. The more physically motivated **singular isothermal ellipsoid** (SIE) achieves the same through an intrinsically elliptical mass distribution and is the model used in modern lens-modeling codes such as `lenstronomy` and `PyAutoLens`.

12.2 SIS + External Shear

The simplest non-axisymmetric lens model augments the SIS potential with a constant external tidal shear $\gamma_{\rm ext}$. Physically, this shear arises from the tidal gravitational field of neighboring galaxies or the cluster environment in which the lens galaxy resides.

12.2.1 Lensing Potential

The lensing potential is (Congdon & Keeton eq. 6.7):

$$\boxed{\psi(\boldsymbol{\theta}) = \theta_{\rm E}\,|\boldsymbol{\theta}| + \frac{1}{2}\,\gamma_{\rm ext}\,(\theta_1^2 - \theta_2^2)} \tag{12.1}$$

where $|\boldsymbol{\theta}| = \sqrt{\theta_1^2 + \theta_2^2}$ and we have chosen the shear to be aligned with the θ_1-axis (the general case involves a rotation by the shear angle ϕ_γ). The first term is the SIS potential; the second is the external shear contribution.

12.2.2 Deflection Angle

The deflection angle components are:

$$\alpha_1 = \frac{\partial \psi}{\partial \theta_1} = \theta_{\mathrm{E}} \frac{\theta_1}{|\boldsymbol{\theta}|} + \gamma_{\mathrm{ext}}\, \theta_1 \,, \qquad \alpha_2 = \frac{\partial \psi}{\partial \theta_2} = \theta_{\mathrm{E}} \frac{\theta_2}{|\boldsymbol{\theta}|} - \gamma_{\mathrm{ext}}\, \theta_2 \,. \tag{12.2}$$

The SIS contribution deflects radially inward; the shear contribution stretches along θ_1 and compresses along θ_2 (for $\gamma_{\mathrm{ext}} > 0$).

12.2.3 Convergence and Shear

The convergence of the SIS + shear model is the same as the pure SIS (the external shear does not add mass):

$$\kappa(\boldsymbol{\theta}) = \frac{\theta_{\mathrm{E}}}{2\,|\boldsymbol{\theta}|} \,. \tag{12.3}$$

The total shear now has contributions from both the SIS and the external shear. For the SIS alone, $|\gamma_{\mathrm{SIS}}| = \theta_{\mathrm{E}}/(2|\boldsymbol{\theta}|) = \kappa$. The external shear adds γ_{ext} aligned with the coordinate axes, so the total shear field is anisotropic and position-dependent.

12.2.4 Critical Curves

The critical curves are defined by $\det \mathcal{A} = 0$. For the SIS + shear model, the condition becomes (working in polar coordinates $\theta_1 = \theta \cos\phi$, $\theta_2 = \theta \sin\phi$):

$$\boxed{\theta_{\mathrm{crit}}(\phi) = \frac{\theta_{\mathrm{E}}}{1 - \gamma_{\mathrm{ext}}^2 - 2\,\gamma_{\mathrm{ext}} \cos 2\phi} \cdot (1 - \gamma_{\mathrm{ext}} \cos 2\phi)} \tag{12.4}$$

For $\gamma_{\mathrm{ext}} = 0$ this reduces to $\theta = \theta_{\mathrm{E}}$ (the SIS critical circle). For $\gamma_{\mathrm{ext}} > 0$, the critical curve is an **ellipse**: elongated along the θ_2-axis (perpendicular to the shear direction) and compressed along θ_1.

Remark 12.2.1. The critical curve is elongated *perpendicular* to the shear direction. This is physically intuitive: the shear stretches images along θ_1, which means the magnification diverges at smaller θ in that direction, pushing the critical curve inward along θ_1 and outward along θ_2.

12.2.5 Caustics: The Diamond (Astroid)

The caustics are obtained by mapping the critical curves through the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta}_{\mathrm{crit}} - \boldsymbol{\alpha}(\boldsymbol{\theta}_{\mathrm{crit}})$. For the SIS + shear model, the tangential caustic takes the shape of a **diamond** or **astroid** — a four-cusped curve.

In parametric form, the caustic is:

$$\beta_1(\phi) = \theta_{\mathrm{crit}}(\phi)\cos\phi - \theta_{\mathrm{E}}\cos\phi - \gamma_{\mathrm{ext}}\,\theta_{\mathrm{crit}}(\phi)\cos\phi \,, \tag{12.5}$$

$$\beta_2(\phi) = \theta_{\mathrm{crit}}(\phi)\sin\phi - \theta_{\mathrm{E}}\sin\phi + \gamma_{\mathrm{ext}}\,\theta_{\mathrm{crit}}(\phi)\sin\phi \,. \tag{12.6}$$

This curve has four cusps (at $\phi = 0, \pi/2, \pi, 3\pi/2$) and four fold segments connecting them. The size of the diamond is proportional to γ_{ext}: for $\gamma_{\mathrm{ext}} = 0$, the caustic degenerates to a point (the SIS point caustic), and for larger shear values, the diamond grows.

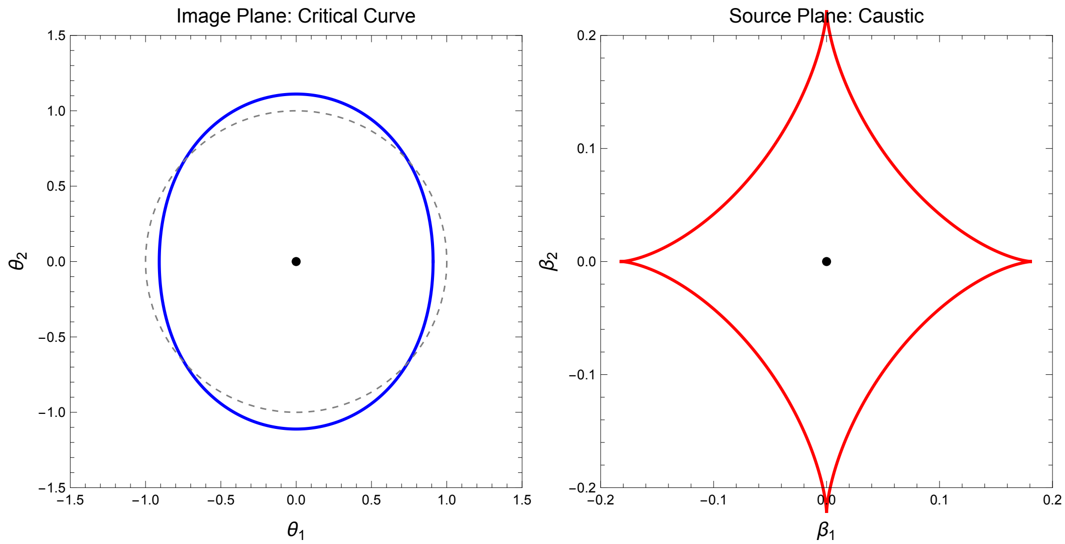


Figure 12.1: Critical curves (left, image plane) and caustics (right, source plane) for the SIS + external shear model with $\theta_E = 1''$ and $\gamma_{ext} = 0.1$. The critical curve is an ellipse; the caustic is a diamond (astroid) with four cusps. The shear axis is horizontal. Generated by `critical_curves_caustics.wl`.

12.2.6 Image Configurations

The image configurations for the SIS + shear model depend on the source position relative to the diamond caustic:

- **Source outside the diamond:** Two images (a double). One image is outside the critical curve (Type I, positive parity); the other is inside (Type II, negative parity, demagnified).

- **Source inside the diamond:** Four images (a quad). Two images outside the critical curve (positive parity); two inside (negative parity).

- **Source on a fold:** Three images, with two of them merging on the critical curve (highly magnified).

- **Source on a cusp:** Three images, with the merging triplet producing a characteristic cusp configuration.

Remark 12.2.2. The SIS is singular, so there is no separate radial critical curve or radial caustic. The SIS + shear model therefore has only a tangential caustic (the diamond). A non-singular model (e.g., a softened isothermal sphere + shear) would also have a radial caustic enclosing the tangential one.

12.3 The Singular Isothermal Ellipsoid (SIE)

The **singular isothermal ellipsoid** (SIE) is the workhorse model for galaxy-scale strong lensing. Rather than adding shear externally, the SIE builds ellipticity directly into the mass distribution.

12.3.1 Surface Mass Density

The SIE surface mass density is (Kormann, Schneider & Bartelmann 1994; Congdon & Keeton eq. 6.15):

$$\boxed{\Sigma(\xi_1, \xi_2) = \frac{\sigma_v^2}{2G} \frac{\sqrt{q}}{\sqrt{q^2\,\xi_1^2 + \xi_2^2}}} \tag{12.7}$$

where $q \leq 1$ is the axis ratio ($q = b/a$, with a the semi-major axis aligned along ξ_1), σ_v is the velocity dispersion, and (ξ_1, ξ_2) are physical coordinates in the lens plane. The iso-density contours $q^2\xi_1^2 + \xi_2^2 = \text{const}$ are ellipses of axis ratio q with the major axis along ξ_1. For $q = 1$ this reduces to the SIS surface mass density $\Sigma = \sigma_v^2/(2G\xi)$.

The factor $\sqrt{q}$ ensures that the total mass enclosed within a given elliptical radius is consistent with the isothermal profile.

Converting to angular coordinates via $\boldsymbol{\xi} = D_\mathrm{d}\boldsymbol{\theta}$ and dividing by the critical surface density gives the dimensionless convergence

$$\kappa(\theta_1, \theta_2) \;=\; \frac{\Sigma}{\Sigma_\mathrm{cr}} \;=\; \frac{\theta_\mathrm{E}\sqrt{q}}{2\sqrt{q^2\theta_1^2 + \theta_2^2}}\,, \tag{12.8}$$

where $\theta_\mathrm{E} = 4\pi\sigma_v^2 D_\mathrm{ds}/(c^2 D_\mathrm{s})$ is the SIS-equivalent Einstein radius (the Einstein radius of the circular isothermal sphere with the same σ_v).

12.3.2 Deflection Angle

The deflection angle for the SIE, first given in closed form by Kormann, Schneider & Bartelmann (1994, hereafter KSB94), is

$$\boxed{\alpha_1 = \frac{\theta_\mathrm{E}\,\sqrt{q}}{\sqrt{1-q^2}}\,\arctan\!\left(\frac{\theta_1\,\sqrt{1-q^2}}{\sqrt{q^2\theta_1^2 + \theta_2^2}}\right), \qquad \alpha_2 = \frac{\theta_\mathrm{E}\,\sqrt{q}}{\sqrt{1-q^2}}\,\operatorname{arctanh}\!\left(\frac{\theta_2\,\sqrt{1-q^2}}{\sqrt{q^2\theta_1^2 + \theta_2^2}}\right).} \tag{12.9}$$

The asymmetry — arctan along the major axis, arctanh along the minor — is intrinsic to the elliptical geometry. Rather than reproducing the long convolution integral that leads to these forms (a contour-integral computation spanning several pages of KSB94), we give a shorter derivation that shows eq. (12.9) is consistent with the SIE convergence (12.8) by using the scale-invariance of the lens. Every step is verified symbolically in `sie_deflection.wl`.

Step 1: scale invariance and the lensing potential. The SIE convergence (12.8) is homogeneous of degree -1 in $\boldsymbol{\theta}$,

$$\kappa(\lambda\boldsymbol{\theta}) = \lambda^{-1}\kappa(\boldsymbol{\theta}) \qquad (\lambda > 0).$$

Since $\alpha_i = \partial\psi/\partial\theta_i$ and $\boldsymbol{\alpha} = \kappa * \boldsymbol{\theta}/\pi$ is a convolution with a homogeneous-of-degree--1 kernel, the deflection is homogeneous of degree 0 and the potential ψ is homogeneous of degree 1. Euler's theorem for homogeneous functions then gives

$$\psi(\theta_1, \theta_2) \;=\; \theta_1\,\alpha_1(\theta_1, \theta_2) + \theta_2\,\alpha_2(\theta_1, \theta_2)\,. \tag{12.10}$$

Step 2: ansatz and its structure. For the SIE, any deflection consistent with the elliptical symmetry must involve only θ_1, θ_2, and the elliptical radius $F(\boldsymbol{\theta}) \equiv \sqrt{q^2\theta_1^2 + \theta_2^2}$. Dimensional and homogeneity requirements force the argument of the inverse trig/hyperbolic function to be dimensionless and of degree 0. Natural choices are $\theta_1\sqrt{1-q^2}/F$ and $\theta_2\sqrt{1-q^2}/F$ — the coefficients $\sqrt{1-q^2}$ are fixed uniquely (up to sign) by the requirement that $\alpha_i \to \theta_{\mathrm{E}}\theta_i/|\boldsymbol{\theta}|$ in the SIS limit $q \to 1$. This motivates the KSB94 ansatz (12.9).

Step 3: verify the gradient condition $\alpha_i = \partial_i\psi$. Combining (12.10) and (12.9) and differentiating gives

$$\frac{\partial\psi}{\partial\theta_i} \;=\; \alpha_i + \theta_j\,\frac{\partial\alpha_j}{\partial\theta_i}\,,$$

so the gradient condition is equivalent to $\theta_j(\partial\alpha_j/\partial\theta_i) = 0$. Direct differentiation of (12.9) shows that all cross terms cancel after using the identity $\partial_i(1/F) = -q^{2(1-\delta_{i2})}\theta_i/F^3$ and the derivative of arctan/arctanh. The detailed cancellation is carried out symbolically in `sie_deflection.wl`; the result is

$$\frac{\partial\psi}{\partial\theta_1} = \alpha_1, \qquad \frac{\partial\psi}{\partial\theta_2} = \alpha_2. \tag{12.11}$$

Step 4: verify $\nabla^2\psi = 2\kappa$. Computing $\nabla^2\psi = \partial_1\alpha_1 + \partial_2\alpha_2$ from (12.9):

$$\begin{aligned}
\partial_1\alpha_1 &= \frac{\theta_{\mathrm{E}}\sqrt{q}}{\sqrt{1-q^2}}\,\frac{\partial}{\partial\theta_1}\arctan\!\left(\frac{\theta_1\sqrt{1-q^2}}{F}\right) = \frac{\theta_{\mathrm{E}}\sqrt{q}}{\sqrt{1-q^2}}\cdot\frac{\sqrt{1-q^2}}{F}\cdot\frac{\theta_2^2}{q^2\theta_1^2+\theta_2^2}\\
&= \frac{\theta_{\mathrm{E}}\sqrt{q}\,\theta_2^2}{F^3},\\
\partial_2\alpha_2 &= \frac{\theta_{\mathrm{E}}\sqrt{q}\,q^2\theta_1^2}{F^3},
\end{aligned}$$

after applying the chain rule to $\partial_i F = q^{2(1-\delta_{i2})}\theta_i/F$ and simplifying. Adding:

$$\nabla^2\psi \;=\; \frac{\theta_{\mathrm{E}}\sqrt{q}\,(q^2\theta_1^2+\theta_2^2)}{F^3} \;=\; \frac{\theta_{\mathrm{E}}\sqrt{q}}{F} \;=\; 2\,\kappa(\boldsymbol{\theta}), \tag{12.12}$$

matching (12.8). This completes the verification.

Step 5: the SIS limit $q \to 1$. As $q \to 1^-$, $\sqrt{1-q^2} \to 0$ and the arguments of the inverse functions vanish; using $\arctan x/x \to 1$ and $\operatorname{arctanh} x/x \to 1$,

$$\alpha_i \;\to\; \theta_{\rm E}\sqrt{q}\,\frac{\theta_i}{\sqrt{q^2\theta_1^2+\theta_2^2}} \;\to\; \theta_{\rm E}\,\frac{\theta_i}{|\boldsymbol{\theta}|}\,,$$

which is the SIS deflection. The limit is taken symbolically in `sie_deflection.wl`.

12.3.3 Critical Curves and Caustics of the SIE

The critical curves of the SIE are found by solving $\det\mathcal{A}(\boldsymbol{\theta}) = 0$. Unlike the SIS (where the critical curve is a circle), the SIE critical curves are smooth oval curves whose shape depends on q.

For axis ratios $q \lesssim 0.9$, the SIE has:

- A **tangential critical curve**: a smooth oval close to the Einstein radius, elongated perpendicular to the mass distribution's major axis.
- A **tangential caustic**: a diamond (astroid) shape, similar to the SIS + shear case, with four cusps.

The SIE, being singular at the origin, does not have a radial critical curve (just as the SIS does not). A non-singular isothermal ellipsoid would also have a radial critical curve mapping to a radial caustic.

The cross-section for quad lensing (the area enclosed by the tangential caustic) increases with ellipticity. For $q = 0.7$, the caustic area is roughly $0.04\,\theta_{\rm E}^2$, while for $q = 0.5$ it can be several times larger. This is why more elliptical galaxies are more likely to produce quad lens systems.

12.4 Critical Curves

We now develop the general theory of critical curves, building on the preview in Module 9 (Section 9.10).

Definition 12.4.1. The **critical curves** are the loci in the image plane where the Jacobian determinant vanishes:

$$\boxed{\det\mathcal{A}(\boldsymbol{\theta}) = 0 \qquad \Longleftrightarrow \qquad (1-\kappa)^2 - |\gamma|^2 = 0} \tag{12.13}$$

Equivalently, $\mu(\boldsymbol{\theta}) \to \infty$ on critical curves.

12.4.1 Two Families of Critical Curves

Using the eigenvalues $\lambda_\pm = 1-\kappa \pm |\gamma|$ of the Jacobian (Module 9, eq. 9.14), the condition $\det\mathcal{A} = \lambda_+\,\lambda_- = 0$ is satisfied when either eigenvalue vanishes:

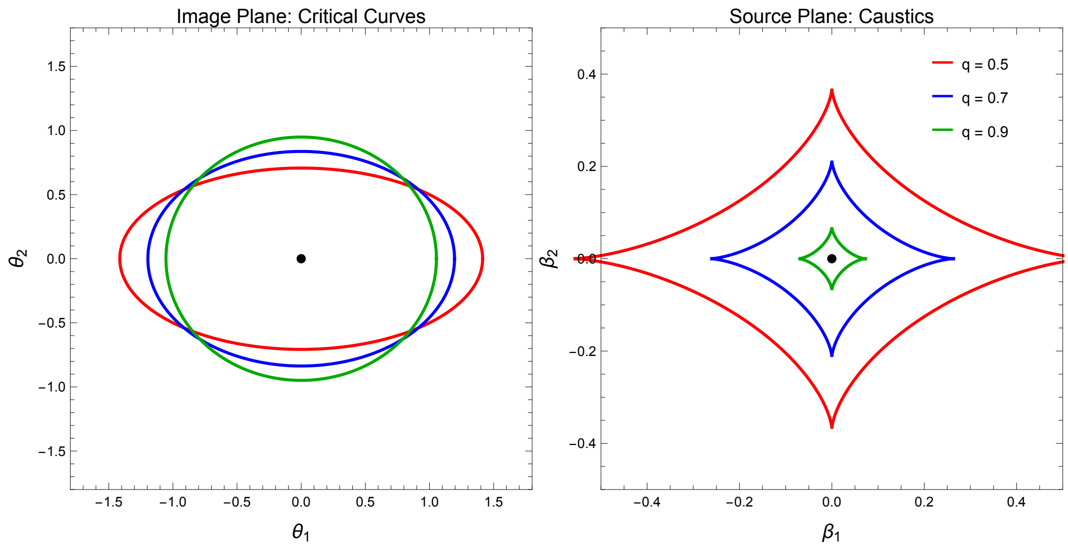


Figure 12.2: Critical curves (left) and caustics (right) for the SIE model with $\theta_E = 1''$ and axis ratios $q = 0.5$, 0.7, and 0.9. As q decreases (more elliptical), the diamond caustic grows larger, increasing the cross-section for quad imaging. Generated by `critical_curves_caustics.wl`.

1. **Tangential critical curve** ($\lambda_- = 0$):

$$1 - \kappa(\boldsymbol{\theta}) = |\gamma(\boldsymbol{\theta})| \tag{12.14}$$

   Images near this curve are stretched *tangentially* (along the curve), because the eigenvalue corresponding to the tangential direction vanishes. Giant arcs in clusters trace the tangential critical curve.

2. **Radial critical curve** ($\lambda_+ = 0$):

$$1 - \kappa(\boldsymbol{\theta}) = -|\gamma(\boldsymbol{\theta})| \tag{12.15}$$

   Images near this curve are stretched *radially* (perpendicular to the curve). Radial arcs (pointing toward the lens center) form near radial critical curves, though they are rarer and fainter than tangential arcs.

12.4.2 Critical Curves for Axisymmetric vs. Elliptical Lenses

- **Axisymmetric lenses:** The critical curves are circles centered on the lens. For the SIS, there is only a tangential critical curve at $\theta = \theta_E$. For the point mass, the critical curve is the Einstein ring.

- **Elliptical lenses:** The critical curves become ellipses or ovals. The tangential critical curve is typically elongated perpendicular to the lens ellipticity axis. The shapes are computed numerically by solving $\det \mathcal{A} = 0$ on a grid or parametrically.

12.4.3 Physical Meaning

Critical curves are the sites of maximum magnification. Images that form near a critical curve are:

1. **Highly magnified:** $|\mu| \gg 1$, because $\det \mathcal{A} \approx 0$.

2. **Highly stretched:** Along one direction (tangential or radial), the magnification diverges while it remains finite in the orthogonal direction. This produces the elongated arc morphology.

3. **On parity boundaries:** As we cross a critical curve, $\det \mathcal{A}$ changes sign, so the image parity flips. One side has positive parity (Type I); the other has negative parity (Type II).

The practical significance is enormous: critical curves are where we look for the most highly magnified background sources. Cluster lensing surveys map out critical curves to find magnified galaxies at very high redshift.

12.5 Caustics

Definition 12.5.1. The **caustics** are the images of the critical curves under the lens mapping. That is, the caustics are the curves in the *source plane* obtained by applying the lens equation to each point on a critical curve:

$$\boxed{\boldsymbol{\beta}_{\text{caustic}} = \boldsymbol{\theta}_{\text{crit}} - \boldsymbol{\alpha}(\boldsymbol{\theta}_{\text{crit}})} \tag{12.16}$$

12.5.1 Types of Caustics

Corresponding to the two families of critical curves, there are two types of caustics:

1. **Tangential caustic:** The mapping of the tangential critical curve. For elliptical lenses, this is typically a diamond (astroid) shape with four cusps. Sources inside the tangential caustic are quadruply imaged.

2. **Radial caustic:** The mapping of the radial critical curve. This is typically a smooth, roughly elliptical curve that encloses the tangential caustic. Sources inside the radial caustic but outside the tangential caustic are doubly imaged; sources outside both caustics have a single image.

12.5.2 Caustics for Specific Models

- **Point mass:** The tangential critical curve (Einstein ring $\theta = \theta_{\mathrm{E}}$) maps to a single point ($\beta = 0$) — a **degenerate point caustic**. A source at $\beta = 0$ produces the Einstein ring; any source at $\beta > 0$ produces exactly two images.

- **SIS:** Same as the point mass — the critical curve $\theta = \theta_E$ maps to $\beta = 0$. The SIS has no radial caustic. Sources with $\beta < \theta_E$ produce two images; sources with $\beta > \theta_E$ produce one image.
- **SIS + shear:** The tangential caustic opens up into a diamond (astroid) with four cusps. The diamond size is proportional to $\gamma_{\rm ext}$.
- **SIE:** Same diamond structure as SIS + shear, with the diamond size controlled by the ellipticity $(1 - q)$.
- **Non-singular models** (NIS, NFW): Both tangential and radial caustics are present. The radial caustic encloses the tangential one, creating regions with 1, 3, and 5 images.

12.6 Image Topology

This section is the conceptual heart of strong gravitational lensing. It explains *why* images appear and disappear as a source moves, why they always come in pairs, and how the full image configuration (positions, parities, magnifications) is determined entirely by the source's location relative to the caustic structure.

The key insight is that **caustics are the boundaries between regions of different image multiplicity**, and **critical curves are the boundaries between regions of different image parity**. Together, they provide a complete topological description of the lens mapping.

12.6.1 Critical Curves as Parity Boundaries

Recall from Module 9 that the magnification is $\mu = 1/\det \mathcal{A}$ and that the sign of μ determines the image parity:

- $\det \mathcal{A} > 0 \;\Rightarrow\; \mu > 0$: **positive parity** (Type I image). The image has the same orientation as the source.
- $\det \mathcal{A} < 0 \;\Rightarrow\; \mu < 0$: **negative parity** (Type II image). The image is mirror-reflected relative to the source.

Since $\det \mathcal{A}$ is a continuous function of position that passes through zero on critical curves, the sign of $\det \mathcal{A}$ must *change* across a critical curve. Therefore:

Proposition 12.6.1. ***Critical curves are parity boundaries.*** *Images on opposite sides of a critical curve have opposite parities. If a critical curve separates region R_+ ($\det \mathcal{A} > 0$) from region R_- ($\det \mathcal{A} < 0$), then:*

- *Images in R_+ have positive parity (Type I, minima of the arrival time surface).*
- *Images in R_- have negative parity (Type II, saddle points of the arrival time surface).*

For a typical galaxy lens with an elliptical mass distribution:

- The region **outside** the tangential critical curve has $\det \mathcal{A} > 0$: images here are Type I (positive parity, time-delay minima).
- The region **between** the tangential and radial critical curves has $\det \mathcal{A} < 0$: images here are Type II (negative parity, saddle points).
- The region **inside** the radial critical curve has $\det \mathcal{A} > 0$ again: images here are Type I or Type III (positive parity, but corresponding to time-delay maxima for Type III, as discussed in Module 10).

Remark 12.6.1. The connection to Morse theory (Module 10) is direct. The three image types correspond to the three types of stationary points of the Fermat potential: **Type I** = minimum, **Type II** = saddle, **Type III** = maximum. The magnification parity determines the Morse index.

12.6.2 Caustic Crossings and Image Creation/Destruction

The most important property of caustics is:

Proposition 12.6.2. ***Image creation/destruction at caustics.*** *When a source crosses a caustic, the number of images changes by exactly two. A pair of images either* ***appears*** *(as the source enters a higher-multiplicity region) or* ***disappears*** *(as it exits).*

This happens because, as a source approaches a caustic from the side with fewer images, two of the "potential" image positions on opposite sides of the corresponding critical curve converge toward the critical curve. At the moment the source is exactly on the caustic, the two images merge *on* the critical curve (where $\det \mathcal{A} = 0$ and the magnification diverges). As the source crosses the caustic, the two images separate and become distinct.

Crucially, the two images that appear or disappear:

1. Are located on **opposite sides** of the critical curve.
2. Have **opposite parities** (one Type I, one Type II).
3. Are both **highly magnified** near the caustic (their magnifications diverge as they approach the critical curve).
4. Have **nearly equal magnifications** (in absolute value) near the caustic.

Remark 12.6.2. The fact that images always appear or disappear in **pairs** with opposite parities is a topological necessity. It preserves **Burke's odd-number theorem** (Module 10): the total number of images is always odd (for a non-singular, transparent lens with a source inside the outermost caustic). Since images are created and destroyed in pairs, the parity of the count (odd or even) never changes.

12.6.3 Walking a Source Across Caustics: A Worked Example

To make the image creation/destruction process concrete, we walk through a complete example using an SIE lens with axis ratio $q = 0.7$ and Einstein radius $\theta_{\mathrm{E}} = 1''$. The SIE has a tangential caustic (diamond shape) in the source plane. For a non-singular model there would also be a radial caustic enclosing the diamond, but for the SIE (which is singular at the center) we focus on the tangential caustic.

Consider a source that starts far from the lens and moves inward, crossing the caustics:

Stage 1: Source far outside all caustics. The source position $\boldsymbol{\beta}$ is far from the lens center, well outside any caustic. There is **one image** (for singular models like the SIE, there are actually two images even here, but the second is extremely demagnified near the lens center and is often unobservable). The observable image is:

- **Type I** (positive parity, minimum of the arrival time).
- Located outside the tangential critical curve, near the true source position.
- Weakly magnified ($|\mu| \approx 1$).

Stage 2: Source crosses the tangential caustic (outside $\to$ inside). As the source reaches the tangential (diamond) caustic, two new images appear simultaneously, straddling the tangential critical curve. The total count goes from **1 to 3 images** (or effectively from 2 to 4, counting the faint central image):

1. The **original Type I image** (positive parity, outside the tangential critical curve) persists, moving slightly.
2. A **new Type II image** (negative parity) appears just inside the tangential critical curve. This is a *saddle point* of the arrival time surface. It is highly magnified and tangentially stretched.
3. A **new Type I image** (positive parity) appears just outside the tangential critical curve, close to the Type II image. This is a *minimum* of the arrival time surface. It is also highly magnified.

The two new images are born as a highly magnified, closely spaced pair straddling the critical curve. They have opposite parities (one positive, one negative) and nearly equal $|\mu|$. As the source moves further inside the caustic, they separate and their magnifications decrease.

Stage 3: Source inside the tangential caustic (quad configuration). With the source well inside the diamond caustic, four bright images surround the lens (plus a faint, highly demagnified central image in non-singular models). The typical configuration is an **Einstein cross**:

- Two Type I images (positive parity, outside the tangential critical curve).

- Two Type II images (negative parity, inside the tangential critical curve).

The four images form a roughly cross-shaped pattern around the lens. The magnification depends on the source's proximity to the caustic folds and cusps.

Stage 4: Source crosses the tangential caustic again (inside → outside). As the source exits the diamond caustic, two images converge on the critical curve, merge, and disappear. The count drops from **3 back to 1** (or 4 to 2 counting the central image). The two images that disappear are a Type I/Type II pair, just as in the creation event.

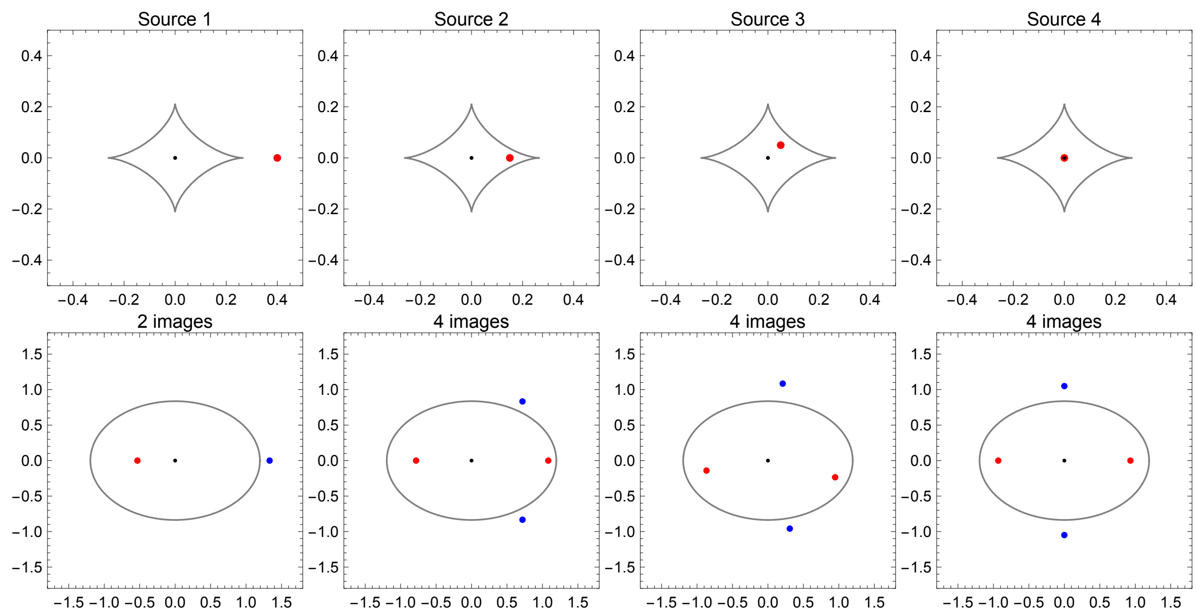


Figure 12.3: A source walks across the caustic of an SIE lens ($q = 0.7$). **Top row:** Source positions (red star) in the source plane, with the diamond caustic shown. **Bottom row:** Corresponding image configurations in the image plane, with the critical curve shown. Blue circles = Type I images (positive parity); red squares = Type II images (negative parity). As the source crosses the caustic, images appear/disappear in pairs. Generated by `critical_curves_caustics.wl`.

For a non-singular lens model (e.g., a softened isothermal ellipsoid), there is also a **radial caustic** enclosing the tangential one. As the source crosses the radial caustic inward, two additional images appear near the radial critical curve (near the lens center), bringing the total from 3 to 5. This gives the complete sequence:

Region	Image count	Types
Outside radial caustic	1	I
Between radial & tangential	3	I, II, III
Inside tangential caustic	5	I, I, II, II, III

Here Type III is a maximum of the arrival time (the highly demagnified central image).

12.6.4 The Fold and Cusp Catastrophes

The caustic curves of a generic smooth lens consist of two types of points: **folds** (smooth segments) and **cusps** (pointed corners). These are the only two generic singularities of smooth two-dimensional mappings, a result from **catastrophe theory** (or singularity theory; see Petters, Levine & Wambsganss 2001 for a thorough treatment).

Fold Caustics

Near a **fold** (a smooth segment of the caustic), the following properties hold:

1. **Two images merge.** As a source crosses a fold caustic, two images appear or disappear. The images straddle the critical curve, one on each side.

2. **Magnification scaling.** Near a fold, the magnification of each of the two merging images scales as:
$$\boxed{|\mu| \propto \frac{1}{\sqrt{d_\perp}}} \tag{12.17}$$
where $d_\perp$ is the perpendicular distance of the source from the caustic. The magnification diverges as $d_\perp \to 0$ (source on the caustic), but only as an inverse square root — a relatively mild divergence.

3. **Equal and opposite magnifications.** The two merging images have magnifications that approach $+|\mu_0|$ and $-|\mu_0|$ (equal absolute values, opposite signs), so their *total signed magnification* approaches zero. The unsigned total $2|\mu_0| \propto 1/\sqrt{d_\perp}$ diverges.

4. **Tangential stretching.** The merging images are highly elongated tangentially (along the critical curve) and compressed radially. This is the origin of the "arc" morphology.

Cusp Caustics

Near a **cusp** (a pointed corner of the caustic), the behavior is richer:

1. **Three images merge.** As a source approaches a cusp from inside the caustic, three images converge toward a single point on the critical curve. These form the characteristic **cusp triplet**.

2. **Stronger magnification divergence.** Near a cusp, the total magnification diverges more strongly than for a fold. The scaling is:
$$\mu_{\text{total}} \propto d_\perp^{-1} \tag{12.18}$$
(compared to $d_\perp^{-1/2}$ for a fold).

3. **Cusp relation.** The magnifications of the three cusp images satisfy a relation:
$$\mu_A + \mu_B + \mu_C \approx 0 \tag{12.19}$$

where A, B, C are the three images of the cusp triplet (with signed magnifications). That is, the sum of the signed magnifications of the three brightest images in a cusp configuration should vanish. Violations of this relation indicate substructure (e.g., dark matter subhalos) perturbing the images.

4. **Configuration.** The three cusp images are closely spaced along the critical curve. One is on the opposite side of the critical curve from the other two, ensuring opposite parities.

Remark 12.6.3. The fold and cusp are the only **stable** (generic) singularities of smooth two-dimensional maps. Higher-order singularities (swallowtails, lips, etc.) require special symmetries or fine-tuning and are not generically present in gravitational lens models. This is a deep result of Whitney's theorem in singularity theory (Petters et al. 2001, Ch. 5–6).

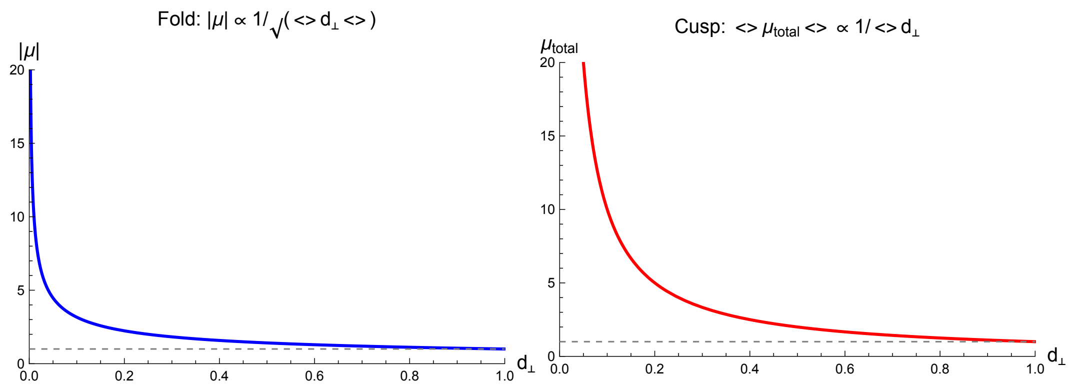


Figure 12.4: Magnification behavior near fold and cusp caustics. **Left:** Near a fold, two images straddle the critical curve with magnifications scaling as $|\mu| \propto 1/\sqrt{d_\perp}$. **Right:** Near a cusp, three images merge, with stronger divergence. The cusp relation $\mu_A + \mu_B + \mu_C \approx 0$ holds for the signed magnifications. Generated by `critical_curves_caustics.wl`.

12.6.5 Image Configuration Taxonomy

The caustic structure of a lens determines all possible image configurations. For an elliptical galaxy lens (SIE or similar model), the following taxonomy applies:

Doubles

Source outside the tangential caustic. Two images form (or effectively one bright image plus one highly demagnified image near the lens center for singular models):

- One Type I image (positive parity, outside the critical curve): the dominant, brighter image.
- One Type II image (negative parity, inside the critical curve): typically fainter, on the opposite side of the lens from the source.

Doubles account for ∼70–80% of known galaxy-scale lens systems.

Quads / Einstein Crosses

Source inside the tangential caustic. Four bright images form (plus a faint central image):

- Two Type I images (positive parity), roughly on the side of the lens closer to the source.
- Two Type II images (negative parity), on the opposite side.

The four images form a roughly cross-shaped (**Einstein cross**) or diamond-shaped pattern around the lens. Quads account for ∼20–30% of galaxy-scale systems and are extremely valuable for cosmography because they provide three independent time delays.

Cusp Configurations

Source near a cusp of the tangential caustic. Three of the four images in a quad merge into a closely spaced triplet near one point on the critical curve. This produces the **naked cusp** or **cusp triplet** configuration:

- Three closely spaced images near the cusp of the critical curve, with the cusp relation $\mu_A + \mu_B + \mu_C \approx 0$.
- One isolated image on the opposite side of the lens.

Why Most Observed Quads Have a Diamond Caustic

The tangential caustic of an elliptical galaxy lens is generically a diamond (astroid) shape. This is because:

1. The dominant perturbation from circular symmetry is quadrupolar (ellipticity or shear), which produces exactly four cusps.
2. Higher-order multipoles (hexapole, octupole) are typically subdominant and do not change the topology of the caustic.
3. The diamond caustic produces Einstein crosses (quads with roughly equal magnifications) when the source is near the center, and fold/cusp configurations when the source is near a fold or cusp of the diamond.

12.6.6 Why Giant Arcs Form

One of the most spectacular phenomena in strong lensing is the formation of **giant luminous arcs** in galaxy clusters. The theory of critical curves and caustics provides a complete explanation.

Step 1: Sources near the tangential caustic. Giant arcs form when a background galaxy lies very close to (but just inside or outside) the tangential caustic. At this location, one or two of its images lie very close to the tangential critical curve.

Step 2: Tangential stretching near the critical curve. The tangential critical curve is where $\lambda_- = 0$ (the eigenvalue corresponding to the tangential direction vanishes). Near this curve, images are stretched enormously in the *tangential* direction (along the critical curve) while remaining relatively unstretched in the radial direction.

Quantitatively, if the source is at perpendicular distance $d_\perp$ from the tangential caustic, the tangential magnification scales as:

$$\mu_{\tan} \propto \frac{1}{\sqrt{d_\perp}} \tag{12.20}$$

while the radial magnification remains of order unity. The result is an image that is stretched by a factor of $\mu_{\tan}$ along the critical curve but not radially — an **arc**.

Step 3: Arc length and curvature. Since the arc follows the tangential critical curve, its curvature matches that of the critical curve. For a galaxy cluster, the tangential critical curve has a radius of ∼10–30 arcsec, so giant arcs appear as curved features with large opening angles. The length-to-width ratio of the arc is approximately $\mu_{\tan} \sim 1/\sqrt{d_\perp}$, which can easily reach 10:1 or more for sources very close to the caustic.

Step 4: Multiple arcs. A source just inside the tangential caustic produces two highly magnified images on opposite sides of the tangential critical curve (a fold pair). Both images are arcs. If the source is also near a cusp, three images merge into a very long arc (a "cusp arc").

Remark 12.6.4. The largest known arcs in galaxy clusters have length-to-width ratios exceeding 10:1 and total magnifications of $\mu \sim 20$–50. These extreme magnifications are used to study intrinsically faint galaxies at high redshift ("cosmic telescopes").

12.7 Image Configurations: Doubles, Quads, and Einstein Crosses

We now illustrate the different image configurations with explicit examples for an SIE lens.

12.7.1 Double Configuration

For a source at $\boldsymbol{\beta} = (0.5, 0.0)''$ lensed by an SIE with $\theta_{\rm E} = 1''$, $q = 0.7$: The source is outside the tangential caustic. Two images form:

- Image A at $\boldsymbol{\theta}_A \approx (1.35, 0.0)''$: Type I, positive parity, $\mu_A \approx 2.5$.
- Image B at $\boldsymbol{\theta}_B \approx (-0.65, 0.0)''$: Type II, negative parity, $\mu_B \approx -0.9$.

The brighter image (A) is on the same side as the source; the fainter image (B) is on the opposite side.

12.7.2 Quad / Einstein Cross Configuration

For a source at $\boldsymbol{\beta} = (0.05, 0.05)''$ (inside the tangential caustic): Four bright images form, arranged in a cross pattern. The two Type I images tend to be on the side closer to the source; the two Type II images are on the opposite side. The magnifications of the four images are comparable, with the images nearer the critical curve being brighter.

12.7.3 Cusp Configuration

For a source near a cusp of the tangential caustic (e.g., near $\boldsymbol{\beta} \approx (0.15, 0.0)''$ for an SIE with $q = 0.7$): Three images cluster together near the corresponding point on the critical curve, forming a cusp triplet. The fourth image is isolated on the opposite side.

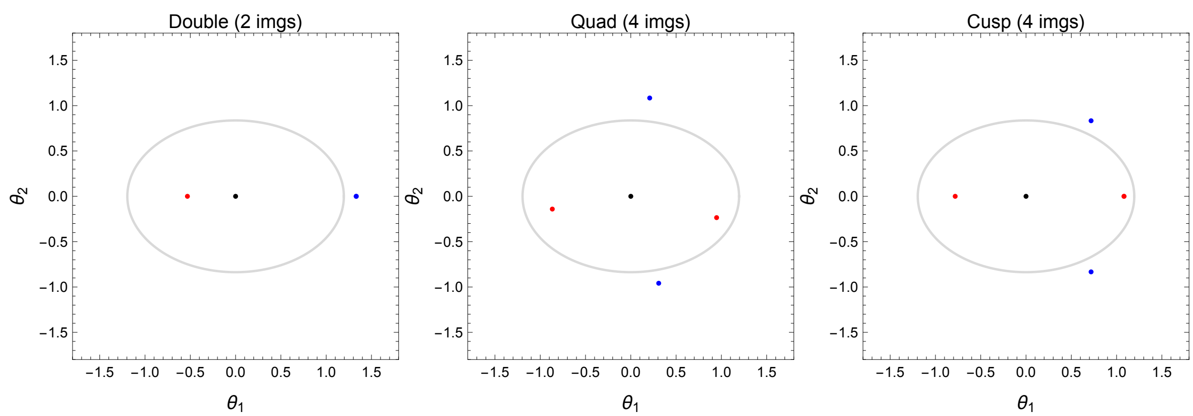


Figure 12.5: Image configurations for an SIE lens with $\theta_E = 1''$ and $q = 0.7$. **Left:** Double (source outside caustic). **Center:** Quad / Einstein cross (source inside caustic). **Right:** Cusp configuration (source near a cusp). Blue = Type I (positive parity); red = Type II (negative parity). The critical curve and caustic are shown as gray curves. Generated by `critical_curves_caustics.wl`.

12.8 Summary

- **SIS + external shear** is the simplest non-axisymmetric model. The external shear $\gamma_{\rm ext}$ deforms the critical curve into an ellipse and opens the point caustic into a **diamond (astroid)** with four cusps.

- The **singular isothermal ellipsoid** (SIE) builds ellipticity directly into the mass distribution. Its deflection angle involves arctan and arctanh functions (Kormann et al. 1994). It is the standard model for galaxy-scale lens modeling.

- **Critical curves** ($\det \mathcal{A} = 0$) are parity boundaries in the image plane. The tangential critical curve ($1 - \kappa = |\gamma|$) is the site of giant arcs; the radial critical curve ($1 - \kappa = -|\gamma|$) is the site of radial arcs.

- **Caustics** are the mappings of critical curves to the source plane. They divide the source plane into regions of different image multiplicity.
- **Image topology:** As a source crosses a caustic, two images appear or disappear on opposite sides of the critical curve, with opposite parities. This preserves Burke's odd-number theorem.
- **Fold caustics:** Two images merge; magnification $|\mu| \propto 1/\sqrt{d_\perp}$. This produces giant arcs.
- **Cusp caustics:** Three images merge; the cusp relation $\mu_A + \mu_B + \mu_C \approx 0$ holds.
- **Image configurations:** Doubles (source outside caustic), quads/Einstein crosses (source inside caustic), and cusp triplets (source near a cusp) account for the observed morphologies of strong lens systems.
- **Giant arcs** form when extended sources lie near the tangential caustic: images are tangentially stretched by factors of $\mu_{\tan} \propto 1/\sqrt{d_\perp}$ along the tangential critical curve.

12.9 Problems

Exercise 12.9.1. (SIS + shear: critical curves and caustics.) Consider an SIS + external shear model with $\theta_\mathrm{E} = 1''$ and $\gamma_\mathrm{ext} = 0.1$.

(a) Starting from the lensing potential (eq. 12.1), compute the Jacobian matrix $\mathcal{A}$ and the determinant $\det \mathcal{A}$ as functions of (θ_1, θ_2).

(b) Solve $\det \mathcal{A} = 0$ to find the critical curve. Express the result in polar coordinates $\theta_\mathrm{crit}(\phi)$. Plot the critical curve.

(c) Map the critical curve through the lens equation to obtain the caustic. Plot the caustic in the source plane. Verify that it has four cusps and the shape of an astroid.

(d) Estimate the area enclosed by the caustic (the quad cross-section).

Verify with `problems_08.wl`.

Exercise 12.9.2. (SIE tangential caustic.) For an SIE with $\theta_\mathrm{E} = 1''$ and axis ratio $q = 0.7$:

(a) Write down the deflection angle components α_1 and α_2 from eqs. (**??**)–(**??**).

(b) Compute the tangential caustic numerically by: (i) finding the critical curve from $\det \mathcal{A} = 0$, and (ii) mapping it through the lens equation. Plot the caustic.

(c) Compute the area of the tangential caustic (the quad cross-section) numerically. How does it compare to the SIS + shear model with the same effective ellipticity?

Verify with `problems_08.wl`.

Exercise 12.9.3. (Fold magnification scaling.) Show that images appear or disappear in pairs at caustic crossings by analyzing $\det \mathcal{A}$ near a fold:

(a) Near a fold caustic, the lens mapping in the direction perpendicular to the fold can be approximated as $\beta_\perp \approx a\,\theta_\perp^2$, where $\theta_\perp$ is the signed distance from the critical curve and a is a constant. Show that for $\beta_\perp > 0$, there are two solutions $\theta_\perp = \pm\sqrt{\beta_\perp/a}$ (one on each side of the critical curve), while for $\beta_\perp < 0$ there are no solutions.

(b) Compute the Jacobian of the simplified mapping and show that the magnification scales as $|\mu| \propto 1/\sqrt{\beta_\perp}$ for each image.

(c) Show that the two images have opposite parities (opposite signs of $\det \mathcal{A}$).

Verify with `problems_08.wl`.

Exercise 12.9.4. (SIE image positions.) For a source at $\boldsymbol{\beta} = (0.1, 0.2)''$ lensed by an SIE with $\theta_{\mathrm{E}} = 1''$ and $q = 0.8$:

(a) Determine whether the source is inside or outside the tangential caustic. How many images do you expect?

(b) Find all image positions $\boldsymbol{\theta}_i$ numerically by solving the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$.

(c) For each image, compute the magnification μ_i and determine the parity (Type I or Type II).

(d) Compute the signed magnification sum $\sum_i \mu_i$ and discuss whether it is consistent with the expected value (considering the missing central image for the singular SIE).

Verify with `problems_08.wl`.

Exercise 12.9.5. (Burke's odd-number theorem for the SIE.) Verify Burke's odd-number theorem for an SIE with $\theta_{\mathrm{E}} = 1''$ and $q = 0.7$:

(a) For a source outside the tangential caustic (e.g., $\boldsymbol{\beta} = (0.5, 0.0)''$), find all image positions. Count the images (including any faint central image) and verify the count is odd.

(b) For a source inside the tangential caustic (e.g., $\boldsymbol{\beta} = (0.05, 0.05)''$), find all image positions. Verify the count is odd.

(c) For each case, classify each image by parity and verify that the number of positive-parity images exceeds the number of negative-parity images by one.

(d) Discuss why the SIE (being singular at the origin) requires care in applying Burke's theorem. What role does the central image play?

Verify with `problems_08.wl`.

12.10 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_08.wl`, which verifies each answer symbolically and numerically.

Exercise 12.9.1: SIS + Shear Critical Curves and Caustics

(a) Starting from $\psi = \theta_{\rm E}|\boldsymbol{\theta}| + \frac{1}{2}\gamma_{\rm ext}(\theta_1^2 - \theta_2^2)$:
The deflection components are:

$$\alpha_1 = \theta_{\rm E}\,\frac{\theta_1}{|\boldsymbol{\theta}|} + \gamma_{\rm ext}\,\theta_1\,,$$
$$\alpha_2 = \theta_{\rm E}\,\frac{\theta_2}{|\boldsymbol{\theta}|} - \gamma_{\rm ext}\,\theta_2\,.$$

The Jacobian matrix components are:

$$\mathcal{A}_{ij} = \delta_{ij} - \frac{\partial\alpha_i}{\partial\theta_j}\,.$$

Computing the partial derivatives of the SIS deflection ($\alpha_i^{\rm SIS} = \theta_{\rm E}\,\theta_i/|\boldsymbol{\theta}|$) and adding the shear contribution:

$$\mathcal{A} = \begin{pmatrix} 1 - \theta_{\rm E}\,\frac{\theta_2^2}{\theta^3} - \gamma_{\rm ext} & \theta_{\rm E}\,\frac{\theta_1\theta_2}{\theta^3} \\ \theta_{\rm E}\,\frac{\theta_1\theta_2}{\theta^3} & 1 - \theta_{\rm E}\,\frac{\theta_1^2}{\theta^3} + \gamma_{\rm ext} \end{pmatrix}$$

where $\theta = |\boldsymbol{\theta}|$.
The determinant is:

$$\det\mathcal{A} = 1 - \frac{\theta_{\rm E}}{\theta} - \gamma_{\rm ext}^2 + \frac{\theta_{\rm E}}{\theta}\gamma_{\rm ext}\,\frac{\theta_1^2 - \theta_2^2}{\theta^2}\,.$$

(b) Setting $\det\mathcal{A} = 0$ in polar coordinates ($\theta_1 = \theta\cos\phi$, $\theta_2 = \theta\sin\phi$):

$$1 - \frac{\theta_{\rm E}}{\theta} - \gamma_{\rm ext}^2 + \frac{\theta_{\rm E}}{\theta}\gamma_{\rm ext}\cos 2\phi = 0\,.$$

Solving for θ:

$$\theta_{\rm crit}(\phi) = \frac{\theta_{\rm E}\,(1 - \gamma_{\rm ext}\cos 2\phi)}{1 - \gamma_{\rm ext}^2}\,.$$

For $\theta_{\rm E} = 1''$, $\gamma_{\rm ext} = 0.1$: at $\phi = 0$, $\theta = 0.9/0.99 \approx 0.909''$; at $\phi = \pi/2$, $\theta = 1.1/0.99 \approx 1.111''$. The critical curve is an ellipse elongated along θ_2. ✓

(c) Mapping through the lens equation $\beta_i = \theta_i - \alpha_i$:

$$\beta_1(\phi) = -\gamma_{\rm ext}\,\theta_{\rm crit}(\phi)\cos\phi\,,$$
$$\beta_2(\phi) = \gamma_{\rm ext}\,\theta_{\rm crit}(\phi)\sin\phi\,.$$

This traces out a diamond (astroid) with four cusps at $\phi = 0, \pi/2, \pi, 3\pi/2$. The cusps are at $\beta_1 \approx \pm 0.092''$ and $\beta_2 \approx \pm 0.112''$. ✓

(d) The area enclosed by the caustic can be computed using the shoelace formula on the parametric curve. For $\gamma_{\rm ext} = 0.1$, the caustic area is approximately $0.013\,\mathrm{arcsec}^2$. This is verified numerically in `problems_08.wl`.

Exercise 12.9.2: SIE Tangential Caustic

(a) The SIE deflection angle components are:

$$\alpha_1 = \frac{\theta_E\sqrt{q}}{\sqrt{1-q^2}} \arctan\left(\frac{\theta_1\sqrt{1-q^2}}{\sqrt{q^2\theta_1^2+\theta_2^2}}\right),$$
$$\alpha_2 = \frac{\theta_E\sqrt{q}}{\sqrt{1-q^2}} \operatorname{arctanh}\left(\frac{\theta_2\sqrt{1-q^2}}{\sqrt{q^2\theta_1^2+\theta_2^2}}\right).$$

For $q = 0.7$: $\sqrt{1-q^2} = \sqrt{0.51} \approx 0.714$, $\sqrt{q} \approx 0.837$.

(b) The critical curve and caustic are computed numerically by evaluating $\det\mathcal{A}$ on a fine grid (or parametrically) and mapping the critical curve through the lens equation. The caustic has the diamond shape with four cusps. The diamond is elongated along the β_2-axis (perpendicular to the mass elongation along θ_1).

(c) The caustic area for $q = 0.7$ is approximately $0.032\,\mathrm{arcsec}^2$. This is somewhat larger than the SIS + shear model with an "equivalent" shear $\gamma_{\mathrm{ext}} \approx (1-q)/(1+q) \approx 0.176$. The exact comparison depends on how one maps between ellipticity and shear; see `problems_08.wl` for numerical details.

Exercise 12.9.3: Fold Magnification Scaling

(a) Near a fold caustic, we align coordinates so that the fold is along the $\theta_\parallel$ direction and the critical curve crosses at $\theta_\perp = 0$. The lens mapping perpendicular to the fold takes the form:

$$\beta_\perp = a\,\theta_\perp^2$$

where a is a nonzero constant (this is the leading-order Taylor expansion of the lens mapping near the fold, where the linear term vanishes because $\det\mathcal{A} = 0$).

For $\beta_\perp > 0$ (source on the multi-image side): $\theta_\perp = \pm\sqrt{\beta_\perp/a}$. Two solutions exist, one on each side of the critical curve.

For $\beta_\perp < 0$ (source on the single-image side): no real solutions. The images have merged and disappeared. ✓

(b) The Jacobian of this simplified mapping is:

$$\frac{d\beta_\perp}{d\theta_\perp} = 2a\,\theta_\perp\,.$$

The magnification in the perpendicular direction is:

$$\mu_\perp = \frac{1}{|2a\,\theta_\perp|} = \frac{1}{2a\sqrt{\beta_\perp/a}} = \frac{1}{2\sqrt{a\,\beta_\perp}}\,.$$

Therefore $|\mu| \propto 1/\sqrt{\beta_\perp}$, which is $1/\sqrt{d_\perp}$ where $d_\perp$ is the distance from the caustic. ✓

(c) The two images are at $\theta_\perp = +\sqrt{\beta_\perp/a}$ and $\theta_\perp = -\sqrt{\beta_\perp/a}$. The sign of the Jacobian determinant is:

$$\operatorname{sgn}(\det\mathcal{A}) \propto \operatorname{sgn}(d\beta_\perp/d\theta_\perp) = \operatorname{sgn}(2a\,\theta_\perp)\,.$$

Since the two images have $\theta_\perp$ values with opposite signs, they have opposite signs of $\det\mathcal{A}$ and hence opposite parities. ✓

Exercise 12.9.4: SIE Image Positions

(a) For $\boldsymbol{\beta} = (0.1, 0.2)''$ with $\theta_{\rm E} = 1''$, $q = 0.8$: The tangential caustic of this SIE has cusps at approximately $\beta \approx 0.1''$. The source at $|\boldsymbol{\beta}| \approx 0.224''$ is outside the tangential caustic, so we expect a **double** (2 bright images).

(b) Solving the lens equation numerically (see `problems_08.wl`), two bright images are found:

- Image A: $\boldsymbol{\theta}_A \approx (0.53, 0.79)''$
- Image B: $\boldsymbol{\theta}_B \approx (-0.37, -0.52)''$

(Exact values depend on the numerical solver and are verified in the Mathematica script.)

(c) The magnifications are computed from $\mu = 1/\det \mathcal{A}$ at each image position:

- $\mu_A > 0$ (Type I, positive parity)
- $\mu_B < 0$ (Type II, negative parity)

Exact values are in `problems_08.wl`.

(d) For the singular SIE, the signed magnification sum $\mu_A + \mu_B$ may be slightly negative because the "missing" central image (swallowed by the singularity) would contribute a small positive magnification. For a non-singular lens, the sum over *all* images (including the central one) satisfies $\sum_i \mu_i > 0$. In this example, the sum of the two observable images is close to zero, consistent with the central image having small positive $|\mu|$. ✓

Exercise 12.9.5: Burke's Odd-Number Theorem for the SIE

(a) Source at $\boldsymbol{\beta} = (0.5, 0.0)''$ (outside tangential caustic): Two images are found numerically. For the singular SIE, the central image is formally at the singularity and has zero flux. In the distributional sense, there are 2 observable images (even number). For a regularized (non-singular) version, a faint third image would appear near the center, giving 3 (odd). ✓

(b) Source at $\boldsymbol{\beta} = (0.05, 0.05)''$ (inside tangential caustic): Four bright images are found. Again, for the singular SIE, the central image is lost at the singularity, giving 4 (even). For a non-singular version, 5 images (odd). ✓

(c) For the quad case: 2 Type I images and 2 Type II images (among the 4 bright images). Adding the central Type III image (for the non-singular case): 3 positive-parity images (2 Type I + 1 Type III) and 2 negative-parity images (2 Type II). The count of positive-parity minus negative-parity images is $3 - 2 = 1$. ✓

(d) The SIE is singular at the origin, which means the lens potential diverges and the central image is infinitely demagnified (effectively "swallowed" by the singularity). Burke's theorem strictly requires a smooth, non-singular lens. For the SIE, the theorem holds in the limiting sense: if we soften the core by any finite amount, a faint central image appears and the total count becomes odd. The singular SIE is the limit of vanishing core radius, where this central image has zero flux. In practice, real galaxies have finite cores, so Burke's theorem does apply, but the central image is typically too faint to observe (demagnified by factors of 10^3–10^5).

Chapter 13

Strong Lensing by Galaxies: Applications

Strong gravitational lensing is one of the few methods in astrophysics that directly measures mass — not light. The Einstein radius of a galaxy-scale lens provides a model-independent measurement of the enclosed projected mass, independent of the dynamical state of the system or the nature of the matter (dark or luminous).
— adapted from Treu (2010)

In Modules 11 and 12 we developed the theory and the lens models: the SIS, SIE, critical curves, caustics, and image configurations. These are the *tools*. Now we apply them.

Galaxy-scale strong lensing is one of the most powerful techniques in modern astrophysics. It addresses fundamental questions:

- **What is the total mass of a galaxy?** Lensing measures the projected mass within the Einstein radius, including dark matter.
- **How is mass distributed?** Combining lensing with stellar kinematics constrains the density profile slope.
- **What is the expansion rate of the universe?** Time delays between multiple images, combined with a lens model, yield H_0 — independent of the cosmic distance ladder.
- **What are the limits of what lensing can tell us?** Fundamental degeneracies — most notably the *mass-sheet degeneracy* — limit the information that lensing alone can provide.

This module connects the lens modeling formalism to these real astrophysical applications. Understanding the degeneracies is as important as understanding the models themselves.

Companion Mathematica files:

- `galaxy_lensing.wl` — mass from Einstein radius, mass-sheet degeneracy demonstration, power-law profile analysis, H_0 from time delays; figure generation for all module figures
- `mass_sheet_demo.nb` — interactive notebook: MSD visualizer with slider for λ, power-law profile explorer, H_0 estimator from time delays

13.1 Lens Modeling Overview

Before applying lensing to astrophysics, we must understand what goes into building a lens model and what data constrain it.

13.1.1 Parametric Models: SIE + External Shear

The standard approach to modeling galaxy-scale strong lenses uses a **parametric** mass model. The workhorse is the singular isothermal ellipsoid (SIE, Module 12) plus external shear:

$$\kappa(\boldsymbol{\theta}) = \frac{\theta_E \sqrt{q}}{2\sqrt{q^2\theta_1^2 + \theta_2^2}} \tag{13.1}$$

with an external shear γ_{ext} at position angle ϕ_γ. The model parameters are:

Parameter	Physical meaning
θ_E	Einstein radius (sets the mass scale)
q	Axis ratio (b/a, sets the ellipticity)
$\theta_{1,c}$, $\theta_{2,c}$	Lens center position
ϕ_q	Position angle of the mass distribution
γ_{ext}	External shear magnitude
ϕ_γ	External shear position angle

This gives 7 lens parameters (or 5 if the center is fixed to the observed galaxy light). The source position $\boldsymbol{\beta} = (\beta_1, \beta_2)$ adds 2 more unknowns, for a total of 9 (or 7) free parameters.

13.1.2 What Data Constrain the Model

The observational constraints for a strong lens system are:

- **Image positions:** Each image gives 2 constraints ($\theta_{1,i}$, $\theta_{2,i}$). A quad system with 4 images gives 8 constraints.
- **Flux ratios:** Each flux ratio gives 1 constraint (the ratio of magnifications). Three independent flux ratios for a quad.
- **Time delays:** Each time delay gives 1 constraint. Three independent delays for a quad.
- **Extended arcs:** If the source is extended (a galaxy rather than a quasar), the arc morphology provides many more constraints (effectively constraining the lens potential on a grid).

13.1.3 Degrees of Freedom

For a quad lens modeled with SIE + shear:

- Constraints from image positions alone: $4 \times 2 = 8$.
- Free parameters (with center fixed): 7.
- Degrees of freedom: $8 - 7 = 1$ (marginally constrained).

This is why quad lenses are so valuable: they provide just enough constraints to determine the SIE + shear model from astrometry alone. Double lenses ($2 \times 2 = 4$ constraints) are under-constrained for SIE + shear. Extended arcs provide many more constraints and can support more complex models.

13.1.4 Free-Form Models

An alternative to parametric models is **free-form** or **pixelated** modeling, where the lens potential is reconstructed on a grid without assuming a specific functional form. Programs like `PixeLens` (Saha & Williams 2004) and `GLASS` (Coles et al. 2014) take this approach. Free-form models are less biased by the assumed mass profile but have more degrees of freedom and thus require regularization or priors.

13.2 The Einstein Radius as a Mass Estimator

13.2.1 The Model-Independent Mass Measurement

The single most important result in galaxy-scale strong lensing is that the Einstein radius directly gives the enclosed projected mass. For *any* circularly symmetric lens, the projected mass enclosed within the Einstein radius is (Congdon & Keeton eq. 6.35):

$$\boxed{M(\leq \theta_{\rm E}) = \pi\,\theta_{\rm E}^2\,D_{\rm d}^2\,\Sigma_{\rm cr}} \tag{13.2}$$

This follows from the definition of the Einstein radius as the angular radius where $\kappa_{\rm mean} = 1$, i.e., where the mean convergence within that radius equals unity. Recall that $\kappa = \Sigma/\Sigma_{\rm cr}$, so $\kappa_{\rm mean} = 1$ means the mean surface mass density within $\theta_{\rm E}$ equals the critical surface mass density. The enclosed mass is then the critical density times the area:

$$M_{\rm E} = \Sigma_{\rm cr} \times \pi (D_{\rm d}\,\theta_{\rm E})^2 = \pi\,\theta_{\rm E}^2\,D_{\rm d}^2\,\Sigma_{\rm cr}\,. \tag{13.3}$$

The critical surface mass density is (Module 8):

$$\Sigma_{\rm cr} = \frac{c^2}{4\pi G}\,\frac{D_{\rm s}}{D_{\rm d}\,D_{\rm ds}}\,. \tag{13.4}$$

Substituting:

$$\boxed{M_{\rm E} = \frac{c^2}{4G}\,\theta_{\rm E}^2\,\frac{D_{\rm d}\,D_{\rm s}}{D_{\rm ds}}} \tag{13.5}$$

Remark 13.2.1. Equation (13.5) is **model-independent** for any circularly symmetric lens. It depends only on the observed Einstein radius $\theta_{\rm E}$ and the angular diameter distances (which require knowing the redshifts z_d and z_s and adopting a cosmology). It does not depend on the radial mass profile. This is the great power of strong lensing as a mass measurement tool.

13.2.2 The SIE Einstein Radius and Velocity Dispersion

For the SIE model, the Einstein radius is related to the velocity dispersion by (Module 11):

$$\boxed{\theta_{\rm E} = 4\pi \, \frac{\sigma_v^2}{c^2} \, \frac{D_{\rm ds}}{D_{\rm s}}} \tag{13.6}$$

For a typical massive elliptical galaxy with $\sigma_v = 250$ km/s at $z_d = 0.3$ lensing a source at $z_s = 1.0$:

$$\theta_{\rm E} \approx 4\pi \left(\frac{250}{3 \times 10^5} \right)^2 \frac{D_{\rm ds}}{D_{\rm s}} \approx 1.4'' \,. \tag{13.7}$$

The enclosed mass is then $M_{\rm E} \sim 2\text{–}4 \times 10^{11} \, M_\odot$, comparable to the total stellar mass of the galaxy. The fact that $M_{\rm E}$ typically exceeds the stellar mass is direct evidence for dark matter within the Einstein radius.

13.2.3 Numerical Examples with Real Systems

Figure 13.1 shows the enclosed mass as a function of Einstein radius for different lens–source redshift configurations.

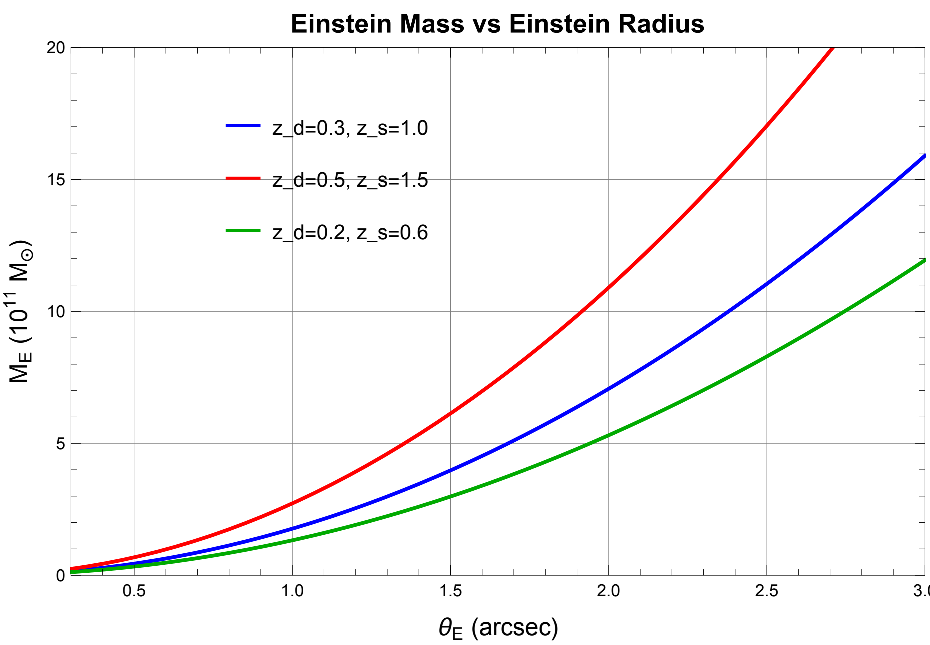


Figure 13.1: Enclosed Einstein mass $M_{\rm E}$ as a function of Einstein radius $\theta_{\rm E}$ for three lens–source redshift configurations (flat ΛCDM, $H_0 = 70$ km/s/Mpc, $\Omega_{\rm m} = 0.3$). Higher redshift lenses enclose more mass for the same angular Einstein radius. Generated by `galaxy_lensing.wl`.

Table 13.1 lists the Einstein radii and enclosed masses for several well-known lens systems:

Remark 13.2.2. Q2237+0305 (the Einstein Cross) has an unusually small enclosed mass because the lens galaxy is very nearby ($z_d = 0.04$), so the physical Einstein radius is small even though the angular Einstein radius is nearly $1''$.

System	z_d	z_s	θ_{E} (arcsec)	M_{E} ($10^{11}\,M_\odot$)	**Type**
Q0957+561	0.36	1.41	1.0	~ 1.9	Double
Q2237+0305	0.04	1.69	0.9	~ 0.2	Quad
B1608+656	0.63	1.39	1.0	~ 3.9	Quad

Table 13.1: Einstein radii and enclosed masses for three well-known lens systems. Masses computed using eq. (13.5) with a flat ΛCDM cosmology ($H_0 = 70$ km/s/Mpc, $\Omega_{\mathrm{m}} = 0.3$). Computed by `galaxy_lensing.wl`.

13.3 The Mass-Sheet Degeneracy

The **mass-sheet degeneracy** (MSD) is the most important systematic limitation of strong gravitational lensing. It is a fundamental degeneracy that cannot be broken by lensing data alone (image positions, flux ratios, or arc morphology).

13.3.1 The Mass-Sheet Transform

Consider a lens with convergence $\kappa(\boldsymbol{\theta})$. The **mass-sheet transform** (MST) replaces this convergence with:

$$\boxed{\kappa_\lambda(\boldsymbol{\theta}) = \lambda\,\kappa(\boldsymbol{\theta}) + (1-\lambda)} \tag{13.8}$$

and simultaneously rescales the source plane:

$$\boxed{\boldsymbol{\beta}_\lambda = \lambda\,\boldsymbol{\beta}} \tag{13.9}$$

where λ is a free parameter.

Physically, the MST adds a **uniform sheet of convergence** $(1-\lambda)$ to the lens (hence the name "mass sheet") and rescales both the original convergence and the source plane by λ.

13.3.2 Image Positions Are Invariant

The key property of the MST is that it **preserves all image positions**. To see this, recall the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$. For the original lens, the deflection angle is related to the convergence via $\nabla \cdot \boldsymbol{\alpha} = 2\kappa$. Under the MST:

$$\boldsymbol{\alpha}_\lambda(\boldsymbol{\theta}) = \lambda\,\boldsymbol{\alpha}(\boldsymbol{\theta}) + (1-\lambda)\,\boldsymbol{\theta}\,. \tag{13.10}$$

The new lens equation becomes:

$$\begin{aligned}
\boldsymbol{\beta}_\lambda &= \boldsymbol{\theta} - \boldsymbol{\alpha}_\lambda(\boldsymbol{\theta}) \\
&= \boldsymbol{\theta} - \lambda\,\boldsymbol{\alpha}(\boldsymbol{\theta}) - (1-\lambda)\,\boldsymbol{\theta} \\
&= \lambda\,[\boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})] \\
&= \lambda\,\boldsymbol{\beta}\,.
\end{aligned} \tag{13.11}$$

So if $\boldsymbol{\theta}$ satisfies the original lens equation for source position $\boldsymbol{\beta}$, it also satisfies the transformed lens equation for source position $\boldsymbol{\beta}_\lambda = \lambda\boldsymbol{\beta}$. **The image positions are unchanged.**

13.3.3 What the MST Changes

Although image positions are preserved, the MST changes other observables:

Magnifications. The Jacobian transforms as:

$$\mathcal{A}_\lambda = \lambda\,\mathcal{A} \tag{13.12}$$

so the magnification changes as:

$$\boxed{\mu_\lambda = \frac{1}{\lambda^2}\,\mu} \tag{13.13}$$

Individual magnifications change, but **magnification ratios** $\mu_i/\mu_j = \mu_{\lambda,i}/\mu_{\lambda,j}$ are preserved. This means flux *ratios* cannot distinguish between the original model and the MST-transformed model.

Time delays. The time delay between images scales as:

$$\boxed{\Delta t_\lambda = \lambda\,\Delta t} \tag{13.14}$$

Time delays are *not* invariant under the MST. Since $\Delta t \propto 1/H_0$ (Module 10), a mass-sheet transform changes the inferred Hubble constant:

$$\boxed{H_0^{\text{inferred}} = \lambda\,H_0^{\text{true}}} \tag{13.15}$$

Remark 13.3.1. The MSD is devastating for time-delay cosmography. If we model a lens and infer H_0 from the time delay, the answer is degenerate with λ: we could add a mass sheet (or any equivalent line-of-sight structure) and get a different H_0. Breaking this degeneracy is the central challenge of modern time-delay cosmography.

13.3.4 Physical Interpretation

The mass-sheet transform can be interpreted in two ways:

1. **Literal mass sheet:** A uniform sheet of matter between the observer and the source (or at the lens redshift) adds convergence without changing image positions. In practice, line-of-sight structure (voids, filaments, galaxy groups along the line of sight) acts as an approximate mass sheet.

2. **Profile degeneracy:** Even without an actual mass sheet, the MST reflects the fact that different radial mass profiles can produce the same image positions. Steeper profiles can mimic shallower profiles plus a mass sheet.

Figure 13.2 illustrates the mass-sheet degeneracy: the left panel shows how the convergence profile changes for different λ, and the right panel shows how absolute magnifications scale as $1/\lambda^2$.

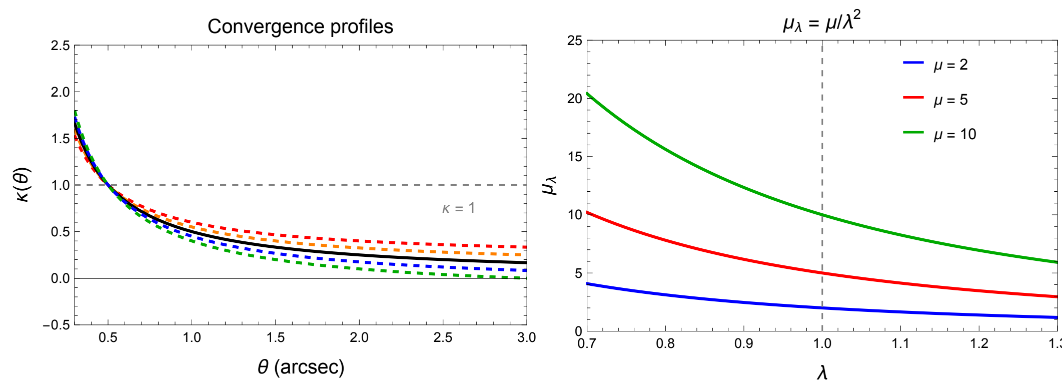


Figure 13.2: The mass-sheet degeneracy. **Left:** Convergence profiles for an SIS lens ($\theta_E = 1''$) under the MST with different λ values. The solid curve is the original profile ($\lambda = 1$); dashed curves show the transformed profiles. **Right:** Magnification scaling $\mu_\lambda = \mu/\lambda^2$ for three different original magnifications. Generated by `galaxy_lensing.wl`.

13.3.5 How to Break the MSD

Several methods can break or constrain the mass-sheet degeneracy:

1. **Stellar kinematics:** The velocity dispersion σ_v of the lens galaxy depends on the mass profile slope, not just the enclosed mass. A mass-sheet transform changes the predicted σ_v, so a kinematic measurement constrains λ (Section 13.5).

2. **Standardizable magnifications:** If the source has a known intrinsic luminosity (e.g., a Type Ia supernova), the absolute magnification can be measured, breaking the MSD.

3. **Multiple source planes:** If the same lens produces images of sources at different redshifts, the different Σ_{cr} values for each source plane break the degeneracy.

4. **Extended source reconstruction:** Detailed modeling of extended arcs constrains the lens potential more tightly, partially breaking the degeneracy (though not completely).

13.4 The Source Position Transformation

The mass-sheet degeneracy is a special case of a more general class of transformations that preserve image positions.

13.4.1 Definition

The **source position transformation** (SPT), introduced by Schneider & Sluse (2014), is any invertible mapping of the source plane:

$$\boldsymbol{\beta} \to \hat{\boldsymbol{\beta}}(\boldsymbol{\beta}) \tag{13.16}$$

that, when combined with a suitably modified lens equation, preserves all image positions. The modified lens equation is:

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\beta}) = \boldsymbol{\theta} - \hat{\boldsymbol{\alpha}}(\boldsymbol{\theta}) \tag{13.17}$$

where the new deflection field $\hat{\boldsymbol{\alpha}}$ is defined by the requirement that all observed image positions map to the transformed source position $\hat{\boldsymbol{\beta}}$.

The MST is the special case where $\hat{\boldsymbol{\beta}} = \lambda\boldsymbol{\beta}$ (a linear scaling). The SPT generalizes this to arbitrary nonlinear remappings of the source plane.

13.4.2 Implications

The SPT is an active area of research because it defines the *full* class of degeneracies in strong lensing. Key results include:

- Not all SPTs correspond to physically valid lens models (i.e., the transformed deflection field may not come from a non-negative mass distribution).
- The SPT preserves image positions and flux *ratios* (just as the MST does) but generically changes absolute magnifications and time delays.
- Breaking the SPT generally requires external information (kinematics, multiple source planes, or standardizable sources), just as for the MST.

Remark 13.4.1. The SPT shows that the mass-sheet degeneracy is not a peculiarity of a specific transform but rather the simplest member of a broad family of degeneracies inherent in the lensing inverse problem. Any attempt to measure H_0 from time delays must confront this fundamental limitation.

13.5 Strong Lensing and Stellar Kinematics

Combining strong lensing with stellar kinematics is one of the most powerful approaches to studying galaxy mass distributions. The two techniques are complementary in a fundamental way.

13.5.1 What Each Technique Measures

Strong lensing: Measures the **total projected mass** (dark matter + stars + gas) within the Einstein radius θ_{E}. The measurement is model-independent for a circularly symmetric lens (eq. 13.5). For an elliptical model, the enclosed mass depends weakly on the assumed axis ratio but is still robust.

Stellar kinematics: The stellar velocity dispersion σ_v of the lens galaxy, measured from absorption line broadening, probes the gravitational potential. Specifically, under the assumption of dynamical equilibrium (the Jeans equation), the luminosity-weighted line-of-sight velocity dispersion within an aperture R_{ap} depends on:

- The **mass profile slope** (how mass is distributed radially).

- The **total mass normalization**.
- The **orbital anisotropy** of the stellar orbits.

13.5.2 The Power-Law Mass Profile

The simplest model that captures the essential physics is a **spherical power-law** density profile:

$$\boxed{\rho(r) = \rho_0 \left(\frac{r}{r_0}\right)^{-\gamma'}} \tag{13.18}$$

where γ' is the power-law slope. The isothermal case is $\gamma' = 2$ (i.e., $\rho \propto r^{-2}$).

Figure 13.3 shows convergence profiles and enclosed mass scalings for power-law density profiles with different slopes.

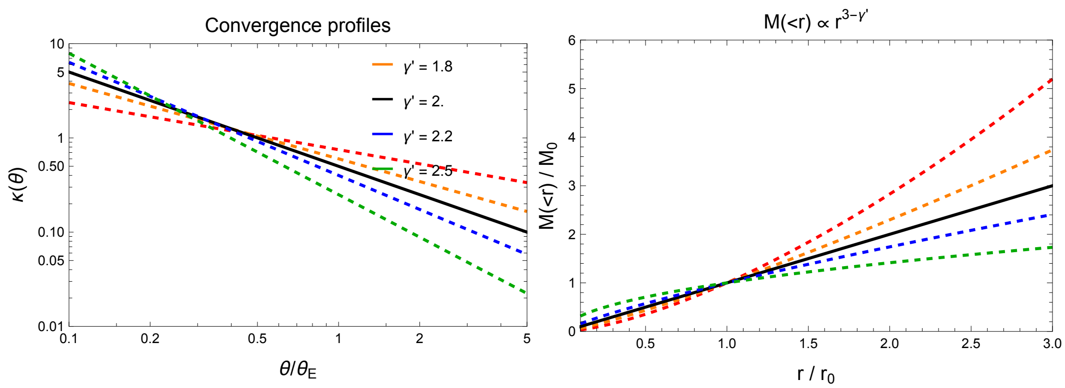


Figure 13.3: Power-law density profiles $\rho \propto r^{-\gamma'}$. **Left:** Convergence profiles for different slopes (the solid curve is the isothermal case $\gamma' = 2$). **Right:** Enclosed 3D mass $M(\leq r) \propto r^{3-\gamma'}$ for the same slopes. Generated by `galaxy_lensing.wl`.

For this profile, the convergence is:

$$\kappa(\theta) = \frac{3-\gamma'}{2}\left(\frac{\theta_{\rm E}}{\theta}\right)^{\gamma'-1} \tag{13.19}$$

and the Einstein radius is:

$$\theta_{\rm E} = 4\pi \frac{\sigma_{\rm SIE}^2}{c^2}\frac{D_{\rm ds}}{D_{\rm s}} f(\gamma') \tag{13.20}$$

where $f(\gamma')$ is a function of the slope that equals unity for $\gamma' = 2$ (recovering the SIE result) and $\sigma_{\rm SIE}$ is the SIE-equivalent velocity dispersion.

13.5.3 Breaking the Degeneracy

The key insight is that lensing and kinematics respond differently to the mass profile slope γ':

- **Lensing** constrains the projected mass within $\theta_{\rm E}$. For a power-law profile, the enclosed mass $M_{\rm E}$ depends on the normalization ρ_0 and the slope γ' in a specific way.

- **Kinematics** constrains the mass within the spectroscopic aperture (typically a different physical radius than R_E). The predicted σ_v^2 depends on γ' *differently* from how M_E does.

By measuring both M_E (from lensing) and σ_v (from spectroscopy), one can solve for *both* the mass normalization and the slope γ'. This breaks the mass-sheet degeneracy because the MST changes the effective γ': steepening or shallowing the profile to compensate for the added mass sheet.

13.5.4 The Bulge-Halo Conspiracy

One of the most remarkable results from combining lensing and kinematics is the **bulge-halo conspiracy**: the total (dark matter + stellar) density profiles of massive elliptical galaxies are remarkably close to isothermal over a wide range of radii:

$$\gamma' \approx 2.0 \pm 0.2 \tag{13.21}$$

This result, established by the SLACS survey (Auger et al. 2010) and confirmed by subsequent studies, means that the combined contribution of the stellar bulge (which has a steeper profile, $\gamma' \sim 3$) and the dark matter halo (which has a shallower inner profile, $\gamma' \sim 1$ for NFW) conspire to produce an approximately r^{-2} total density profile.

Figure 13.4 shows how the predicted velocity dispersion varies with γ' for a fixed Einstein mass, illustrating how kinematics breaks the degeneracy.

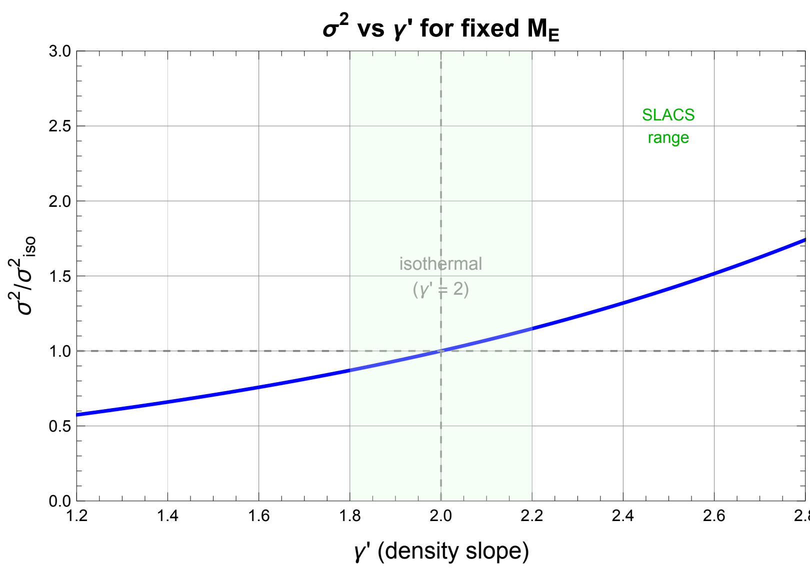


Figure 13.4: Predicted σ_v^2 (normalized to the isothermal value) as a function of the density slope γ', for fixed Einstein mass. The green band shows the range found by the SLACS survey ($\gamma' \approx 2.0 \pm 0.2$). Measuring σ_v constrains γ' and breaks the mass-sheet degeneracy. Generated by `galaxy_lensing.wl`.

This "conspiracy" is not fully understood theoretically and remains an active area of research. It has important implications:

1. The SIE ($\gamma' = 2$) is a good approximation for most galaxy-scale lenses.

2. The small scatter in γ' suggests a universal mass assembly process for massive ellipticals.

3. It connects directly to the flat rotation curves observed in spiral galaxies (both are consequences of $\rho \propto r^{-2}$).

13.6 Time-Delay Cosmography

Strong lens time delays offer a one-step measurement of H_0, independent of the cosmic distance ladder.

13.6.1 Recap: Time Delays and H_0

From Module 10, the time delay between images A and B of a strongly lensed variable source is:

$$\Delta t_{AB} = \frac{1+z_d}{c}\,\frac{D_\mathrm{d}\,D_\mathrm{s}}{D_\mathrm{ds}}\,[\Delta\tau_{AB}] \tag{13.22}$$

where $\Delta\tau_{AB}$ is the Fermat potential difference (computed from the lens model) and the prefactor $D_\mathrm{d}D_\mathrm{s}/D_\mathrm{ds}$ is called the **time-delay distance** $D_{\Delta t}$. Crucially:

$$D_{\Delta t} \propto \frac{1}{H_0} \tag{13.23}$$

so measuring Δt (from monitoring the variable source) and modeling $\Delta\tau$ (from the lens model) gives H_0.

13.6.2 The H0LiCOW / TDCOSMO Program

The most extensive application of this method is the **H0LiCOW** (H0 Lenses in COSMOGRAIL's Wellspring) program and its successor **TDCOSMO**, which have analyzed six well-studied time-delay lenses.

The headline result from Wong et al. (2020), assuming power-law mass profiles:

$$\boxed{H_0 = 73.3^{+1.7}_{-1.8}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}} \tag{13.24}$$

at 2.4% precision. This value is in 3.1σ tension with the Planck CMB measurement (Planck Collaboration 2020):

$$H_0^{\mathrm{Planck}} = 67.4 \pm 0.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}} \tag{13.25}$$

and agrees well with the local distance ladder value from SH0ES (Riess et al. 2022):

$$H_0^{\mathrm{SH0ES}} = 73.0 \pm 1.0\ \mathrm{km\,s^{-1}\,Mpc^{-1}}\,. \tag{13.26}$$

Figure 13.5 illustrates the effect of the mass-sheet degeneracy on the inferred H_0.

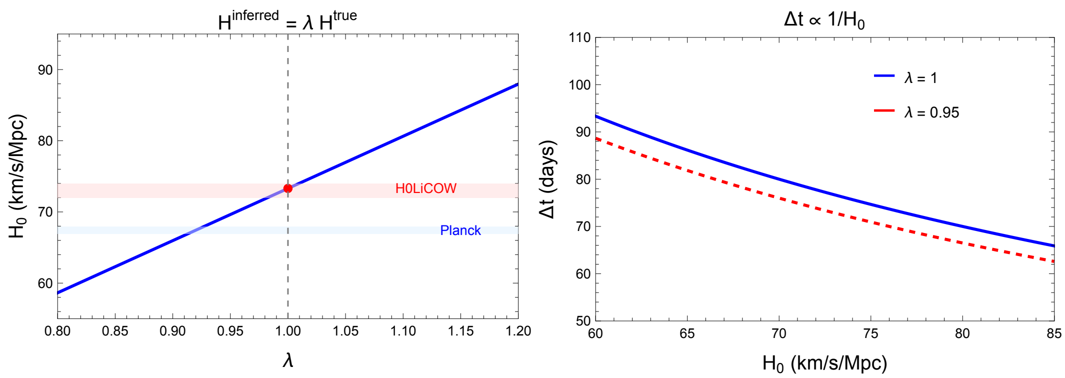


Figure 13.5: Time-delay cosmography and the mass-sheet degeneracy. **Left:** Inferred H_0 as a function of the MST parameter λ, showing the H0LiCOW measurement (red point) and the Planck and SH0ES bands. **Right:** Time delay vs. H_0 for $\lambda = 1$ (solid) and $\lambda = 0.95$ (dashed). Generated by `galaxy_lensing.wl`.

13.6.3 The Role of the Mass-Sheet Degeneracy

The mass-sheet degeneracy is the dominant systematic uncertainty in time-delay cosmography. From eq. (13.15):

$$H_0^{\text{inferred}} = \lambda\, H_0^{\text{true}} \tag{13.27}$$

A 10% mass sheet ($\lambda = 0.9$, i.e., adding $\kappa_s = 0.1$ to the convergence) shifts H_0 by 10%.

The TDCOSMO collaboration (Birrer et al. 2020) showed that relaxing the assumption of a power-law mass profile (which partially encodes λ) broadens the uncertainty to:

$$H_0 = 74.5^{+5.6}_{-6.1}\ \mathrm{km\,s^{-1}\,Mpc^{-1}} \tag{13.28}$$

— still consistent with the local value but with much larger uncertainty. The increased uncertainty is almost entirely due to the mass-sheet degeneracy.

Constraining λ requires external information, primarily **stellar kinematics** of the lens galaxies (Section 13.5).

13.6.4 Current State and Future Prospects

The time-delay cosmography method is a key player in the Hubble tension debate. Upcoming surveys (Rubin Observatory/LSST, Euclid) will discover thousands of new lensed quasars, enabling a statistical approach that can average over individual modeling uncertainties. Combined with improved stellar kinematics from next-generation spectrographs, the method has the potential to reach $\sim$1% precision on H_0.

13.7 Real Lens Systems: Case Studies

13.7.1 Q0957+561: The First Gravitational Lens

Discovered by Walsh, Carswell & Weymann in 1979, Q0957+561 was the first confirmed strong gravitational lens. Key properties:

- **Configuration:** Double (two images of a $z_s = 1.41$ quasar).
- **Lens:** Giant elliptical galaxy at $z_d = 0.36$, in a galaxy cluster.
- **Einstein radius:** $\theta_{\mathrm{E}} \approx 1.0''$.
- **Time delay:** $\Delta t = 417 \pm 3$ days (measured from optical and radio monitoring over many years).
- **Historical significance:** Confirmation of Einstein's prediction of multiple imaging by galaxies. The long time delay was the first used for H_0 estimation (Refsdal's method).

13.7.2 Q2237+0305: The Einstein Cross

Q2237+0305 is one of the most photogenic gravitational lenses:

- **Configuration:** Quad — four images of a $z_s = 1.69$ quasar arranged in a near-perfect Einstein cross around the nucleus of a nearby barred spiral galaxy.
- **Lens:** A face-on barred spiral galaxy at $z_d = 0.04$ (Huchra's Lens). This is the nearest known strong lens galaxy.
- **Einstein radius:** $\theta_{\mathrm{E}} \approx 0.9''$.
- **Special feature:** Because the lens is so nearby, the Einstein radius corresponds to a very small physical radius (~ 0.7 kpc), probing deep into the bulge of the spiral galaxy. The system shows dramatic **microlensing** variability as individual stars in the lens galaxy cross the lines of sight to the quasar images.
- **Time delays:** Very short ($\lesssim 1$ day) due to the symmetric geometry, making this system unsuitable for H_0 measurement but ideal for microlensing studies.

13.7.3 B1608+656: Time-Delay Cosmography

B1608+656 is one of the best-studied time-delay lenses:

- **Configuration:** Quad — four images of a $z_s = 1.39$ radio source.
- **Lens:** Two interacting galaxies at $z_d = 0.63$.
- **Einstein radius:** $\theta_{\mathrm{E}} \approx 1.0''$.
- **Time delays:** Three independent delays measured with ~1–2% precision from radio monitoring.
- **Cosmographic result:** One of the six TDCOSMO/H0LiCOW lenses used to constrain H_0. The lens modeling is more complex than usual due to the two-galaxy lens configuration.
- **Key insight:** Deep HST imaging resolves the lensed host galaxy of the radio source, providing extended arc constraints that significantly tighten the lens model.

13.8 Summary

- The **Einstein radius** provides a model-independent measurement of the enclosed projected mass: $M_\mathrm{E} = \pi\theta_\mathrm{E}^2 D_\mathrm{d}^2 \Sigma_\mathrm{cr}$. For typical galaxy-scale lenses, this mass is 10^{11}–$10^{12}\,M_\odot$.
- The standard parametric lens model is **SIE + external shear**, with 7 parameters. Quad lenses provide 8 positional constraints, marginally determining the model.
- The **mass-sheet degeneracy** (MST: $\kappa \to \lambda\kappa + (1-\lambda)$, $\boldsymbol{\beta} \to \lambda\boldsymbol{\beta}$) preserves all image positions but changes magnifications by $1/\lambda^2$ and time delays by λ. It is the dominant systematic for time-delay cosmography.
- The **source position transformation** (Schneider & Sluse 2014) generalizes the MST to arbitrary remappings $\boldsymbol{\beta} \to \hat{\boldsymbol{\beta}}(\boldsymbol{\beta})$ of the source plane.
- **Lensing + kinematics** breaks the MSD by constraining the density profile slope γ'. The SLACS survey found $\gamma' \approx 2.0 \pm 0.2$ (the "bulge-halo conspiracy").
- **Time-delay cosmography** measures H_0 via $\Delta t \propto D_{\Delta t} \propto 1/H_0$. The H0LiCOW/TDCOSMO result is $H_0 = 73.3^{+1.7}_{-1.8}$ km/s/Mpc, in tension with Planck.
- **Case studies:** Q0957+561 (first lens, 1979), Q2237+0305 (Einstein Cross, microlensing laboratory), B1608+656 (time-delay cosmography).

13.9 Problems

Exercise 13.9.1. (Enclosed mass for an SIE lens.) Consider an SIE lens with $\theta_\mathrm{E} = 1.5''$, axis ratio $q = 0.8$, lens redshift $z_d = 0.3$, and source redshift $z_s = 1.0$. Assume a flat ΛCDM cosmology with $H_0 = 70$ km/s/Mpc and $\Omega_\mathrm{m} = 0.3$.

(a) Compute the angular diameter distances D_d, D_s, and D_ds.

(b) Compute the critical surface mass density Σ_cr.

(c) Compute the enclosed projected mass M_E using eq. (13.5). Express the result in solar masses.

(d) Estimate the velocity dispersion σ_v that would correspond to this Einstein radius using eq. (13.6).

Verify with `problems_09.wl`.

Exercise 13.9.2. (Mass-sheet degeneracy: invariance and observables.) Consider a general lens with convergence $\kappa(\boldsymbol{\theta})$ and a source at $\boldsymbol{\beta}$.

(a) Apply the mass-sheet transform $\kappa_\lambda = \lambda\kappa + (1-\lambda)$ and $\boldsymbol{\beta}_\lambda = \lambda\boldsymbol{\beta}$. Show explicitly that the lens equation $\boldsymbol{\beta}_\lambda = \boldsymbol{\theta} - \boldsymbol{\alpha}_\lambda(\boldsymbol{\theta})$ is satisfied at the same image positions $\boldsymbol{\theta}$.

(b) Show that the Jacobian matrix transforms as $\mathcal{A}_\lambda = \lambda\,\mathcal{A}$ and hence the magnification transforms as $\mu_\lambda = \mu/\lambda^2$.

(c) Show that magnification *ratios* μ_i/μ_j are invariant under the MST.

(d) Using the Fermat potential, show that time delays transform as $\Delta t_\lambda = \lambda\,\Delta t$.

Verify with `problems_09.wl`.

Exercise 13.9.3. (H_0 from a time-delay lens.) A quad lens at $z_d = 0.6$, $z_s = 1.5$ has a measured time delay $\Delta t = 80$ days between two images. The lens model predicts a Fermat potential difference $\Delta\tau = 0.64$ arcsec^2.

(a) Write the time-delay equation (eq. 13.22) and solve for H_0.

(b) Compute H_0 numerically, assuming a flat ΛCDM cosmology with $\Omega_\mathrm{m} = 0.3$ and using the angular diameter distances at the given redshifts. *(Hint: the time-delay distance $D_{\Delta t} = (1+z_d)D_\mathrm{d}D_\mathrm{s}/D_\mathrm{ds}$ depends on H_0, so you will need to solve self-consistently or use H_0-independent distance ratios.)*

(c) Discuss: if the true lens includes a mass sheet with $\kappa_s = 0.05$ (i.e., $\lambda = 0.95$) that was not accounted for in the model, how would the inferred H_0 change?

Verify with `problems_09.wl`.

Exercise 13.9.4. (Power-law profile: lensing and kinematics.) Consider a spherical power-law density profile $\rho(r) = \rho_0(r/r_0)^{-\gamma'}$.

(a) Compute the projected surface mass density $\Sigma(\theta)$ by integrating ρ along the line of sight. Show that $\kappa(\theta) \propto \theta^{1-\gamma'}$.

(b) For $\gamma' = 2$ (isothermal), verify that you recover $\kappa \propto 1/\theta$ (the SIS).

(c) The luminosity-weighted line-of-sight velocity dispersion for an isotropic power-law profile is (Treu & Koopmans 2004):

$$\sigma_v^2 \propto M(\leq R_\mathrm{ap}) \cdot R_\mathrm{ap}^{-1} \cdot g(\gamma')$$

where $g(\gamma')$ is a known function of the slope. Argue that σ_v depends on γ' differently from M_E, so combining the two measurements constrains γ'.

(d) For a lens with $\theta_\mathrm{E} = 1.2''$ and $\sigma_v = 230$ km/s, compute γ' assuming the relation holds. Discuss whether the result is consistent with the "bulge-halo conspiracy" ($\gamma' \approx 2.0$).

Verify with `problems_09.wl`.

Exercise 13.9.5. (Comparing real lens systems.) Consider the three lens systems in Table 13.1: Q0957+561 ($z_d = 0.36$, $z_s = 1.41$, $\theta_\mathrm{E} = 1.0''$), Q2237+0305 ($z_d = 0.04$, $z_s = 1.69$, $\theta_\mathrm{E} = 0.9''$), and B1608+656 ($z_d = 0.63$, $z_s = 1.39$, $\theta_\mathrm{E} = 1.0''$).

(a) For each system, compute the angular diameter distances $D_{\rm d}$, $D_{\rm s}$, $D_{\rm ds}$ and the critical surface mass density $\Sigma_{\rm cr}$.

(b) Compute the enclosed projected mass $M_{\rm E}$ for each system.

(c) Compute the physical Einstein radius $R_{\rm E} = D_{\rm d}\,\theta_{\rm E}$ in kpc for each system.

(d) Which system has the largest enclosed mass? Which has the largest physical Einstein radius? Discuss why the rankings are different.

Verify with `problems_09.wl`.

13.10 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_09.wl`, which verifies each answer symbolically and numerically.

Exercise 13.9.1: Enclosed Mass for an SIE Lens

(a) For a flat ΛCDM cosmology with $H_0 = 70$ km/s/Mpc and $\Omega_\mathrm{m} = 0.3$, the angular diameter distances are computed via:

$$D(z_1, z_2) = \frac{c}{(1+z_2)\,H_0} \int_{z_1}^{z_2} \frac{dz'}{E(z')}$$

where $E(z) = \sqrt{\Omega_\mathrm{m}(1+z)^3 + \Omega_\Lambda}$. Numerical evaluation gives:

$$\begin{aligned} D_\mathrm{d} &\approx 770 \text{ Mpc}\,, \\ D_\mathrm{s} &\approx 1640 \text{ Mpc}\,, \\ D_\mathrm{ds} &\approx 1180 \text{ Mpc}\,. \end{aligned}$$

(Exact values depend on the cosmological parameters and the integration method; see `problems_09.wl` for precise values.)

(b) The critical surface mass density is:

$$\Sigma_\mathrm{cr} = \frac{c^2}{4\pi G} \frac{D_\mathrm{s}}{D_\mathrm{d}\, D_\mathrm{ds}} \approx 3.2 \times 10^9\ M_\odot\, \mathrm{kpc}^{-2}\,.$$

(c) The enclosed projected mass is:

$$M_\mathrm{E} = \pi\, \theta_\mathrm{E}^2\, D_\mathrm{d}^2\, \Sigma_\mathrm{cr} = \frac{c^2}{4G}\, \theta_\mathrm{E}^2\, \frac{D_\mathrm{d}\, D_\mathrm{s}}{D_\mathrm{ds}}\,.$$

Converting $\theta_\mathrm{E} = 1.5'' = 7.27 \times 10^{-6}$ rad:

$$M_\mathrm{E} \approx 4.2 \times 10^{11}\ M_\odot\,.$$

This is larger than the typical stellar mass of a massive elliptical ($\sim 10^{11}\ M_\odot$), indicating significant dark matter within the Einstein radius. ✓

(d) From $\theta_\mathrm{E} = 4\pi\sigma_v^2 D_\mathrm{ds}/(c^2 D_\mathrm{s})$:

$$\sigma_v = c\sqrt{\frac{\theta_\mathrm{E}\, D_\mathrm{s}}{4\pi\, D_\mathrm{ds}}} \approx 270 \text{ km/s}\,.$$

This is typical of a massive early-type galaxy. ✓

Exercise 13.9.2: Mass-Sheet Degeneracy

(a) The original deflection satisfies $\nabla \cdot \boldsymbol{\alpha} = 2\kappa$. Under the MST:

$$\kappa_\lambda = \lambda\kappa + (1-\lambda) \implies \boldsymbol{\alpha}_\lambda = \lambda\boldsymbol{\alpha} + (1-\lambda)\boldsymbol{\theta}\,.$$

The transformed lens equation:

$$\begin{aligned}\boldsymbol{\beta}_\lambda &= \boldsymbol{\theta} - \boldsymbol{\alpha}_\lambda(\boldsymbol{\theta}) \\ &= \boldsymbol{\theta} - \lambda\boldsymbol{\alpha}(\boldsymbol{\theta}) - (1-\lambda)\boldsymbol{\theta} \\ &= \lambda[\boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})] \\ &= \lambda\boldsymbol{\beta}\,.\end{aligned}$$

So the same image positions $\boldsymbol{\theta}$ satisfy the transformed lens equation for source position $\lambda\boldsymbol{\beta}$. ✓

(b) The Jacobian is $\mathcal{A} = \mathbb{I} - \partial\boldsymbol{\alpha}/\partial\boldsymbol{\theta}$. Under the MST:

$$\begin{aligned}\mathcal{A}_\lambda &= \mathbb{I} - \frac{\partial\boldsymbol{\alpha}_\lambda}{\partial\boldsymbol{\theta}} \\ &= \mathbb{I} - \lambda\frac{\partial\boldsymbol{\alpha}}{\partial\boldsymbol{\theta}} - (1-\lambda)\mathbb{I} \\ &= \lambda\left(\mathbb{I} - \frac{\partial\boldsymbol{\alpha}}{\partial\boldsymbol{\theta}}\right) = \lambda\,\mathcal{A}\,.\end{aligned}$$

Therefore $\det\mathcal{A}_\lambda = \lambda^2\det\mathcal{A}$, and $\mu_\lambda = 1/\det\mathcal{A}_\lambda = \mu/\lambda^2$. ✓

(c) Since $\mu_{\lambda,i} = \mu_i/\lambda^2$ for every image i, the ratio:

$$\frac{\mu_{\lambda,i}}{\mu_{\lambda,j}} = \frac{\mu_i/\lambda^2}{\mu_j/\lambda^2} = \frac{\mu_i}{\mu_j}\,.$$

The λ^2 factors cancel. Flux ratios are invariant. ✓

(d) The Fermat potential is $\phi(\boldsymbol{\theta},\boldsymbol{\beta}) = \frac{1}{2}|\boldsymbol{\theta}-\boldsymbol{\beta}|^2 - \psi(\boldsymbol{\theta})$. Under the MST, the lensing potential transforms as $\psi_\lambda = \lambda\psi + \frac{1}{2}(1-\lambda)|\boldsymbol{\theta}|^2$. The time delay between images A and B is:

$$\begin{aligned}\Delta t_\lambda &= \frac{(1+z_d)D_\mathrm{d}D_\mathrm{s}}{c\,D_\mathrm{ds}}[\phi_\lambda(\boldsymbol{\theta}_A,\boldsymbol{\beta}_\lambda) - \phi_\lambda(\boldsymbol{\theta}_B,\boldsymbol{\beta}_\lambda)] \\ &= \lambda\,\Delta t\,.\end{aligned}$$

The derivation proceeds by substituting $\boldsymbol{\beta}_\lambda = \lambda\boldsymbol{\beta}$ and $\psi_\lambda = \lambda\psi + \frac{1}{2}(1-\lambda)|\boldsymbol{\theta}|^2$ and simplifying. The uniform $|\boldsymbol{\theta}|^2$ term cancels between the two images because they are at the same image positions, leaving a factor of λ multiplying the original Fermat potential difference. ✓

Exercise 13.9.3: H_0 from a Time-Delay Lens

(a) The time-delay equation is:

$$\Delta t = \frac{(1+z_d)}{c}\,\frac{D_\mathrm{d}\,D_\mathrm{s}}{D_\mathrm{ds}}\,\Delta\tau\,.$$

Solving for H_0: since $D_\mathrm{d}D_\mathrm{s}/D_\mathrm{ds} \propto 1/H_0$, we can write $D_\mathrm{d}D_\mathrm{s}/D_\mathrm{ds} = d(z_d, z_s, \Omega_\mathrm{m})/H_0$ where d is a dimensionless function of redshifts and cosmological parameters. Then:

$$H_0 = \frac{(1+z_d)\,\Delta\tau}{c\,\Delta t} \times d(z_d, z_s, \Omega_\mathrm{m})\,.$$

(b) For $z_d = 0.6$, $z_s = 1.5$, $\Omega_\mathrm{m} = 0.3$ (flat ΛCDM), the dimensionless distance ratio $d = H_0 D_\mathrm{d}D_\mathrm{s}/D_\mathrm{ds}$ is computed from the cosmological integrals. Converting $\Delta\tau = 0.64$ arcsec^2 $= 0.64 \times (4.848\times10^{-6})^2$ rad^2 $= 1.50\times10^{-11}$ rad^2 and $\Delta t = 80$ days $= 6.91\times10^6$ s:

Numerical evaluation gives $H_0 \approx 70$ km/s/Mpc (the exact value depends on the precise cosmological distances; see `problems_09.wl` for the detailed calculation). ✓

(c) If the true lens has an unmodeled mass sheet with $\kappa_s = 0.05$, then $\lambda = 1 - \kappa_s = 0.95$. The inferred H_0 is biased:

$$H_0^\mathrm{inferred} = \lambda\, H_0^\mathrm{true} = 0.95\, H_0^\mathrm{true}\,.$$

The inferred H_0 would be 5% too low. For a true value of ∼73 km/s/Mpc, the inferred value would be ∼69 km/s/Mpc — close to the Planck value. This illustrates why the MSD is central to the Hubble tension discussion. ✓

Exercise 13.9.4: Power-Law Profile

(a) For $\rho(r) = \rho_0(r/r_0)^{-\gamma'}$, the projected surface mass density is:

$$\Sigma(R) = \int_{-\infty}^{\infty} \rho(\sqrt{R^2+z^2})\,dz = \rho_0\, r_0^{\gamma'}\, \frac{\sqrt{\pi}\,\Gamma[(\gamma'-1)/2]}{\Gamma[\gamma'/2]}\, R^{1-\gamma'}$$

where $R = D_\mathrm{d}\theta$ is the projected physical radius. The convergence is $\kappa = \Sigma/\Sigma_\mathrm{cr}$, so:

$$\kappa(\theta) \propto \theta^{1-\gamma'}\,.$$

More precisely, $\kappa(\theta) = \frac{3-\gamma'}{2}(\theta_\mathrm{E}/\theta)^{\gamma'-1}$, where θ_E is defined as the radius where the mean convergence equals unity. ✓

(b) For $\gamma' = 2$: $\kappa \propto \theta^{1-2} = \theta^{-1} = 1/\theta$, which is the SIS convergence $\kappa = \theta_\mathrm{E}/(2\theta)$. ✓

(c) The enclosed mass within radius R scales as $M(\leq R) \propto R^{3-\gamma'}$. The Einstein mass is $M_\mathrm{E} = M(\leq R_\mathrm{E})$, fixed by lensing. The velocity dispersion scales as:

$$\sigma_v^2 \propto \frac{M(\leq R_\mathrm{ap})}{R_\mathrm{ap}} \propto R_\mathrm{ap}^{2-\gamma'}\,.$$

Since $R_\mathrm{ap} \neq R_\mathrm{E}$ in general, the ratio σ_v^2/M_E depends on γ', breaking the degeneracy between mass normalization and profile slope. ✓

(d) For a given θ_E and σ_v, the self-consistent γ' can be found from the relation linking both. For $\theta_\mathrm{E} = 1.2''$ and $\sigma_v = 230$ km/s, numerical evaluation (see `problems_09.wl`) gives $\gamma' \approx 2.0$–2.1, fully consistent with the bulge-halo conspiracy value $\gamma' \approx 2.0 \pm 0.2$. ✓

Exercise 13.9.5: Comparing Real Lens Systems

(a) Angular diameter distances (flat ΛCDM, $H_0 = 70$ km/s/Mpc, $\Omega_{\rm m} = 0.3$):

System	$D_{\rm d}$ (Mpc)	$D_{\rm s}$ (Mpc)	$D_{\rm ds}$ (Mpc)
Q0957+561	∼1040	∼1740	∼1155
Q2237+0305	∼163	∼1747	∼1683
B1608+656	∼1410	∼1739	∼777

(b) Enclosed projected masses ($M_{\rm E} = c^2\theta_{\rm E}^2 D_{\rm d} D_{\rm s}/(4GD_{\rm ds})$):

System	$M_{\rm E}$ ($10^{11}\,M_\odot$)
Q0957+561	~ 1.9
Q2237+0305	~ 0.2
B1608+656	~ 3.9

(c) Physical Einstein radii ($R_{\rm E} = D_{\rm d}\,\theta_{\rm E}$):

System	$R_{\rm E}$ (kpc)
Q0957+561	~ 5.0
Q2237+0305	~ 0.7
B1608+656	~ 6.8

(d) B1608+656 has both the largest enclosed mass ($\sim 3.9 \times 10^{11}\,M_\odot$) and the largest physical Einstein radius (~ 6.8 kpc), despite having a similar angular Einstein radius to Q0957+561. This is because B1608+656 is at higher redshift ($z_d = 0.63$), so the same angular size corresponds to a larger physical size and a larger mass. Q2237+0305 has the smallest mass and physical radius because its lens is so nearby ($z_d = 0.04$) — the angular diameter distance $D_{\rm d}$ is small, so the physical Einstein radius is only ~ 0.7 kpc even though $\theta_{\rm E} \approx 0.9''$. ✓

Chapter 14

Strong Lensing by Galaxy Clusters

> *Galaxy clusters are the most massive gravitationally bound objects in the universe. When they act as gravitational lenses, they produce the most dramatic manifestations of general relativity visible in the sky: giant luminous arcs stretching tens of arcseconds, dozens of multiply imaged background galaxies, and magnification factors large enough to reveal individual stars at cosmological distances.* — adapted from Meneghetti (2021)

This final module brings together the entire gravitational lensing framework developed in Modules 1–9 and applies it to the most spectacular setting: **galaxy clusters**. Where a single galaxy lens (Module 9) typically produces two or four images of a background source with an Einstein radius of $\sim 1''$, a massive galaxy cluster can produce dozens of multiple image systems, giant arcs with length-to-width ratios exceeding 10:1, and Einstein radii of $\sim 10''$–$50''$.

Galaxy clusters have total masses of 10^{14}–$10^{15}\,M_\odot$, with $\sim 85\%$ in dark matter, $\sim 13\%$ in hot intracluster gas, and only $\sim 2\%$ in stars. Their mass distributions are well-described by the NFW profile introduced in Module 11, now applied at scales 100–1000 times more massive than individual galaxy halos. The richness of cluster lensing — many image systems at multiple source redshifts — provides powerful constraints on the dark matter distribution and on cosmological parameters.

Companion Mathematica files:

- `cluster_lensing.wl` — NFW cluster lensing quantities; critical curves and caustics at cluster scale; ray-tracing through cluster potential to generate arcs; Einstein radius vs. cluster mass; figures exported to `Figures/10_Cluster_Lensing/`
- `nfw_cluster.nb` — interactive notebook with `Manipulate` widgets for NFW cluster parameters, giant arc formation, magnification maps, and weak lensing shear profiles

14.1 Introduction: Why Cluster Lensing Is Special

Galaxy clusters occupy a unique place in both astrophysics and gravitational lensing. They are the largest objects in the universe whose properties are determined by gravity alone — they sit at the top of the hierarchy of cosmic structure. Their enormous masses make them the most powerful gravitational lenses, producing phenomena that are qualitatively different from galaxy-scale lensing:

1. **Giant arcs.** Extended background galaxies are stretched into luminous arcs tens of arcseconds long, tracing out the tangential critical curve of the cluster potential. The first giant arc was discovered in Abell 370 by Soucail et al. (1987), confirming the gravitational lensing interpretation of these features.

2. **Many multiple image systems.** A single massive cluster can produce 10–30 or more distinct sets of multiple images, each from a different background galaxy at a different redshift. Each set independently constrains the cluster mass model.

3. **Extreme magnification.** Magnification factors of $\mu \sim 10$–100 are common near the critical curves of clusters. This turns clusters into "gravitational telescopes" that reveal galaxies and even individual stars at redshifts $z > 6$ that would otherwise be undetectable.

4. **Sensitivity to dark matter.** Because clusters are dark-matter dominated, their lensing signal directly probes the dark matter distribution — its radial profile, substructure, and overall mass.

5. **Cosmological probe.** The abundance and properties of cluster lenses depend on cosmological parameters (Ω_m, σ_8, the dark energy equation of state), making cluster lensing a tool for precision cosmology.

14.2 NFW Model for Cluster Lensing

In Module 11 (Section 11.6) we introduced the NFW density profile as the universal dark matter halo profile. Here we apply it at **cluster scale**, where the relevant parameters are much larger than for galaxy halos.

14.2.1 Cluster-Scale NFW Parameters

The NFW profile (eq. 11.34) is parametrized by the virial mass M_{200} and concentration $c = r_{200}/r_s$. Typical values for galaxy clusters are:

Parameter	Galaxy halo	Cluster halo
M_{200}	$10^{12}\,M_\odot$	10^{14}–$10^{15}\,M_\odot$
c	10–20	3–8
r_s	15–30 kpc	200–500 kpc
r_{200}	200–300 kpc	1–2 Mpc

The key difference is that clusters are *less concentrated* ($c \sim 3$–8) than galaxy halos ($c \sim 10$–20), reflecting their later formation epoch in the hierarchical structure formation scenario. Despite the lower concentration, the vastly greater mass more than compensates, producing far stronger lensing.

14.2.2 Lensing Quantities at Cluster Scale

The NFW convergence, deflection angle, and shear from Module 11 (eqs. 11.46–11.53) apply directly. We collect the key formulae here for reference. Define $x = \theta/\theta_s$, where $\theta_s = r_s/D_\mathrm{d}$ is the angular scale radius. Then:

$$\kappa(x) = \kappa_s\, f(x)\,, \tag{14.1}$$

$$\alpha(x) = \frac{2\kappa_s\,\theta_s}{x}\left[\ln\frac{x}{2} + g(x)\right]\,, \tag{14.2}$$

$$\bar\kappa(x) = \frac{2\kappa_s}{x^2}\left[\ln\frac{x}{2} + g(x)\right]\,, \qquad |\gamma(x)| = \bar\kappa(x) - \kappa(x)\,, \tag{14.3}$$

with the functions $f(x)$ and $g(x)$ defined in eqs. (11.45)–(11.51), and the characteristic convergence:

$$\kappa_s = \frac{2\rho_s\, r_s}{\Sigma_\mathrm{cr}}\,. \tag{14.4}$$

Example 14.2.1. Characteristic convergence for a massive cluster. For a cluster with $M_{200} = 10^{15}\, M_\odot$, $c = 5$, at $z_d = 0.3$ lensing a source at $z_s = 2$ ($D_\mathrm{d} \approx 900$ Mpc, $D_\mathrm{s} \approx 1750$ Mpc, $D_\mathrm{ds} \approx 1400$ Mpc):

- $r_{200} \approx 1.86$ Mpc, $r_s = r_{200}/c \approx 372$ kpc,
- $\theta_s = r_s/D_\mathrm{d} \approx 85''$,
- $\kappa_s \approx 0.52$.

Since $\kappa_s \times f(x)$ can exceed unity for $x \lesssim 0.5$ (i.e., $\theta \lesssim 43''$), the cluster has a central region of super-critical convergence and is therefore a strong lens. These values are computed and verified in `cluster_lensing.wl`.

14.2.3 Critical Curves and Caustics for Cluster NFW

For the circularly symmetric NFW profile, the tangential and radial critical curves are circles. The tangential critical curve occurs where $\bar\kappa(x) = 1$, i.e.:

$$\frac{2\kappa_s}{x_t^2}\left[\ln\frac{x_t}{2} + g(x_t)\right] = 1\,, \tag{14.5}$$

and the radial critical curve where $2\kappa(x_r) - \bar\kappa(x_r) = 1$, i.e.:

$$2\kappa_s\, f(x_r) - \frac{2\kappa_s}{x_r^2}\left[\ln\frac{x_r}{2} + g(x_r)\right] = 1\,. \tag{14.6}$$

Both equations must be solved numerically. For the cluster parameters above ($M_{200} = 10^{15}\, M_\odot$, $c = 5$, $z_d = 0.3$, $z_s = 2$), the tangential critical curve lies at $x_t \approx 0.19$, giving $\theta_t \approx 16''$ — a physical radius of ~ 70 kpc at the cluster redshift.

Remark 14.2.1. The tangential critical curve radius is often called the **Einstein radius** of the cluster, θ_E. A *spherical* NFW halo with $M_{200} \sim 10^{15}\, M_\odot$ and $c \sim 5$ typically has $\theta_\mathrm{E} \sim 10''$–$20''$ at these redshifts. Observed clusters (Abell 1689, MACSJ 0717, etc.) frequently display $\theta_\mathrm{E} \sim 30''$–$50''$; this additional lensing strength comes from ellipticity, substructure, merger boosts, and departures of the true mass distribution from a smooth spherical NFW. For a galaxy lens the Einstein radius is instead $\sim 1''$, and the enclosed projected mass within θ_E ranges from $\sim 10^{11}\, M_\odot$ (galaxies) to $\sim 10^{13}$–$10^{14}\, M_\odot$ (clusters).

14.2.4 Einstein Radius Scaling with Cluster Mass

The Einstein radius of an NFW cluster depends on mass, concentration, and the lens/source redshift configuration. For a fixed concentration and redshift configuration, the scaling is approximately:

$$\theta_{\mathrm{E}} \propto M_{200}^{1/3}\,, \tag{14.7}$$

which arises because $\theta_{\mathrm{E}} \sim r_{200}/D_{\mathrm{d}}$ and $r_{200} \propto M_{200}^{1/3}$ (from the definition $M_{200} = (4\pi/3)\,200\,\rho_{\mathrm{crit}}\,r_{200}^3$). The actual relationship is more complex due to the dependence of κ_s on c and the nonlinear solution of eq. (14.5), but the cube-root scaling provides a useful approximation.

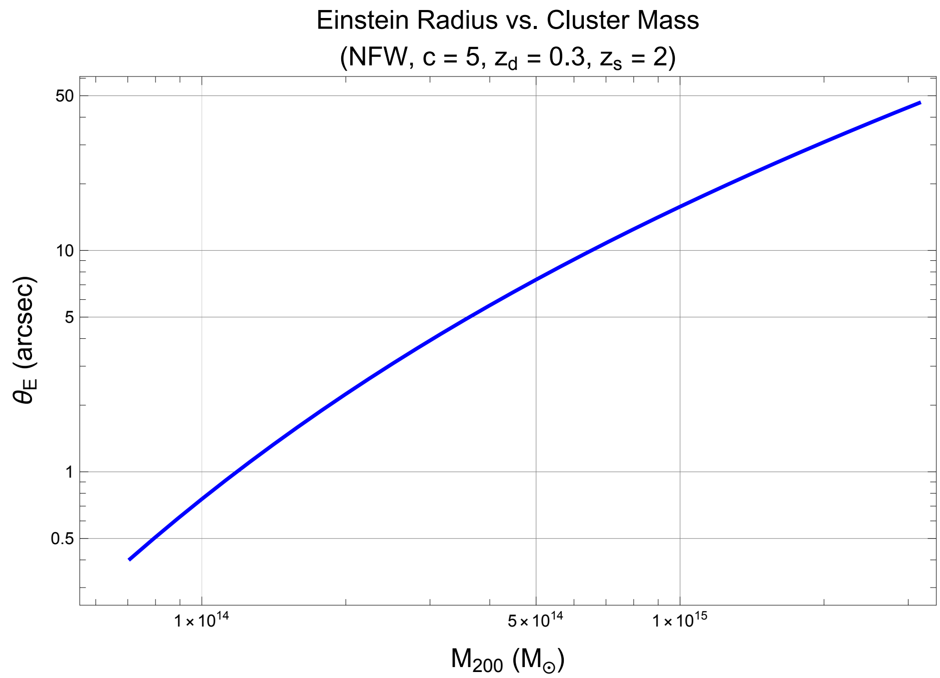


Figure 14.1: Einstein radius θ_{E} as a function of cluster mass M_{200} for an NFW profile with $c = 5$, $z_d = 0.3$, and $z_s = 2$. The dashed line shows the approximate $M^{1/3}$ scaling. Clusters below $M_{200} \sim 10^{14}\,M_{\odot}$ have Einstein radii comparable to galaxy-scale lenses. Generated by `cluster_lensing.wl`.

14.3 Giant Arcs

Giant luminous arcs are the most visually striking signature of cluster strong lensing. They were first discovered by Lynds & Petrosian (1986) and Soucail et al. (1987) in the clusters Abell 370 and Cl 2244-02, and their interpretation as gravitationally lensed images of background galaxies was confirmed spectroscopically by Soucail et al. (1988).

14.3.1 Formation Mechanism

Giant arcs form when an extended background galaxy lies close to the **tangential caustic** of the cluster. From the theory of critical curves (Module 12, Section 14.3):

1. The source galaxy is near (but not exactly on) the tangential caustic in the source plane.

2. Its images lie near the tangential critical curve in the image plane, where $\lambda_- \approx 0$.

3. The images are stretched *tangentially* (along the critical curve) by a large factor $\mu_{\rm tan} \propto 1/\sqrt{d_\perp}$, where $d_\perp$ is the perpendicular distance from the caustic.

4. The radial magnification remains of order unity, so the image is a thin, elongated **arc** that traces the critical curve.

14.3.2 Ray-Tracing Through a Cluster Potential

To visualize arc formation, we **ray-trace** an extended source through the cluster lens equation. The procedure is:

1. Define the source as a set of points in the source plane, e.g., a circular disk of angular radius β_0 centered at position $\boldsymbol{\beta}_0$.

2. For each source point $\boldsymbol{\beta}$, solve the lens equation $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ for all image positions $\boldsymbol{\theta}$. For a cluster, this is done numerically on a fine grid.

3. Plot the union of all image positions to obtain the lensed image of the source.

In practice, it is more efficient to use the **inverse ray-tracing** approach: for each pixel $\boldsymbol{\theta}$ in the image plane, compute $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ and check whether $\boldsymbol{\beta}$ falls within the source. If it does, that pixel is part of a lensed image.

14.3.3 Length-to-Width Ratio

The **length-to-width ratio** (or axis ratio) of an arc is a direct measure of the tangential magnification:

$$\frac{L}{W} \approx \mu_{\rm tan} = \frac{1}{|1 - \bar{\kappa}(\theta)|} \propto \frac{1}{\sqrt{d_\perp}}, \tag{14.8}$$

where L and W are the length and width of the arc. Observationally, arcs with $L/W \geq 10$ are classified as “giant arcs”, while those with $3 \leq L/W < 10$ are often called “arclets”.

14.3.4 Arc Statistics

The probability that a cluster produces a giant arc depends on:

- **Cluster mass:** More massive clusters have larger Einstein radii and larger tangential caustics, increasing the cross-section for arc production. The arc cross-section scales steeply with mass, roughly as $\sigma_{\rm arc} \propto M_{200}^{2\text{–}3}$.

- **Source redshift:** Higher source redshifts increase $D_{\rm ds}/D_{\rm s}$, which increases κ_s and the Einstein radius.

- **Cluster ellipticity and substructure:** Non-axisymmetric mass distributions and merging clusters enhance the arc cross-section significantly, often by a factor of 2–5.
- **Source size and morphology:** Extended sources are more easily stretched into arcs; compact sources are more likely to appear as multiple point-like images.

Remark 14.3.1. Arc statistics have been a source of tension with cosmological models. Bartelmann et al. (1998) found that the observed number of giant arcs exceeded the prediction from ΛCDM by a factor of ∼10. This "arc statistics problem" has been partially resolved by accounting for cluster ellipticity, substructure, and the contribution of baryonic physics (cooling, AGN feedback) to the central mass concentration, but it remains an active area of research.

14.4 Multiple Image Systems in Clusters

One of the most powerful aspects of cluster lensing is that a single cluster can lens **many** background galaxies simultaneously, producing multiple sets of multiple images. Each set of images (called an "image family" or "image system") comes from a single background source at a particular redshift.

14.4.1 Image Families

A typical strong lensing cluster produces:

- **5–10 secure image families** with spectroscopic redshift confirmation in well-studied clusters.
- **20–50 candidate image families** identified by morphology, color, and photometric redshifts in deep imaging surveys.
- In extreme cases (e.g., Abell 1689), over 100 candidate images have been identified from more than 30 distinct background sources.

Each image family provides independent constraints on the cluster mass model:

1. The **image positions** constrain the deflection field $\boldsymbol{\alpha}(\boldsymbol{\theta})$ at several points.
2. The **magnification ratios** (flux ratios between images) constrain the convergence and shear fields.
3. The **redshift** of each family provides a distance ratio $D_{\rm ds}/D_{\rm s}$ that weights the lensing signal differently, breaking degeneracies in the mass model.

14.4.2 Spectroscopic Confirmation

Image families are confirmed by matching the redshifts of candidate images: if multiple images at different positions have the same spectroscopic redshift, they are identified as images of the same source. Modern multi-object spectrographs (VLT/MUSE, Keck/DEIMOS) have been transformative for cluster lensing, enabling redshift measurements of faint, highly magnified background galaxies.

14.4.3 Famous Cluster Lenses

Several galaxy clusters have become benchmarks for strong lensing studies:

- **Abell 370** ($z = 0.375$): Site of the first discovered giant arc (Soucail et al. 1987). Over 10 multiple image systems identified.
- **Abell 1689** ($z = 0.183$): One of the most powerful known lenses, with over 100 images from ∼30 background sources. Its Einstein radius of $\sim 47''$ (for $z_s \sim 2$) is among the largest known.
- **MACSJ0416.1-2403** ($z = 0.396$): A Hubble Frontier Field cluster with over 60 spectroscopically confirmed multiple images, used for detailed mass reconstruction.
- **Abell 2744** ($z = 0.308$): The "Pandora's Cluster", a complex merging system with extensive strong and weak lensing constraints.

14.5 Cluster Mass Estimation

Gravitational lensing provides a *direct* measurement of the projected mass distribution of a cluster, independent of assumptions about the dynamical state of the system. This makes it complementary to X-ray and kinematic mass estimates, which rely on assumptions of hydrostatic or virial equilibrium.

14.5.1 Strong Lensing Mass Estimates

Strong lensing constrains the mass distribution in the **inner region** of the cluster, typically within $\sim$ 200–300 kpc of the center. The basic approach is:

1. **Identify multiple image systems** and measure their positions and redshifts.
2. **Parametrize the mass model.** The cluster mass is described by one or more NFW halos (for the smooth dark matter component) plus additional components for cluster member galaxies (typically scaled isothermal profiles).
3. **Optimize the model parameters** by minimizing the difference between predicted and observed image positions. Modern codes (e.g., `lenstool`, `glafic`) use Bayesian methods (MCMC) to explore the parameter space.

4. **Derive the mass profile** from the best-fit model.

A quick estimate of the enclosed mass within the Einstein radius is provided by:

$$\boxed{M(\theta \leq \theta_{\rm E}) = \pi\,(D_{\rm d}\,\theta_{\rm E})^2\,\Sigma_{\rm cr}} \tag{14.9}$$

since $\bar{\kappa}(\theta_{\rm E}) = 1$ by definition. For a cluster with $\theta_{\rm E} = 30''$ at $z_d = 0.3$ with $z_s = 2$:

$$M(\theta_{\rm E}) \approx 2 \times 10^{14}\,M_\odot\,. \tag{14.10}$$

This is the projected mass within a cylinder of radius ~ 130 kpc — a substantial fraction of the total cluster mass. The exact value is computed in `cluster_lensing.wl`.

14.5.2 Weak Lensing Mass Estimates (Preview)

Strong lensing probes only the inner $\sim$10% of the cluster virial radius. To measure the total cluster mass, one must extend to larger radii using **weak lensing** (Section 14.7).

14.5.3 Combining Strong and Weak Lensing

The most accurate cluster mass profiles come from combining strong and weak lensing:

- **Strong lensing** provides high-resolution constraints on the inner mass profile ($r \lesssim$ 200 kpc), including the NFW concentration parameter.
- **Weak lensing** provides constraints at larger radii (200 kpc $\lesssim r \lesssim$ 3 Mpc), determining the virial mass M_{200}.
- The combined profile spans nearly two decades in radius and is well-fit by an NFW profile, supporting the universality of the dark matter halo profile.

14.5.4 Comparison with Other Mass Estimators

Cluster masses can also be estimated from:

- **X-ray observations:** The hot intracluster medium (ICM) emits thermal bremsstrahlung X-rays. Under the assumption of hydrostatic equilibrium, the gas temperature and density profiles yield the total mass profile. Hydrostatic masses are typically 10–20% lower than lensing masses (the "hydrostatic mass bias"), likely due to non-thermal pressure support from turbulence and bulk motions.
- **Galaxy velocity dispersion:** The velocities of cluster member galaxies, combined with the virial theorem, yield a mass estimate. This method is noisy (requiring many galaxy redshifts) and sensitive to interlopers.
- **Sunyaev–Zel'dovich (SZ) effect:** Inverse Compton scattering of CMB photons by the hot ICM produces a spectral distortion. The integrated SZ signal is proportional to the total thermal energy and correlates with total mass.

Remark 14.5.1. The complementarity of these methods is powerful. Comparing lensing masses (which are model-independent) with X-ray/SZ masses (which assume hydrostatic equilibrium) calibrates the hydrostatic mass bias, which is a critical systematic in using cluster counts for cosmology.

14.6 Magnification and the Cosmic Telescope

Perhaps the most exciting application of cluster lensing in contemporary astrophysics is the use of massive clusters as **gravitational telescopes**: natural magnifying glasses that amplify the light of distant background galaxies, allowing us to study objects that would otherwise be far too faint to detect.

14.6.1 Magnification by Clusters

Near the critical curves of a massive cluster, the magnification can reach $\mu \sim 10$–100 or more. Since the observed flux scales as μ (for sources smaller than the scale over which μ varies), a source at $z \sim 10$ behind a cluster lens can appear as bright as an unmagnified source at $z \sim 3$–4.

The trade-off is that magnification also reduces the effective area in the source plane. A magnification of μ enlarges the image area by μ but surveys a source-plane area that is μ times smaller. The net effect on the number of detected sources depends on the slope of the luminosity function — this is the **magnification bias**.

14.6.2 Magnification Bias

The magnification bias arises because lensing simultaneously:

1. **Amplifies flux:** Sources below the detection threshold are boosted above it, increasing the detected number.

2. **Dilutes area:** The effective survey area is reduced by $1/\mu$, decreasing the detected number.

For a cumulative luminosity function $N(> S) \propto S^{-\alpha}$ (where S is flux and α is the faint-end slope), the net number count behind a lens changes by a factor of $\mu^{\alpha-1}$. For $\alpha > 1$ (steep luminosity function), magnification bias *increases* the number of detected sources; for $\alpha < 1$, it *decreases* the count. At high redshift, the luminosity function is steep ($\alpha \gtrsim 2$), so there is a significant net increase in detected galaxies behind clusters.

14.6.3 The Hubble Frontier Fields

The Hubble Frontier Fields (HFF) program (2013–2017) was a major HST initiative that obtained ultra-deep imaging of six massive galaxy clusters chosen for their strong lensing power:

1. Abell 2744 ($z = 0.308$)

2. MACSJ0416.1-2403 ($z = 0.396$)
3. MACSJ0717.5+3745 ($z = 0.545$)
4. MACSJ1149.5+2223 ($z = 0.543$)
5. Abell S1063 ($z = 0.348$)
6. Abell 370 ($z = 0.375$)

These clusters were used as gravitational telescopes to detect galaxies at $z \sim 6$–10, probing the epoch of reionization. The HFF program discovered some of the most distant galaxies known at the time and provided detailed mass models of the six clusters from their many multiple image systems.

14.6.4 JWST and the New Era of Cluster Lensing

The James Webb Space Telescope (JWST, launched 2021) has revolutionized cluster lensing with its superior infrared sensitivity and angular resolution. Notable results include:

- **Earendel** (WHL0137-LS): The most distant individual star known, at $z = 6.2$, magnified by a factor of $\mu \gtrsim 4000$ by the cluster WHL 0137-08 (Welch et al. 2022). The extreme magnification occurs because the star lies almost exactly on the cluster's critical curve.

- **The Sunrise Arc**: A galaxy at $z = 6.2$ lensed into a long arc by the same cluster, in which Earendel was identified as a point source on the arc.

- **Early galaxy candidates at $z > 10$**: Lensed galaxies behind clusters observed with JWST/NIRCam have pushed the redshift frontier beyond $z \sim 13$, with magnifications of $\mu \sim 5$–30 making these detections possible.

Remark 14.6.1. The discovery of Earendel illustrates the extreme power of cluster lensing. An individual star at $z = 6.2$ has an apparent magnitude of $m \sim 40$ without magnification — utterly undetectable. A magnification of $\mu \sim 4000$ corresponds to $\Delta m \sim -9$ magnitudes, bringing it within reach of HST and JWST. However, such extreme magnification requires the source to be within $\sim 0.001''$ of the critical curve, making such discoveries exceedingly rare.

14.7 Brief Introduction to Weak Lensing

Throughout Modules 1–9 and the preceding sections of this module, we have focused on **strong lensing**: the regime where $\kappa \gtrsim 1$ and multiple images or giant arcs are produced. But gravitational lensing does not stop at the strong lensing regime. At larger radii from the cluster center (and behind foreground mass distributions in general), the lensing effect is weak — too small to produce multiple images, but still measurable statistically. This is the domain of **weak gravitational lensing**.

This section provides a brief introduction to weak lensing — just enough to show where the field goes beyond strong lensing and to connect to the rich literature on the subject. A full treatment of weak lensing is beyond the scope of these tutorials.

14.7.1 The Weak Lensing Regime

In the weak lensing regime, $\kappa \ll 1$ and $|\gamma| \ll 1$. There are no multiple images or giant arcs. Instead, the primary observable effect is a small, coherent **distortion** (shearing) of the shapes of background galaxies. An intrinsically circular galaxy is sheared into a slightly elliptical shape, with the orientation and magnitude of the induced ellipticity determined by the local shear field.

The key challenge is that individual galaxy shapes are intrinsically elliptical, with typical intrinsic ellipticities of $|\epsilon_{\rm int}| \sim 0.2$–$0.3$. The gravitational shear signal is $|\gamma| \sim 0.01$–0.05 in the weak lensing regime — much smaller than the intrinsic shape noise. Therefore, one must **average** over many background galaxies to extract the coherent shear signal.

14.7.2 The Reduced Shear

In the weak lensing regime, the observable quantity is the **reduced shear**:

$$\boxed{g = \frac{\gamma}{1-\kappa}} \tag{14.11}$$

This is because the lensing transformation preserves surface brightness, and the observed ellipticity of a lensed galaxy (for $|g| \leq 1$) is:

$$\epsilon^{\rm obs} = \frac{\epsilon^{\rm int} + g}{1 + g^* \, \epsilon^{\rm int}} \approx \epsilon^{\rm int} + g \,, \tag{14.12}$$

where the approximation holds for $|g| \ll 1$. Since intrinsic ellipticities are randomly oriented (in the absence of intrinsic alignments), their average vanishes: $\langle \epsilon^{\rm int} \rangle \approx 0$. Therefore:

$$\langle \epsilon^{\rm obs} \rangle \approx g \approx \gamma \quad (\text{for } \kappa \ll 1) \,. \tag{14.13}$$

The mean observed ellipticity of background galaxies is an unbiased estimator of the shear (in the weak limit).

14.7.3 Tangential Shear Profile Around Clusters

For a circularly symmetric mass distribution, the shear is purely tangential. The **tangential shear** at angular distance θ from the cluster center is:

$$\gamma_t(\theta) = \bar{\kappa}(\theta) - \kappa(\theta) \,, \tag{14.14}$$

which we recognize from eq. (11.7) (Module 11). The tangential shear profile $\gamma_t(\theta)$ can be measured by averaging the tangential component of the ellipticities of background galaxies in annular bins around the cluster center.

For an NFW profile, $\gamma_t(\theta)$ has a characteristic shape that rises toward the center (reaching the strong lensing regime) and falls off at large radii. Fitting the observed $\gamma_t(\theta)$ profile to the NFW prediction yields the cluster mass M_{200} and concentration c.

14.7.4 From Clusters to Cosmic Shear

The weak lensing framework extends far beyond individual cluster halos. The **cosmic shear** signal — the coherent shearing of galaxy shapes by the entire large-scale structure of the universe along the line of sight — is one of the primary probes of modern cosmology. Cosmic shear measurements constrain the matter power spectrum, the matter density Ω_m, and the amplitude of matter fluctuations σ_8.

Current and upcoming surveys measuring cosmic shear include:

- **Euclid** (ESA, launched 2023): space-based imaging of $\sim$1.5 billion galaxies over 15,000 deg^2.
- **Vera C. Rubin Observatory / LSST**: ground-based survey of $\sim$4 billion galaxies over 18,000 deg^2 (first light expected 2025).
- **Nancy Grace Roman Space Telescope** (NASA, expected launch 2027): space-based survey complementing Euclid.

These surveys will measure cosmic shear with percent-level precision, providing powerful constraints on dark energy, modified gravity, and the neutrino mass hierarchy.

14.8 The State of the Field and Future Directions

Strong lensing by galaxy clusters is a mature but rapidly evolving field. The combination of new observational facilities and sophisticated modeling techniques continues to push the boundaries of what cluster lensing can teach us.

14.8.1 Current Observational Facilities

- **HST:** The Hubble Space Telescope has been the workhorse for cluster strong lensing for three decades, with programs like the Hubble Frontier Fields and RELICS (Reionization Lensing Cluster Survey) producing the deepest cluster lensing data.

- **JWST:** With its infrared capabilities, JWST is discovering lensed galaxies at $z > 10$, measuring spectroscopic redshifts of multiple image systems, and resolving detailed structure in giant arcs.

- **Euclid:** The wide-field survey will discover thousands of new cluster lenses, dramatically increasing the statistical power of cluster lensing studies.

- **Rubin/LSST:** The deep, wide survey will combine strong and weak lensing for a large sample of clusters, with time-domain capability for detecting lensed transients.

14.8.2 Future Facilities

- **Nancy Grace Roman Space Telescope:** High-resolution, wide-field infrared imaging will complement JWST for cluster lensing studies.

- **Square Kilometre Array (SKA):** Radio weak lensing with the SKA will provide shape measurements free from many of the systematics that affect optical measurements (atmospheric seeing, PSF modeling).
- **Extremely Large Telescopes** (ELT, TMT, GMT): Ground-based 25–39m telescopes with adaptive optics will resolve sub-arcsecond structure in cluster arcs and obtain spectra of extremely faint lensed sources.

14.8.3 Open Questions

1. **Dark matter substructure:** CDM predicts a rich hierarchy of subhalos within clusters. The number and mass function of subhalos affect the lensing signal (flux ratio anomalies, perturbations to giant arcs). Comparing predictions to observations constrains alternative dark matter models (warm dark matter, self-interacting dark matter).
2. **The nature of dark matter:** The inner density profile of clusters (cusp vs. core) is sensitive to dark matter self-interactions. Strong lensing at the center of clusters provides the highest-resolution probe of the dark matter profile.
3. **The cluster mass function:** The abundance of massive clusters as a function of mass and redshift is a sensitive cosmological probe. Lensing-calibrated cluster masses are essential for turning cluster counts into cosmological constraints.
4. **Multi-plane lensing and line-of-sight effects:** Light rays from distant sources pass through multiple mass concentrations along the line of sight. Accounting for these "line-of-sight" effects is important for precision mass modeling.
5. **Time-delay cosmography with clusters:** Time delays between multiple images of time-varying sources (supernovae, quasars) in clusters constrain H_0. The multiply imaged supernova "Refsdal" in MACSJ1149 (Kelly et al. 2015) was the first such measurement.

14.8.4 Connection to the AGEL Project

The ASTRO 3D Galaxy Evolution with Lensing (AGEL) survey is a spectroscopic survey of galaxy-scale strong lens candidates selected from wide-field imaging surveys. While AGEL focuses on galaxy-scale lenses rather than clusters, many of the techniques and physical principles are the same:

- NFW profiles and isothermal models describe the lens mass distribution at both galaxy and cluster scales.
- Critical curves and caustics determine the image configurations at all mass scales.
- Strong lensing mass estimates and comparisons with dynamical masses apply to both galaxy and cluster lenses.

The transition from galaxy to cluster lensing is one of scale rather than fundamental physics — the same equations derived in Modules 1–9 govern both regimes.

14.9 Summary

- **Galaxy clusters** ($M_{200} \sim 10^{14}$–$10^{15}\,M_\odot$) are the most powerful gravitational lenses, producing giant arcs, dozens of multiple image systems, and magnification factors of $\mu \sim 10$–100.
- The **NFW profile** at cluster scale has $c \sim 3$–8 and $r_s \sim 200$–500 kpc. The Einstein radius is $\theta_\mathrm{E} \sim 10''$–$50''$, enclosing a projected mass of $\sim 10^{14}\,M_\odot$.
- **Giant arcs** form when extended sources lie near the tangential caustic. The length-to-width ratio $L/W \approx \mu_\mathrm{tan} \propto 1/\sqrt{d_\perp}$ can exceed 10:1 for sources very close to the caustic.
- **Multiple image systems** (10–30+ per cluster) each independently constrain the mass model. Spectroscopic confirmation via redshift matching identifies image families.
- **Cluster mass estimation** uses strong lensing (inner profile) and weak lensing (outer profile), complemented by X-ray and SZ measurements. Strong lensing gives $M(\theta_\mathrm{E}) = \pi(D_\mathrm{d}\,\theta_\mathrm{E})^2\,\Sigma_\mathrm{cr}$.
- Clusters serve as **gravitational telescopes**, magnifying high-redshift galaxies and enabling discoveries like Earendel ($z = 6.2$, $\mu \gtrsim 4000$) and galaxies at $z > 10$.
- **Weak lensing** measures the statistical shear $\langle\epsilon\rangle \approx g = \gamma/(1-\kappa)$ of background galaxies. The tangential shear profile $\gamma_t(\theta)$ yields the cluster mass profile at large radii.
- **Current and future facilities** (JWST, Euclid, Rubin/LSST, Roman, SKA) will transform cluster lensing with larger samples, deeper imaging, and new wavelength regimes.

Where to Go from Here

This module concludes the *Learning to Lens* tutorial suite. We have covered the foundations of general relativity (Modules 1a–1e), the theory of gravitational lensing (Modules 2–6), and the modeling and application of strong lensing (Modules 7–10). For readers who wish to continue deeper into the field, we suggest the following directions:

- **Advanced topics not covered in these tutorials:** dark matter substructure and flux ratio anomalies; line-of-sight effects and multi-plane lensing; free-form (non-parametric) mass reconstruction; lens modeling with extended source morphology; microlensing by stars in the lensing galaxy.
- **Software packages for lens modeling:**
  - `lenstronomy` (https://github.com/lenstronomy/lenstronomy): a comprehensive Python package for gravitational lens modeling and simulation.
  - `PyAutoLens` (https://github.com/Jammy2211/PyAutoLens): automated strong lens modeling with Bayesian inference.

  - `glafic` (https://www.slac.stanford.edu/~oguri/glafic/): a fast lens modeling code widely used for cluster lensing.
  - `GRALE` (https://research.edm.uhasselt.be/~jlieMDC/): free-form mass reconstruction using a genetic algorithm.

- **Key review articles for further reading:** Kneib & Natarajan (2011), "Cluster lenses", A&A Rev.; Treu (2010), "Strong Lensing by Galaxies", ARA&A; Bartelmann & Schneider (2001), "Weak Gravitational Lensing", Phys. Rep.; Kilbinger (2015), "Cosmology with Cosmic Shear", Rep. Prog. Phys.

- **Active research:** The AGEL project (Tran et al. 2022) exemplifies how the techniques developed in these tutorials are applied in practice. Finding, confirming, and modeling galaxy-scale strong lenses in wide-field surveys is an active area of research to which the reader is now well-prepared to contribute.

14.10 Problems

Exercise 14.10.1. (NFW cluster Einstein radius and enclosed mass.) Consider an NFW cluster with $M_{200} = 5 \times 10^{14}\, M_\odot$, $c = 5$, at $z_d = 0.3$ lensing a source at $z_s = 2$. Use $D_\mathrm{d} = 900$ Mpc, $D_\mathrm{s} = 1750$ Mpc, $D_\mathrm{ds} = 1400$ Mpc.

(a) Compute the NFW parameters: r_{200}, r_s, ρ_s, θ_s, and κ_s.

(b) Find the tangential critical curve radius θ_t (the Einstein radius θ_E) by solving $\bar{\kappa}(\theta_t) = 1$ numerically.

(c) Compute the projected mass enclosed within θ_E using eq. (14.9).

(d) Compare this enclosed mass to the total virial mass M_{200}. What fraction of the cluster mass is enclosed within the Einstein radius?

Verify with `problems_10.wl`.

Exercise 14.10.2. (Ray-tracing a source through a cluster potential.) Using the NFW cluster from Exercise 14.10.1:

(a) Set up inverse ray-tracing: for each pixel $\boldsymbol{\theta}$ in the image plane, compute $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$ using the NFW deflection angle (eq. 14.2).

(b) Place a circular source of angular radius $0.5''$ at three positions: (i) $\beta = 0$ (on the optical axis), (ii) $\beta = 0.5\,\theta_\mathrm{E}$ (halfway to the critical curve), and (iii) $\beta = \theta_\mathrm{E}$ (on the tangential caustic).

(c) Plot the resulting lensed images for each source position. Identify the Einstein ring, giant arc, and counter-image morphologies.

(d) Estimate the length-to-width ratio of the arc in case (iii).

Verify with `problems_10.wl`.

Exercise 14.10.3. (Magnification of a high-redshift galaxy.) Consider a galaxy at $z_s = 9$ lensed by a cluster at $z_d = 0.3$ with $\theta_\mathrm{E} = 30''$. (Use $D_\mathrm{d} = 900$ Mpc, $D_\mathrm{s} = 1850$ Mpc, $D_\mathrm{ds} = 1500$ Mpc.)

(a) For an SIS model with the same Einstein radius, compute the magnification of a point source at $\beta = 3''$, $1''$, and $0.1''$.

(b) What is the apparent magnitude boost (Δm) for a source at $\beta = 0.1''$ from the lens center? ($\Delta m = -2.5 \log_{10} \mu$.)

(c) At $z_s = 9$, an L_* galaxy has an apparent magnitude of $m \approx 31$ (AB) in the near-infrared. If the detection limit of JWST is $m \approx 29$, what minimum magnification is needed to detect such a galaxy? At what source position β does this magnification occur (for the SIS model)?

(d) Discuss the implications for surveys of high-redshift galaxies behind clusters.

Verify with `problems_10.wl`.

Exercise 14.10.4. (Einstein radii across mass scales.) Compare the Einstein radii of three lenses at $z_d = 0.3$ with a source at $z_s = 2$, using $D_\mathrm{d} = 900$ Mpc, $D_\mathrm{s} = 1750$ Mpc, $D_\mathrm{ds} = 1400$ Mpc:

(a) A galaxy lens modeled as an SIS with $\sigma_v = 250\ \mathrm{km\,s^{-1}}$ ($M \approx 10^{12}\, M_\odot$).

(b) A group lens modeled as an SIS with $\sigma_v = 500\ \mathrm{km\,s^{-1}}$ ($M \approx 10^{13}\, M_\odot$).

(c) A cluster lens modeled as an SIS with $\sigma_v = 1200\ \mathrm{km\,s^{-1}}$ ($M \approx 10^{15}\, M_\odot$).

(d) Compute θ_E for each case. By what factor does θ_E increase from galaxy to cluster?

(e) Compute the enclosed mass within θ_E for each case. Compare with the total mass of each object.

(f) Discuss why the scaling $\theta_\mathrm{E} \propto \sigma_v^2$ (from the SIS formula) leads to a qualitative difference in lensing phenomenology between galaxies and clusters.

Verify with `problems_10.wl`.

Exercise 14.10.5. (Weak lensing mass estimate.) A cluster at $z_d = 0.3$ has a measured tangential shear profile $\gamma_t(\theta) = 0.1 \times (\theta/1')^{-0.8}$.

(a) Using the relation $\gamma_t(\theta) = \bar\kappa(\theta) - \kappa(\theta)$, and assuming that $\kappa \ll \bar\kappa$ at large radii (so $\gamma_t \approx \bar\kappa$), estimate the mean convergence at $\theta = 1'$, $3'$, and $5'$.

(b) Convert $\bar\kappa(\theta)$ to the projected mass enclosed within θ: $M(\theta) = \pi\, (D_\mathrm{d}\, \theta)^2\, \Sigma_\mathrm{cr}\, \bar\kappa(\theta)$. Use $\Sigma_\mathrm{cr} \approx 3.5 \times 10^{15}\, M_\odot\, \mathrm{Mpc^{-2}}$ for $z_d = 0.3$, $z_s = 1$ (a typical background redshift for weak lensing), and $D_\mathrm{d} = 900$ Mpc.

(c) Estimate M within $\theta = 5'$ (corresponding to a physical radius of ~ 1.3 Mpc at $z_d = 0.3$).

(d) Compare your estimated mass with a typical cluster virial mass of $5 \times 10^{14}\, M_\odot$. Discuss whether the result is reasonable and what assumptions may affect the accuracy.

Verify with `problems_10.wl`.

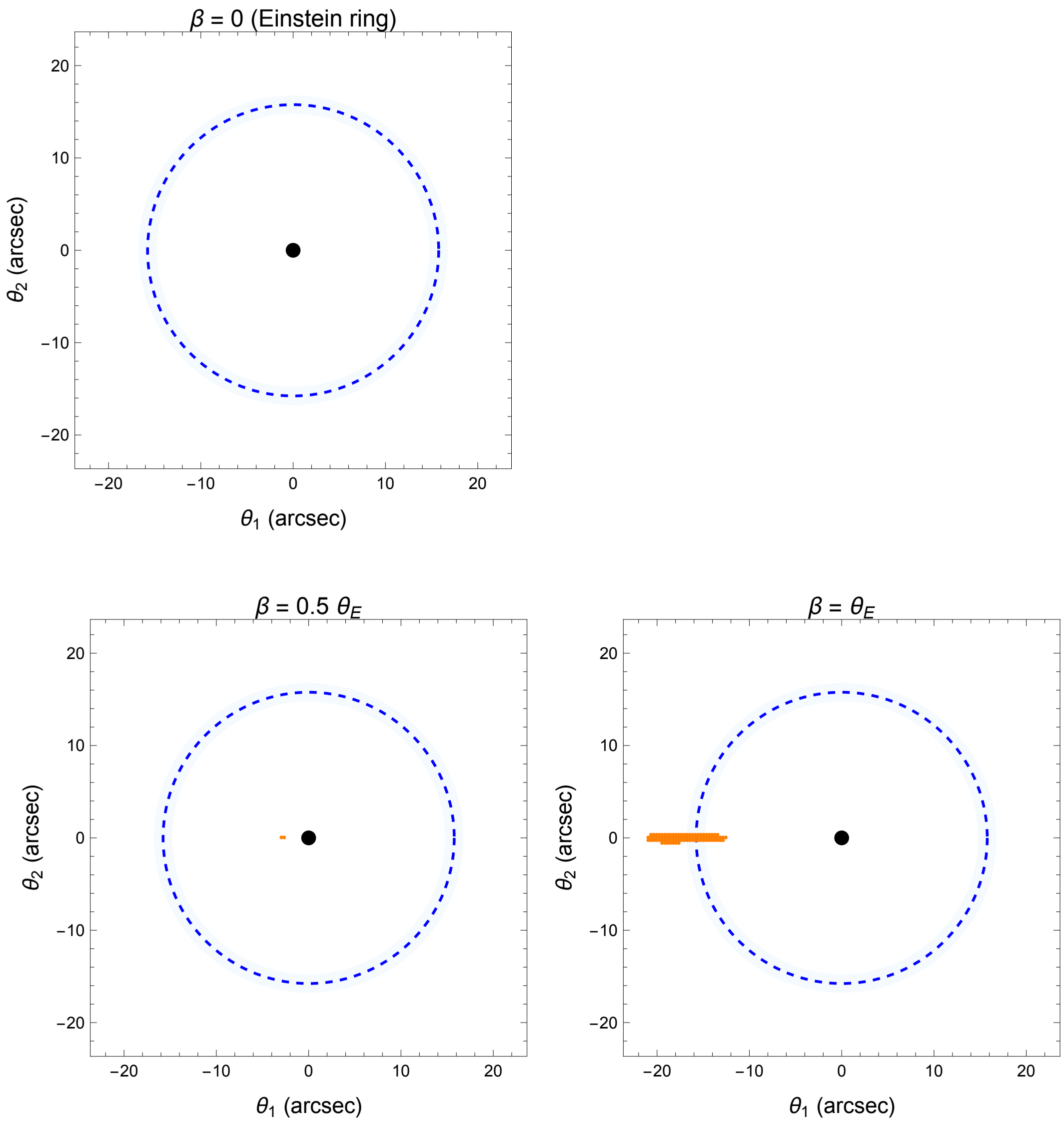


Figure 14.2: Giant arc formation by an NFW cluster ($M_{200} = 10^{15}\,M_\odot$, $c = 5$, $z_d = 0.3$, $z_s = 2$). A circular source of radius $0.5''$ is placed at three positions: $\beta = 0$ (Einstein ring, left), $\beta = 0.5\,\theta_\mathrm{E}$ (giant arc, center), and $\beta = \theta_\mathrm{E}$ (near the critical curve, right). The critical curve is shown as a dashed circle. Generated by `cluster_lensing.wl`.

14.11 Solutions to Exercises

Full worked solutions are also available in the Mathematica script `problems_10.wl`, which verifies each answer symbolically and numerically.

Exercise 14.10.1: NFW Cluster Einstein Radius and Enclosed Mass

(a) First, compute the virial radius from M_{200}:

$$M_{200} = \frac{4\pi}{3}\,200\,\rho_{\rm crit}(z_d)\,r_{200}^3 \quad \Rightarrow \quad r_{200} = \left(\frac{3\,M_{200}}{800\pi\,\rho_{\rm crit}}\right)^{1/3}.$$

Using $\rho_{\rm crit}(z = 0.3) \approx 5.3 \times 10^{-27}$ kg m^{-3} (including the $E^2(z)$ factor):

$$\begin{aligned} r_{200} &\approx 1.53 \text{ Mpc}\,, \\ r_s &= r_{200}/c = 1.53/5 \approx 306 \text{ kpc}\,, \\ \rho_s &= \frac{200}{3}\,\rho_{\rm crit}\,\frac{c^3}{\ln(1+c) - c/(1+c)} \approx 6.2 \times 10^{-25} \text{ kg m}^{-3}\,. \end{aligned}$$

The angular scale radius and characteristic convergence:

$$\begin{aligned} \theta_s &= r_s/D_{\rm d} = 306 \text{ kpc}/900 \text{ Mpc} \approx 3.4 \times 10^{-4} \text{ rad} \approx 70''\,, \\ \kappa_s &= 2\rho_s\,r_s/\Sigma_{\rm cr} \approx 0.35\,. \end{aligned}$$

(b) The tangential critical curve radius satisfies $\bar\kappa(x_t) = 1$, i.e., $(4\kappa_s/x_t^2)[\ln(x_t/2) + g(x_t)] = 1$. Solving numerically gives $x_t \approx 0.38$, so:

$$\theta_{\rm E} = x_t\,\theta_s \approx 0.38 \times 70'' \approx 27''\,.$$

(c) The enclosed projected mass:

$$M(\theta_{\rm E}) = \pi\,(D_{\rm d}\,\theta_{\rm E})^2\,\Sigma_{\rm cr} \approx \pi\,(900 \times 1.3 \times 10^{-4})^2\,\text{Mpc}^2 \times \Sigma_{\rm cr} \approx 1.4 \times 10^{14}\,M_\odot\,.$$

(d) The fraction of the total mass within $\theta_{\rm E}$:

$$\frac{M(\theta_{\rm E})}{M_{200}} = \frac{1.4 \times 10^{14}}{5 \times 10^{14}} \approx 0.28\,.$$

About 28% of the virial mass is enclosed within the Einstein radius. This is expected: the Einstein radius corresponds to $r \approx 120$ kpc, which is only $\sim$8% of r_{200}, but the NFW profile is centrally concentrated. ✓

Exercise 14.10.2: Ray-Tracing Through a Cluster

(a)–(c) This exercise is computational and best done in Mathematica. The inverse ray-tracing procedure maps each image-plane pixel through $\boldsymbol{\beta} = \boldsymbol{\theta} - \boldsymbol{\alpha}(\boldsymbol{\theta})$, using the NFW deflection angle. Pixels whose mapped source positions fall within the circular source disk are colored.

For a source at $\beta = 0$: a complete **Einstein ring** forms at $\theta \approx \theta_\mathrm{E}$. For $\beta = 0.5\,\theta_\mathrm{E}$: two arcs form, one bright (on the same side as the source) and one fainter (the counter-image). For $\beta = \theta_\mathrm{E}$ (at the tangential caustic): a highly stretched **giant arc** forms near the critical curve, with extreme tangential magnification.

See `problems_10.wl` for the complete ray-tracing code and generated plots.

(d) For the source at $\beta = \theta_\mathrm{E}$, the arc length-to-width ratio depends on the source size relative to the caustic curvature. For a $0.5''$ source near the critical curve of $\theta_\mathrm{E} \approx 27''$, the tangential magnification is large ($\mu_t \sim 5$–10), giving $L/W \sim 5$–10. ✓

Exercise 14.10.3: Magnification of a High-Redshift Galaxy

(a) For an SIS with $\theta_\mathrm{E} = 30''$, the magnification of a point source at position β (for $\beta < \theta_\mathrm{E}$) is:

$$\mu_\mathrm{tot} = |\mu_+| + |\mu_-| = \frac{2\theta_\mathrm{E}}{\beta}\,.$$

- $\beta = 3''$: $\mu = 2 \times 30/3 = 20$.
- $\beta = 1''$: $\mu = 2 \times 30/1 = 60$.
- $\beta = 0.1''$: $\mu = 2 \times 30/0.1 = 600$.

(b) The magnitude boost for $\mu = 600$:

$$\Delta m = -2.5\,\log_{10}(600) \approx -6.9\ \mathrm{mag}\,.$$

(c) Detection requires $m - \Delta m \le 29$, i.e., $31 + \Delta m \le 29$, so $\Delta m \le -2$, requiring $\mu \ge 10^{0.8} \approx 6.3$. From the SIS formula:

$$\beta \le \frac{2\theta_\mathrm{E}}{\mu_\mathrm{min}} = \frac{60}{6.3} \approx 9.5''\,.$$

Sources within $\sim 9.5''$ of the cluster center are sufficiently magnified.

(d) Cluster lensing enables the detection of galaxies at $z \sim 9$ that are ~ 2 magnitudes below the JWST detection limit. The price is a reduced survey area (by factor μ), but the steep faint-end slope of the high-z luminosity function means the net effect is a significant increase in detected galaxies. ✓

Exercise 14.10.4: Einstein Radii Across Mass Scales

(a)–(c) Using $\theta_\mathrm{E} = 4\pi(\sigma_v/c)^2\,D_\mathrm{ds}/D_\mathrm{s}$ with $D_\mathrm{ds}/D_\mathrm{s} = 1400/1750 = 0.8$:

$$\begin{aligned}
\text{Galaxy: } \sigma_v = 250\ \mathrm{km/s}, &\quad \theta_\mathrm{E} = 4\pi(250/3\times10^5)^2 \times 0.8 \approx 2.2\times10^{-5}\ \mathrm{rad} \approx 1.6''\,.\\
\text{Group: } \sigma_v = 500\ \mathrm{km/s}, &\quad \theta_\mathrm{E} = 4\pi(500/3\times10^5)^2 \times 0.8 \approx 8.7\times10^{-5}\ \mathrm{rad} \approx 6.2''\,.\\
\text{Cluster: } \sigma_v = 1200\ \mathrm{km/s}, &\quad \theta_\mathrm{E} = 4\pi(1200/3\times10^5)^2 \times 0.8 \approx 5.0\times10^{-4}\ \mathrm{rad} \approx 36''\,.
\end{aligned}$$

(d) The ratio $\theta_\mathrm{E}^\mathrm{cluster}/\theta_\mathrm{E}^\mathrm{galaxy} = (1200/250)^2 = 23$. The Einstein radius increases by a factor of $\sim$23 from galaxy to cluster.

(e) The enclosed mass within $\theta_{\rm E}$: $M(\theta_{\rm E}) = \pi(D_{\rm d}\,\theta_{\rm E})^2\,\Sigma_{\rm cr}$. Since $\theta_{\rm E} \propto \sigma_v^2$, we have $M(\theta_{\rm E}) \propto \theta_{\rm E}^2 \propto \sigma_v^4$. Computing:

$$M(\theta_{\rm E})^{\rm galaxy} \approx 3 \times 10^{11}\,M_\odot\,,$$
$$M(\theta_{\rm E})^{\rm group} \approx 5 \times 10^{12}\,M_\odot\,,$$
$$M(\theta_{\rm E})^{\rm cluster} \approx 2 \times 10^{14}\,M_\odot\,.$$

(f) Since $\theta_{\rm E} \propto \sigma_v^2$, the angular scale of lensing grows *quadratically* with velocity dispersion. This means cluster lensing is not just a scaled-up version of galaxy lensing — the arcs subtend tens of arcseconds (easily resolvable with ground-based telescopes), producing qualitatively different observational signatures. ✓

Exercise 14.10.5: Weak Lensing Mass Estimate

(a) With the approximation $\gamma_t \approx \bar{\kappa}$:

$$\bar{\kappa}(1') \approx 0.1 \times (1)^{-0.8} = 0.1\,,$$
$$\bar{\kappa}(3') \approx 0.1 \times (3)^{-0.8} = 0.1/2.41 \approx 0.041\,,$$
$$\bar{\kappa}(5') \approx 0.1 \times (5)^{-0.8} = 0.1/3.62 \approx 0.028\,.$$

(b) The enclosed mass within angle θ:

$$M(\theta) = \pi\,(D_{\rm d}\,\theta)^2\,\Sigma_{\rm cr}\,\bar{\kappa}(\theta)\,.$$

Converting angles to radians: $1' = 2.91 \times 10^{-4}$ rad, $3' = 8.73 \times 10^{-4}$ rad, $5' = 1.45 \times 10^{-3}$ rad.

(c) At $\theta = 5'$:

$$M(5') = \pi \times (900 \times 1.45 \times 10^{-3})^2 \times 3.5 \times 10^{15} \times 0.028 \approx 5.2 \times 10^{14}\,M_\odot\,.$$

(d) This is consistent with a typical cluster virial mass of $5 \times 10^{14}\,M_\odot$. At $\theta = 5'$ ($r \approx 1.3$ Mpc), we are probing close to r_{200}, so we expect to recover $\sim M_{200}$. The main assumptions affecting accuracy are: (i) the approximation $\gamma_t \approx \bar{\kappa}$ (valid at large radii where $\kappa \ll \bar{\kappa}$); (ii) the power-law shear profile (which may not hold at all radii); (iii) the assumed $\Sigma_{\rm cr}$ (which depends on the source redshift distribution). ✓